Participatory It from Bit: Selective Communication, Inference, and Effective Physical Structure

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Abstract

Wheeler's participatory it-from-bit program motivates a theory of how local records, communication and learned descriptions can support a shared physical world. This monograph develops a classical probabilistic framework in which agents are local sections of an associated statistical bundle over a continuous, possibly curved base. At every point of overlap, a network of selective matrix-valued channels communicates only part of each agent's information. A normalized population model yields an exact variational free energy; a separate channel-consensus objective describes agreement when no such generative identification has been established. One joint statistical fiber accommodates both current-state beliefs and uncertain models. Independent invertible interactions are represented by latent Lie-algebra generators with a normalized Gaussian-based loop prior. Their recognition flow is first-order Fisher descent; a separately specified episode-history prior gives Euclidean history inertia and a controlled Proca-type linearized link equation. Prediction and conditional expectation derive cross-scale priors and messages once a process law and retained variables are specified. Exact coarse-graining produces effective interactions and often memory, while a conditional four-agent selection theorem illustrates what formation dynamics can establish. Worked constructions distinguish Fisher stiffness, dissipative response and metastable persistence from mechanical mass. Information clocks and collective distinguishability are developed with explicit covariance and gauge conventions. A complex two-dimensional message sector supplies the full proper orthochronous Lorentz group through Hermitian determinant geometry; identifying this internal cone with physical spacetime requires a further soldering structure and dynamics. The resulting program minimizes independent constructions while retaining the assumptions that normalization, inference and coarse-graining cannot remove. It establishes conditional mathematical mechanisms, not a derivation of quantum theory, gravity or matter from information alone.

Keywords: Participatory realism · It from bit · Gauge theory · Information geometry · Variational free energy · Multi-agent inference

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Part I: The Construction

1 Introduction

Wheeler's participatory program asks how registered distinctions acquire the significance of physical facts and how many situated observers come to describe one world [1]. A theory addressing that question needs more than a metaphor between information and matter. It needs laws for what an observer can register, how different descriptions can be compared, how collective descriptions form and what they predict. This monograph develops a probabilistic and geometric framework for those tasks, then examines what further work is required for physical geometry, persistence and time.

The starting point is local. An agent carries a probability law over a domain of an underlying continuous base. Where domains overlap, agents interact through selective links. Different points of the overlap carry different values of the same section fields and of the associated attention network. The base is not a graph of agents and is not presumed flat. Its ultimate physical interpretation remains open: the theory describes agents' records and models of underlying reality rather than presupposing that their descriptions exhaust it.

Two distinctions organize the construction. First, a coordinate identification between cognitive frames is not a communication channel. Sending may expose only a quotient of the sender's informational space, while receiving places that signal into a different subspace. Matrix-valued links, including rank-deficient links, represent this selective interaction. Invertible flat frame identifications are a special case. Second, the variational free energy of a normalized joint law is distinct from a composite objective assembled from pairwise divergences. Both can be studied, but they imply different probabilistic and dynamical claims.

We take the normalized joint as the foundation. Its collective evidence bound accommodates correlated beliefs, source-selection attention, model learning and observations that can be shared without being counted twice. Belief and model descriptions can live within one associated statistical bundle: a law over state and model coordinates admits a conditional state belief and a model marginal. This removes unnecessary duplication of primitive bundles. It does not explain time-scale separation by definition. We therefore examine actual mechanisms for accumulated memory, projected dynamics and slowly changing predictions.

Independent relations can themselves be inferred. At one base point, an invertible interaction is written as a frame identification with an exponential twist whose Lie-algebra generator is latent. A proper prior on the generators and normalized record kernels put their recognition law inside the same collective evidence bound as the agents. Section 6 states the resulting prediction-error current, Fisher recognition flow, and a separate law over link histories across learning episodes. Its near-zero, weak-current history equation has the positive mass, curl-curl and episode-difference structure of a Euclidean Proca operator. The episode index, increment precision and agent-graph operator still require a physical identification.

At larger scales, a normalized coarse channel pushes fine laws to parent laws. Conditional expectation supplies the upward and reverse operators; integration generates new interactions and normalizing potentials; prediction supplies later priors. These constructions give precise versions of cross-scale influence without postulating an additional shadow field. Their remaining input is the retained variable or channel. Dynamical formation must explain why a particular retention becomes coherent, persistent and predictive, and why it can support the next formation episode.

Section 14 states a concrete selection result, proved in Appendix J: at sufficiently small fixed temperature, slow source-prior learning on a homogeneous four-agent cycle selects one matching from almost every Gaussian initial mean, whereas a frozen-prior control has a symmetric long-time pairing

law. The supplied initial mean provides the asymmetry. The result is a step toward endogenous organization, not a proof of general recursive hierarchy. Here its assumptions are stated alongside the exact parent laws and the remaining memory and closure obligations.

The physical part separates positive statistical geometry from Lorentz geometry. Fisher length measures distinguishability along belief histories, and Fisher pullbacks measure sensitivity to changes over the base. They are positive semidefinite. A different exact construction uses Hermitian two-by-two tensors: determinant polarization gives Lorentz signature, and determinant-preserving complex frame changes cover the proper orthochronous Lorentz group. Selective complex messages can generate these tensors classically. The mathematical representation is exact; selecting the required channel, a nondegenerate soldering to the base, physical scale and causal dynamics remains a physical research problem.

Persistence also admits a constructive treatment. Eliminating hidden first-order modes produces memory and colored fluctuations. It does not generally create Newtonian inertia. Gaussian calculations separate stiffness, friction, precision and covariance response, while a metastable example shows how a long residence time can arise without a kinetic-mass postulate. These calculations replace an identification of a rock with a high-precision belief by explicit response and lifetime questions.

Participatory question	Mathematical construction	Remaining identification
What is registered?	Normalized record channels; expected information gain.	A physical registration mechanism and its irreversibility.
How do perspectives meet?	Selective covariant links and common observables.	Empirical meaning and calibration of the channels.
How do collectives form?	Learned selection, exact retained laws and memory.	Recursively inherited formation under general interfaces.
How do scales act back?	Conditional inference, learned parameters and prediction.	A dynamical choice of retained variables and boundary laws.
How does physical geometry arise?	Fisher geometry; conditional determinant-cone Lorentz kinematics.	Soldering, scale, causal characteristics and field equations.

Table 1: Wheeler's motivation and the corresponding research obligations. The third column is not implied by the second.

Interpretive stance. The motivating stance is participatory and relational: physical descriptions are made through interactions, and collective descriptions should be accountable to the records of their constituents. This stance does not require treating a prior as a belief about nothing, denying an underlying reality, or assigning consciousness to every mathematical section. Calling a boundary mechanism an agent changes no probability law unless a criterion of agency changes its predictions. The monograph therefore asks what agency can explain rather than using universal agency as an explanatory premise.

The philosophical ambition is retained: stable things might be understood through persistent patterns of constrained participation. The mathematical work establishes conditional mechanisms and identifies the quantities they preserve. Whether those mechanisms describe matter is a separate question to be answered through dynamics and observation.

Reading route. Part I develops the continuous-base setting, the collective variational principle, learning and memory, and exact coarse-graining at one base point. Part II examines information clocks, induced geometry, the Lorentz construction and physical interpretation. The appendices

retain full Gaussian calculations and normalized hierarchical examples. They are specializations whose assumptions are stated explicitly, not an implementation specification.

2 Assumptions, Results and Open Obligations

The framework has a small set of primitive choices: a continuous base and local domains; a gauge action on a statistical sample space; normalized priors, record kernels and interaction channels; a recognition family; and, when dynamics are discussed, a metric and mobility or a stochastic generator. Coarse-graining additionally requires a retained channel. An episode-history law for latent links is an additional normalized model with a declared increment penalty. Its Euclidean stationarity does not supply a real-time kinetic action, Lorentz soldering, thermodynamic interpretation or physical ontology.

Most mathematical results are conditional identities or constructions. An evidence bound follows from a normalized joint, source attention follows from an explicit label model, and coarse actions follow from integration. None of these identities selects the joint, the source alphabet or the retention by itself. The distinction lets the theory use exact results without claiming that the variational principle derives every modeling ingredient.

Claim	Established scope	Remaining obligation
Selective communication	Endpoint covariance, rank bounds, normalized noisy channels.	Learned interface identifiability and predictive meaning.
Collective VFE and attention	Exact posterior identity and conditional or restricted source rows.	A generative model for a target system.
Latent interaction links	Normalized generator prior, Fisher recognition flow and controlled Euclidean Proca-type history equation.	Coarse link-law closure, rate separation and physical identification.
One statistical bundle	Conditional state–model decomposition and exact VFE accounting.	Emergent model coordinates and their slow kinetics.
Persistent parents	Exact pushforwards, descent conditions and memory equations.	Finite predictive closure for the actual evolving network.
Formation	Fixed-temperature four-cycle pairing selection in a declared model (proof in Appendix J).	General selective-interface and recursive formation.
Information clocks	Fisher length, dissipation identity and fixed-channel contraction.	Physical time calibration and causal order.
Lorentz structure	Hermitian determinant form and the $SL(2,\mathbb{C})$ double cover.	Dynamic channel selection, soldering and physical propagation.
Mass and persistence	Response, stiffness, memory and metastable lifetimes in stated models.	A derived inertial mass law and physical units.

Table 2: The distinction between proved conditional mathematics and the research program.

The technical appendices sometimes use the short status codes P for a proved conditional claim, D for an explicit derivation, S for a structural correspondence, C for a conjecture or open obligation, and E for empirical evidence. These codes classify statements rather than certify a whole section. No empirical realization of the physical program is reported here.

2.1 Scope and limitations

Formation and inheritance are analyzed at one fixed base point. The smooth base remains the ambient setting; conditional base geometry appears only when a derivative or soldering is explicitly supplied. Gaussian families, fixed links, compact gauge reductions, prescribed partitions and imposed rate separation are named special cases. They are not assumptions of every result.

A normalizer matters whenever parameters are varied. A posterior from one inference problem cannot be silently reused as a likelihood in another. A small marginal divergence need not control derivatives, force errors or path laws. A successful first parent does not establish the next formation episode. These are the principal sources of overextension the revised construction avoids.

2.2 Location of the longer proofs

Results are stated with their hypotheses where they are used, and the longer proofs are collected in the appendices. The fixed-temperature pairing-selection theorem is proved in Appendix J. The latent-link propositions, the measure-theoretic conditions and the coarse link laws are proved in Appendix K, and the remaining supporting results are proved in Appendix L. The text does not treat its prototypes as the only possible agents, nor an unresolved prototype as a general impossibility theorem.

3 Relation to Existing Programs

Wheeler's participatory program asks how registered distinctions acquire physical meaning and how many participants can belong to one world [1, 2]. The present construction treats this as a question about records, conditional laws and the compatibility of local descriptions. It begins with classical probability. It does not infer a Hilbert space, the Born rule or quantum complementarity from Bayesian normalization. Relational and personalist interpretations of quantum theory provide philosophical context [3, 4], but their physical postulates are not supplied by the present classical model.

Variational inference supplies the central identity between free energy, evidence and posterior KL. Active inference additionally specifies a generative process, policy variables and preferences or decision criteria [5–7]. Its distinction between inference about states, learning parameters and selecting models is directly relevant to the question of fast beliefs and slow models. These can be coordinates or marginals of one joint law. Their differing time scales still require dynamics, information accumulation or a controlled elimination, as Section 8 shows.

The Friston program has also pursued structure learning and multiscale generative models. Supervised structure learning learns aspects of a model within a specified representational scheme [8]. Renormalizing generative models construct hierarchical descriptions by spatial and temporal grouping, compression and transition learning [9]. That construction supplies grouping and time blocking; its tractable latent hierarchy assumes that children are not shared by several parents. This concerns the generative graph, not a general prohibition on overlapping spatial input patches. It is a precedent for learned effective models, not a proof that every smooth overlap network develops an autonomous hierarchy without further assumptions.

Gauge formulations of neuronal free energy relate local representation changes, prediction errors and precision [10]. Here the geometric distinction is explicit: a coordinate transition is invertible, whereas a message need not be. The interaction $L_{ij} = R_i S_j$ can erase directions and is represented probabilistically by a normalized channel. Flat reference maps Ω_{ij} remain a useful specialization. Independent invertible interaction links and general matrix links carry different notions of loop

structure. This distinction connects the framework to geometric consensus while keeping statistical laws, rather than only vectors, as the objects being compared [11, 12].

Markov blankets enter only when a specified probability law has the relevant conditional independence. Geometric overlap alone supplies no such independence. Bayesian-mechanics constructions and critiques clarify why a graphical or dynamical separation must not be substituted for a statement about the joint law [13–15]. In this monograph, a local agent is defined geometrically and statistically; a blanket is an additional property to prove in a particular generative model.

Statistical distance and thermodynamic length motivate the use of Fisher length to quantify distinguishable change [16–18]. The resulting information clock depends on the chosen record and state description. It is not automatically physical proper time. Likewise a Fisher pullback gives a positive-semidefinite distinguishability tensor; its positivity prevents it from being a Lorentz metric without additional structure. Section 20 therefore uses the standard determinant geometry of two-dimensional complex Hermitian forms, familiar from polarization and spinor descriptions [19, 20]. The additional step identifying that internal geometry with spacetime is stated separately.

Finally, exact marginalization and projected dynamics are established tools of statistical physics. They generate effective interactions and, in dynamics, memory kernels [21, 22]. They do not generally preserve the form of the fine model. Here these issues arise for normalized multi-agent laws, selective interfaces and retained states. This monograph develops their physical interpretation and finite worked constructions, including where closure, recursive formation and a physical identification remain to be established.

4 Agents over a Continuous Base and Selective Communication

An agent is a local section, and an interaction relates section values on an overlap. The network seen at one base point is a slice of this construction. The base is not the set of agents, a communication link is not a transition function of the bundle, and a change of coordinates is not an exchange of information.

4.1 The base and the networks above its points

Definition 1 (Base and agent domains)**.** *Let $\mathcal{C}$ be a smooth manifold and $\mathcal{U}_i \subseteq \mathcal{C}$ the open domain of agent i. At c, the available agents form $V(c) = \{i : c \in \mathcal{U}_i\}$. An interaction $j \to i$ is defined on $\mathcal{U}_i \cap \mathcal{U}_j$.*

The base is the domain over which the fields are organized. It need not be directly observable, flat, or identical to physical spacetime. Its topology, dimension and differentiable structure are inputs to this description. It may support curved geometry, but no particular metric is assumed in the pointwise inference theory. Agents have records and models of underlying reality rather than direct access to its observer-independent coordinates. This epistemic restriction does not assert that the underlying reality is absent.

On an overlap containing $(c - \delta c, c + \delta c)$, both $c - \delta c/2$ and $c + \delta c/2$ carry an attention network with the same available agents if no domain boundary is crossed. Beliefs, model descriptions, frames, links and weights generally differ at these two points and vary smoothly within the declared regularity class. Fixing $c = c_*$ for formation analysis retains this ambient construction; it does not collapse the domains or discretize the base. Variation with c is distinct from evolution with inference parameter t.

A density $d\nu(c)$ must be specified before integrating an energy over $\mathcal{C}$. Smoothness alone supplies neither a canonical integration measure nor a physical metric. A base-integrated functional is an

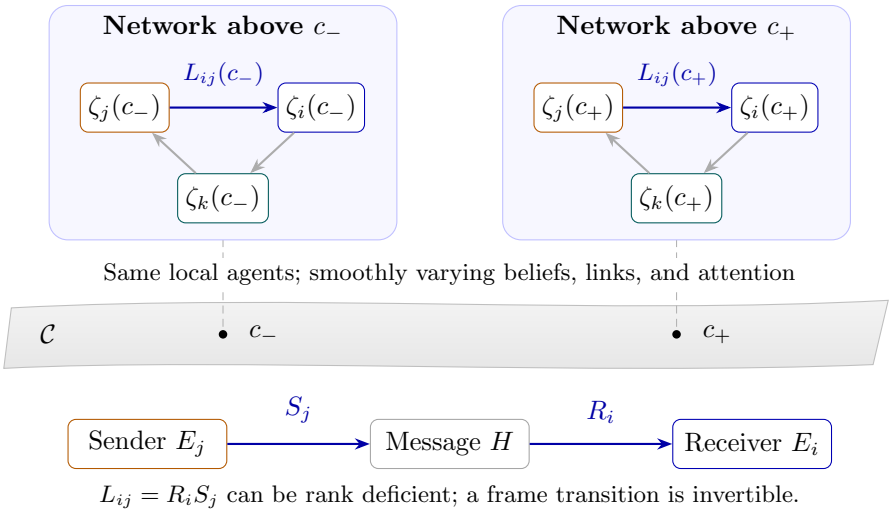


Figure 1: **Local sections and selective communication.** Two points of the same overlap carry networks with the same agents and generally different section values. The drawing is a local schematic, not a discretization of the base or a connection between its points. At each point a sender exposes information through S_j and the receiver reconstructs it through R_i. Their composition is an interaction, not a bundle transition.

optional field model; without base-derivative terms its equations separate pointwise. Coupling different base points requires further transport and regularity assumptions.

4.2 One statistical bundle

Let $P \to \mathcal{C}$ be a principal G-bundle. Choose a standard Borel sample space Y with a measurable G-action and a regular family $\mathcal{B} \subseteq \mathcal{P}(\mathsf{Y})$ stable under pushforward, with a smooth induced action when differential bundle statements are used. The associated statistical bundle is

$$\mathcal{S} = P \times_G \mathcal{B}, \qquad \zeta_i \in \Gamma(\mathcal{U}_i, \mathcal{S}). \tag{1}$$

Smoothness refers to a declared smooth statistical family or a specified differentiable structure on laws. It is not automatic for arbitrary probability measures. General measurable laws suffice for the evidence and coarse-graining identities; the differential calculations use regular finite-dimensional families.

When model parameters are retained, take $\mathsf{Y} = \mathsf{K} \times \mathsf{M}$ with standard Borel factors, and a G-action respecting the projections. A joint section has disintegration

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm), \qquad q_i(dk) = \int q_i(dk \mid m) s_i(dm). \tag{2}$$

Here q_i is a state belief, m a model presentation and s_i the belief about it. A model presentation becomes predictive through an evaluator $m \mapsto K_{i,m}$ assigning a normalized kernel. The state prior is p_i and the model-parameter prior is r_i; neither is synonymous with a generative model. The full generative probability law is $\mathbb{P}$.

This gives one primitive statistical bundle. Its marginal bundles are induced descriptions, not independently postulated agents. The product restriction $\zeta_i = q_i \otimes s_i$ discards state–model dependence. Calling one marginal slow is a dynamical assertion, examined in Section 8; this

repackaging alone does not make models emerge. One may instead begin with an unsplit Y and seek predictive coarse variables within its dynamics.

Definition 2 (Agent and prior)**.** *An agent is a local statistical section ζ_i, its access to records, a frame presentation, and normalized mechanisms for interaction and updating. Its prior is a law available before the current record is assimilated: an initial boundary law, a prediction from earlier records, or a conditional law in a specified joint hierarchy.*

A support domain is not a Markov blanket. A blanket is a conditional independence in a joint probability law; overlap explains where interaction is possible but does not prove probabilistic screening. Smooth statistical sections alone do not establish biological agency, consciousness or physical objecthood.

4.3 Frames and reference identification

On a common trivializing neighborhood let U_i present the reference sample fiber in agent i's coordinates. In a vector realization a passive coordinate change $x_i' = g_i x_i$ gives

$$\mu_i' = g_i \mu_i, \qquad \Sigma_i' = g_i \Sigma_i g_i^\top, \qquad U_i' = g_i U_i. \tag{3}$$

The representations on K and M can differ while sharing one principal frame. The parametrization $U_i = \exp \phi_i$ is local and need not cover an entire general linear group.

The invertible reference identification

$$\Omega_{ij} = U_i U_j^{-1} : E_j \longrightarrow E_i \tag{4}$$

expresses the same reference vector in different coordinates and obeys $\Omega_{ij}' = g_i \Omega_{ij} g_j^{-1}$. It does not specify which sender directions are communicated.

Lemma 1 (Flat identification)**.** *Where the frames share a reference trivialization,*

$$\Omega_{ij}\Omega_{jk} = \Omega_{ik}, \qquad \Omega_{i_0 i_1} \cdots \Omega_{i_{r-1} i_0} = I. \tag{5}$$

Proof. Intermediate frames cancel in the products. □

Bundle transition functions obey cocycle identities because they relate coordinate charts. Independent interactions need not obey them.

Proposition 1 (Coordinate invariance of relative entropy)**.** *For a bimeasurable bijection g,*

$$D_{\mathrm{KL}}(g_{\#}q \| g_{\#}p) = D_{\mathrm{KL}}(q \| p) \tag{6}$$

in $[0, \infty]$.

Proof. Absolute continuity is preserved in both directions and $d(g_{\#}q)/d(g_{\#}p) = (dq/dp) \circ g^{-1}$ almost everywhere. Integration proves the identity; failure of absolute continuity gives infinity on both sides. □

The same pushforward preserves the Fisher metric of a regular family,

$$G_{ab} = \mathbb{E}_\zeta[(\partial_a \log \zeta)(\partial_b \log \zeta)]. \tag{7}$$

Coordinate covariance makes comparisons meaningful but is not, by itself, a dynamical restriction.

Proposition 2 (Removability of reference frames)**.** *For* $\bar{q}_i = (U_i^{-1})_{\#} q_i$,

$$D_{\mathrm{KL}}(q_i \| (\Omega_{ij})_{\#} q_j) = D_{\mathrm{KL}}(\bar{q}_i \| \bar{q}_j). \tag{8}$$

Proof. Use coordinate invariance with $g = U_i^{-1}$. □

This removes only the reference frames. A general link becomes $U_i^{-1} L_{ij} U_j$ and can retain rank loss, learned content and nontrivial composition. Gauge covariance must not be used to infer that these interactions are removable.

4.4 Selective sending and receiving

The order ij means sender j to receiver i: a message from i to j has index ji. Choose a message space H and maps

$$S_j : E_j \longrightarrow H, \qquad R_i : H \longrightarrow E_i, \qquad L_{ij} = R_i S_j. \tag{9}$$

Maps and message spaces may more generally depend on the edge. An intermediate transmission map may be inserted. For a message-coordinate change h,

$$S'_j = h S_j g_j^{-1}, \qquad R'_i = g_i R_i h^{-1}, \qquad L'_{ij} = g_i L_{ij} g_j^{-1}. \tag{10}$$

Multiplication proves covariance and gives

$$\ker S_j \subseteq \ker L_{ij}, \quad \operatorname{im} L_{ij} \subseteq \operatorname{im} R_i, \quad \operatorname{rank} L_{ij} \leq \min(\operatorname{rank} S_j, \operatorname{rank} R_i). \tag{11}$$

Neither map must be orthogonal or idempotent. A low-rank link can read every input coordinate and write into every output coordinate. Rank limits transmitted linear variation, not coordinate support.

For a statement such as *cats are cute*, speech and gesture expose selected features of a sender's state, while hearing and remembered encounters shape their placement in the receiver's state. The construction formalizes selective access and reconstruction. Rank and covariance do not establish shared meaning; that requires specified records and predictions.

The attention realization $S_j = T U_j^{-1}$, $R_i = U_i O$ has $L_{ij} = U_i O T U_j^{-1}$. Even with flat reference frames,

$$L_{ij} L_{jk} - L_{ik} = R_i (S_j R_j - I_H) S_k. \tag{12}$$

Composition is defect-free if $S_j R_j = I_H$. With injective R_i and surjective S_k the converse follows. A sum of routed heads can have greater rank than each head. Loop products transform by conjugation but may be singular. They are communication loop products; group holonomy is reserved for invertible connection-type links.

4.5 Normalized communication channels

A linear map alone specifies no noise law or likelihood. The probabilistic object is a normalized kernel $\mathcal{K}_{ij} : E_j \rightsquigarrow E_i$. Sending, transmission and receiving kernels compose to a normalized kernel [23], with

$$\mathcal{K}'_{ij}(g_j x, A) = \mathcal{K}_{ij}(x, g_i^{-1} A), \qquad q'_j \mathcal{K}'_{ij} = (g_i)_{\#}(q_j \mathcal{K}_{ij}). \tag{13}$$

For a Gaussian source $q_j = \mathcal{N}(\mu_j, \Sigma_j)$ and a linear Gaussian realization,

$$\mathcal{K}_{ij}(x, \cdot) = \mathcal{N}(L_{ij} x, N_{ij}), \quad N_{ij} \succ 0, \qquad q_j \mathcal{K}_{ij} = \mathcal{N}(L_{ij} \mu_j, L_{ij} \Sigma_j L_{ij}^{\top} + N_{ij}). \tag{14}$$

The output covariance is positive definite at every link rank; its transformation is $N'_{ij} = g_i N_{ij} g_i^\top$. With zero noise the kernel still exists but may output a singular measure, to which an ambient Gaussian density formula cannot be applied.

Alternatively compare $A_{ij\#}q_i$ and $B_{ij\#}q_j$ in a common measured space. Full-row-rank maps give positive-definite reduced covariances. Comparison interfaces determine attention and content maps determine the message; they coincide only under an additional modeling relation.

For nonsingular Gaussian laws,

$$D_{\mathrm{KL}}(\mathcal{N}(\mu,\Sigma)\|\mathcal{N}(\nu,V)) = \frac{1}{2}\left[\mathrm{tr}(V^{-1}\Sigma) + (\mu-\nu)^\top V^{-1}(\mu-\nu) - K + \log\frac{\det V}{\det\Sigma}\right]. \tag{15}$$

For a selective channel $V = L_{ij}\Sigma_j L_{ij}^\top + N_{ij}$; for a reduced comparison the formula lives in the measured space. Substituting a singular noise-free covariance would not generalize the formula.

4.6 Agreement in public variables and private information

For a deterministic interface T on standard Borel spaces, the chain rule gives

$$D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(T_\# q\|T_\# p) + \int D_{\mathrm{KL}}(q(\cdot \mid T=z)\|p(\cdot \mid T=z))(T_\# q)(dz). \tag{16}$$

The identity follows by disintegrating $(X, T(X))$, under the standard absolute-continuity and extended-real conventions. Perfect communicated agreement can coexist with different private conditional laws. It therefore establishes neither complete agreement nor a closed parent dynamics. Stochastic channels satisfy the corresponding data-processing inequality [24], which is proved for normalized kernels after Eq. (75).

4.7 Independent group links and base connections

Regime I names only the flat identification $\Omega_{ij} = U_i U_j^{-1}$. Regime II supplies independent invertible links Θ_{ij} obeying

$$\Theta'_{ij} = g_i \Theta_{ij} g_j^{-1}. \tag{17}$$

An unoriented edge representing reversible identification may impose $\Theta_{ji} = \Theta_{ij}^{-1}$. Selective communication imposes no such inverse relation. Network holonomy

$$H_i = \Theta_{ii_1}\cdots\Theta_{i_r i}, \qquad H'_i = g_i H_i g_i^{-1} \tag{18}$$

can be nontrivial at c_*. Agreement around a loop forces $H_{i\#}q_i = q_i$, not $H_i = I$ unless the law identifies the group element uniquely.

Latent exponential twists. For the dynamic-link model choose one orientation E_+ from each reversal pair and a reference frame at c_*. The common vector representation writes

$$\boxed{\Theta_e = U_i \exp(\mathring{\omega}_e) U_j^{-1}, \qquad \mathring{\omega}_e \in \mathfrak{g}, \qquad \mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e, \quad e: j \to i.} \tag{19}$$

For another representation the exponential is $\exp(\mathrm{d}\rho_x \mathring{\omega}_e)$; state and model channels may carry different generator families. The zero generator gives the flat identification. A passive change $U'_i = g_i U_i$ changes no generator and gives the endpoint law of Eq. (17). Changing the common reference to frames $U_i h$ instead sends all generators to $\mathrm{Ad}_{h^{-1}}\mathring{\omega}_e$. These two coordinate actions are distinct (Lemma 19).

An ordered loop product of exponential twists need not be the exponential of their sum. For $\mathrm{GL}(K,\mathbb{R})$, the exponential image lies in $\mathrm{GL}^+(K)$ but does not cover it when $K \geq 2$; products of chart links can also leave that image. Thus the chart is a declared family of independent group links, not a parametrization of every invertible interaction. Nor does it replace the possibly singular selective content map L_{ij}: an invertible latent transmission can be composed with the sending and receiving maps of Eq. (9). The interaction graph and the frames remain structural; its reversal pairs do not require reciprocal source-attention rows. Section 5.2 supplies the generator prior and Section 6 its recognition and history equations.

Between different base points, transport requires an additional connection. A frame gradient $A_\mu^{(i)} = U_i^{-1}\partial_\mu U_i$ instead obeys

$$F_{\mu\nu}^{(i)} = \partial_\mu A_\nu^{(i)} - \partial_\nu A_\mu^{(i)} + [A_\mu^{(i)}, A_\nu^{(i)}] = 0 \tag{20}$$

by Maurer–Cartan. This says nothing about arbitrary independent interaction curvature and supplies no physical conservation law. Formation below uses no base paths; conditional base-geometry calculations identify their derivative separately.

4.8 Working scope

Normalized laws and channels are the general setting. Real Gaussian examples use $\mathrm{GL}^+(K)$ with full covariance matrices; other representations are admitted when their types are supplied. The Gaussian technical appendices examine an invertible full-state consensus specialization. Their formulas do not extend to singular links by a change of symbols. The geometry fixes neither a physical spacetime nor a universal agent ontology.

5 One Collective Variational Free Energy

The probabilistic foundation is one normalized law for the population, not a sum of separately asserted evidence bounds. The same foundation accommodates selective communication, attention and subsequent coarse-graining. A composite consensus potential remains useful as a worked model, but we write it as $\mathcal{E}$ to distinguish it from the collective variational free energy $\mathcal{F}_V$. The distinction matters when priors, interfaces or other agents' beliefs move.

5.1 A normalized population model

At c_* fix a finite agent set V and let $y = (y_i)_{i\in V}$, with $y_i = (k_i, m_i)$ when model parameters are explicit. Let X collect structural parameters: available edges, interface maps, noise parameters and frame presentations. Choose a proper prior $P_0(dy \mid X)$, a source-label prior $\Pi(dJ \mid X)$ and normalized record kernels. For a finite collection of conditionally independent records with declared scopes,

$$\mathbb{P}(dy, dJ, do \mid X) = P_0(dy \mid X)\Pi(dJ \mid X)\prod_a K_a(do_a \mid y_{\partial a}, J, X). \tag{21}$$

The product is normalized by integrating the records and then the prior. A correlated record kernel can replace the product. Every record appears once; giving several agents access to one record does not create several independent likelihood factors. Loops in the interaction graph do not cause a normalization problem for record kernels conditional on the whole latent population.

At a record o, assume a common dominating reference and finite positive evidence $Z_o(X) = \int \ell_o(y, J; X)P_0(dy \mid X)\Pi(dJ \mid X)$. For a recognition law $Q(dy, dJ)$ with the required absolute

continuity and integrability,

$$\begin{aligned}\mathcal{F}_V[Q; X, o] &= D_{\mathrm{KL}}(Q\|P_0\Pi) - \mathbb{E}_Q \log \ell_o \\ &= D_{\mathrm{KL}}(Q\|\mathbb{P}(\cdot \mid o, X)) - \log Z_o(X).\end{aligned} \tag{22}$$

The identity follows by substituting the posterior density ℓ_o/Z_o. The evidence bound and its equality case are exact. Where an energy-minus-entropy expression has indeterminate infinite terms, the posterior-relative-entropy formulation is the definition. Precisely, the second line, with $D_{\mathrm{KL}} = +\infty$ off absolute continuity, defines $\mathcal{F}_V[Q; X, o] \in [-\log Z_o, +\infty]$ for every recognition law, and it equals the first line in $(-\infty, +\infty]$ whenever $D_{\mathrm{KL}}(Q\|P_0\Pi) < \infty$ and the positive part of $\log \ell_o$ is Q-integrable. When Q and the prior have densities against a common σ-finite reference, the split into expected log densities holds once the logarithms of the recognition density and of the joint density are Q-integrable. Appendix K.1 proves these statements.

The local sections ζ_i are marginals of Q, with dependence on the records restricted by their access maps if required. A product $Q_Y = \prod_i \zeta_i$ is an optional mean-field family. Marginals alone do not determine Q_Y, and information discarded by a product restriction is not zero merely because the marginal notation omits it.

The generative law reads latent coordinates and declared parameters, not the live recognition law Q. A model evaluator may read a latent model presentation m_i. Replacing a generative factor by another agent's current posterior changes the model during inference and requires a separate predictive, conditional or learning interpretation.

5.2 Latent links share the evidence bound

The held-link model extends by treating the generators of Eq. (19) as latent coordinates. For one channel put $\mathcal{T} = \mathfrak{g}^{E_+}$ and choose an inner product, Gaussian anchors $\mathring{\alpha}_e$ and positive-definite prior covariances $\mathring{\Sigma}_e$ on its summands. Define

$$\begin{aligned}\nu^{\mathcal{T}} &= \bigotimes_{e \in E_+} \mathcal{N}(\mathring{\alpha}_e, \mathring{\Sigma}_e), \\ \mathcal{A}_{\mathrm{loop}}(\mathring{\omega}) &= \sum_{p \in \mathbb{L}} c_p \mathfrak{Q}_p(\mathring{\omega}), \qquad c_p \geq 0, \\ P_{\mathrm{gen}}(d\mathring{\omega}) &= Z_{\mathrm{loop}}^{-1} e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega})} \nu^{\mathcal{T}}(d\mathring{\omega}), \\ Z_{\mathrm{loop}} &= \mathbb{E}_{\nu^{\mathcal{T}}} e^{-\mathcal{A}_{\mathrm{loop}}}.\end{aligned} \tag{23}$$

Here $\mathbb{L}$ is a supplied finite family of closed walks, and $\mathfrak{Q}_p$ measures the defect of their represented reference holonomy. For a real vector representation ρ and a positive-definite fiber matrix $\mathsf{G}^{\mathrm{fib}}$ at the reference, take

$$\begin{aligned}\mathfrak{Q}_p &= \tfrac{1}{2}\|I - \rho(\mathring{\Theta}_p)\|_{\mathsf{G}}^2, \qquad \mathring{\Theta}_p = \prod_{e \in p}^{\text{ordered}} \exp(\mathring{\omega}_e), \\ \|A\|_{\mathsf{G}}^2 &= \mathrm{tr}[(\mathsf{G}^{\mathrm{fib}})^{-1} A^\top \mathsf{G}^{\mathrm{fib}} A].\end{aligned} \tag{24}$$

The reversed-edge convention determines each factor. The action is finite, nonnegative and Gaussian-integrable, giving $0 < e^{-\mathbb{E}_{\nu^{\mathcal{T}}} \mathcal{A}_{\mathrm{loop}}} \leq Z_{\mathrm{loop}} \leq 1$. This supplies a proper generator prior for noncompact groups without invoking a finite Haar law. Passive frame changes leave the defect unchanged; a reference change must carry the fiber metric and the Gaussian data. Holding the metric matrix fixed leaves only the narrower reference symmetry acting by multiples of its isometries: these reference changes preserve every defect, and when $\mathrm{d}\rho(\mathfrak{g})$ is the full matrix algebra of the fiber and a closed

walk traverses some reversal pair exactly once, no other reference change preserves its defect. On $\mathrm{GL}(K)$ a continuous conjugacy-class function cannot distinguish every unipotent holonomy from identity, so the metric is substantive modeling data. Appendix K.2 proves the normalization bounds and these covariance statements.

Let $\mathbb{P}_{\mathring{\omega}}(dy, dJ, do \mid X)$ be the normalized population law conditional on $\Theta(\mathring{\omega})$. Structural data X contain the graph, frames and interfaces, while anchors and covariances are fixed generative parameters during recognition. Then

$$\mathbb{P}(d\mathring{\omega}, dy, dJ, do \mid X) = P_{\mathrm{gen}}(d\mathring{\omega})\mathbb{P}_{\mathring{\omega}}(dy, dJ, do \mid X). \tag{25}$$

It has unit mass by conditional integration, even with cyclic interaction records. For a joint recognition law $Q(d\mathring{\omega}, dy, dJ) = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy, dJ)$ the conditional relative-entropy chain rule yields

$$\boxed{\begin{aligned} \mathcal{F}_V[Q; X, o] &= D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) + \mathbb{E}_R\mathcal{F}_V[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o], \\ D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) &= D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) + \mathbb{E}_R\mathcal{A}_{\mathrm{loop}} + \log Z_{\mathrm{loop}}. \end{aligned}} \tag{26}$$

The first identity has the extended-real meaning under bounded record densities and positive conditional evidence; the second uses finite generator divergence and action expectation. It is the latent-link collective bound, proved together with the unit mass of the joint in Appendix K.3. In particular, the loop term enters with its normalizer. The identity admits generator–matter correlations; the product $R \otimes Q_Y$ used in the Gaussian stationarity calculation below is a recognition restriction.

Making generators latent does not convert a comparison between live recognition laws into a fixed generative likelihood. The separate consensus potential below retains its original status. Held-link calculations remain conditional calculations; Gaussian matter conditional on a generator can have a non-Gaussian marginal. Scale the prior covariance as $\mathring{\Sigma} = \mathring{\sigma}^2\mathring{\Sigma}_0$ at a held anchor $\mathring{\alpha}$, and let $\mathbf{F}(\mathring{\omega})$ be the infimum of the held-link free energy at $\Theta(\mathring{\omega})$ over a class $\mathcal{Q}'$ of matter recognition laws. Suppose that the record densities are bounded with positive conditional evidence at every generator, that $\mathbf{F}$ is finite and lower semicontinuous at the anchor, and that a subclass of $\mathcal{Q}'$ with the same infimum at the anchor consists of laws whose held-link free energy is continuous at the anchor and bounded above by $ce^{c\|\mathring{\omega}\|}$ for a constant c depending on the law. Then, as $\mathring{\sigma} \to 0$, narrowing the generator prior to its anchor recovers the held-link free-energy infimum $\mathbf{F}(\mathring{\alpha})$, both over product recognition laws and over laws whose conditional matter recognition lies in $\mathcal{Q}'$. Narrowing recognition alone against a fixed-width prior instead gives infinite divergence in the point-mass limit. Appendix K.4 proves these statements and gives the non-Gaussian marginal.

5.3 Selective records place the links inside the functional

For a linear Gaussian comparison record, take

$$o_i = A_i k_i - B_{ij}k_j + \epsilon_i \quad \text{when } J_i = j, \qquad \epsilon_i \sim \mathcal{N}(0, \tau_i\Gamma_i), \quad \tau_i > 0, \quad \Gamma_i \succ 0. \tag{27}$$

The maps have a common output space within row i, and Γ_i is the same for its candidate sources. The full-state receiving form has $A_i = I$, $B_{ij} = L_{ij}$; a lower-dimensional interface has rectangular A_i, B_{ij}. At the selected zero comparison record define

$$D_{ij}(y) = \frac{1}{2}(A_i k_i - B_{ij}k_j)^\top \Gamma_i^{-1}(A_i k_i - B_{ij}k_j), \qquad \ell_i(0 \mid y, J_i = j) = c_i e^{-D_{ij}(y)/\tau_i}. \tag{28}$$

Here $c_i = (2\pi\tau_i)^{-\dim H_i/2}|\Gamma_i|^{-1/2}$. The Gaussian is normalized as a law of o_i for every state and every link rank. For fixed message-record coordinates, $A'_i = A_i g_i^{-1}$ and $B'_{ij} = B_{ij}g_j^{-1}$ leave the density

unchanged. If the record coordinates also change by $o'_i = h_i o_i$, then $A'_i = h_i A_i g_i^{-1}$, $B'_{ij} = h_i B_{ij} g_j^{-1}$ and $\Gamma'_i = h_i \Gamma_i h_i^\top$. The probability kernel is covariant. Its density is invariant against the transported reference measure; against fresh Lebesgue measure it acquires $|\det h_i|^{-1}$. The evidence density has the same factor and $\mathcal{F}_V$ shifts by $\sum_i \log|\det h_i|$, while posterior and source probabilities are unchanged. This distinction matters when $A_i = I$ and record coordinates follow the receiver. If source-dependent covariances are admitted on a common observation space, their normalizers enter the source energies and cannot be omitted.

For a correlated Gaussian recognition law with cross covariance $\Sigma_{ij} = \mathrm{Cov}(k_i, k_j)$,

$$\begin{aligned} 2\mathbb{E}_Q D_{ij} = {} & \|A_i\mu_i - B_{ij}\mu_j\|^2_{\Gamma_i^{-1}} \\ & + \operatorname{tr}\left[\Gamma_i^{-1}(A_i\Sigma_{ii}A_i^\top + B_{ij}\Sigma_{jj}B_{ij}^\top - A_i\Sigma_{ij}B_{ij}^\top - B_{ij}\Sigma_{ji}A_i^\top)\right]. \end{aligned} \tag{29}$$

Expanding the square proves the formula. The last two terms vanish under independence; they can also vanish when the interfaces annihilate the relevant cross covariances. This is one place where a collective law conveys information that separate marginal beliefs miss.

The interfaces are generative parameters unless made latent with a proper prior. Learning them requires differentiating the complete normalized model, including covariance normalizers. If the observations or structural constraints fail to identify their scale or rank, minimizing VFE alone need not select an informative interface. The variational principle optimizes a supplied model; it does not supply that model's missing identifiability conditions.

5.4 Attention as source inference

Suppose the labels are prior-independent with rows $\pi_{ij} > 0$ on the available source sets, and no record other than row i reads J_i. Conditional on the latent state, Bayes' rule gives

$$\mathbb{P}(J_i = j \mid y, o, X) = \frac{\pi_{ij} e^{-D_{ij}(y)/\tau_i}}{\sum_k \pi_{ik} e^{-D_{ik}(y)/\tau_i}}. \tag{30}$$

A recognition restriction $Q(dy, dJ) = Q_Y(dy)\prod_i \beta_i(dJ_i)$ instead gives

$$\beta^*_{ij} = \frac{\pi_{ij}\exp[-\mathbb{E}_{Q_Y} D_{ij}/\tau_i]}{\sum_k \pi_{ik}\exp[-\mathbb{E}_{Q_Y} D_{ik}/\tau_i]}. \tag{31}$$

These are different rows: a softmax of averaged costs is generally not an average of conditional softmaxes. The restriction discards label–state dependence.

More generally, for fixed finite costs E_{ij} and $\tau > 0$, set

$$f_i(\beta_i) = \sum_j \beta_{ij}E_{ij} + \tau D_{\mathrm{KL}}(\beta_i\|\pi_i), \qquad Z_i = \sum_j \pi_{ij} e^{-E_{ij}/\tau}. \tag{32}$$

The exact algebraic identity

$$f_i(\beta_i) = -\tau\log Z_i + \tau D_{\mathrm{KL}}(\beta_i\|\beta^*_i), \quad \beta^*_{ij} = \pi_{ij}e^{-E_{ij}/\tau}/Z_i \tag{33}$$

proves uniqueness, the reduced value and the correct sign of the entropy term. In the normalized record model its row contribution to $\mathcal{F}_V$ is $f_i/\tau - \log c_i$ with $E_{ij} = \mathbb{E}_Q D_{ij}$. In a consensus model E_{ij} can instead be a divergence between law points. The same optimization formula does not make these two probabilistic levels identical.

For the conditional-row family, optimization before averaging gives $-\mathbb{E}_Q \log\sum_j \pi_{ij} e^{-D_{ij}(y)/\tau_i}$. Restricting to a constant row gives $-\log\sum_j \pi_{ij} e^{-\mathbb{E}_Q D_{ij}/\tau_i}$. The constant-row value minus the conditional-row value is nonnegative by the variational inclusion, or by concavity of the negative log-sum-exp. Equality requires a common optimizing row almost everywhere.

5.5 A separate consensus potential with general channels

If the intended dynamics compare beliefs directly, define

$$E_{ij}^{\mathcal{K}} = D_{\mathrm{KL}}(q_i \| q_j \mathcal{K}_{ij}), \qquad E_{ij}^{m} = D_{\mathrm{KL}}(s_i \| s_j \mathcal{K}_{ij}^{m}). \tag{34}$$

The corresponding pointwise composite potential is

$$\begin{aligned} \mathcal{E}(c) = & \sum_i D_{\mathrm{KL}}(q_i \| p_i) + \lambda_h \sum_i D_{\mathrm{KL}}(s_i \| r_i) - \sum_i \mathbb{E}_{q_i \otimes s_i} \log \ell_i(o_i \mid k_i, m_i) \\ & + \sum_i \left[\sum_j \beta_{ij} E_{ij}^{\mathcal{K}} + \tau D_{\mathrm{KL}}(\beta_i \| \pi_i) \right] + \sum_i \left[\sum_j \gamma_{ij} E_{ij}^{m} + \tau_\gamma D_{\mathrm{KL}}(\gamma_i \| \pi_i^{m}) \right]. \end{aligned} \tag{35}$$

All quantities here are evaluated at c. Source rows range over agents present on the relevant overlaps. Reduced-space divergences can replace the costs when comparison is defined in message space. Equation (35) is a well-defined scalar potential where its terms are finite. It is not asserted to be the evidence bound of Eq. (21).

For a fixed source-template family, the normalized row mixture

$$P_i(dk, J_i = j) = \pi_{ij}(q_j \mathcal{K}_{ij})(dk), \qquad Q_i(dk, J_i = j) = q_i(dk)\beta_{ij} \tag{36}$$

satisfies at unit temperature

$$D_{\mathrm{KL}}(Q_i \| P_i) = \sum_j \beta_{ij} E_{ij}^{\mathcal{K}} + D_{\mathrm{KL}}(\beta_i \| \pi_i). \tag{37}$$

This is exact with fixed templates. When q_j evolves, P_i evolves with the variational state; the identity does not turn the simultaneous population potential into inference under a single fixed state-level joint. The technical appendices analyze this distinction and a normalized lift to a different space of law configurations.

For a private joint state–model law ζ_i and independent prior $p_i \otimes r_i$,

$$D_{\mathrm{KL}}(\zeta_i \| p_i \otimes r_i) = D_{\mathrm{KL}}(q_i \| p_i) + D_{\mathrm{KL}}(s_i \| r_i) + I_{\zeta_i}(K; M). \tag{38}$$

Thus the two self terms and product observation term in the consensus potential agree with a private mean-field VFE block only at $\lambda_h = 1$. A different scalar weight is an additional objective choice, not a consequence of the ordinary ELBO.

5.6 Fields, reduction and the worked Gaussian specialization

With a declared base density and support functions, the field potential is

$$\mathcal{E}_{\mathcal{C}} = \int_{\mathcal{C}} \mathcal{E}(c) d\nu(c), \tag{39}$$

where absent agents and unavailable edges are omitted, and the terms must be integrable. Smooth interior support weights may be used for differential calculations. Sharp domain boundaries or formation changes need their own regularity treatment. This integral is not a claim that an uncountable continuum of records has an independent product likelihood.

Substituting optimal rows gives a reduced potential $\mathcal{E}_{\mathrm{red}}$ and

$$\mathcal{E}_{\mathrm{full}} = \mathcal{E}_{\mathrm{red}} + \tau \sum_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*) + \tau_\gamma \sum_i D_{\mathrm{KL}}(\gamma_i \| \gamma_i^*). \tag{40}$$

At their optima,

$$\mathcal{E}_{\text{full}} = \mathcal{E}_{\text{red}}, \qquad d\mathcal{E}_{\text{red}} = d\mathcal{E}_{\text{self+obs}} + \sum_{ij} \beta^*_{ij} dE^{\mathcal{K}}_{ij} + \sum_{ij} \gamma^*_{ij} dE^{m}_{ij}, \tag{41}$$

for fixed temperatures and source priors. Changing priors contributes $-\tau \sum_{ij} \beta^*_{ij} d\log \pi_{ij}$ and its model counterpart; changing the channels also contributes through the costs. These terms cannot be discarded while learning the corresponding parameters.

The invertible noiseless specialization $\mathcal{K}_{ij}(x, \cdot) = \delta_{\Omega_{ij} x}$ recovers the old transported Gaussian consensus model. Sections 7 and 9, and the Gaussian appendices, retain its detailed calculations under their stated restrictions. The principal construction of the monograph is the normalized collective model with selective interactions. Its consequences and those of the consensus specialization are kept separate.

6 Lie-Algebra Links as Latent Relations

The latent extension of Section 5.2 makes the relation between agents an inferred quantity within the same normalized population model. At the fixed base point c_*, an independent invertible link has an exponential generator $\mathring{\omega}_e \in \mathfrak{g}$, and R recognizes the generator family on $\mathcal{T} = \mathfrak{g}^{E_+}$. This section uses one common channel with representation ρ; separate belief and model channels have their own generator blocks. The ring accent distinguishes these coordinates from agent means and covariances. The Gaussian anchor, loop action and episode law are supplied generative structure. Inference adjusts their recognition without deriving their choice from the evidence bound.

Two meanings of a dynamic link must be distinguished. Recognition can relax as a record is assimilated through a first-order inference flow. A generative law across learning episodes can instead penalize link increments and assign a most probable history. The second construction supplies a history stationarity equation with inertia; it does not turn recognition into a second-order flow. Both constructions concern relations among agents at c_* and introduce no transport between base points. The stationarity conditions, flows and history equations below are stated with their hypotheses and proved in Appendix K.

6.1 The expected link field equation

Hold the agents' recognition Q_Y and restrict the joint recognition to $R \otimes Q_Y$, with $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ and $\mathring{\Lambda} \succ 0$. Let $\mathring{\alpha}$ and $\mathring{\Sigma} \succ 0$ be the generator anchor and prior covariance, and write $\mathcal{A}_{\text{loop}}$ for the nonnegative loop action. For the complete held-link joint density define $H_o(\mathring{\omega}, y) = -\log p(o, y \mid X, \Theta(\mathring{\omega}))$. Assume bounded record densities with positive conditional evidence at every generator, a matter recognition law Q_Y with a density q_Y against a σ-finite reference that dominates every conditional matter prior, $\mathbb{E}_{Q_Y} |\log q_Y| < \infty$, and H_o twice continuously differentiable in $\mathring{\omega}$ with $\mathbb{E}_{Q_Y} \sup_{\|\mathring{\omega}\| \le r} (|H_o| + \|\nabla H_o\| + \|\nabla^2 H_o\|) \le c_1 e^{c_2 r}$ for every $r \ge 0$. Then the free energy of $R \otimes Q_Y$ is finite and continuously differentiable in $(\mathring{\mu}, \mathring{\Lambda})$, and Gaussian mean and precision stationarity are

$$\boxed{\begin{aligned} \mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha}) + \mathbb{E}_R \nabla \mathcal{A}_{\text{loop}} &= -\mathbb{E}_{R \otimes Q_Y} \nabla H_o, \\ \mathring{\Lambda} &= \mathring{\Sigma}^{-1} + \mathbb{E}_R \nabla^2 \mathcal{A}_{\text{loop}} + \mathbb{E}_{R \otimes Q_Y} \nabla^2 H_o. \end{aligned}} \tag{42}$$

An edge-block or diagonal Gaussian family projects the precision equation onto its admitted blocks. These are stationarity conditions within that family (Appendix K.5). Every link-dependent part of the held joint belongs in H_o, including a latent law or normalizer when present.

Put $M_e = \exp(\mathrm{d}\rho\mathring{\omega}_e)$ in reference coordinates. When H_o reads $\mathring{\omega}_e$ only through M_e, differentiably on an open set of matrices containing the represented twists and with Q_Y-integrable gradient, define $\mathbf{J}_e = -\mathbb{E}_{Q_Y}\partial H_o/\partial M_e$. With the matrix trace pairing and its generator adjoint, the edge component is

$$\begin{aligned}(\mathring{\Sigma}_e)^{-1}(\mathring{\mu}_e - \mathring{\alpha}_e) + \mathbb{E}_R\nabla_e\mathcal{A}_{\mathrm{loop}} &= \mathbb{E}_R\left[(\mathrm{d}\rho)^\top\mathbf{L}\big((\mathrm{d}\rho\mathring{\omega}_e)^\top, \mathbf{J}_e\big)\right],\\ \mathbf{L}(B,T) &= \int_0^1 e^{uB}Te^{(1-u)B}du.\end{aligned} \tag{43}$$

The Fréchet derivative pulls the matrix current back through the exponential; averaging gives the stress balance of the prior. Evaluating this only at $\mathring{\mu}$ would be a further approximation.

For a comparison record with energy $\frac{1}{2}(k_i - M_e k_j)^\top W_a(k_i - M_e k_j)$ and $W_a \succ 0$ that is the only term of H_o reading $\mathring{\omega}_e$, and belief marginals with finite second moments, the current includes mean mismatch and uncertainty:

$$\mathbf{J}_e = W_a\left[(\mu_i - M_e\mu_j)\mu_j^\top + \Sigma_{ij} - M_e\Sigma_j\right]. \tag{44}$$

Here Σ_{ij} is the cross covariance under Q_Y, zero for product matter. The error outer product has the delta-rule form, while the remaining terms retain dependence and source uncertainty. Selective source records insert their recognized weight and comparison maps, with a generator-independent recognition row in the displayed specialization; Appendix K.5 derives both currents. This form does not identify a fast-weight architecture. A point mass at $\mathring{\mu}$ has infinite D_{KL} from a nondegenerate Gaussian generator prior; a most probable link is a generative mode rather than a recognition optimum of zero width (Appendices K.4 and K.5).

6.2 Fisher inference and a finite learning step

Let Q_λ vary with the agents' recognition coordinates, using the product metric $\mathsf{G} = g^F \oplus \mathsf{G}_Y$ with the Gaussian Fisher metric g^F of R. At fixed generator-prior parameters, define, up to a recognition-independent constant,

$$\mathfrak{i}_\lambda(\mathring{\omega}) = \mathcal{F}_V[Q_\lambda; X, \Theta(\mathring{\omega}), o] + \mathcal{A}_{\mathrm{loop}}(\mathring{\omega}) + \frac{1}{2}\langle\mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha})\rangle. \tag{45}$$

Assume bounded record densities with positive conditional evidence at every generator, let λ range over a finite-dimensional C^2 manifold with a C^1 positive-definite metric G_Y, and let the free energy of $R \otimes Q_\lambda$ be C^2 and each $\mathfrak{i}_\lambda$ be C^2 with $|\mathfrak{i}_\lambda| + \|\nabla\mathfrak{i}_\lambda\| + \|\nabla^2\mathfrak{i}_\lambda\| \le c_1 e^{c_2\|\mathring{\omega}\|}$. For $\mathcal{Q} = R \otimes Q_\lambda$ the joint inference law is $\dot{\mathcal{Q}} = -\operatorname{grad}^{\mathsf{G}}\mathcal{F}_V$, with $d\mathcal{F}_V/dt = -\|\operatorname{grad}^{\mathsf{G}}\mathcal{F}_V\|^2_{\mathsf{G}}$. Its generator Euler step in Gaussian natural parameters, rewritten as mean and precision, is the Gaussian Bayesian learning rule:

$$\boxed{\begin{aligned}\mathring{\Lambda}' &= (1-\Delta t)\mathring{\Lambda} + \Delta t\,\mathrm{pr}_{\mathring{\Xi}}\,\mathbb{E}_R\nabla^2\mathfrak{i}_\lambda,\\ \mathring{\mu}' &= \mathring{\mu} - \Delta t\mathring{\Lambda}'^{-1}\mathbb{E}_R\nabla\mathfrak{i}_\lambda.\end{aligned}} \tag{46}$$

The projection selects full, edge-block or diagonal precision. Admissibility requires $\mathring{\Lambda}' \succ 0$. Finite-step descent additionally requires a relative curvature bound: on the segment joining the expectation parameters of the two Gaussian laws, the Hessian of the free energy in these coordinates is at most c_{G} times the Fisher metric of the family in the same coordinates, and the step satisfies $\Delta t \le 1/c_{\mathsf{G}}$ (Appendix K.6). A unit step gives the Gaussian minimizer for a positive quadratic integrand in the full family. General convergence and automatically slower link recognition are not established. The continuous law is first order and dissipative; its Fisher metric acts as friction in this force balance.

6.3 A normalized episode law and history inertia

For a finite horizon $n = 0, \dots, n_1$, hold frames static and specify the history law

$$\begin{aligned} \mathbb{P}^{\mathrm{hist}}(d\mathring{\omega}^{(0:n_1)}) &= Z_{\mathrm{hist}}^{-1} e^{-\mathcal{A}_{\mathrm{loop,inc}}(\mathring{\omega}^{(0:n_1)})} \prod_{n=0}^{n_1} \nu^{\mathcal{T}}(d\mathring{\omega}^{(n)}), \\ \mathcal{A}_{\mathrm{loop,inc}}(\mathring{\omega}^{(0:n_1)}) &= \sum_{n=0}^{n_1} \mathcal{A}_{\mathrm{loop}}(\mathring{\omega}^{(n)}) + \mathbf{m} \sum_{n<n_1} \mathfrak{U}(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}). \end{aligned} \tag{47}$$

Here $\nu^{\mathcal{T}}$ is the Gaussian base law and $\mathbf{m} > 0$. Either the Gaussian increment $\mathfrak{U} = \frac{1}{2}\|\mathring{\omega}' - \mathring{\omega}\|^2$ or the nonnegative metric loop defect of $\exp(-\mathring{\omega}'_e)\exp(\mathring{\omega}_e)$ is admitted. A finite nonnegative action gives $0 < Z_{\mathrm{hist}} \leq 1$. The exact forward kernels include backward normalizers depending on the horizon; normalizing each increment independently gives the same law only when the last backward normalizer is almost everywhere constant, which fails already for one scalar Gaussian generator. Links between copies of one agent across episodes are identities, with no extra latent variable (Appendix K.7).

Let $\mathfrak{V}_n$ include the anchor quadratic, loop action and any declared episode matter term. Gaussian increments give the exact interior stationarity equation

$$\boxed{\begin{gathered} \mathbf{m}\big(2\mathring{\omega}^{(n)} - \mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n-1)}\big) + \nabla\mathfrak{V}_n(\mathring{\omega}^{(n)}) = 0, \\ 0 < n < n_1. \end{gathered}} \tag{48}$$

Free endpoints have one-sided equations; held endpoints impose boundary values. A minimizing history exists when each $\mathfrak{V}_n$ is continuous and $\sum_n \mathfrak{V}_n(\mathring{\omega}^{(n)}) \to \infty$ as the history norm diverges, as when each $\mathfrak{V}_n$ is at least the anchor quadratic minus a constant. With unit episode spacing the central difference has the Euclidean sign $\mathbf{m}\ddot{\mathring{\omega}} = +\nabla\mathfrak{V}$, corresponding to Newtonian potential $-\mathfrak{V}$. Increment precision acts as mass in this history equation without establishing a physical mass or real-time dynamics in $+\mathfrak{V}$. When $\mathrm{d}\rho$ is injective and generators carry the inner product pulled back from the fiber trace norm, the time-like loop defect gives a generator-dependent increment tensor, agreeing with the Gaussian equation only at its controlled near-zero order (Appendix K.7).

6.4 The Proca-type operator and its domain of validity

Assume $\mathrm{d}\rho$ is injective and give generators the inner product pulled back from the fiber trace norm; all adjoints and increment norms below use this metric. For zero anchor, fixed product Gaussian matter, and quadratic comparison records of the form above on edge copies $e : j \to i$ of E_+ or their reversals, entering with generator-independent weights $\beta_a \in [0, 1]$, the history equation to first order jointly in generators and individual currents is

$$\boxed{\begin{aligned} &\mathbf{K}\mathring{\omega} = \mathbf{j}, \\ &\quad \mathbf{K} = \mathring{\Sigma}^{-1} + \sum_p c_p \mathbf{C}_p^{\top}\mathbf{C}_p + \mathbf{m}(-\Delta_n) + \mathring{\Phi}, \\ &\quad \mathbf{K} \succeq c_{\mathring{\Sigma}} I, \\ &\mathring{\omega}_{\mathrm{lin}} = \mathbf{K}^{-1}\mathbf{j}. \end{aligned}} \tag{49}$$

The cycle map $\mathbf{C}_p\mathring{\omega} = \sum_{e\in p} \mathring{\omega}_e$ counts each traversal with its orientation. The positive episode Laplacian $-\Delta_n$ has the central difference in the interior and one-sided differences at free endpoints.

The anchor precision acts episode by episode, with smallest eigenvalue $c_{\mathring{\Sigma}} > 0$. Define $B_{a,n} = \mathrm{d}\rho\mathring{\omega}_e^{(n)}$ and $M_{j,n} = \Sigma_j^{(n)} + \mu_j^{(n)}\mu_j^{(n)\top}$. The nonnegative matter operator and source have the quadratic and linear forms

$$\begin{aligned}\langle\mathring{\omega}, \mathring{\Phi}\mathring{\omega}\rangle &= \sum_{n,a} \beta_a \operatorname{tr}(W_a B_{a,n} M_{j,n} B_{a,n}^\top),\\ \langle\mathbf{j}, \mathring{\omega}\rangle &= \sum_{n,a} \beta_a \operatorname{tr}(\mathbf{J}_a^\top B_{a,n}),\\ \mathbf{J}_a &= W_a[(\mu_i^{(n)} - \mu_j^{(n)})\mu_j^{(n)\top} - \Sigma_j^{(n)}].\end{aligned} \tag{50}$$

The cycle term has graph curl-curl form, and the anchor and matter terms supply bare and screening masses. This is a massive Euclidean vector operator of Proca type on agent edges. Its episode index remains an inference index; no physical Proca field, Lorentzian propagation law or base connection has been identified.

The exact zero-generator Hessian is $\mathbf{K} + \mathbf{N}_0$, where

$$\langle\mathring{\omega}, \mathbf{N}_0\mathring{\omega}\rangle = -\sum_{n,a} \beta_a \operatorname{tr}(\mathbf{J}_a^\top B_{a,n}^2). \tag{51}$$

Thus linearizing generators at fixed currents retains $\mathbf{N}_0$. A small total source does not ensure small individual-current corrections. Write the full history objective as $\mathcal{A}^{\mathrm{hist}} = \sum_n \mathfrak{V}_n + \mathbf{m}\sum_n \mathfrak{U}$. If $c_0 = \|\mathbf{N}_0\|$, $c_r = \sup_{\|\mathring{\omega}\|\le r} \|D^3\mathcal{A}^{\mathrm{hist}}\|$, $c_0 + c_r r \le c_{\mathring{\Sigma}}/2$ and $\|\mathbf{j}\| \le c_{\mathring{\Sigma}} r/4$, there is exactly one critical history in that ball, with a controlled second-order discrepancy from $\mathbf{K}^{-1}\mathbf{j}$. This local conclusion establishes neither global uniqueness of nonlinear histories nor a universal link timescale. Appendix K.8 proves these statements and bounds the discrepancy explicitly.

7 Precision, Observation, and Symmetry

Precision has three distinct meanings in this construction: the inverse covariance of a law, a parameter of a normalized observation kernel, and a weight multiplying a divergence in a consensus objective. Confusing them changes the dynamics. A coefficient of one on $D_{\mathrm{KL}}(q_i\|p_i)$ does not require a spatially uniform or isotropic prior.

7.1 Adaptive anchoring as a declared penalty

For a finite divergence $D_i = D_{\mathrm{KL}}(q_i\|p_i)$, consider the optional consensus sector

$$\mathcal{E}_i^{\mathrm{adaptive}} = \alpha_i D_i + R(\alpha_i), \qquad R(\alpha_i) = b_0\alpha_i - c_0\log\alpha_i, \quad b_0, c_0 > 0. \tag{52}$$

Its unique minimum on $\alpha_i > 0$ is

$$\alpha_i^* = \frac{c_0}{b_0 + D_i}, \qquad \mathcal{E}_i^{\mathrm{adaptive},*} = c_0 + c_0\log\frac{b_0 + D_i}{c_0}. \tag{53}$$

Indeed, the first derivative vanishes there and the second is $c_0/\alpha_i^2 > 0$; the objective diverges at either endpoint. This yields a concave penalty in D_i, with decreasing slope. It is a choice of robust anchoring, not an inferred law that agents must trust an increasingly inaccurate prior less.

For a rate-parameterized Gamma density of shape $c_0 + 1$,

$$-\log p(\alpha_i) = b_0\alpha_i - c_0\log\alpha_i + \log\Gamma(c_0 + 1) - (c_0 + 1)\log b_0. \tag{54}$$

The omitted constants depend on the hyperparameters:

$$\partial_{b_0}[-\log p] = \alpha_i - \frac{c_0+1}{b_0}, \qquad \partial_{c_0}[-\log p] = -\log\alpha_i + \psi(c_0+1) - \log b_0. \tag{55}$$

Multiplying an entire KL divergence by α_i is not equivalent to placing this prior on a Gaussian precision. For the normalized model $k_i \mid \alpha_i \sim \mathcal{N}(\mu_{p,i}, \Sigma_{p,i}/\alpha_i)$, the corresponding expected negative-log-density optimum instead is

$$\alpha_i^{\mathrm{MAP}} = \frac{c_0 + K_q/2}{b_0 + \frac{1}{2}\mathbb{E}_{q_i}[(k_i-\mu_{p,i})^\top \Sigma_{p,i}^{-1}(k_i-\mu_{p,i})]}. \tag{56}$$

The numerator contains the Gaussian normalizer. The denominator contains a quadratic error, not a KL divergence. Learning precision inside $\mathcal{F}_V$ therefore requires the normalized likelihood or prior from which its update is derived.

Coordinate-dependent gains. For product Gaussian laws in a fixed chart, with variances $v_{q,k}, v_{p,k} > 0$,

$$D_i = \sum_k D_{i,k}, \qquad D_{i,k} = \frac{1}{2}\left[\frac{v_{q,k} + (\mu_{q,k}-\mu_{p,k})^2}{v_{p,k}} - 1 + \log\frac{v_{p,k}}{v_{q,k}}\right]. \tag{57}$$

Separate penalties $\sum_k[\alpha_{i,k}D_{i,k} + b_k\alpha_{i,k} - c_k\log\alpha_{i,k}]$ give

$$\alpha_{i,k}^* = \frac{c_k}{b_k + D_{i,k}}. \tag{58}$$

This refinement singles out a coordinate factorization. General $\mathrm{GL}(K_q)$ changes mix the factors, so diagonal gains alone do not define an invariant construction under that whole group. A covariant extension must transform the relevant precision tensor and retain the resulting non-diagonal family.

7.2 What the envelope theorem removes

Differentiating the complete minimized penalty gives

$$d\alpha_i^* = -\frac{(\alpha_i^*)^2}{c_0} dD_i, \qquad d[\alpha_i^* D_i] = \frac{b_0(\alpha_i^*)^2}{c_0} dD_i, \tag{59}$$

but

$$d[\alpha_i^* D_i + R(\alpha_i^*)] = \alpha_i^* dD_i. \tag{60}$$

The regularizer supplies the missing chain-rule contribution. The attention sector behaves in the same way: at fixed π_i and τ, the minimized row is

$$\mathcal{E}_{i,\mathrm{row}}^* = -\tau\log\sum_j \pi_{ij} e^{-E_{ij}/\tau}, \qquad d\mathcal{E}_{i,\mathrm{row}}^* = \sum_j \beta_{ij}^* dE_{ij}. \tag{61}$$

These formulas hold at smooth interior optima with finite energies [25]. Optimizing a temperature or a source prior introduces their explicit derivatives. The envelope theorem does not authorize suppressing those dependencies.

For the state-channel specialization of $\mathcal{E}$, freeze other sectors and let x_i be a mean or covariance parameter. With fixed presence weights, the alignment derivative is

$$\partial_{x_i}\mathcal{E}^*_{\text{align}} = \chi_i \sum_j \beta^*_{ij}\partial_{x_i}E_{ij} + \sum_{k\neq i}\chi_k\beta^*_{ki}\partial_{x_i}E_{ki}, \tag{62}$$

where the first sum includes the full derivative if a self edge is permitted. The second sum records the agent's role as a source. For a parameter ϑ shared with the model law, the general rule is

$$\partial_\vartheta\mathcal{E}^*_{\text{align}} = \sum_{ij}\chi_i\beta^*_{ij}\partial_\vartheta E_{ij}, \tag{63}$$

with every dependence of the joint law and channel included. Independent marginal updates are an additional product-family specialization.

7.3 Environmental records and likelihood kernels

An observation is naturally represented by a normalized kernel $K(do \mid y)$ in the population model. Re-expressing the likelihood as an environmental boundary factor can be useful, but calling that factor an agent neither removes the observation nor supplies a posterior law for the environment.

Proposition 3 (Gaussian-sensor equivalence)**.** *Let $o = Hk + \epsilon$, $\epsilon \sim \mathcal{N}(0, R)$, $R \succ 0$, and suppose H has full column rank. Define $\Lambda_o = H^\top R^{-1}H$, $m_o = \Lambda_o^{-1}H^\top R^{-1}o$ and $q_e = \mathcal{N}(m_o, \Lambda_o^{-1})$. For Gaussian $q = \mathcal{N}(\mu, \Sigma)$, the expected negative log-likelihood and $D_{\mathrm{KL}}(q\|q_e)$ have the same mean gradient. Their covariance gradients differ by $-\Sigma^{-1}/2$. The cross-entropy $-\mathbb{E}_q \log q_e$ differs from the expected negative log-likelihood only by a constant independent of q.*

Proof. Completing the square gives

$$-\mathbb{E}_q \log p(o \mid k) = \frac{1}{2}[(\mu - m_o)^\top\Lambda_o(\mu - m_o) + \operatorname{tr}(\Lambda_o\Sigma)] + \text{const.} \tag{64}$$

The KL adds $-\frac{1}{2}\log\det\Sigma$ and another constant. Hence the mean gradients agree, while their covariance gradients are respectively $\Lambda_o/2$ and $(\Lambda_o - \Sigma^{-1})/2$. □

Thus the exact observation substitution is

$$-\mathbb{E}_q \log q_e = D_{\mathrm{KL}}(q\|q_e) + H(q), \tag{65}$$

not a new KL factor alone. Adding one KL for every observed record would repeat the receiving entropy and change covariance learning.

For rank-deficient H, the likelihood remains normalized in observation space but its full latent-space Gaussian surrogate has singular precision and is not a probability density. Retain the observation kernel. Nonlinear sensors also need not have a Gaussian surrogate. A Dirac environmental law is no remedy: an absolutely continuous belief generally has infinite KL relative to it. No limit of infinite sensory precision is needed for the general formulation.

7.4 Coordinate freedom and physical symmetry

A passive frame change re-expresses beliefs, priors, interaction channels and record kernels together. It leaves predictions unchanged. A physical symmetry of a fixed model is a transformation that

preserves its specified priors, likelihoods and dynamics. The first statement does not imply the second.

In the flat reference specialization, a common right action $U_i \mapsto U_i g$ leaves every $\Omega_{ij} = U_i U_j^{-1}$ unchanged. It changes the common reference labeling. A simultaneous left action can be a symmetry of an observation-free model only if all its fixed structural data are invariant under that action. Co-transforming a fixed prior establishes covariance between presentations, not a symmetry of that same prior.

An external sensor or asymmetric prior can explicitly distinguish otherwise equivalent configurations. This is the ordinary external-field meaning of symmetry breaking. It does not break the freedom to change coordinates. Spontaneous symmetry breaking would additionally require an invariant law, a specified limit or selection mechanism, and a fluctuation analysis. The finite formation theorem of Section 14 states its initial asymmetry and limiting regime instead of silently invoking that mechanism.

8 Belief, Predictive Memory, and Separated Timescales

The single statistical bundle of Section 4.2 allows an agent to carry one joint law while distinguishing its current-state predictions from the information retained across observations. This removes the need to postulate two different kinds of probabilistic substance. It does not, by itself, make one part of the law slow. We first derive the exact conditional decomposition and the inheritance of priors, then give two mechanisms for persistent predictive memory. A declared separation of kinetic rates remains a useful specialization, with its assumptions stated separately.

8.1 Conditional Belief and Model Laws Within One Statistical State

At a fixed base point, write the agent's joint recognition law as

$$\zeta_i(dk, dm) = s_i(dm) q_i(dk \mid m), \qquad q_i(dk) = \int q_i(dk \mid m) s_i(dm). \tag{66}$$

The conditional and marginal meanings of q_i are distinguished by their arguments. On standard Borel sample spaces this disintegration exists, up to s_i-null sets. It is an exact description of the joint law. The familiar product $q_i \otimes s_i$ adds the restriction that the state belief is independent of the model sample. Two marginals alone do not determine their coupling, so identifying one bundle with two marginal bundles is exact only after that restriction is declared. In a joint Gaussian family the discarded information includes the off-diagonal state–model covariance.

To see what inference requires, suppress i and fix the normalized joint law

$$P(dm, dk, do) = r(dm) p(dk \mid m) \ell(o \mid k, m) \nu_O(do), \qquad Z_o = \int r(dm) p(dk \mid m) \ell(o \mid k, m), \tag{67}$$

where r is a proper prior over model samples, $p(\cdot \mid m)$ is a normalized state-prior kernel, and ℓ is an observation density normalized against ν_O for every (k, m). The observed o is admitted only when $0 < Z_o < \infty$. Every factor is fixed while recognition varies. In the population, the same calculation applies to a conditional block with the outside configuration held fixed; it does not assert independence of agents or add their private bounds to manufacture a collective one.

Let $P_0(dm, dk) = r(dm) p(dk \mid m)$. When $\zeta \ll P_0$ and both logarithms below are integrable, the

defining expectation and its decomposition are

$$\begin{aligned}\mathcal{F}_{\mathrm{loc}}(\zeta; o) &= \mathbb{E}_\zeta \left[\log \frac{d\zeta}{dP_0}(K, M) - \log \ell(o \mid K, M) \right] \\ &= D_{\mathrm{KL}}(s \| r) + \int D_{\mathrm{KL}}\big(q(\cdot \mid m) \| p(\cdot \mid m)\big) s(dm) - \mathbb{E}_\zeta \log \ell(o \mid K, M). \end{aligned} \tag{68}$$

Without these integrability assumptions we use the extended definition $\mathcal{F}_{\mathrm{loc}} = -\log Z_o + D_{\mathrm{KL}}(\zeta \| P(\cdot \mid o))$. This avoids subtracting infinite entropy and energy terms. With

$$\begin{aligned} Z_o(m) &= \int \ell(o \mid k, m) p(dk \mid m), \\ s^*(dm) &= \frac{Z_o(m) r(dm)}{Z_o}, \qquad q^*(dk \mid m) = \frac{\ell(o \mid k, m) p(dk \mid m)}{Z_o(m)}. \end{aligned} \tag{69}$$

on $0 < Z_o(m) < \infty$, with conditional versions on null sets irrelevant, the KL chain rule gives

$$\mathcal{F}_{\mathrm{loc}}(\zeta; o) + \log Z_o = D_{\mathrm{KL}}(s \| s^*) + \int D_{\mathrm{KL}}\big(q(\cdot \mid m) \| q^*(\cdot \mid m)\big) s(dm). \tag{70}$$

The proof is the factorization of the joint density ratio into its marginal and conditional ratios, followed by integration. Each term is nonnegative. For fixed s, unrestricted conditional inference sets $q(\cdot \mid m) = q^*(\cdot \mid m)$ almost everywhere; subsequent unrestricted marginal inference sets $s = s^*$. For a fixed conditional family, put $d(m) = D_{\mathrm{KL}}(q(\cdot \mid m) \| q^*(\cdot \mid m))$. Its optimal model marginal is

$$s_q(dm) = \frac{e^{-d(m)} s^*(dm)}{\int e^{-d(v)} s^*(dv)}, \tag{71}$$

provided the denominator is positive. The objective becomes a KL divergence to s_q plus a constant, which proves normalization, positivity on its supported part, and optimality. Equivalently, normalized variations $q_\epsilon = (1 + \epsilon v) q$ with $\mathbb{E}_q v = 0$ make the first variation $\mathbb{E}_q[v \log(dq/dq^*)]$ vanish exactly at the conditional target in the interior. These are optimizer statements, not identities for a finite sequence of updates.

For independent priors the state–model mutual information gives the exact additional complexity in Eq. (38). Actually replacing a correlated law by $\bar{\zeta} = q \otimes s$ also changes the expectation of every coupled likelihood. When the terms are finite and P_o denotes the fixed generative slice density,

$$\mathcal{F}_{\mathrm{loc}}(\zeta; o) - \mathcal{F}_{\mathrm{loc}}(\bar{\zeta}; o) = I_\zeta(K; M) - \big(\mathbb{E}_\zeta - \mathbb{E}_{\bar{\zeta}}\big) \log P_o(K, M). \tag{72}$$

Thus the cost of a product approximation is not generally the mutual information alone. By the extended definition above, the unrestricted and restricted optima differ by the infimum of $D_{\mathrm{KL}}(q \otimes s \| P(\cdot \mid o))$, and this identity does not require the infimum to be attained (Proposition 32).

8.2 Where Priors Come From

A prior need not be the shadow of an already formed parent. A normalized temporal model supplies a more general origin. Let Y_t be the complete retained state, including any links or records needed to specify its dynamics, and let

$$P(dy_{0:n}, do_{1:n}) = p_0(dy_0) \prod_{t=1}^{n} T_t(dy_t \mid y_{t-1}) R_t(do_t \mid y_t) \tag{73}$$

be an ordered product of normalized kernels and a proper initial law. At every positive-evidence history,

$$p_t^-(dy) = \int T_t(dy \mid y')P(dy' \mid o_{1:t-1}), \qquad P(dy \mid o_{1:t}) = \frac{\ell_t(o_t \mid y)p_t^-(dy)}{\int \ell_t(o_t \mid v)p_t^-(dv)}. \tag{74}$$

The previous posterior becomes the next prior through prediction. A deterministic parent-to-child channel gives the pushforward used in a cross-scale shadow, while a noisy channel gives a mixture. These are consequences of a specified joint law. An instantaneous circular assignment of priors needs its own normalized joint and does not follow from the recursion above.

Replacing the previous posterior by recognition Q_{t-1} makes the predictive prior approximate, with the support-aware bound

$$D_{\mathrm{KL}}(Q_{t-1}T_t \| P_{t-1}T_t) \le D_{\mathrm{KL}}(Q_{t-1} \| P_{t-1}). \tag{75}$$

Indeed, the two input–output joints have the divergence on the right because they share T_t; disintegration over their output leaves the left side plus a nonnegative conditional divergence. This controls prediction, not the error at every possible subsequent observation. The initial prior, a hyperprior, or a separately justified invariant law remains a boundary datum. Neither Bayesian conditioning nor VFE determines it without additional structure.

8.3 Predictive Memory from Accumulating Evidence

There is a precise sense in which the predictive part of one statistical state becomes slow without an imposed small learning rate. Let m be a categorical probability vector with a proper Dirichlet prior $r = \mathrm{Dir}(\alpha_1, \dots, \alpha_d)$, where every $\alpha_a > 0$, and draw categorical records $K_1, K_2, \dots$ conditionally independently given m. Against Lebesgue measure on the simplex and counting measure on the records, the normalized joint density is the Dirichlet density times $\prod_t m_{K_t}$. If $N_{n,a}$ counts occurrences of category a and $A = \sum_a \alpha_a$, exact conditioning gives

$$s_n = \mathrm{Dir}(\alpha + N_n), \qquad p_{n+1}^-(a) = \bar{m}_{n,a} = \frac{\alpha_a + N_{n,a}}{A + n}. \tag{76}$$

Subtracting successive fractions proves

$$\bar{m}_n - \bar{m}_{n-1} = \frac{e_{K_n} - \bar{m}_{n-1}}{A + n}, \qquad \|\bar{m}_n - \bar{m}_{n-1}\|_{\mathrm{TV}} \le \frac{1}{A + n}. \tag{77}$$

Current-category beliefs can change by order one, whereas the predictive model changes by at most $1/(A + n)$ per new record. The slowing concerns this predictive distribution. It is not a bound in every metric on the full parameter posterior, and it presupposes a persistent latent parameter and its evaluator $m \mapsto \mathrm{Cat}(m)$. Hidden categories generally produce a mixture of Dirichlet posteriors; replacing their counts by expected counts requires an explicit variational restriction.

Friston and colleagues already treat state inference and parameter learning as probabilistic belief updates [6]. The discrete-state synthesis derives learning by accumulating sufficient statistics under an explicit factorization [7, Sec. 8 and App. A.1]. Structure learning also admits changes to latent-state and parameter representations, within supplied candidate families and priors [8, Secs. 1–3]. This literature supports a joint statistical description, while leaving the emergence of a finite smooth model sector in the present network to be established.

One can define a model operationally without introducing a separate parameter sample. For a process with a finite record alphabet, let $S(h)$ be the conditional law of all future records given a

positive-probability history h. Histories are predictively equivalent when their continuation laws agree. For a new record a of positive conditional probability and any future event B,

$$S(ha)(B) = \frac{S(h)(\{a\} \times B)}{S(h)(\{a\} \times \mathsf{O}^{\mathbb{N}})}. \tag{78}$$

Conditioning therefore supplies a closed update of the predictive equivalence class. This gives a derived predictive model of the history. Its realization can be infinite-dimensional, and its changes need not be slow. A finite-dimensional, smooth, gauge-equivariant realization requires a further sufficiency or closure theorem.

8.4 What Can Select a Slow Sector?

A slow direction cannot be justified solely by a small Euclidean Hessian eigenvalue. Natural-gradient dynamics depend on the Hessian relative to the metric. The exact-posterior case gives a useful restriction.

Proposition 4 (Local Fisher relaxation at an exact posterior)**.** *Let $\{Q_\theta\}$ be a twice continuously differentiable, dominated, identifiable statistical family on common support, with positive-definite continuously differentiable Fisher metric G. Suppose an interior θ_* satisfies $Q_{\theta_*} = \Pi$, and differentiation twice under the normalization and KL integrals is valid. For $F(\theta) = D_{\mathrm{KL}}(Q_\theta \| \Pi) - \log Z$, unit Fisher natural-gradient descent has linearization $-I$ at θ_*.*

Proof. Writing q_θ for the density and using $\int \partial_a q_\theta = 0$ gives

$$\partial_a F = \int (\partial_a q_\theta) \log(q_\theta/\pi), \qquad \partial_b \partial_a F = \int (\partial_{ab} q_\theta) \log(q_\theta/\pi) + \int \frac{\partial_a q_\theta \partial_b q_\theta}{q_\theta}. \tag{79}$$

At θ_* the gradient and first Hessian term vanish, while the second term is G_{ab}. Differentiating $-G^{-1}\nabla F$ therefore gives $-G^{-1}G = -I$. A coordinate change conjugates this linearization and preserves its spectrum. □

A common rate η gives $-\eta I$. For example, $Q_\mu = \mathcal{N}(\mu, H^{-1})$ and $\Pi = \mathcal{N}(0, H^{-1})$ give $F = \mu^\top H \mu/2$, $G = H$, and $\dot{\mu} = -\mu$, independently of the condition number of H. The proposition does not cover a misspecified variational optimum, singular boundary, constrained kinetic operator, moving generative parameter, or nonsmooth switching point. These are possible sources of separated rates, but each requires its own argument.

A positive kinetic example starts with one finite statistical state. Let a reversible Markov generator L have unit jump rates on $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$, rate $\epsilon > 0$ on $1 \leftrightarrow 3$ and $2 \leftrightarrow 4$, and no other edges. Its invariant law is uniform. Direct multiplication gives eigenvectors

$$(1,1,1,1), \quad (1,1,-1,-1), \quad (1,-1,1,-1), \quad (1,-1,-1,1) \tag{80}$$

with eigenvalues $0, -2\epsilon, -2, -2(1+\epsilon)$, respectively. The label of the cells $\{1,2\}$ and $\{3,4\}$ is exactly Markov with flip rate ϵ, whereas the within-cell contrasts relax at rates at least two. An observation that reads this label has persistent predictive information. Two clocks have thus been constructed within one law. The weak-edge kinetics supply their ratio: the stationary law and its KL objective are identical for every ϵ. This is kinetic emergence of a retained slow coordinate, not derivation of kinetics from VFE alone.

8.5 Elimination Produces Memory Before It Produces an Autonomous Model

The exact price of retaining only part of a statistical state is visible in finite dimensions. Let L be a finite irreducible Markov generator reversible under $\pi > 0$, and let $r_t = dQ_t/d\pi$. Then $\dot{r} = \mathsf{L}r$ in $L^2(\pi)$, and

$$\frac{d}{dt} D_{\mathrm{KL}}(Q_t \| \pi) = -\frac{1}{2} \sum_{x,y} \pi_x \mathsf{L}_{xy} (r_x - r_y)(\log r_x - \log r_y) \le 0. \tag{81}$$

The identity follows by symmetrizing $\sum_x \pi_x (\mathsf{L}r)_x \log r_x$ using detailed balance. It applies directly at positive times; boundary densities are handled by continuity. This is a specified kinetic descent, distinct from the Fisher flow of Proposition 4.

For a partition statistic T, let $\mathsf{P}f = \mathbb{E}_\pi[f \mid T]$, $\mathsf{R} = I - \mathsf{P}$, $a = \mathsf{P}r$, and $b = \mathsf{R}r$. On the respective subspaces write

$$\mathsf{A} = \mathsf{PLP}, \quad \mathsf{B} = \mathsf{PLR}, \quad \mathsf{C} = \mathsf{RLP}, \quad \mathsf{D} = \mathsf{RLR}. \tag{82}$$

Solving $\dot{b} = \mathsf{C}a + \mathsf{D}b$ and substituting into $\dot{a} = \mathsf{A}a + \mathsf{B}b$ gives

$$\dot{a}(t) = \mathsf{A}a(t) + \mathsf{B}e^{t\mathsf{D}} b(0) + \int_0^t \mathsf{B}e^{(t-u)\mathsf{D}} \mathsf{C}a(u)\,du. \tag{83}$$

This is an exact memory equation. The partition gives closed Markov dynamics for every initial law exactly when $\mathsf{C} = 0$: applying L to each cell indicator shows that this condition means equal total jump rates to each other cell from every state in a given cell. Reversibility gives $\mathsf{B} = \mathsf{C}^*$, so both residual terms then vanish.

When $\mathsf{D} \preceq -\gamma I$ with $\gamma > 0$, the initial residual is bounded by $\|\mathsf{B}\| e^{-\gamma t} \|b(0)\|$. Put $M_t = \sup_{0 \le u \le t} \|\dot{a}(u)\|$. The local replacement of the memory integral has error

$$\left\| \int_0^t \mathsf{B}e^{(t-u)\mathsf{D}} \mathsf{C}a(u)\,du + \mathsf{B}\mathsf{D}^{-1}\mathsf{C}a(t) \right\| \le \|\mathsf{B}\| \|\mathsf{C}\| \left(\frac{e^{-\gamma t}}{\gamma} \|a(t)\| + \frac{M_t}{\gamma^2} \right). \tag{84}$$

To obtain the bound, replace $a(t-v)$ by $a(t)$, extend the integral in v to infinity, and use $\|e^{v\mathsf{D}}\| \le e^{-\gamma v}$ and $\|a(t-v) - a(t)\| \le v M_t$. Positivity of the approximate operator $\mathsf{A} - \mathsf{B}\mathsf{D}^{-1}\mathsf{C}$ as a probability generator must still be checked. A small memory residual, statistical sufficiency, and a slow spectrum are distinct conditions for an autonomous predictive model (Proposition 33).

8.6 Declared Rate Separation and the Adiabatic Hypothesis

When a useful fast–slow split has been identified or deliberately imposed, write its coordinates as (f, w). Let $\mathcal{V}$ denote either the normalized collective free energy $\mathcal{F}_V$ or the separately declared consensus potential $\mathcal{E}$; the choice is held fixed throughout a calculation. With positive-definite block metrics, a declared rate separation in slow-time units is

$$\epsilon \dot{f} = -\mathsf{G}_f^{-1} \partial_f \mathcal{V}, \qquad \dot{w} = -\mathsf{G}_w^{-1} \partial_w \mathcal{V}, \tag{85}$$

which satisfies

$$\frac{d\mathcal{V}}{dt} = -\epsilon^{-1} (\partial_f \mathcal{V})^\top \mathsf{G}_f^{-1} \partial_f \mathcal{V} - (\partial_w \mathcal{V})^\top \mathsf{G}_w^{-1} \partial_w \mathcal{V} \le 0. \tag{86}$$

A full correlated recognition family generally requires its full Fisher metric; arbitrary block preconditioning is a declared kinetic metric. A product family with independently parameterized factors gives the usual Fisher blocks. With latent links, these include the Gaussian generator recognition block of Section 6; moving its mean and precision is recognition, while moving an anchor or prior

covariance is generative-parameter learning. Assigning generators to the slow coordinates w is an additional rate assumption, not a consequence of their being relations. Priors and generative parameters remain fixed during a recognition update; parameter learning changes the fixed joint between evaluations, with its own acceptance rule: a new parameter value is accepted only if it does not increase $\mathcal{F}_V$ at the fixed recognition law, as in the free-energy reading of expectation maximization [26].

If attention rows are optimized exactly, their entropy and prior terms must remain in the objective. The envelope identity permits holding an optimal row fixed when differentiating the reduced scalar. Dropping that entropy and then differentiating the row's state dependence defines a different scalar. Similarly, eliminating $q(\cdot \mid m)$ from Eq. (70) is an optimizer identity and does not assert that a finite-rate conditional law has already equilibrated.

Suppose $\mathcal{V}$ is twice continuously differentiable, the block metrics are continuously differentiable and positive definite, and G_w is independent of f. On a compact slow region assume a continuously differentiable branch $f^*(w)$ of conditional minimizers with the uniform fast-curvature bound

$$\partial_{ff}^2 \mathcal{V}(f^*(w), w) \succeq c\mathsf{G}_f(f^*(w), w), \qquad c > 0. \tag{87}$$

Assume the reduced solution stays in the interior of this region, the initial fast state lies in the basin of the frozen fast solution, and the vector fields have the local regularity and bounded neighborhoods required for existence throughout the time interval considered. Then the standard finite-time singular-perturbation conclusion is convergence of w_ϵ to the reduced solution on that interval and convergence of f_ϵ to $f^*(w)$ away from the initial layer [27]. With sufficiently smooth data, a compact normally hyperbolic portion has a nearby invariant slow manifold [28]. The envelope identity gives

$$d\mathcal{V}_{\mathrm{red}} = \partial_w \mathcal{V}(f^*(w), w)dw, \qquad \mathcal{V}_{\mathrm{red}}(w) = \mathcal{V}(f^*(w), w). \tag{88}$$

At finite ϵ the conditional-minimum graph is generally not invariant: the fast vector field vanishes there while motion of w would require the tangent velocity $Df^*(w)\dot{w}$.

The stated hypotheses are conditional, not established for every selective-link network. After a Gibbs attention row is eliminated, its curvature is

$$\nabla^2 \left[-\tau \log \sum_j \pi_j e^{-E_j/\tau} \right] = \sum_j \beta_j^* \nabla^2 E_j - \tau^{-1} \operatorname{Cov}_{\beta^*}(\nabla E_j). \tag{89}$$

The negative covariance term can defeat the required positive curvature even when each frozen-weight Hessian is positive. At folds, changing supports, and source jumps a separate analysis is needed. Held-sheet results and supplied slow-model mobilities do not establish general recursive formation; a slow-model memory can produce persistence and delay without creating a second restoring interaction (Appendices B.2 and B.3).

8.7 A collective slow coordinate in one statistical state

A joint statistical description can support fast internal adjustment and a slower collective belief without introducing a second bundle or a mechanical momentum. The following exact example specifies where this separation comes from. Work at one base point with a fixed connected undirected network of $N \geq 2$ nodes. Let $W_{ij} = W_{ji} \geq 0$, $W_{ii} = 0$, and suppose every weighted degree is the same positive number $d = \sum_j W_{ij}$. Its graph Laplacian is $L = dI - W$, with eigenvalues $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_N$. For supplied constants $\epsilon, k > 0$, set

$$H = \epsilon I + kL, \qquad p(z) = \frac{\det(H)^{1/2}}{(2\pi)^{N/2}} \exp\left(-\frac{1}{2} z^T H z \right), \qquad z \in \mathbb{R}^N. \tag{90}$$

Thus $p = \mathcal{N}(0, H^{-1})$ is a normalized, correlated prior. There are no observations in this example. Equivalently, the normalized VFE below is the complexity contribution of an unobserved model. The graph and the weak collective curvature ϵ are inputs, not inferred here.

Restrict the joint belief to the node-factorized family

$$q(z) = \prod_{i=1}^{N} \mathcal{N}(z_i; \mu_i, s_i), \qquad (\mu, s) \in \mathbb{R}^N \times (0, \infty)^N. \tag{91}$$

For this single joint state, the exact normalized VFE is

$$\mathcal{F}_V = \mathrm{KL}(q\|p) = \frac{1}{2}\left[\mu^T H \mu + \sum_i H_{ii} s_i - N - \log\det H - \sum_i \log s_i\right]. \tag{92}$$

Use continuous natural-gradient descent with one common rate $\eta > 0$ and the Fisher metric of the specified product family. Its mean and variance blocks are $G_{\mu_i\mu_i} = s_i^{-1}$ and $G_{s_i s_i} = (2s_i^2)^{-1}$, with zero cross-block terms. In particular, the covariance convention is $\dot{\Sigma} = -2\eta\Sigma(\partial_\Sigma \mathcal{F}_V)\Sigma$ before restriction to diagonal coordinates. Differentiating the displayed VFE gives the exact equations

$$\dot{\mu} = -\eta\,\mathrm{diag}(s_i) H \mu, \qquad \dot{s}_i = \eta(s_i - H_{ii} s_i^2). \tag{93}$$

There is no stochastic forcing or temporal discretization in this construction. Its time parameter is the supplied learning time. Along its solutions,

$$\frac{d\mathcal{F}_V}{dt} = -\eta \sum_i s_i (\partial_{\mu_i} \mathcal{F}_V)^2 - 2\eta \sum_i s_i^2 (\partial_{s_i} \mathcal{F}_V)^2 \le 0. \tag{94}$$

Regularity of the graph makes every diagonal entry $H_{ii} = \alpha := \epsilon + kd$ equal. The precision variables $h_i = s_i^{-1}$ solve

$$\dot{h}_i = \eta(\alpha - h_i), \qquad h_i(t) = \alpha + [h_i(0) - \alpha] e^{-\eta t}. \tag{95}$$

For every positive initial variance this remains positive for all $t \ge 0$, and hence defines a global positive variance solution. The mean equation is then a linear equation with continuous bounded coefficients on every finite time interval, so the full solution exists uniquely for all $t \ge 0$. The covariance manifold $s_i = \alpha^{-1}$ is invariant. On that manifold,

$$\dot{\mu} = -\frac{\eta}{\alpha}(\epsilon I + kL)\mu. \tag{96}$$

Choose an orthonormal eigenbasis v_j of L, with $v_1 = N^{-1/2}\mathbf{1}$. Each coordinate $a_j = v_j^T \mu$ then satisfies the exact law

$$a_j(t) = a_j(0) \exp\left[-\frac{\eta(\epsilon + k\lambda_j)}{\epsilon + kd} t\right]. \tag{97}$$

In particular, the collective mean $X = N^{-1}\sum_i \mu_i$ obeys

$$\dot{X} = -\frac{\eta\epsilon}{\epsilon + kd} X, \qquad \frac{\tau_{\text{collective}}}{\tau_{\text{internal}}} = \frac{\epsilon + k\lambda_2}{\epsilon}. \tag{98}$$

Here $\tau_{\text{internal}} = (\epsilon + kd)/[\eta(\epsilon + k\lambda_2)]$ is the longest relaxation time among mean modes orthogonal to $\mathbf{1}$. Consequently, for any requested factor $R > 1$, the separation is at least R whenever $k\lambda_2 \ge (R-1)\epsilon$.

Equal initial variances away from the invariant manifold retain these eigenvectors with a time-dependent common rate. Unequal variances generally prevent an autonomous equation for X during the covariance transient. On an irregular graph the diagonal fixed variances differ, so the simple arithmetic-mean law above is not claimed.

The separation also depends on the variational family. If instead q is an unrestricted full Gaussian and its covariance starts at $\Sigma = H^{-1}$, then

$$\dot{\Sigma} = \eta(\Sigma - \Sigma H \Sigma) = 0, \qquad \dot{\mu} = -\eta \Sigma H \mu = -\eta \mu. \tag{99}$$

Every mean direction now relaxes at rate η, despite the identical correlated target. In the factorized family, local Fisher resistance cannot cancel the target's off-diagonal curvature; the weaker collective restoring direction therefore remains slow. This supplies a conditional mechanism for a persistent collective belief within one statistical state. It does not select the network, the factorization, or the collective observable, establish recursive agent formation, or derive Newtonian inertia. Those require further structure and evidence.

8.8 Gauge Frames and Independent Interaction Links

Passive changes of frame re-express a law and every incident channel together; they do not evolve an agent. In the frame-factorized special case $\Omega_{ij} = U_i U_j^{-1}$, common-pushforward invariance removes the frames from alignment terms by using $(U_i^{-1})_{\#} q_i$. If an observation channel is held fixed while a frame is changed, that operation changes a relative coupling and is an active model change. Independent L_{ij} links carry interaction data beyond the choice of frame. Their dynamics and metric must be specified in addition to the statistical objective. No frame-gradient term or transport along the base is required for this pointwise statement.

8.9 Gaussian Natural Gradients and Frame Retractions

The Gaussian formulas remain useful within the single-bundle description. For a joint Gaussian section $\zeta_i = \mathcal{N}(\mu_i, \Sigma_i)$, or a declared Gaussian marginal block, the Fisher metric is

$$g_{\mu\mu} = \Sigma_i^{-1}, \qquad g_{\Sigma\Sigma}[V, W] = \tfrac{1}{2} \operatorname{tr}(\Sigma_i^{-1} V \Sigma_i^{-1} W), \tag{100}$$

with zero mean–covariance block [29]. The full joint covariance includes state–model correlations. If frames are evolved under a separately specified metric, the local equations can be written

$$\dot{\mu}_i = -\eta_\mu \Sigma_i \nabla_{\mu_i} \mathcal{V}, \tag{101}$$

$$\dot{\Sigma}_i = -2\eta_\Sigma \Sigma_i (\nabla_{\Sigma_i} \mathcal{V}) \Sigma_i, \tag{102}$$

$$\dot{U}_i = -\eta_\phi U_i \xi_i. \tag{103}$$

For unequal η_μ and η_Σ, this is natural descent in a rate-weighted metric; it is a common-rate Fisher flow only when the rates agree. A covariance retraction must preserve positive definiteness. Continuous local descent alone supplies no global existence claim.

In a regular exponential chart $U_i = \exp \phi_i$, the chart vector and the left-trivialized group velocity are related by

$$v_{\phi_i} = \mathcal{G}_{\text{pull}}^{-1}(\phi_i) \nabla_{\phi_i} \mathcal{V}, \qquad \xi_i = \operatorname{dexp}_{\phi_i}^{L}(v_{\phi_i}), \qquad \operatorname{dexp}_{\phi}^{L}(v) = e^{-\phi} D \exp_{\phi}[v]. \tag{104}$$

One possible choice is the pullback of the ambient Frobenius metric,

$$\begin{aligned}\mathcal{G}_{\mathrm{pull},ab}(\phi) &= \langle D\exp_\phi[T_a], D\exp_\phi[T_b]\rangle_F \\ &= \mathrm{tr}\left[\Psi_R(\mathrm{ad}_\phi)(T_a)^\top \Psi_R(\mathrm{ad}_\phi)(T_b) e^{\phi} e^{\phi^\top}\right], \qquad \Psi_R(z) = \frac{e^z - 1}{z},\end{aligned} \tag{105}$$

with the value at $z = 0$ defined continuously. This metric is a coordinate construction, not Fisher information of the agent law.

Proposition 5 (Regular exponential charts)**.** *The operator* dexp_ϕ^L *is invertible exactly when the spectrum of* ad_ϕ *contains no nonzero element of* $2\pi i\mathbb{Z}$*. The pullback metric is positive definite on precisely this locus.*

Proof. The integral formula $\mathrm{dexp}_\phi^L = \int_0^1 e^{-u\,\mathrm{ad}_\phi} du$ is the analytic matrix function $(1 - e^{-z})/z$ of ad_ϕ. Its zeros are exactly the stated nonzero imaginary multiples. The determinant of an analytic matrix function is the product of its values on the eigenvalues, including multiplicities. The Gram form in Eq. (105) is nondegenerate exactly when this differential is injective. □

The corresponding group retraction is

$$U_i^{t+1} = U_i^t \exp(-\eta_\phi \xi_i^t). \tag{106}$$

It remains in G and agrees with the chart update through first order. For the full $\mathrm{GL}^+(K)$ chart, the ambient pullback gives $\xi_i = U_i^{-1}\nabla_{U_i}\mathcal{V}$, hence ambient Euclidean descent to first order. A fixed Frobenius inner product on the body velocity instead defines a left-invariant group metric and gives $\dot{U}_i = -\eta_\phi U_i U_i^\top \nabla_{U_i}\mathcal{V}$. A right-invariant choice gives its right-trivialized counterpart. These are different kinetic assumptions.

For $K \geq 2$, no fixed positive-definite inner product on $\mathfrak{gl}(K)$ is invariant under every adjoint action: conjugation by $\mathrm{diag}(e^t, e^{-t}, 1, \ldots, 1)$ multiplies E_{12} by e^{2t}, contradicting invariance of its positive norm. Thus a fixed bi-invariant Riemannian group metric is unavailable in this case. This does not rule out law-dependent covariant metrics on the complete agent state. Any claim about frame relaxation must specify the actual metric and transformation of all coupled fields; scalar invariance of the objective alone is insufficient.

9 Statistical Precision as Stiffness and Friction

For the interaction-Hessian calculations in this section, all edges join distinct agents. If a self edge is retained elsewhere, its mean Hessian is computed from the combined residual $(I - L_{ii})\mu_i$, giving $(I - L_{ii})^\top W_{ii}(I - L_{ii})$ rather than two separate endpoint blocks.

In the full-state Gaussian consensus specialization, the natural-gradient flow of Eq. (103) descends the consensus potential $\mathcal{E}$ of Section 5, which is therefore the potential of the agent dynamics. This section computes the second variation of $\mathcal{E}$ in the Gaussian belief sector and derives the two roles of statistical precision in that first-order dynamics (status D, Table 2). The Hessian is a stiffness tensor whose blocks are built from precisions: the prior term contributes the prior precision $\bar{\Lambda}_{p_i}$, and the interaction terms contribute transported neighbor precisions. The Fisher metric, which in the mean sector is the belief precision, is the friction tensor of the flow, and no term of the dynamics carries an inertia. The only velocity-quadratic form in the dynamics is the Rayleigh dissipation function $\mathcal{R}$, which is half the free-energy dissipation rate and fixes the rate of the information clock of Section 18 through $(d\vartheta/dt)^2 = 2\eta\mathcal{R}$ (Section 9.6). An inertial reading of precision needs a separate kinetic-metric postulate with its own second-order equation of motion, stated in Section 9.8.

Stiffness, friction, and inertia. For a damped oscillator $m\ddot{x} + \gamma\dot{x} + kx = 0$ the three coefficients are independent data: the Hessian of the potential gives the stiffness k, the dissipation gives the friction γ, and the kinetic energy gives the mass m. The Hessian of $\mathcal{E}$ derived below supplies k, the Fisher metric of the natural-gradient flow supplies γ, and neither $\mathcal{E}$ nor the flow supplies m. We hold the prior parameters $(\bar{\mu}_i, \bar{\Sigma}_i)$ fixed, the quasi-static approximation in which they evolve slowly relative to the beliefs (μ_i, Σ_i).

Scope of this calculation. The detailed covariance Hessians below concern the legacy consensus potential $\mathcal{E}$, with invertible full-state transports and fixed Gaussian prior and observation laws at one base point. They are not formulas for an arbitrary normalized collective VFE $\mathcal{F}_V$. In particular, the inverse-link formulas cannot be used with a rank-deficient selective communication map. The independent invertible-link case also need not be flat: every additional use of common transported consensus must be checked against its cycle holonomy. A later subsection gives the mean stiffness for general Gaussian communication channels without an inverse link. All these claims concern continuous dynamics; neither a numerical step nor physical time is specified by the potential alone.

9.1 Setup and Notation

Each agent i maintains a belief distribution $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ anchored to a prior $p_i = \mathcal{N}(\bar{\mu}_i, \bar{\Sigma}_i)$ and receives observations o_i through the identity readout $p(o_i \mid k_i, m_i) = \mathcal{N}(o_i; k_i, \Sigma_{o_i})$, whose likelihood does not depend on the model parameter m_i and is written $p(o_i \mid k_i)$ below. We define precision matrices

$$\Lambda_{q_i} = \Sigma_i^{-1} \qquad \text{(belief precision)} \tag{107}$$

$$\bar{\Lambda}_{p_i} = \bar{\Sigma}_i^{-1} \qquad \text{(prior precision)} \tag{108}$$

$$\Lambda_{o_i} = \Sigma_{o_i}^{-1} \qquad \text{(observation precision)}, \tag{109}$$

with transported quantities $\tilde{\mu}_k = \Omega_{ik}\mu_k$ and $\tilde{\Sigma}_k = \Omega_{ik}\Sigma_k\Omega_{ik}^\top$, where $\Omega_{ik} \in \mathrm{GL}(K_q)$ is the transport of the interaction term: in Regime I the frame quotient $e^{\phi_i}e^{-\phi_k}$, and in Regime II an independent link on the interaction graph with orientation reversal $\Omega_{ki} = \Omega_{ik}^{-1}$. The block formulas below use only the transports and hold in both regimes. The transported precision is the inverse of the transported covariance,

$$\tilde{\Lambda}_{q_k} := (\Omega_{ik}\Sigma_k\Omega_{ik}^\top)^{-1} = \Omega_{ik}^{-\top}\Lambda_{q_k}\Omega_{ik}^{-1}, \tag{110}$$

which differs from the covariance transport law because precision is the inverse of covariance. For orthogonal transport $\Omega \in \mathrm{O}(K_q)$ the two laws coincide ($\Omega^{-\top} = \Omega$), and Eq. (110) reduces to $\Omega\Lambda\Omega^\top$; for general $\mathrm{GL}(K_q)$ the dual transport is essential, and we use Eq. (110) throughout. The boxed identity $\Omega_{ji}^\top\tilde{\Lambda}_{q_i}^{(j)}\Omega_{ji} = \Lambda_{q_i}$ used below holds in $\mathrm{GL}(K_q)$ because $\Omega^\top\Omega^{-\top} = I$ algebraically; the secondary simplifications $\tilde{\Lambda}_k\Omega_{ik} = \Omega_{ik}\Lambda_k$ and $\Omega_{ki}^\top\tilde{\Lambda}_i = \Lambda_i\Omega_{ki}^\top$ hold only under $\mathrm{O}(K_q)$; the general-linear identity is given below.

9.2 The Gaussian Consensus Potential

The extended working functional at one base point, the fixed-β^* block-coordinate form of Eq. (39) with explicit sensory evidence (the attention-entropy block is omitted as constant in the beliefs; this is not the entropy-suppressed surrogate of Section 5.6), is

$$\mathcal{E}[\{q_i\}] = \sum_i D_{\mathrm{KL}}(q_i \| p_i) + \sum_{i,k} \beta_{ik} D_{\mathrm{KL}}(q_i \| \Omega_{ik}[q_k]) - \sum_i \mathbb{E}_{q_i}[\log p(o_i \mid k_i)]. \tag{111}$$

The three terms represent prior anchoring, interaction through the declared transports, and sensory evidence. Here p_i is supplied before the update. A parent predictive law is one possible source of such a prior, conditional on a specified parent model and observation history; a parent is not required for this calculation. Under the product recognition law $q_i \otimes s_i$ and the identity readout, the jointly typed observation term $-\mathbb{E}_{q_i \otimes s_i}[\log p(o_i \mid k_i, m_i)]$ of Eq. (39) reduces to the q_i-expectation shown, and at fixed s_i the model-fiber terms are constant in (μ_i, Σ_i) and drop out of every variation below. Under the frozen-β^* convention adopted for this section (the differentiation-convention paragraph below), the omitted attention-entropy term $\tau\beta_{ik}\log(\beta_{ik}/\pi_{ik})$ of the canonical functional is an additive constant in (μ, Σ), so the first and second variations taken under that convention coincide with the frozen-β variations of Eq. (39).

9.3 Component Free Energies for Gaussians

For $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$, the relative entropy is given by the closed-form expression of Eq. (15), applied with the substitution $(\mu_1, \Sigma_1, \mu_2, \Sigma_2) \to (\mu_q, \Sigma_q, \mu_p, \Sigma_p)$.

For the Gaussian likelihood $p(o_i \mid k_i) = \mathcal{N}(o_i; k_i, \Sigma_{o_i})$, the expected log-likelihood evaluates to

$$-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)] = \frac{1}{2}(o_i - \mu_i)^\top \Lambda_{o_i}(o_i - \mu_i) + \frac{1}{2}\operatorname{tr}(\Lambda_{o_i}\Sigma_i) + \text{const.} \tag{112}$$

9.4 First Variations (Gradient)

Prior Term

$$\frac{\partial D_{\mathrm{KL}}(q_i \| p_i)}{\partial \mu_i} = \bar{\Lambda}_{p_i}(\mu_i - \bar{\mu}_i), \tag{113}$$

$$\frac{\partial D_{\mathrm{KL}}(q_i \| p_i)}{\partial \Sigma_i} = \frac{1}{2}(\bar{\Lambda}_{p_i} - \Lambda_{q_i}). \tag{114}$$

Interaction Term With respect to receiver i:

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \mu_i} = \tilde{\Lambda}_{q_k}(\mu_i - \tilde{\mu}_k), \tag{115}$$

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \Sigma_i} = \frac{1}{2}(\tilde{\Lambda}_{q_k} - \Lambda_{q_i}). \tag{116}$$

With respect to sender k (using the general-linear precision transport of Eq. (110)):

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \mu_k} = \Lambda_{q_k}\Omega_{ik}^{-1}(\tilde{\mu}_k - \mu_i), \tag{117}$$

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \Sigma_k} = \frac{1}{2}\Omega_{ik}^\top \left[\tilde{\Lambda}_{q_k} - \tilde{\Lambda}_{q_k}\big(\Sigma_i + \tilde{d}_{ik}\tilde{d}_{ik}^\top\big)\tilde{\Lambda}_{q_k}\right]\Omega_{ik}, \tag{118}$$

where $\tilde{d}_{ik} := \tilde{\mu}_k - \mu_i = \Omega_{ik}\mu_k - \mu_i$ is the receiver-frame mean residual. The quadratic-form contribution $-\tilde{\Lambda}_{q_k}\tilde{d}_{ik}\tilde{d}_{ik}^\top\tilde{\Lambda}_{q_k}$ originates in differentiating the Mahalanobis term $\tilde{d}_{ik}^\top\tilde{\Lambda}_{q_k}\tilde{d}_{ik}$ with respect to $\tilde{\Sigma}_k$ via $\partial\tilde{\Lambda}_{q_k}/\partial\tilde{\Sigma}_k = -\tilde{\Lambda}_{q_k} \otimes \tilde{\Lambda}_{q_k}$, then chain-ruled through $\tilde{\Sigma}_k = \Omega_{ik}\Sigma_k\Omega_{ik}^\top$. The outer $\Omega_{ik}^\top(\cdot)\Omega_{ik}$ structure follows from the same chain rule applied to the leading $\tilde{\Lambda}_{q_k}$ and $\tilde{\Lambda}_{q_k}\Sigma_i\tilde{\Lambda}_{q_k}$ contributions; for the mean gradient, the same chain rule yields the explicit inverse Ω_{ik}^{-1}, which for orthogonal Ω would equal $\Omega_{ik}^\top$. The $\tilde{d}_{ik}\tilde{d}_{ik}^\top$ term vanishes at consensus ($\tilde{d}_{ik} = 0$), so the boxed stiffness blocks of Eqs. (123)–(130), evaluated at consensus, are unchanged. Off consensus this is a rank-one contribution to the covariance gradient in the precision-weighted residual direction; a second-variation calculation must differentiate this contribution as well.

Sensory Term

$$\frac{\partial}{\partial \mu_i}\left[-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)]\right] = \Lambda_{o_i}(\mu_i - o_i), \tag{119}$$

$$\frac{\partial}{\partial \Sigma_i}\left[-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)]\right] = \frac{1}{2}\Lambda_{o_i}. \tag{120}$$

Total Gradient

$$\frac{\partial \mathcal{E}}{\partial \mu_i} = \bar{\Lambda}_{p_i}(\mu_i - \bar{\mu}_i) + \sum_k \beta_{ik}\tilde{\Lambda}_{q_k}(\mu_i - \tilde{\mu}_k) + \sum_j \beta_{ji}\Lambda_{q_i}\Omega_{ji}^{-1}(\tilde{\mu}_i^{(j)} - \mu_j) + \Lambda_{o_i}(\mu_i - o_i), \tag{121}$$

where $\tilde{\mu}_i^{(j)} = \Omega_{ji}\mu_i$ is agent i's mean transported into agent j's frame.

9.5 Second Variations: The Stiffness Tensor

The frozen-β Hessian of the free energy, $M = \partial^2\mathcal{E}/\partial\xi\partial\xi$ with the attention weights held at β^* (see the differentiation-convention paragraph below), is a symmetric bilinear form on belief configuration space, the stiffness tensor of the potential $\mathcal{E}$. It is positive definite in the covariance sector at the reciprocal-consensus point established below, and it is not in general a metric: the reduced-functional correction and the state-dependent prior weight discussed below both add negative-semidefinite terms away from consensus. The Hessian is taken in the flat affine chart $(\mu_i, \Sigma_i) \in \mathbb{R}^{K_q} \times \mathrm{Sym}(K_q)$. A coordinate Hessian is a tensor, independent of the chart, only at a critical point of $\mathcal{E}$, and consensus is a critical point only when the prior and sensory terms are also stationary there. Away from critical points the chart matters: for $f(\Sigma) = D_{\mathrm{KL}}(\mathcal{N}(0,\Sigma)\|\mathcal{N}(0,\Sigma_p)) + \frac{1}{2}\operatorname{tr}(\Lambda_o\Sigma)$ the second variations along the same tangent vectors in the Σ chart and in the precision chart $\Lambda = \Sigma^{-1}$ agree at the critical point but differ by 49 percent at a noncritical point. A chart-free alternative is the covariant Hessian of the Fisher Levi-Civita connection, which differs from the coordinate Hessian by a Christoffel-symbol term proportional to the gradient and therefore coincides with it at critical points. The full state vector is $\xi = (\mu_1, \ldots, \mu_N, \Sigma_1, \ldots, \Sigma_N)$, yielding the block structure

$$M = \begin{pmatrix} M_{\mu\mu} & C^{\mu\Sigma} \\ (C^{\mu\Sigma})^\top & M_{\Sigma\Sigma} \end{pmatrix}, \tag{122}$$

where each block is itself an $N \times N$ matrix of sub-blocks.

Notation hierarchy. We use three related but distinct objects throughout: (i) $M_{\mu\mu}$ (a matrix), the full Hessian of the mean sector defined in (122); (ii) $\bar{\Lambda}_{p_i} = \bar{\Sigma}_i^{-1}$ (a matrix), the prior precision of agent i, which is the whole of $[M_{\mu\mu}]_{ii}$ when inter-agent coupling and observation likelihoods are absent ($\beta_{ij} = 0$, $\Lambda_{o_i} = 0$); and (iii) the scalar stiffness $\bar{M}_i := \operatorname{tr}([M_{\mu\mu}]_{ii})/K_q$, used when a one-number summary is needed. When unqualified, "the stiffness tensor" refers to (i) and "stiffness" as a scalar refers to (iii).

Differentiation convention. The second-variation expressions below treat the attention weights β_{ij} as fixed at their softmax-equilibrium values β_{ij}^*. The envelope theorem licenses this frozen-β convention at first order only: for the reduced free energy with β eliminated, the true Hessian adds the correction $-\tau^{-1}\mathrm{Cov}_{\beta^*}(\nabla E_{ij}, \nabla E_{ij})$ to the frozen-β blocks (the reduced-Hessian identity (363) of Appendix I), a negative-semidefinite term that vanishes exactly at consensus, where the per-edge gradients ∇E_{ij} coincide, but persists at generic equilibria in which prior, interaction, and sensory pulls

balance with nonzero per-edge gradients, so the frozen-β blocks overstate the curvature of the reduced functional away from consensus. Differentiating through the softmax rows produces these cross terms through $\partial\beta_{ij}/\partial\mu$ and $\partial\beta_{ij}/\partial\Sigma$. The correction is bounded: writing $D_i = \max_{j,k} \|\nabla E_{ij} - \nabla E_{ik}\|$ for the per-edge gradient diameter, the identity $\mathrm{Cov}_\beta(X) = \frac{1}{2}\sum_{j,k} \beta_j\beta_k(X_j - X_k)(X_j - X_k)^\top$ gives the spectral bound $\|\tau^{-1}\mathrm{Cov}_{\beta^*}(\nabla E_{ij})\| \le D_i^2/(2\tau)$. At nondegenerate energy gaps the covariance decays exponentially in the gap over τ, so the correction vanishes in the sharp-attention limit away from ties and is large precisely near attention switching points, where per-edge energies nearly coincide at small τ; this is the rapid-reorganization regime in which the frozen-β blocks are unreliable. The boxed forms below are written for general $\beta^*_{ij} \ge 0$ under this convention and reduce to the isolated-agent limit by setting $\beta^*_{ij} = 0$ in the sums. The first variations of the preceding subsection need no such convention, since by the envelope theorem the frozen-β gradient equals the gradient of the reduced functional at $\beta = \beta^*$; the convention is stated here because it first matters at second order.

Mean Sector

Diagonal blocks $(i = k)$**:** From the prior we obtain $\partial^2 D_{\mathrm{KL}}(q_i\|p_i)/\partial\mu_i\partial\mu_i^\top = \bar{\Lambda}_{p_i}$. From the interaction term as receiver, $\partial^2 D_{\mathrm{KL}}(q_i\|\tilde{q}_k)/\partial\mu_i\partial\mu_i^\top = \tilde{\Lambda}_{q_k}$. From the interaction term as sender to agent j, $\partial^2 D_{\mathrm{KL}}(q_j\|\tilde{q}_i)/\partial\mu_i\partial\mu_i^\top = \Omega_{ji}^\top \tilde{\Lambda}_{q_i}^{(j)} \Omega_{ji} = \Lambda_{q_i}$. From sensory evidence, $\partial^2[-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)]]/\partial\mu_i\partial\mu_i^\top = \Lambda_{o_i}$. The total diagonal stiffness block is

$$\boxed{[M_{\mu\mu}]_{ii} = \bar{\Lambda}_{p_i} + \sum_k \beta_{ik}\tilde{\Lambda}_{q_k} + \sum_j \beta_{ji}\Lambda_{q_i} + \Lambda_{o_i}.} \tag{123}$$

Off-diagonal blocks $(i \neq k)$**:** From $D_{\mathrm{KL}}(q_i\|\tilde{q}_k)$, using Eq. (110),

$$\frac{\partial^2 D_{\mathrm{KL}}(q_i\|\tilde{q}_k)}{\partial\mu_i\partial\mu_k^\top} = -\tilde{\Lambda}_{q_k}\Omega_{ik} = -\Omega_{ik}^{-\top}\Lambda_{q_k}, \tag{124}$$

where the second equality follows from $\tilde{\Lambda}_{q_k}\Omega_{ik} = \Omega_{ik}^{-\top}\Lambda_{q_k}\Omega_{ik}^{-1}\Omega_{ik} = \Omega_{ik}^{-\top}\Lambda_{q_k}$. From $D_{\mathrm{KL}}(q_k\|\tilde{q}_i)$ (if k also attends to i),

$$\frac{\partial^2 D_{\mathrm{KL}}(q_k\|\tilde{q}_i)}{\partial\mu_i\partial\mu_k^\top} = -\Lambda_{q_i}\Omega_{ki}^{-1}. \tag{125}$$

The sensory term does not couple different agents. Therefore

$$\boxed{[M_{\mu\mu}]_{ik} = -\beta_{ik}\Omega_{ik}^{-\top}\Lambda_{q_k} - \beta_{ki}\Lambda_{q_i}\Omega_{ki}^{-1} \quad (i \neq k).} \tag{126}$$

Under $\mathrm{O}(K_q)$ this reduces to $-\beta_{ik}\Omega_{ik}\Lambda_{q_k} - \beta_{ki}\Lambda_{q_i}\Omega_{ki}^\top$.

Off-diagonal block caveats. As the Hessian of the scalar functional $\mathcal{E}$, the block matrix $M_{\mu\mu}$ is automatically symmetric in the bilinear-form sense $M_{\mu\mu}^\top = M_{\mu\mu}$, equivalently $[M_{\mu\mu}]_{ik} = [M_{\mu\mu}]_{ki}^\top$, by equality of mixed partial derivatives (Schwarz's theorem). The block-transpose identity is exactly the condition for $M_{\mu\mu}$ to define a quadratic form $\frac{1}{2}\delta\mu^\top M_{\mu\mu}\delta\mu$ on the stacked variable $(\mu_i)_i$, the second-order term of $\mathcal{E}$; no stronger blockwise-equality condition is required, and an individual block $[M_{\mu\mu}]_{ik}$ need not be symmetric. Where the block-transpose identity appears not to hold in displayed expressions, the cause is that the displayed $[M_{\mu\mu}]_{ik}$ has been restricted to a subset of the

contributions to the Hessian (for example a single sender-only or receiver-only contribution), and the symmetry is restored once both contributions are summed.

Row-asymmetric attention does not by itself destroy the scalar potential: fixed asymmetric weights still define the scalar $\sum_{ij} \beta_{ij} E_{ij}$, and state-dependent Gibbs weights in the canonical entropy-retaining objective reduce to $-\tau \sum_i \log Z_i$. In both cases the full receiver-plus-sender Hessian is symmetric, and the full flow is a gradient flow for every fixed β, reciprocal or not. A nonpotential running rule appears only after the additional receiver-only truncation of Eq. (344), which discards the sender-row derivatives. For that optional field the bilinear form $M_{\mu\mu}$ is not the linearization of the dynamics; the linearization is the Jacobian of the truncated field, which is not symmetric and can have complex eigenvalues. The boxed forms of this section describe the full scalar $\mathcal{E}$ and do not describe the receiver-only field.

Covariance Sector For matrix-valued variables, we use the directional derivative convention

$$\frac{\partial^2 f}{\partial \Sigma \partial \Sigma}[V, W] = \lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[\left. \frac{\partial f}{\partial \Sigma} \right|_{\Sigma + \epsilon W} - \left. \frac{\partial f}{\partial \Sigma} \right|_{\Sigma} \right][V], \tag{127}$$

with the identity $\partial(\Sigma^{-1})/\partial\Sigma = -\Sigma^{-1} \otimes \Sigma^{-1}$.

Diagonal blocks ($i = k$)**:** From the prior,

$$\frac{\partial^2 D_{\mathrm{KL}}(q_i \| p_i)}{\partial \Sigma_i \partial \Sigma_i}[V, W] = \frac{1}{2} \operatorname{tr}[\Lambda_{q_i} V \Lambda_{q_i} W]. \tag{128}$$

In tensor notation, $\partial^2 D_{\mathrm{KL}}(q_i \| p_i)/\partial\Sigma_i \partial\Sigma_i = \frac{1}{2}(\Lambda_{q_i} \otimes \Lambda_{q_i})$. From the interaction term as receiver and as sender, identical contributions arise *at consensus* (see the remark below). The sensory term $\frac{1}{2} \operatorname{tr}(\Lambda_{o_i} \Sigma_i)$ is linear in Σ_i, so its second derivative vanishes. Therefore

$$\boxed{[M_{\Sigma\Sigma}]_{ii} = \frac{1}{2}(\Lambda_{q_i} \otimes \Lambda_{q_i}) \cdot \left(1 + \sum_k \beta_{ik} + \sum_j \beta_{ji} \right) \quad \text{at consensus.}} \tag{129}$$

The sensory precision Λ_{o_i} contributes to the mean-sector stiffness but not to the covariance-sector stiffness.

Remark on the consensus simplification. The factor $(1 + \sum_k \beta_{ik} + \sum_j \beta_{ji})$ in Eq. (129) arises because the sender contribution from $D_{\mathrm{KL}}(q_j \| \tilde{q}_i)$ involves both a negative term $-\frac{1}{2}\Lambda_{q_i} \otimes \Lambda_{q_i}$ from $\partial^2 \ln |\tilde{\Sigma}_i| / \partial \Sigma_i^2$ and a positive trace-term contribution that, away from consensus, equals $\frac{1}{2}\partial^2 \operatorname{tr}((\Omega_{ji} \Sigma_i \Omega_{ji}^\top)^{-1} \Sigma_j)/\partial \Sigma_i^2$ and couples Σ_i to Σ_j. At consensus ($\Sigma_j = \Omega_{ji} \Sigma_i \Omega_{ji}^\top$) this trace-term contribution simplifies to $+\Lambda_{q_i} \otimes \Lambda_{q_i}$, and the sender net contribution collapses to $+\frac{1}{2}\Lambda_{q_i} \otimes \Lambda_{q_i}$, recovering Eq. (129). Off consensus, the sender contribution carries an explicit Σ_j-dependent cross-coupling and the diagonal block does not factor into a multiple of $\Lambda_{q_i} \otimes \Lambda_{q_i}$; the boxed form above is the consensus-point restriction.

Off-diagonal blocks ($i \neq k$)**:** From $D_{\mathrm{KL}}(q_i \| \tilde{q}_k)$, varying both Σ_i and Σ_k, with $\tilde{\Lambda}_{q_k}$ defined by Eq. (110),

$$\boxed{\begin{aligned} [M_{\Sigma\Sigma}]_{ik} = &-\frac{1}{2}\beta_{ik} \left(\Omega_{ik}^{-\top} \Lambda_{q_k} \otimes \Omega_{ik}^{-\top} \Lambda_{q_k} \right) \\ &-\frac{1}{2}\beta_{ki} \left(\Omega_{ki}^{-\top} \Lambda_{q_i} \otimes \Omega_{ki}^{-\top} \Lambda_{q_i} \right)^\top \quad (i \neq k). \end{aligned}} \tag{130}$$

The block is a pure Kronecker square of the two mean-sector transport factors. Each contribution is independent of the native covariance of the agent whose divergence produces it, a structural property any candidate form of this block must satisfy: agent i's native covariance Σ_i enters $D_{\mathrm{KL}}(q_i\|\tilde{q}_k)$ only linearly through the trace $\frac{1}{2}\,\mathrm{tr}(\tilde{\Lambda}_{q_k}\Sigma_i)$, so differentiating in Σ_i removes Σ_i entirely and leaves $\frac{1}{2}\tilde{\Lambda}_{q_k}$, and the subsequent differentiation in Σ_k through the precision transport of Eq. (110) produces the Kronecker square $\Omega_{ik}^{-\top}\Lambda_{q_k}\otimes\Omega_{ik}^{-\top}\Lambda_{q_k}$, which depends on Σ_k through Λ_{q_k} but carries no residual Σ_i. By the same argument the sender contribution is independent of Σ_k and depends on Σ_i through Λ_{q_i}, so the assembled block depends on both precisions while each term is free of its own agent's covariance. The two-term structure mirrors the mean-sector off-diagonal block of Eq. (126): the first term is the receiver contribution from $D_{\mathrm{KL}}(q_i\|\tilde{q}_k)$ when i attends to k, built from $A_{ik}=\Omega_{ik}^{-\top}\Lambda_{q_k}$; the second is the sender contribution from $D_{\mathrm{KL}}(q_k\|\tilde{q}_i)$ when k also attends to i, built from $B_{ki}=\Omega_{ki}^{-\top}\Lambda_{q_i}$. The transpose on the second Kronecker factor is essential. The sender mixed partial is naturally assembled with the index ordering (Σ_k,Σ_i), whereas the block $[M_{\Sigma\Sigma}]_{ik}$ requires the ordering (Σ_i,Σ_k); since $\Omega_{ki}^{-\top}\Lambda_{q_i}\otimes\Omega_{ki}^{-\top}\Lambda_{q_i}$ is not symmetric, the index swap is exactly the vec-space transpose, and it is what makes the assembled block satisfy the Schwarz block-transpose identity $[M_{\Sigma\Sigma}]_{ik}=[M_{\Sigma\Sigma}]_{ki}^{\top}$. As in the mean sector, this identity holds for every fixed β, reciprocal or not, while an individual block $[M_{\Sigma\Sigma}]_{ik}$ need not be symmetric. Under $\mathrm{O}(K_q)$ the factors reduce to $\Omega_{ik}\Lambda_{q_k}\otimes\Omega_{ik}\Lambda_{q_k}$.

At the reciprocal-consensus point, the covariance-sector Hessian restricted to the symmetric tangent space on which Σ lives is positive definite for all $\beta\geq 0$ and arbitrary precision heterogeneity. Take Regime I transports $\Omega_{ik}=\exp(\phi_i)\exp(-\phi_k)$, so that the cocycle holds and the transported neighbor $\tilde{q}_k$ coincides with q_i, with reciprocal β and each prior equal to the native belief. Every relative entropy then sits at its global zero, contributing a positive-semidefinite Hessian, while the strictly convex prior term $\frac{1}{2}(\Lambda_{q_i}\otimes\Lambda_{q_i})$ on each diagonal block makes the total strictly positive definite. Four scope conditions are required. The result is established only at the consensus point; off consensus the Σ_j cross-coupling re-enters and positive definiteness is not established there. It uses the frozen-β (envelope-theorem) convention of the differentiation paragraph above. It requires a global consensus through the cocycle, not merely pairwise reciprocity; the full $K_q^2\times K_q^2$ vec operator can be indefinite on the non-symmetric complement, which is not the physical object, since Σ is constrained to the symmetric subspace. And the point is a critical point of $\mathcal{E}$ only when the sensory term is absent: the sensory term adds no covariance curvature but contributes the constant covariance gradient $\frac{1}{2}\Lambda_{o_i}$, so with sensory evidence the statement concerns the flat-chart second variation at a noncritical point, with the chart dependence described at the start of this subsection.

Cross Mean-Covariance Blocks The prior term gives $\partial^2 D_{\mathrm{KL}}(q_i\|p_i)/\partial\mu_i\partial\Sigma_i=0$. The sensory free energy decomposes into a term quadratic in μ_i and a term linear in Σ_i; these are independent, so $[C^{\mu\Sigma}]_{ii}^{\mathrm{sensory}}=0$.

For the interaction term, from $\partial D_{\mathrm{KL}}(q_i\|\tilde{q}_k)/\partial\mu_i=\tilde{\Lambda}_{q_k}(\mu_i-\tilde{\mu}_k)$ with $\tilde{\Lambda}_{q_k}=\Omega_{ik}^{-\top}\Lambda_{q_k}\Omega_{ik}^{-1}$ (Eq. (110)), varying Σ_k gives $\delta\tilde{\Lambda}_{q_k}=-\Omega_{ik}^{-\top}\Lambda_{q_k}V\Lambda_{q_k}\Omega_{ik}^{-1}$ for $V=\delta\Sigma_k$, and therefore

$$\frac{\partial^2 D_{\mathrm{KL}}(q_i\|\tilde{q}_k)}{\partial\mu_i\partial\Sigma_k}[V]=-\Omega_{ik}^{-\top}\Lambda_{q_k}V\Lambda_{q_k}\Omega_{ik}^{-1}(\mu_i-\tilde{\mu}_k). \tag{131}$$

In components,

$$\frac{\partial^2 D_{\mathrm{KL}}}{\partial(\mu_i)_a\partial(\Sigma_k)_{bc}}=-[\Omega_{ik}^{-\top}\Lambda_{q_k}]_{ab}[\Lambda_{q_k}\Omega_{ik}^{-1}(\mu_i-\tilde{\mu}_k)]_c. \tag{132}$$

Under $\mathrm{O}(K_q)$ this reduces to $-\Omega_{ik}\Lambda_{q_k}V\Lambda_{q_k}\Omega_{ik}^{\top}(\mu_i-\tilde{\mu}_k)$. The cross block vanishes at consensus $(\mu_i=\tilde{\mu}_k)$:

$$\boxed{[C^{\mu\Sigma}]_{ik}=0\quad\text{when }\mu_i=\tilde{\mu}_k.} \tag{133}$$

9.6 Precision as Stiffness and the Fisher Metric as Friction

The mean-sector stiffness of agent i, in the frozen-β convention, is

$$M_i = \underbrace{\bar{\Lambda}_{p_i}}_{\text{prior}} + \underbrace{\sum_k \beta_{ik}\tilde{\Lambda}_{q_k}}_{\text{receiver}} + \underbrace{\sum_j \beta_{ji}\Lambda_{q_i}}_{\text{sender}} + \underbrace{\Lambda_{o_i}}_{\text{sensory}} . \tag{134}$$

The prior block resists deviation from the prior p_i, the receiver block resists deviation from the transported neighbors that agent i attends to, the sender block resists moves that would disturb the agents attending to i, and the sensory block anchors the agent to its observations. The stiffness is dimensionless on the representational space; no identification with a physical mass is made.

Constant versus state-dependent self-coupling. The boxed stiffness blocks of this section take the prior self-coupling weight at its canonical default $\alpha_i = 1$; the prior term of Eq. (111) carries unit weight. Under the state-dependent generalization of Section 7.1, where $\alpha_i^*(c) = c_0/(b_0 + D_{\mathrm{KL}}(q_i\|p_i))$ is recomputed from the current belief at each inner step, the prior block scales as $\alpha_i^*\bar{\Lambda}_{p_i}$. Because α_i^* co-varies with the belief, the second variation acquires a rank-one residual that the first-order envelope cancellation of Eq. (60) does not remove,

$$[M_{\mu\mu}]_{ii}^{\text{prior}} = \alpha_i^*\bar{\Lambda}_{p_i} - \frac{(\alpha_i^*)^2}{c_0}\left(\bar{\Lambda}_{p_i}\Delta\mu_i\right)\left(\bar{\Lambda}_{p_i}\Delta\mu_i\right)^\top, \qquad \Delta\mu_i = \mu_i - \bar{\mu}_i, \tag{135}$$

which vanishes only at $\mu_i = \bar{\mu}_i$. This is a prior-sector Hessian correction. It is distinct from the residual-dependent covariance-gradient term retained in the interaction calculation above. For a mean displacement at matched covariance, where $D_{\mathrm{KL}}(q_i\|p_i) = \frac{1}{2}\Delta\mu_i^\top\bar{\Lambda}_{p_i}\Delta\mu_i$, the regularized prior block stays positive definite until $D_{\mathrm{KL}} = b_0$, whereas the unregularized weight $\alpha_i^* D_{\mathrm{KL}}$ alone would lose definiteness already at $D_{\mathrm{KL}} = b_0/3$; the log-barrier delays it. Beyond that bound the prior contribution to the stiffness is indefinite. The boxed forms of this section are therefore the quasi-static $\alpha_i = 1$ specialization.

Friction and relaxation. The flow of Eq. (103) is first order. In the Gaussian sector it reads

$$\eta_b^{-1}G_b\dot{\xi}_b = -\nabla_{\xi_b}\mathcal{E}, \qquad G_\mu = \Lambda_q, \qquad G_\Sigma[V,W] = \tfrac{1}{2}\operatorname{tr}(\Sigma^{-1}V\Sigma^{-1}W), \tag{136}$$

for each block $b \in \{\mu, \Sigma\}$ with its rate η_b, where G_b is the Fisher metric of the block. This is the form $\gamma\dot{q} = -\nabla V$ of an overdamped system with friction tensor $\gamma = \eta^{-1}G$ and mobility ηG^{-1}, and it is the Rayleigh form $\partial\mathcal{R}/\partial\dot{\xi} = -\nabla\mathcal{E}$ with dissipation function

$$\mathcal{R}(\dot{\xi}) = \sum_b \frac{1}{2\eta_b}G_b(\dot{\xi}_b, \dot{\xi}_b), \qquad \frac{d\mathcal{E}}{dt} = \nabla\mathcal{E}\cdot\dot{\xi} = -2\mathcal{R}. \tag{137}$$

With a common rate $\eta_b = \eta$ the squared Fisher speed is $G(\dot{\xi},\dot{\xi}) = (d\vartheta/dt)^2 = 2\eta\mathcal{R}$, which is the clock-dissipation identity of Eq. (177) for an autonomous single-rate flow. The block-descent law for arbitrary positive block rates and the properties of the Fisher clock used here, including its reading along a flow with unequal block rates, are proved in Proposition 34. Linearized about a critical point ξ^*, the flow is $\delta\dot{\xi} = -\eta G^{-1}M\delta\xi$. Since G is positive definite and M is symmetric, $G^{-1}M$ is similar to $G^{-1/2}MG^{-1/2}$, so its spectrum is real and the linearized flow has no oscillatory modes; the optional receiver-only truncation of Eq. (344) is one way to leave this gradient-flow conclusion. This spectral claim is local at a critical point and does not cover a driven trajectory or

an added antisymmetric flow. For an isolated agent at $q_i = p_i$ the stiffness and the friction coincide, $[M_{\mu\mu}]_{ii} = \bar{\Lambda}_{p_i} = \Lambda_{q_i} = G_\mu$ and $[M_{\Sigma\Sigma}]_{ii} = \frac{1}{2}\Lambda_{q_i} \otimes \Lambda_{q_i} = G_\Sigma$, so the linearized mean and covariance relaxation rates are exactly η_μ and η_Σ, independent of the precision. Under a Euclidean gradient flow $\dot{\xi} = -\eta\nabla\mathcal{E}$ the rates would instead be η times the eigenvalues of M, so a precise agent would relax faster. In this isolated, matched-prior calculation high precision does not slow either flow; large precision produces large stiffness and small static compliance M^{-1}.

High-precision agents. Consider an agent with very sharp prior self-localization, $\bar{\Sigma}_i = \delta I$ with $\delta \ll 1$ a dimensionless precision scale to which no physical units are assigned (the sharp-belief example of Appendix B.1). In the isolated-agent limit of Eq. (134), where the interaction blocks switch off ($\beta_{ik} = \beta_{ji} = 0$) and the sensory block is negligible ($\Lambda_{o_i} \to 0$), M_i reduces to the prior precision $\delta^{-1}I$; the interaction and sensory terms add further stiffness when the agent is embedded. Such an agent has large stiffness and compliance of order δ, so a bounded static force displaces it by an amount of order δ; under the natural-gradient flow its linear relaxation rate is still η.

The stiffness above runs entirely on the state-fiber prior precision $\bar{\Lambda}_{p_i}$ and is therefore specific to the prior-pullback tier in the sense of Section 19.3. Under the structural-tier reading of Section 22.1, perceived geometry is carried instead by the model-fiber generative-model precision $\Sigma_{s,i}^{-1}$, so the stiffness of this section and the pulled-back geometry are distinct objects at that tier; relating them would require a dynamical link between $\bar{\Lambda}_{p_i}$ and $\Sigma_{s,i}^{-1}$ that the present construction does not supply.

9.7 Stiffness for selective communication

Let j send to i through $L_{ij} = R_i S_j$, with received covariance $C_{ij} = L_{ij}\Sigma_j L_{ij}^\top + D_{ij}$ and $D_{ij} \succ 0$. Define $W_{ij} = C_{ij}^{-1}$ and $d_{ij} = \mu_i - L_{ij}\mu_j$. For fixed links, noise covariances, and attention weights, an edge energy $E_{ij} = D_{\mathrm{KL}}(q_i \| \mathcal{N}(L_{ij}\mu_j, C_{ij}))$ has derivatives

$$\nabla_{\mu_i} E_{ij} = W_{ij} d_{ij}, \qquad \nabla_{\mu_j} E_{ij} = -L_{ij}^\top W_{ij} d_{ij}, \tag{138}$$

$$\nabla_{\Sigma_i} E_{ij} = \tfrac{1}{2}(W_{ij} - \Sigma_i^{-1}), \qquad \nabla_{\Sigma_j} E_{ij} = \tfrac{1}{2} L_{ij}^\top [W_{ij} - W_{ij}(\Sigma_i + d_{ij}d_{ij}^\top)W_{ij}] L_{ij}. \tag{139}$$

These formulas follow by differentiating $\log\det C_{ij}$ and C_{ij}^{-1} and using $\delta C_{ij} = L_{ij}\delta\Sigma_j L_{ij}^\top$. They require no inverse of L_{ij}. Holding covariances fixed, the complete mean Hessian is

$$[H_{\mu\mu}]_{ii} = \bar{\Lambda}_{p_i} + \Lambda_{o_i} + \sum_j \beta_{ij} W_{ij} + \sum_r \beta_{ri} L_{ri}^\top W_{ri} L_{ri}, \tag{140}$$

$$[H_{\mu\mu}]_{ij} = -\beta_{ij} W_{ij} L_{ij} - \beta_{ji} L_{ji}^\top W_{ji} \quad (i \neq j). \tag{141}$$

Its positivity can be checked without component expansion: for every mean variation $v = (v_i)_i$,

$$v^\top H_{\mu\mu} v = \sum_i v_i^\top (\bar{\Lambda}_{p_i} + \Lambda_{o_i}) v_i + \sum_{i,j} \beta_{ij} (v_i - L_{ij}v_j)^\top W_{ij} (v_i - L_{ij}v_j). \tag{142}$$

Positive prior precisions make this form positive definite. The sender stiffness is restricted to directions visible through its link: it is $L_{ri}^\top W_{ri} L_{ri}$, not the full sender precision. The older sender simplification is recovered only when $D_{ri} = 0$ and L_{ri} is invertible. No analogous positive full covariance Hessian is asserted away from a matching point.

If attention is eliminated by its entropy-regularized optimum, the reduced Hessian still includes $-\tau^{-1}\sum_i \mathrm{Cov}_{\beta_i^*}(\nabla E_{ij})$. This term vanishes exactly when the candidate gradients coincide within each positively weighted row. Arbitrary selective noisy channels need not admit simultaneous zero edge divergence, so that condition cannot be assumed from a full-state consensus picture. Under fixed frame changes, $L'_{ij} = g_i L_{ij} g_j^{-1}$ and $D'_{ij} = g_i D_{ij} g_i^\top$ give $C'_{ij} = g_i C_{ij} g_i^\top$; the displayed derivatives and quadratic form then transform by the corresponding primal and dual laws.

9.8 Conditional kinetic mechanics and its missing input

The Euclidean generator-history inertia of Section 6 is derived from a specified normalized history law. It is separate from the first-order recognition flow considered here. A real-time kinetic extension can also be considered, with its own additional data. Choose a positive C^2 tensor $M_K(\xi)$ independently of the potential and define

$$\mathcal{T} = \tfrac{1}{2}\dot{\xi}^\top M_K(\xi)\dot{\xi}, \qquad L = \mathcal{T} - \mathcal{E}, \qquad \mathcal{R} = \tfrac{1}{2}\dot{\xi}^\top \Gamma(\xi)\dot{\xi}, \quad \Gamma = \eta^{-1}G. \tag{143}$$

The exact Euler–Lagrange equation with this Rayleigh dissipation is

$$(M_K)_{ab}\ddot{\xi}^b + \frac{1}{2}\big[\partial_c(M_K)_{ab} + \partial_b(M_K)_{ac} - \partial_a(M_K)_{bc}\big]\dot{\xi}^b\dot{\xi}^c + \Gamma_{ab}\dot{\xi}^b = -\partial_a\mathcal{E}. \tag{144}$$

Repeated coordinate indices are summed. Differentiating $\partial L/\partial\dot{\xi}^a = (M_K)_{ab}\dot{\xi}^b$ gives the displayed metric-derivative term. It vanishes for a constant kinetic metric and at linear order about a stationary state, but cannot be omitted for a state-dependent Fisher or frame metric. Contracting with $\dot{\xi}$ gives $d(\mathcal{T}+\mathcal{E})/dt = -\dot{\xi}^\top\Gamma\dot{\xi}$ for an autonomous potential. This energy balance verifies the conditional extension, not its derivation from inference.

At an equilibrium with positive stiffness H, the undamped linear modes satisfy $Hv = \omega^2 M_K v$. The local choice $M_K = H$ makes every frequency equal to one in the declared units. Precision-dependent frequencies therefore do not follow from identifying stiffness and mass. For an isolated matched-prior agent with $H = G = M_K$, the damped linear equation is $\ddot{\delta\xi} + \eta^{-1}\dot{\delta\xi} + \delta\xi = 0$; it is overdamped for $\eta < 1/2$ and its slow rate approaches η as $\eta \downarrow 0$. This is a constant-coefficient singular-limit statement. A nonlinear overdamped limit requires an initial-layer estimate and regularity and confinement hypotheses in addition to a small inertia parameter.

When a kinetic tensor couples agents, momentum is the full covector $\pi_i = \sum_j (M_K)_{ij}\dot{\xi}_j$, not a separately assigned diagonal mass times each velocity. Its frame covariance must be imposed on M_K itself; a Frobenius norm on passive frame coordinates does not generally descend to the physical quotient. The scalar potential determines none of this additional kinetic data. Appendix B derives the memory obtained by eliminating internal first-order coordinates and shows why an apparent second-order equation can still be entirely relaxational. That derivation provides a route to persistence without the kinetic postulate, while leaving the identification of mechanical mass open.

10 Meta-Agents as Retained Laws

A parent should first be defined by what it retains and predicts. It need not begin as an average Gaussian, an averaged frame or a threshold crossing. Given a fine law and a normalized coarse channel, its law is induced exactly. Whether this retained description is a useful persistent agent is a further dynamical question.

10.1 The exact parent and its free energy

Let Y be the fine population state, P_o its posterior at the observed record, and $C : Y \rightsquigarrow Z$ a normalized coarse channel held fixed as Q varies. The parent laws are

$$Q^c = QC, \qquad P_o^c = P_oC. \tag{145}$$

On standard Borel spaces form the two augmented laws $Q(dy)C(y,dz)$ and $P_o(dy)C(y,dz)$. Their conditional laws of Y given Z yield

$$\begin{aligned} D_{\mathrm{KL}}(Q\|P_o) &= D_{\mathrm{KL}}(Q^c\|P_o^c) \\ &\quad + \int D_{\mathrm{KL}}(Q(dy \mid z)\|P_o(dy \mid z))Q^c(dz). \end{aligned} \tag{146}$$

The common forward kernel adds no relative entropy; disintegration proves the second line. The same evidence Z_o occurs at both levels, so

$$\mathcal{F}_V(Q) = \mathcal{F}_V^c(Q^c) + \Delta_C(Q), \qquad \Delta_C(Q) \geq 0. \tag{147}$$

This exact reduction is a data-processing statement. The drop measures discarded conditional mismatch, not improvement in the evidence for the world.

With the latent Lie-algebra links of Section 6, the fine state includes the generators: the recognition law is $R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$ on $(\mathring{\omega}, Y)$. A coarse channel that reads the links through tree transports acts on this joint state, and its parent law must be computed from the joint dependence of links and matter. Statements proved with links held fixed apply conditionally on $\mathring{\omega}$ under their stated hypotheses. They pass to the link-marginal law when the property is preserved by mixtures; otherwise transfer requires a separate argument. In particular, Gaussian conditional matter laws need not remain Gaussian after the random links are integrated out (Proposition 35). A Gaussian parent fitted to that marginal is consequently a declared projection unless Gaussian closure is proved.

Consequently a rule that chooses the largest raw VFE drop under arbitrary exact coarse channels can prefer severe information loss. Coarse-graining does not become a formation principle by calling that drop a gain. One must specify what observations, interventions or future laws the parent is required to retain. This obligation is mathematical and physical, not a requirement that a hierarchy save computer memory.

10.2 Selective links descend only under a compatibility condition

Suppose a deterministic linear retention $C_i : E_i \to Z_i$ is surjective and a child interaction is $L_{ij} : E_j \to E_i$. A parent interaction L_{IJ}^c satisfying

$$C_i L_{ij} = L_{IJ}^c C_j \tag{148}$$

exists if and only if $\ker C_j \subseteq \ker(C_i L_{ij})$. Indeed, define $L_{IJ}^c(C_j x) = C_i L_{ij} x$; it is well-defined precisely under this inclusion and is then unique. No inverse of L_{ij} is needed. For affine Gaussian channels, compatible means and pushed noise laws give the corresponding kernel square.

This condition says that discarded source distinctions cannot remain visible to the retained receiver. Equal transmitted beliefs at one instant do not imply it. When the square fails, the parent can retain additional variables or a conditional law of the discarded state. Replacing the child link by an arbitrary relabeling would alter its predictions. With evolving retention maps, the drift also contains $\dot{C}_i x_i$; an instantaneous compatible link is not sufficient for an exact moving quotient.

A many-agent block generally induces several boundary channels, correlated source events and higher-order interactions. Retaining only one matrix for each pair of parents is a closure approximation unless these additional dependencies vanish or are represented.

A holonomy-fixed Gaussian parent is a more restricted construction than the pushforward parent of Eq. (145). For $\mathrm{GL}^+(K)$ acting in its defining representation, suppose a connected block has an internal closed walk with nonzero net traversal of some reversal pair, and its generator

law has a Lebesgue density on the full Lie algebra, including the trace directions. The loop determinant is the exponential of the signed sum of generator traces. The zero-sum condition is a null hyperplane, so almost surely that loop cannot preserve any positive-definite Gaussian covariance. The holonomy-fixed nondegenerate Gaussian family is then empty and its associated coherence score is infinite (Proposition 36). The exact retained law remains defined. The result concerns that invariant Gaussian family and those cycle and density hypotheses; it does not establish a general obstruction to parent formation. Restricted generator support, a different representation, or a different parent-law family requires its own analysis.

10.3 Gaussian summaries with different meanings

A Gaussian summary is useful once the random variable it summarizes is identified. Express child states in a chosen parent frame, writing their means and covariances as $\tilde{\mu}_i, \tilde{\Sigma}_i$, and normalize weights $\bar{w}_i \geq 0$ with $\sum_i \bar{w}_i = 1$. The mixture law $\sum_i \bar{w}_i \tilde{q}_i$ has Gaussian moment projection

$$\mu_I = \sum_i \bar{w}_i \tilde{\mu}_i, \tag{149}$$

$$\Sigma_I = \sum_i \bar{w}_i \left[\tilde{\Sigma}_i + (\tilde{\mu}_i - \mu_I)(\tilde{\mu}_i - \mu_I)^\top \right]. \tag{150}$$

The dispersion term is retained. It describes population heterogeneity, not an avoidable correction.

Proposition 6 (Gaussian moment projection)**.** *For Gaussian $\tilde{q}_i$ with positive-definite $\tilde{\Sigma}_i$, the unique Gaussian minimizer of*

$$\sum_i \bar{w}_i D_{\mathrm{KL}}(\tilde{q}_i \| q_I) \tag{151}$$

is given by the preceding moments.

Proof. Up to a constant, the objective is $\frac{1}{2} \log\det \Sigma_I + \frac{1}{2} \operatorname{tr}[\Sigma_I^{-1} \mathbb{E}(X - \mu_I)(X - \mu_I)^\top]$ for the mixture variable X. Its mean optimum is $\mathbb{E}X$. If S is the resulting positive-definite covariance, subtracting the value at S gives $\frac{1}{2}[\operatorname{tr}(\Sigma_I^{-1} S) - K - \log\det(\Sigma_I^{-1} S)] \geq 0$, with equality exactly at $\Sigma_I = S$. □

This is a distribution of a randomly selected member, followed by moment matching. It differs from the law of a collective average $Z_I = \sum_i \bar{w}_i \tilde{X}_i$, whose covariance is

$$\operatorname{Cov}(Z_I) = \sum_{ij} \bar{w}_i \bar{w}_j \operatorname{Cov}(\tilde{X}_i, \tilde{X}_j). \tag{152}$$

It also differs from a posterior over a shared latent inferred from several records, whose precision can add. Mixing these three covariance conventions produces incorrect claims about collective uncertainty and clock slowdown.

The moment projection is an exact parameter update in a declared Gaussian random-effects model: maximizing $\sum_i \bar{w}_i \mathbb{E}_{q_i} \log \mathcal{N}(\tilde{k}_i; m, S)$ over (m, S) gives these moments. This is a fitted population distribution, not automatically a posterior on a new physical object. A regularized barycenter $\sum_i \bar{w}_i D_{\mathrm{KL}}(\tilde{q}_i \| q_I) + \lambda D_{\mathrm{KL}}(q_I \| p_I)$ is a different optimization problem and does not retain the same closed form for $\lambda > 0$.

Proposition 7 (Parent frame is a convention)**.** *With a chosen parent frame U_I, a retained Gaussian law transforms as $(\mu_I, \Sigma_I) \mapsto (g_I \mu_I, g_I \Sigma_I g_I^\top)$ and every outgoing map transforms at its parent endpoint. Predictions are unchanged. A symmetric, left-equivariant global mean $G^n \to G$ cannot exist when a nonidentity finite-order element of order n cyclically permutes an input tuple.*

Proof. Covariance follows by pushforward and the endpoint law. For the second assertion take the tuple $(I, v, \dots, v^{n-1})$, with $v^n = I$ and $v \neq I$. Symmetry and equivariance would give $M = vM$, impossible for invertible M. □

A frame can simply be chosen. A Riemannian center of mass

$$U_I \in \arg\min_U \sum_i \bar{w}_i d_G(U, U_i)^2 \tag{153}$$

is an optional local convention when its existence and uniqueness hypotheses hold; it is not part of what makes a parent. Averaging Lie-algebra coordinates is generally not an equivariant substitute.

10.4 Coherence, persistence and autonomy

Coherence is agreement in declared communicated observables. Persistence is continued retention of a state or response law over a declared time scale. Autonomy is closure of the retained dynamics, possibly with retained memory. None implies the other two without assumptions. A numerical similarity threshold can diagnose coherence at a resolution; it is not a derivation of an agent.

Complete consensus also need not erase useful private distinctions. Equation (16) separates public agreement from conditional private information. A parent may be coherent in what it exchanges while its members retain distinct internal descriptions. This gives a precise alternative to identifying formation with collapse of all beliefs to one law.

11 Cross-Scale Inference without Additional Shadow Postulates

Once a normalized model and a retained variable are specified, upward and downward inference are conditional probability operations. They need not be introduced as separate physical forces or as copies of a parent's posterior. The direction of a message, its conditioning information and its temporal episode must be stated.

11.1 Upward retention and conditional reconstruction

For a coarse channel C, the upward law is $Q^c = QC$. A reverse conditional $P(dy \mid z, o)$ exists under standard Borel assumptions, but depends on the joint law and the record. It is not a canonical inverse of a lossy map. For a deterministic retention $z = C(y)$, all fine states in the same fiber have the same retained value and can still have different conditional probabilities.

The corresponding operator on observables is

$$(Cf)(y) = \int f(z) C(y, dz), \qquad (C_P^\dagger g)(z) = \mathbb{E}_P[g(Y) \mid Z = z]. \tag{154}$$

These operators are derived from the chosen channel and reference law. Their existence does not uniquely select that channel or law. Composition obeys the tower property for nested retained sigma-algebras. More generally, channels generated as a Markov chain $Y \to Z \to W$ compose by integrating over Z. Arbitrary overlapping summaries of one joint law need not obey this identity.

11.2 A hierarchy within one episode

Let z be a parent latent with prior $r(z)$ and normalized child kernels $p_i(k_i \mid z)$, with each child's record likelihood $\ell_i(o_i \mid k_i)$. The joint

$$p(z, k, o) = r(z) \prod_i p_i(k_i \mid z) \ell_i(o_i \mid k_i) \tag{155}$$

gives exact evidence messages $m_i(z) = \int p_i(k_i \mid z) \ell_i(o_i \mid k_i) dk_i$. The downward message for child i is

$$p_i^{\mathrm{cav}}(dk_i) \propto \int p_i(dk_i \mid z) r(z) \prod_{j \neq i} m_j(z) dz. \tag{156}$$

Multiplying by $\ell_i(o_i \mid k_i)$ then uses the child's record once. Replacing the cavity law by the complete parent posterior would reuse the child's evidence.

Mean-field coordinate ascent is different from this sum-product calculation. Under $q(z) \prod_i q_i(k_i)$, the child coordinate contains $\exp\{\mathbb{E}_{q(z)} \log p_i(k_i \mid z)\}$, not $\mathbb{E}_{q(z)} p_i(k_i \mid z)$. Their difference is not a matter of terminology. The Gaussian calculations in Appendix G display it explicitly.

11.3 Prediction between episodes

Let Y_n be the retained population state after episode n and let $T_n(Y_n, dY_{n+1})$ be a normalized transition independent of the next recognition law. Filtering and prediction give

$$Q_n = \mathbb{P}(dY_n \mid o_{0:n}), \qquad P_{n+1}^{-} = Q_n T_n. \tag{157}$$

A variational approximation to Q_n gives a variational predictive prior instead of an exact filter. No new record is counted twice: the record o_{n+1} has not entered Q_n.

The older cross-scale shadow is one special kernel. If a child is a deterministic transported copy of a parent latent in the previous episode,

$$p_i^{(\ell),n+1} = (\Omega_{iI})_{\#} q_I^{(\ell+1),n}, \qquad r_i^{(\ell),n+1} = (\widetilde{\Omega}_{iI})_{\#} s_I^{(\ell+1),n}. \tag{158}$$

A general selective downward kernel replaces $\Omega_{\#}$ by $\mathcal{K}_{iI}$ and can change dimension or add noise. Equation (158) is therefore derived under a transported-copy transition, not a universal source of priors. The same-time assignment using the full parent posterior is removed from the foundational dynamics.

The posterior predictive law gives a concrete answer to where later priors come from. Initial priors, transition laws and the distinction between observations and latent variables still require specification. An invariant prior may sometimes be derived from a specified ergodic dynamics, but that moves the assumptions into the dynamics; it does not produce a probability law without assumptions.

11.4 Participation as constrained feedback

Upward inference changes what a collective predicts; those predictions shape subsequent child inference through the same normalized model or through a later transition. The feedback is real within the model, but a separately named shadow field adds no mathematics. Wheeler's participatory interpretation concerns what such reciprocal dependence means physically [1]. The probability construction supplies conditional and predictive consistency; it does not by itself show that observers constitute the laws of nature.

12 Exact Coarse-Graining and Induced Interactions

Coarse-graining can generate cross-scale operators without postulating their form. It does so once a fine law and a channel have been supplied. This conditional statement is stronger than a formal analogy with renormalization and weaker than a derivation of a preferred hierarchy.

12.1 The effective action

Keep the coarse channel C fixed when varying the recognition law. Let $\rho(dy)$ be a proper reference law and $m_o(dy) = e^{-H_o(y)}\rho(dy)$ a finite positive evidence measure. Push both through the same channel:

$$\rho^c = \rho C, \qquad m_o^c = m_o C, \qquad H_o^c(z) = -\log \frac{dm_o^c}{d\rho^c}(z). \tag{159}$$

A reverse conditional under $\rho(dy)C(y, dz)$ gives

$$e^{-H_o^c(z)} = \mathbb{E}_\rho[e^{-H_o(Y)} \mid Z = z]. \tag{160}$$

To prove it, integrate both sides against an arbitrary bounded measurable function of z. Both integrals equal $\int f(z)e^{-H_o(y)}\rho(dy)C(y, dz)$. This identifies the Radon–Nikodym derivative. Total mass, hence the evidence, is preserved. Successive channels compose exactly when the conditionals are taken from one joint tower. For a second channel C', associativity of kernel composition gives $(m_oC)C' = m_o(CC')$ and $(\rho C)C' = \rho(CC')$, so uniqueness of Radon–Nikodym derivatives makes the staged and direct effective actions agree $\rho CC'$-almost everywhere. In the conditional form, with every conditional taken from the joint law $\rho(dy)C(y, dz)C'(z, dz')$ of (Y, Z, Z'), the tower property and the conditional independence of Y and Z' given Z give $\mathbb{E}[e^{-H_o(Y)} \mid Z'] = \mathbb{E}[e^{-H_o^c(Z)} \mid Z']$.

When ρ is a latent prior and e^{-H_o} a record likelihood, this is an evidence-preserving coarse model. When ρ is instead a reference on belief configurations and $H = \mathcal{E}/T_{\mathrm{cfg}}$, it is a different probability model over those configurations. Deterministic descent of $\mathcal{E}$ does not sample that Gibbs ensemble. The configuration temperature T_{cfg}, the source temperature τ and physical thermodynamic temperature are distinct.

12.2 What integration generates

For a bounded perturbation V of H_o, restrict to coarse states where $0 < dm_o^c/d\rho^c < \infty$, which holds P_o^c-almost surely. The effective perturbation is

$$H_{o,\lambda}^c - H_o^c = -\log \mathbb{E}_{P_o(dy|z)} e^{-\lambda V(Y)} = \lambda \mathbb{E}[V \mid z] - \frac{\lambda^2}{2} \operatorname{Var}(V \mid z) + O(\lambda^3). \tag{161}$$

The expansion follows by differentiating the conditional moment generating function near zero; boundedness supplies domination and local analyticity. For sums of fine interactions the variance contains cross covariances, and higher cumulants couple additional retained variables. Thus exact coarse-graining generally generates higher-order interactions. Assuming the same pairwise functional at every scale discards these terms unless a closure theorem applies.

An elementary Gaussian calculation makes the induced operators explicit. Suppose the dimensionless quadratic energy is

$$H(x, z) = \frac{1}{2}x^\top Ax + x^\top Bz + \frac{1}{2}z^\top Dz, \qquad A \succ 0. \tag{162}$$

Integration over x with Lebesgue reference gives

$$H^c(z) = \frac{1}{2} z^\top (D - B^\top A^{-1} B) z + \frac{1}{2} \log\det A - \frac{d_x}{2} \log(2\pi). \tag{163}$$

Completing the square proves the formula. For a proper joint Gaussian, the Schur complement is positive definite. If $A = A(z)$, its log determinant is an induced potential on retained states and must remain in the action. Completing the square still works pointwise when the integration is finite, but the result need not be quadratic.

The matrix $B^\top A^{-1} B$ supplies interactions between retained variables that previously communicated only through the eliminated block. It can also generate self terms. A closure family that forbids every newly generated self term is narrower than the family of valid Gaussian effective laws. For instance, eliminating a hidden scalar x from $\frac{1}{2} s x^2 + \frac{1}{2} w (x - z)^2$ with $s, w > 0$ leaves the proper effective energy $\frac{1}{2} \frac{sw}{s+w} z^2$ up to a constant, a self term on z, whereas the fine energy coupled z only through the difference $x - z$. These induced terms are examples of cross-scale structure obtained by integration, not additional forces appended to the fine functional.

Integrating matter has different consequences for the two normalized latent-link models. In the directed record extension of Section 6, integrating all matter and records returns the generator prior; it induces no additional prior link action. A fixed observed record changes the link posterior through its conditional evidence. If generators and undirected matter are instead normalized jointly, let $Z_{X,\Theta(\mathring{\omega})}$ be the matter partition function. The generator marginal acquires the induced action $\mathcal{A}_{\mathrm{ind}}(\mathring{\omega}) = -\log Z_{X,\Theta(\mathring{\omega})}$, and its normalization requires

$$0 < \mathfrak{Z} = \int e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega})} Z_{X,\Theta(\mathring{\omega})} \nu^{\mathcal{T}}(d\mathring{\omega}) < \infty. \tag{164}$$

A proper generator prior does not by itself imply this global integrability condition. The determinant action of Gaussian matter is one realization when the matter precision and the joint integral are admissible: for a centered Gaussian matter potential with precision $\mathbf{P}(\Theta)$ on $\mathbb{R}^d$, $\mathcal{A}_{\mathrm{ind}} = \frac{1}{2} \log\det \mathbf{P}(\Theta(\mathring{\omega})) - \frac{d}{2} \log(2\pi)$, provided $\mathbf{P}(\Theta(\mathring{\omega})) \succ 0$ for almost every generator and $\mathfrak{Z} < \infty$. Appendix K.9 proves these statements and gives a scalar example with a proper generator prior and $\mathfrak{Z} = +\infty$.

12.3 Random links at coarse scales

For two fine links in series, express the retained link in the chosen root frames. Its exact twist is the ordered product

$$\boxed{\mathring{\Theta}^c_e = \exp(\mathring{\omega}_{e'}) \exp(\mathring{\omega}_e).} \tag{165}$$

It is a measurable group-valued function of the fine generators, so its law and its joint law with retained matter exist by pushforward for a fixed partition and fixed trees. Replacing this product by the exponential of the sum is exact in an abelian Lie algebra. For two independent nondegenerate Gaussian generators on $\mathfrak{gl}(K, \mathbb{R})$ with $K \geq 2$, the product has no real logarithm with positive probability. Its full law therefore has no representation as the exponential of a law of real generators. Even on a narrow sector where the principal logarithm exists with high probability, nonzero commutator cumulants can prevent a Gaussian generator description. Precisely, let both generator laws narrow with a width τ, their means scaling as τ and their covariances as τ^2. Outside an event of probability $O(e^{-c/\tau^2})$ the principal logarithm of the product exists and lies in $\mathfrak{g}$, and for all sufficiently small τ no Gaussian generator coincides with it there whenever an explicit cubic commutator form built from the two covariances is not identically zero, or the means vanish and an

explicit quartic form is not identically zero; on $\mathfrak{gl}(K,\mathbb{R})$ both cases occur. Appendix K.10 proves these statements, together with the absence of a real logarithm. The retained group law is the input to the next depth. A finite-dimensional family containing the fine Gaussian exponential laws and closed under these products remains an open problem, stated precisely in Remark 7. Exact coarse-graining is available while that closure question remains open.

12.4 Closure and its residual

If the exact coarse action H^c is replaced by an ansatz $\widehat{H}^c$, write $R = H^c - \widehat{H}^c$. Against the same reference, the exact law satisfies

$$p^c = \frac{e^{-R}\widehat{p}^c}{\mathbb{E}_{\widehat{p}^c} e^{-R}}. \tag{166}$$

For a bounded residual with $\sup R - \inf R \le \delta$,

$$D_{\mathrm{KL}}(p^c \| \widehat{p}^c) = -\mathbb{E}_{p^c} R - \log \mathbb{E}_{\widehat{p}^c} e^{-R} \le \delta. \tag{167}$$

The inequality follows by bounding the logarithmic normalizer between $-\sup R$ and $-\inf R$. Additive constants in the action cancel. This elementary bound is conservative, but it measures a stated approximation; a small visual mismatch of Gaussian means is not an equivalent criterion.

A finite sequence of channels is exact coarse-graining. To obtain a renormalization flow on one coupling space, one must additionally identify retained spaces, specify rescaling and control the generated operator family. Finiteness alone does not prohibit defining comparison maps or fixed points; it means that they are not supplied by decreasing the number of agents. The finite-step theory and the spatial or infinite-system extensions answer different questions. The latter require additional hypotheses.

12.5 Persistent laws versus instantaneous marginals

An exact marginal need not have a closed Markov evolution. If the fine generator is $\mathcal{L}$ and retention is deterministic, a sufficient generator closure is

$$\mathcal{L}(f \circ C) = (\mathcal{L}^c f) \circ C \tag{168}$$

on a determining domain, together with the well-posedness assumptions that turn generators into the stated processes. If fine states sharing $C(y)$ predict different future retained laws, an instantaneous parent state is insufficient.

The conditional law of the hidden fine state given retained history is the canonical exact repair. It is generally a measure-valued state; finite-dimensional closure is a property to prove for the particular system. A generic dimension-counting argument cannot exclude arbitrary measurable encodings, and ordinary path relative entropy may be infinite when candidate processes have different quadratic variation. Approximation must therefore name its class, observations and time horizon. Section 8 and Appendix B develop constructive memory calculations rather than assuming memory away.

13 The Participatory Loop and the Origin of Priors

The image of an ouroboros expresses a research aim: descriptions at one scale are sustained by activity at other scales, while the resulting collective constrains subsequent activity of its members. The probability theory does not require a self-referential prior equal to a posterior that it helped

generate. It provides three precise forms of feedback: conditional inference within one joint, parameter learning across records, and predictive transitions between episodes.

In conditional inference, changing a parent's approximate posterior changes the next child coordinate update because the two are factors of one coupled variational problem. In parameter learning, records change a model parameter or its posterior; the learned object then changes later predictions. In temporal feedback, the posterior of one episode becomes the input to a normalized transition. Each operation has a well-defined objective or law. Combining them requires a single schedule and probability model, not a simultaneous substitution of every posterior into every prior.

13.1 What is and is not removed by closing a loop

A finite directed acyclic generative model has root variables, and their laws must be specified. Replacing that hierarchy by a reciprocal undirected model is possible when its global partition function is finite. It replaces directed roots by a globally normalized reference and interaction law; it does not remove all assumptions. The existence of normalized cyclic models therefore defeats a blanket claim that every probabilistic loop is impossible, but does not establish an assumption-free universe.

The optimal prior in an unrestricted family fitted to a collection of recognition laws is their aggregate. Indeed, for weights $w_a \geq 0$ summing to one,

$$\sum_a w_a D_{\mathrm{KL}}(Q_a \| P) = \sum_a w_a D_{\mathrm{KL}}(Q_a \| \bar{Q}) + D_{\mathrm{KL}}(\bar{Q} \| P), \qquad \bar{Q} = \sum_a w_a Q_a, \tag{169}$$

whenever the terms are finite. The last term identifies the minimizer $P = \bar{Q}$. For sources with finite second moments and finite KL to some nonsingular Gaussian, restricting the prior to nonsingular Gaussians gives moment matching when the aggregate covariance is positive definite. This is an empirical-Bayes coordinate under a specified learning problem, not permission to reuse a current record twice, and not a unique source of initial priors.

13.2 Which structure may emerge

Selective interactions can establish public agreement while retaining private states. A population dynamics may develop long-lived regions, predictive sufficient statistics or coherent modes. Such retained structures can supply parent laws and memory. A hierarchy is recursively composable only if the parent carries the information needed to repeat the same operation: its state, interactions, records, retained source events and, where needed, memory.

This identifies the minimal research target more sharply than a tower drawn in advance. The target is a dynamically selected family of predictive retained states whose induced laws remain usable at the next formation episode. Exact marginalization derives the parent law once the retention is given. It does not select that retention; a metastable partition may be dynamically distinguished without being unique; and a successful first episode does not prove recursive formation.

13.3 Drive, duration and stationarity

For an autonomous Riemannian gradient flow with positive mobility, the potential decreases until stationarity. Under analytic and precompactness assumptions the familiar gradient-convergence arguments apply. A slow mode can preserve a record for a long time without sustaining activity forever. Driven observations, stochastic fluctuations or explicitly nonequilibrium dynamics can maintain continuing changes, but their statistics are additional model data.

The information clock of Section 18 measures change along a supplied history. A participatory loop does not create that history's ordering parameter merely by feeding one conditional law into another. Wheeler's self-excited circuit remains the physical and philosophical motivation [1, 2]; the present mathematics determines which inferential loops are consistent and what each retains.

13.4 Participation as an operational decision

A posterior probability that a link is active is an inference about a supplied routing model. Choosing a link, experiment or action is a decision. Normalization alone cannot select that decision. A minimal illustration uses a finite action set, a strictly positive reference policy $\rho(a)$, and an action-independent predictive prior $p^-(k)$. A normalized observation channel for each action gives expected information gain

$$\mathcal{I}_a = \mathbb{E}_{o|a} D_{\mathrm{KL}}(P(dk \mid o, a) \| p^-(dk)). \tag{170}$$

For finite gains and finite real declared costs $\ell(a)$, weights $\lambda > 0$ and temperature $\tau_A > 0$, the decision objective

$$J(\pi) = \sum_a \pi(a)[\ell(a) - \lambda \mathcal{I}_a] + \tau_A D_{\mathrm{KL}}(\pi \| \rho) \tag{171}$$

has unique optimum

$$\pi^*(a) = \frac{\rho(a) \exp[(\lambda \mathcal{I}_a - \ell(a))/\tau_A]}{\sum_b \rho(b) \exp[(\lambda \mathcal{I}_b - \ell(b))/\tau_A]}. \tag{172}$$

The simplex multiplier yields the displayed expression; subtracting its value leaves $\tau_A D_{\mathrm{KL}}(\pi \| \pi^*)$. If an action changes the latent predictive law, that action-conditioned law replaces p^- in the information gain.

This is an optional decision model, not another consequence of the evidence identity. Its costs, reference policy and valuation of information are visible assumptions. A fair hidden bit makes the distinction concrete. Reading it perfectly gains $\log 2$ nats; receiving a constant blank gains zero. Equal costs and equal reference-policy weights make the information-seeking policy favor reading, whereas minimizing expected observation surprisal alone favors the blank. Wheeler's participant therefore cannot be identified with an indiscriminate minimizer of sensory entropy.

An informational hierarchy also cannot acquire new evidence just by naming a retained level. Appending the same normalized encoder $C(dz \mid y)$ to both a prior and a recognition law leaves their joint KL unchanged. Likewise, if a coarse record is a deterministic copy of an existing record, its correct conditional law is a delta kernel given that record, not an additional independent likelihood. These facts allow feedback and repeated inference while preventing an apparent gain produced only by duplicated observations.

14 What Formation Theory Establishes

This section states a pairing-selection theorem for a declared four-agent model and describes the structure of its proof, which is given in full in Appendix J. The result distinguishes learned from frozen source priors. Its scope must be separated from physical object formation and an indefinitely recursive hierarchy.

14.1 The fixed-temperature selection result

The model has a homogeneous cycle of four agents at c_*. Its configuration consists of four scalar belief coordinates, four model coordinates and one shared scalar translation on each unoriented

edge. With a chosen orientation of edge $e = i \to i+1$, the comparison cost is

$$D_e = \frac{1}{2}\left[(k_{i+1} - k_i - w_e)^2 + (m_{i+1} - m_i - w_e)^2\right]. \tag{173}$$

Both directions of source selection use the same edge variable and cost. A Gaussian reference confines the configuration, a stochastic dynamics evolves it at temperature $T > 0$, and source assignments are refreshed using their conditional law. The product source-prior rows start uniform and learn from the actual recognition law through

$$\dot{\pi}_{ij} = \epsilon_L\big(q_t(J_i = j) - \pi_{ij}\big). \tag{174}$$

Write S for the occupied unoriented edge set induced by the source assignments J. These ingredients, including the reference and kinetics, are part of the model; Definition 4 in Appendix J declares the reference, the source records, the evaluator coupling of model and belief samples, and the kinetics precisely. The statement is not transferable to arbitrary links or arbitrary source updates.

For the initial law $\mathcal{N}(\mathbf{m}_0, I_{12})$ times uniform assignments, the selection theorem states: there is $T_0 > 0$ such that for every fixed $T \in (0, T_0]$ a finite constant C_T exists and, for Lebesgue-almost every fixed $\mathbf{m}_0$, one perfect matching M satisfies

$$\lim_{\epsilon_L \to 0} \sup_{t_L \le t \le 2t_L} q_t^{\epsilon_L}(S \neq M) = 0, \qquad t_L = C_T \epsilon_L^{-1} \log(1/\epsilon_L). \tag{175}$$

At frozen uniform rows, for every $T > 0$ the long-time difference of the two matching probabilities instead tends to zero.

The proof in Appendix J combines uniform geometric ergodicity and entropy contraction at frozen rows, a nonvanishing matching-mode imprint from almost every initial mean, the small-temperature landscape of the reduced row field and controlled escape from its saddle on the logarithmic slow horizon. The initial mean supplies the asymmetry amplified by learning. The statement neither starts from an exactly symmetry-invariant law and spontaneously chooses a deterministic sign, nor takes a joint temperature–mobility limit. It concerns the actual law uniformly over the stated window, not the probability that a single trajectory shows a chosen pair once.

14.2 Selection, inheritance and the next episode

Sample paths show a held pairing with positive probability even when source learning is idle. At frozen rows the initial assignment is a perfect matching with probability $1/8$, no redraw occurs before a fixed time $t > 0$ with probability e^{-t}, and the held Gaussian motion has a nondegenerate law at time t, so the comparison costs of the two attended edges lie below, and those of the two unattended edges above, any prescribed thresholds with positive probability. Selection against a symmetric frozen control therefore tests a stronger mechanism than observing a coherent cluster. The theorem closes that mechanism for its declared family.

Three questions remain distinct. A selected pairing must first admit a retained parent state and outgoing kernels. Its law must then persist or carry the necessary conditional memory during a second episode. Finally, the resulting parent dynamics must itself support another formation step without replacing inherited parameters by newly chosen ones. The theory has conditional parent constructions and supplied finite-depth examples, but those do not settle the general recursive problem. In particular, a slow model record can account for persistence while failing to cause the next coalescence.

Rank-deficient selective communication changes this problem. Public agreement can hide private variables that later affect parent interactions. The descent square (148), generator condition (168)

and conditional-memory state identify what has to be retained. An extension that merely adds low-rank notation to the four-cycle theorem would not prove formation with learned selective interfaces.

14.3 A physical comparison requires an external target

The theorem is a mathematical result about the model. A physical claim would additionally identify a target system, observable records, an inferred state and channel family, parameter uncertainty and a prediction not fitted at the coarse scale. Appropriate contrasts include frozen source learning, altered interface rank, and parent predictions with and without retained memory, provided each comparison preserves the stated model assumptions.

No arbitrary detector threshold, seed count or pass percentage is elevated to a physical law here. A validation design must derive its resolution and uncertainty from the target measurement and the particular claim. Failure to match that target would reject the physical identification or approximation, not the conditional algebra of Bayesian inference. Conversely, reproducing a declared stochastic construction would not establish that nature implements it.

15 Discussion: What the Variational Principle Contributes

The principal gain is a consistent account of how local descriptions are compared, revised and retained. A normalized joint yields an evidence bound. Selective channels make the comparison physically interpretable as partial access. A joint statistical bundle avoids treating beliefs and models as unrelated ontological substances. Conditional expectation and prediction generate cross-scale inference once a retention and dynamics are supplied.

The principle does not eliminate the need for a model. Changing a prior, a measurement interface, a noise law or a mobility changes the question being answered. Some of those ingredients can themselves be inferred or learned under a larger model. Enlarging the model shifts the boundary of what is explained; it does not make the boundary disappear. The productive task is to derive more structure while tracking that shift explicitly.

15.1 Persistence, geometry and mechanics

Precision has several distinguishable roles. It can enter a restoring Hessian, the Fisher metric or an observational likelihood. First-order natural-gradient dynamics turn the metric into dissipative resistance. A separately postulated kinetic metric defines a different dynamics. The exact memory construction of Appendix B demonstrates persistence without conflating these roles.

Likewise, Fisher geometry and Lorentz geometry answer different questions. The former measures statistical sensitivity; the latter arises conditionally from a determinant cone and its transformations. No contradiction is resolved by pretending the positive Fisher form becomes indefinite. The open physical task is to derive the structure that relates these two constructions to records of motion and propagation.

15.2 Participatory interpretation

The mathematical framework is compatible with a participatory reading because records, selective communication and retained collective descriptions affect subsequent predictions. Its stronger claim would be that such dynamics generate the stable, recursively organized objects through which a

physical world is described. The four-cycle selection theorem addresses one part of that claim in one family. Exact parent laws and memory identify the next requirements.

Interpretation does not follow uniquely from the equations. A realist may read them as models of situated access to an underlying world; a structural interpretation may emphasize invariant relations among records. The framework's empirical content must come from specified predictions, not from selecting a philosophical vocabulary.

16 From the Inferential Construction to Physical Questions

The construction now has a single probabilistic center: normalized population laws with selective covariant interactions. The composite Gaussian consensus energy remains an explicit worked specialization. Parent laws, induced interactions and predictive feedback follow by exact operations once their inputs are given. Conditional dynamics produce memory and persistent records; general recursive formation remains open.

The following part asks whether these mathematical structures can carry physical significance. It does not add a physical interpretation to every symbol. Instead it develops the clock and geometry calculations, identifies an exact Lorentz representation, and states what would be needed to make them descriptions of time, space and matter.

Part II: Physical Structure and Participatory Interpretation

17 From the Statistical Theory to a Physical Program

Part I supplies a normalized population model, selective communication, variational inference and conditional constructions of retained agents. Part II asks how structures of those types could acquire physical meaning. The question is stronger than whether a familiar physical equation can be written using information-theoretic symbols. A physical interpretation must connect a state of the model to a reproducible record, derive how that record responds to interventions, and identify which alternative models would give a different result. Wheeler's participatory program motivates this direction [1]; the mathematics does not make its philosophical interpretation compulsory.

Three distinctions organize the development. A mathematical construction specifies an object and proves its properties. Fisher length is a length of a statistical history, the covariant Fisher pullback is a positive semidefinite base tensor, and the Hermitian determinant construction carries the full proper Lorentz group. An operational attachment specifies how such an object is read from records, including the measurement channel and calibration. A dynamical selection result establishes that the updating law produces or stabilizes the object under stated initial conditions. These are separate obligations. A tensor with Lorentzian signature meets the first; it becomes a spacetime candidate only after a nondegenerate collective readout and a compatible propagation law are supplied.

The existing results already reduce the number of independent constructions required. Parent laws arise by retaining variables of a fine law; upward and downward cross-scale operators follow from pushforward and conditional reconstruction; priors for a later episode follow from prediction through a declared transition kernel. Effective interactions arise from integrating out fine variables, and conditional memory records what an instantaneous parent law omits. None of these operations needs a further shadow force or an additional substance. Conversely, exact marginalization alone does not explain why a particular parent forms, why it persists, or why the resulting observable has a physical unit. Those are the next questions, not definitions to be inserted into the answer.

The same economy guides the geometric discussion. The one statistical bundle of Section 4.2 is sufficient for joint, marginal and predictive information geometry. A connection is declared only when comparing section values across the continuous base. A complex rank-two channel is introduced explicitly when deriving the Lorentz representation, and a soldering is required only when attaching that representation to base tangents. We do not add an indefinite frame-derivative block to the Fisher metric or infer a physical dimension count from a threshold chosen for that purpose. Each specialization is stated where it is used rather than collected into a second hierarchy of universal postulates.

The preceding sections provide exact identities, regularity conditions and obstructions for these constructions. Their mathematical scope neither proves nor refutes a proposed physical realization. The task here is to carry their usable structure into a physicist's account while retaining the distinction between a proved conditional result and a physical hypothesis. The following sections develop duration, observer and collective geometry, Lorentz kinematics, persistence, measurement and empirical symmetry in that order. Consciousness is discussed separately as an interpretation on which none of those derivations depends.

18 Time as Registered Change: The Information Clock

A probability law can change by different amounts along histories ordered by the same parameter t. Information geometry supplies an intrinsic length for that change. It does not, by itself, supply the ordering of the history or identify its length with physical proper time. This distinction preserves the useful part of the information-clock idea without building the desired physical conclusion into its definition.

18.1 Fisher length and dissipation

For a regular statistical path $q_{\theta(t)}$ with Fisher metric G,

$$\vartheta(t) = \int_0^t \sqrt{\dot\theta^a(s) G_{ab}(\theta(s)) \dot\theta^b(s)}\, ds. \tag{176}$$

It is invariant under smooth parameter changes and orientation-preserving reparametrizations of the path. It measures length, not signed displacement. If frames vary with t, their pure coordinate motion must be removed by a declared horizontal or quotient derivative; differentiating a moving coordinate presentation as if it were a fixed sample coordinate can create spurious statistical change.

For an autonomous potential $\mathcal{V}$ and single-rate natural-gradient flow $\dot\theta = -\eta G^{-1} d\mathcal{V}$,

$$\dot{\mathcal{V}} = -\eta \|d\mathcal{V}\|^2_{G^{-1}}, \qquad \dot\vartheta^2 = -\eta \dot{\mathcal{V}}. \tag{177}$$

This follows by contraction of the gradient equation. With explicitly changing observations, parameters or drive,

$$\dot\vartheta^2 = -\eta \left(\frac{d\mathcal{V}}{dt} - \partial_t \mathcal{V} \right). \tag{178}$$

The identity uses the metric and mobility actually present. Separate rates on different blocks define a different rate-weighted metric; the single Fisher identity cannot be applied unchanged. A clock stops at a stationary law even when the externally chosen parameter continues.

The closest physical analog is thermodynamic length, which measures distinguishability along statistical state paths under specified thermodynamic assumptions [17, 18]. Entropic-time constructions likewise add a rule for organizing updates [30]. Neither analogy makes Fisher length an experimentally calibrated time coordinate in the present model.

18.2 Continuous change is not a stream of independent bits

The local expansion is

$$D_{\mathrm{KL}}(q_{\theta+d\theta} \| q_\theta) = \frac{1}{2} G_{ab} d\theta^a d\theta^b + o(\|d\theta\|^2). \tag{179}$$

For Gaussian mean changes at fixed covariance it is exactly $\frac{1}{2} d\mu^\top \Sigma^{-1} d\mu$. Dividing by $\ln 2$ expresses the relative entropy in bits. Summing this quadratic quantity over finer subdivisions of a smooth path generally drives the sum to zero, whereas Fisher length remains fixed. A per-step information gain therefore depends on the specified update granularity; it is not a parameter-free continuous bit counter. Full Gaussian Fisher distance between endpoints also need not equal the length of a path artificially restricted to constant covariance.

Bateson's difference that makes a difference is useful here as a motivation for operational distinguishability [31]. It is not a prescription to equate every infinitesimal parameter displacement with one registered bit.

18.3 Binary registrations

Wheeler's yes-or-no registration can instead be represented by an actual binary record channel [1].

Proposition 8 (Information budget of a binary record)**.** *For latent X and binary O,*

$$\mathbb{E}_O D_{\mathrm{KL}}(p(X \mid O) \| p(X)) = I(X;O) \le H(O) \le \ln 2. \tag{180}$$

For binary records $O_1, \dots, O_n$,

$$I(X; O_{1:n}) = \sum_{k=1}^{n} I(X; O_k \mid O_{<k}) \le n \ln 2. \tag{181}$$

No conditional-independence assumption is needed for this last bound.

Proof. The expected posterior-to-prior logarithmic ratio is the mutual-information integral. Since the record is discrete, $I(X;O) = H(O) - H(O \mid X) \le H(O)$. Apply the mutual-information chain rule and the same bound to each conditional record [24]. □

A single rare outcome may change a posterior by more than one bit. The bound is on the expected update. If records are conditionally independent given X, the sharper comparison $I(X; O_k \mid O_{<k}) = I(X;O_k) - I(O_k; O_{<k})$ holds. Registration supplies an inferential account of information gain; choosing questions, irreversible physical records and quantum complementarity are additional problems.

18.4 The scale-dependent clock

A parent clock depends on the parent law, so the covariance convention must be specified. Consider N constituents with common fixed covariance Σ and mean increments $\delta\mu_i = \delta c + \epsilon_i$. Assume zero-mean idiosyncratic increments, no cross correlations between distinct ϵ_i, and no correlation with the common increment. Write

$$A = \mathbb{E}\|\delta c\|^2_{\Sigma^{-1}}, \qquad B = \mathbb{E}\|\epsilon_i\|^2_{\Sigma^{-1}}, \qquad f_{\mathrm{coh}} = \frac{A}{A+B}, \tag{182}$$

with the same second moment B for every i and with $A + B > 0$. The increments may be random diagnostics of a collection of smooth histories; no stochastic path is silently assigned finite Fisher arc length.

Proposition 9 (Conditional collective clock ratio)**.** *For a parent Gaussian with mean $N^{-1}\sum_i \mu_i$ and covariance held at Σ, the expected mean-sector KL increment relative to one constituent obeys*

$$\frac{\mathbb{E}\Delta I_I}{\mathbb{E}\Delta I_i} = f_{\mathrm{coh}} + \frac{1 - f_{\mathrm{coh}}}{N}. \tag{183}$$

For the law of the average of independent constituent variables, whose covariance is Σ/N, the ratio instead is $N f_{\mathrm{coh}} + (1 - f_{\mathrm{coh}})$.

Proof. The parent mean increment is $\delta c + N^{-1}\sum_i \epsilon_i$, with expected squared norm $A + B/N$. Its KL increment is half this value in the first convention and $N/2$ times this value in the second. A constituent has expected increment $(A+B)/2$. □

The first convention is a fixed-covariance summary, not the generic mixture moment projection of Eq. (150). At nonzero between-member dispersion that projection has an additional covariance contribution, and its changes contribute to the clock. Thus there is no universal $1/N$ slowdown of meta-agents. The square root of either ratio compares root-mean-square speeds; it need not equal a ratio of expected path lengths.

A result independent of this choice compares a fixed-channel parent with its full fine law. If $q_\theta^c = q_\theta C$ and differentiation passes through a parameter-independent normalized kernel, the coarse score is

$$s^c(Z) = \mathbb{E}[s(Y) \mid Z], \qquad G - G^c = \mathbb{E}[\mathrm{Cov}(s(Y) \mid Z)] \succeq 0. \tag{184}$$

The conditional expectation identity and variance decomposition prove Fisher contraction. Along the same differentiable path, the parent's Fisher length is no larger than the fine joint length. It need not be smaller than each individual agent's length. If the channel changes with θ or time, its derivative adds information and this fixed-channel argument does not apply.

A physical clock hypothesis would require a retained statistic with reproducible rates, a calibration and an account of how different observers compare its records. Fisher length supplies a candidate measure of internal change. It does not establish relativistic time dilation, a causal order or a universal proper-time standard.

19 Observer-Dependent Geometry: The Pullback Construction

19.1 A continuous base and a declared comparison rule

The contextual base $\mathcal{C}$ is the smooth domain over which agents' statistical sections live. It is not the set of agents or the space of their probability parameters. The same overlap can carry a smoothly varying attention network at neighboring points, and the base may admit curved geometry without a particular physical metric having been supplied. Its topology, dimension and differentiable structure are inputs to the present model. The construction below derives statistical metric data on that base; it does not derive the base itself or assert direct access to an underlying reality beyond the agents' records. This is a limited mathematical contribution to Wheeler's information-to-geometry question [1].

Let $\mathcal{S} = P \times_G \mathcal{B}$ be the statistical bundle of Section 4.2, let ζ_i be an agent section over $\mathcal{U}_i$, and choose a principal connection ω on P. Assume the statistical fiber is a regular identifiable smooth family with Fisher metric g^F, preserved by the represented coordinate action. The connection induces a horizontal distribution on $\mathcal{S}$ and a vertical projection ver^ω. The relevant derivative is

$$D^\omega \zeta_i = \mathrm{ver}^\omega \circ T\zeta_i, \qquad D^\omega \zeta_i : T\mathcal{U}_i \longrightarrow \zeta_i^* V\mathcal{S}. \tag{185}$$

It measures how the law changes after the chosen rule for comparison between neighboring fibers has been accounted for. On a nonlinear statistical family it is a vertical tangent-valued one-form, not matrix subtraction between probability laws. It is unrelated to the inference-time derivative at a fixed context, which is already vertical.

In local coordinates with the convention $z' = gz$, a linear associated connection is $D = d + B$, where $B' = gBg^{-1} - dgg^{-1}$. The induced statistical derivative is $D\zeta = d\zeta + \mathsf{f}_B(\zeta)$, with f_B the fundamental vector field of the action on laws. For Gaussian state marginals this becomes

$$D\mu = d\mu + B\mu, \qquad D\Sigma = d\Sigma + B\Sigma + \Sigma B^\top. \tag{186}$$

Choosing $B = -dUU^{-1}$ recovers the earlier frame-stripped formula, but this is a particular flat comparison rule. Independent interaction links do not determine a base connection or force it to have this form.

19.2 The invariant information tensor

Define

$$G_i^\omega(v,w) = g^F_{\zeta_i(c)}(D^\omega\zeta_i(v), D^\omega\zeta_i(w)). \tag{187}$$

In a regular density chart, if $S_v = (D^\omega\zeta_i(v))/\zeta_i$ is its tangent score, then $G_i^\omega(v,w) = \mathbb{E}_{\zeta_i}[S_v S_w]$. Normalization gives $\mathbb{E}S_v = 0$. This is a semimetric until nondegeneracy is established, rather than a physical distance assigned by definition.

Proposition 10 (Covariance, positivity and rank)**.** *For every regular section and chosen connection as above, G_i^ω is invariant under passive changes of local frame, is positive semidefinite, and satisfies*

$$\operatorname{rad} G_i^\omega = \ker D^\omega\zeta_i, \qquad \operatorname{rank} G_i^\omega = \operatorname{rank} D^\omega\zeta_i. \tag{188}$$

It depends in general on the connection as well as on the statistical section.

Proof. The vertical jet transforms by the tangent action of g, while g^F is invariant under that action, so the two changes cancel in Eq. (187). Positivity follows from $G_i^\omega(v,v) = \mathbb{E}[S_v^2] \geq 0$. Identifiability makes the vertical Fisher metric positive definite; therefore this value vanishes exactly when $D^\omega\zeta_i(v) = 0$, which proves the radical and rank statements. If the connection changes by an adjoint-valued one-form a, write $\mathsf{J}_a(v)$ for its fundamental vertical action along the section. Then

$$\begin{aligned} G_i^{\omega+a}(v,w) - G_i^\omega(v,w) = {} & g^F(\mathsf{J}_a(v), D^\omega\zeta_i(w)) + g^F(D^\omega\zeta_i(v), \mathsf{J}_a(w)) \\ & + g^F(\mathsf{J}_a(v), \mathsf{J}_a(w)), \end{aligned} \tag{189}$$

which need not vanish. □

The distinction is physical as well as mathematical. Re-expressing one chosen connection and one section changes no tensor; choosing a different comparison rule can change the tensor. Even a constant section can acquire a nonzero covariant jet under a connection acting nontrivially on that law. A canonical perceived geometry therefore requires a reason to select both the law and the comparison rule. Gauge covariance alone supplies neither.

If the jet has constant rank, its kernel is a smooth subbundle and G_i^ω induces a positive metric on $T\mathcal{U}_i/\ker D^\omega\zeta_i$. A quotient of vector bundles is not automatically a quotient manifold: interpreting its directions as the tangent directions of an observable space additionally requires integrability and descent along the kernel leaves. These conditions are independent of positivity.

19.3 Joint, marginal and predictive geometry

One statistical bundle can carry all the geometric distinctions previously assigned to separate belief and model bundles. Suppose a joint law disintegrates as $\zeta(dk, dm) = q(dk \mid m)s(dm)$, and the group action and the connection respect the state and model projections. In a regular chart adapted to the connection at the point of evaluation, its score splits as

$$S_v^\zeta = S_v^s + S_v^{q|m}, \qquad \mathbb{E}_{q(\cdot|m)} S_v^{q|m} = 0. \tag{190}$$

The second identity follows by differentiating conditional normalization; it assumes fixed support and enough domination to interchange differentiation and integration. Taking the expected product of scores makes both cross terms vanish and gives

$$G_\zeta^\omega = G_s^\omega + \mathbb{E}_s G_{q|m}^\omega. \tag{191}$$

Thus model uncertainty and conditional state uncertainty contribute to one joint geometry. In the product specialization the conditional state law is independent of m and this becomes a sum of the two marginal tensors. Outside that specialization a sum of marginal tensors discards dependence and is not the joint Fisher geometry.

The prior p_i is a predictive law available before the current record, and its own pullback can be defined when it varies smoothly with context. It is neither the model presentation m_i nor the belief s_i about that presentation. A slowly changing predictive or model statistic can support a stable geometric description while fast posterior beliefs fluctuate. Whether it does is a timescale and closure property to derive from the dynamics, not a consequence of the letter assigned to that statistic. Identifying one of these tensors with experienced ambient space remains a candidate operational interpretation; this section does not privilege the slow marginal by definition.

19.4 Gaussian sections in an arbitrary base chart

For a Gaussian marginal $q_i(c) = \mathcal{N}(\mu_i(c), \Sigma_i(c))$ with $\Sigma_i \succ 0$, its covariant pullback has the explicit form

$$G^{(q)}_{i,\alpha\beta} = (D_\alpha \mu_i)^\top \Sigma_i^{-1} (D_\beta \mu_i) + \tfrac{1}{2} \operatorname{tr}\big(\Sigma_i^{-1} D_\alpha \Sigma_i \Sigma_i^{-1} D_\beta \Sigma_i\big). \tag{192}$$

The indices α, β range over the dimension of the base chart, with no restriction to two dimensions. In a declared reference chart with $B = 0$ and $\Sigma_i = \sigma^2 I_K$, substitution gives

$$G^{(q)}_i = \sigma^{-2}\big(d\mu_i^\top d\mu_i + 2K\, d\sigma \otimes d\sigma\big). \tag{193}$$

Its rank is the rank of the combined map $(d\mu_i, d\sigma)$. The mean and uncertainty gradients both matter; constant means need not give zero geometry when uncertainty varies. This example uses ordinary probability and does not identify the sample dimension K with the dimension of the base.

19.5 The limit of the pullback mechanism

The tensor measures local statistical distinguishability after a connection has been chosen. Its line element is dimensionless in the information convention, while its components transform with the base coordinates. A physical length requires a calibration or an independently derived dimensionful scale. A dimensionless statistical length is not already a number of meters, and changing the logarithm base cannot change a physical prediction.

Positive Fisher geometry also does not generate a negative temporal norm. Adding a gauge-dependent term $\operatorname{tr}(U^{-1}dU \otimes U^{-1}dU)$ would replace this problem with an arbitrary frame choice. Section 20 instead treats Lorentz geometry as a separate representation and soldering construction. Neither a positive tensor nor its Lorentzian extension becomes gravitational dynamics merely by calculating its curvature.

Finally, a readout can influence inference only if the generative law or the updating law actually reads it. In the pointwise theory the contextual pullback is diagnostic. Feeding it back through a normalized record kernel, a population prior, or a specified base-field term defines a further model whose normalizers and variational derivatives must be retained. Such a feedback model can be studied without adding a second primitive bundle, but its existence and physical performance are separate questions from Eq. (187).

20 Lorentz Geometry from a Two-Component Communication Sector

The Lorentzian signature problem has two distinct parts. One is algebraic: can the information-bearing objects of the theory carry the proper, orthochronous Lorentz group $\mathrm{SO}^+(1,3)$? The other is physical: do the variational dynamics select those objects, identify their four components with observable spacetime directions, and give them the propagation laws and units measured in nature? The construction below answers the first question exactly under a specified channel hypothesis. It does not answer the second by renaming an information metric. In particular, the Fisher tensor remains positive semidefinite. For every tangent vector v, its covariant pullback obeys

$$h^{\omega}_q(v,v) = g^F_q(D^{\omega}q(v), D^{\omega}q(v)) \geq 0. \tag{194}$$

Positive weighted sums, regular quotients, and pointwise limits of such tensors remain positive semidefinite, since their value on each v remains nonnegative. A Lorentzian form therefore requires an additional construction with a different mathematical type.

20.1 The channel hypothesis and its geometric types

Suppose a retained communication sector is a complex vector bundle $W \to \mathcal{C}$ of complex rank two. This is a conditional specialization of the general communication model: neither complex rank two nor a preferred such sector has been derived from VFE. For an originally real four-dimensional sector, choosing W means choosing a complex structure J with $J^2 = -I$ and restricting the relevant frame changes to those commuting with J. An arbitrary real $\mathrm{GL}(4,\mathbb{R})$ frame change need not preserve that choice. Selecting a two-component projection from a larger cognitive fiber similarly requires a specified, covariant selection map.

The relevant real bundle is the four-dimensional bundle $\mathcal{H}(W)$ of Hermitian tensors in $W \otimes \overline{W}$. In a complex frame these tensors are Hermitian matrices. They are not assumed to be quantum density operators. If $z \in \mathbb{C}^2$ is an ordinary classical random message with finite second moment, then

$$M = \mathbb{E}[zz^{\dagger}] \in \mathcal{H}(W), \qquad M \succeq 0, \tag{195}$$

is one realization. The probability law of z is normalized; its second moment need not have unit trace. The dagger in this formula denotes conjugate transpose of coordinate arrays. Intrinsically, $zz^{\dagger}$ is $z \otimes \bar{z}$; no invariant positive inner product on W is required.

There is an operationally explicit way to read all four real components. Write $M = \left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$. The four mean intensities

$$\begin{aligned} \mathbb{E}|z_1|^2 &= a, \qquad \mathbb{E}|z_2|^2 = d, \\ \mathbb{E}|z_1 + z_2|^2 &= a + d + 2\,\mathrm{Re}\, b, \\ \mathbb{E}|z_1 + iz_2|^2 &= a + d + 2\,\mathrm{Im}\, b. \end{aligned} \tag{196}$$

recover M by subtraction. These identities follow by expanding the squared moduli. They require access to the two components and a calibrated relative phase; neither capability is implied by arbitrary selective links. Conversely every positive semidefinite M is realizable by a finite classical law: take $z = M^{1/2}\xi$, where the two components of ξ are independent signs with equal probabilities. Then $\mathbb{E}\xi = 0$ and $\mathbb{E}\xi\xi^{\dagger} = I$. Thus the full cone is available when this preparation and readout family is supplied, without invoking quantum probabilities.

Use column coordinates and left actions throughout. A passive frame change sends z_i to $g_i z_i$, where $g_i(c) \in \mathrm{GL}(2,\mathbb{C})$, and consequently

$$X'_i = g_i X_i g_i^{\dagger}, \qquad L'_{ij} = g_i L_{ij} g_j^{-1}, \qquad L_{ij} : W_j \longrightarrow W_i. \tag{197}$$

The first index is the receiver. Independent interaction maps L_{ij} may be singular and are not transition functions of W. Bundle transitions are invertible and satisfy the cocycle identity. At a fixed c_* all formulas concern the attention network above that one point; allowing smooth c-dependence introduces no base-transport equation.

20.2 The determinant and the full Lorentz group

Let $\sigma_0 = I_2$ and let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{198}$$

Every Hermitian matrix has the unique expansion

$$X = x^\mu \sigma_\mu = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}, \qquad x^\mu = \tfrac{1}{2}\operatorname{tr}(\sigma_\mu X) \in \mathbb{R}. \tag{199}$$

Here repeated Greek component indices run from 0 to 3. The determinant gives

$$\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \tag{200}$$

Its negative polarization is the real symmetric bilinear form

$$\eta(X, Y) = \tfrac{1}{2}\big[\operatorname{tr}(XY) - \operatorname{tr}(X)\operatorname{tr}(Y)\big], \qquad \eta(X, X) = -\det X. \tag{201}$$

Thus $\eta(\sigma_\mu, \sigma_\nu) = \operatorname{diag}(-1, 1, 1, 1)_{\mu\nu}$. The trace terms are a coordinate formula for this determinant polarization; neither trace separately is invariant under a general congruence. This matrix realization is classical Lorentz geometry, also realized by optical coherency matrices [19] and used in local quantum filtering [32]. We give the proof to fix the precise group and convention [20, Sec. III].

Theorem 1 (Exact Lorentz and conformal Lorentz representations)**.** *For every $A \in \mathrm{SL}(2, \mathbb{C})$, the real-linear action $X \mapsto AXA^\dagger$ on $\mathcal{H}(\mathbb{C}^2)$ preserves η and the positive cone. It defines a surjective homomorphism*

$$\rho : \mathrm{SL}(2, \mathbb{C}) \longrightarrow \mathrm{SO}^+(1, 3), \qquad \rho(A)^\mu{}_\nu = \tfrac{1}{2}\operatorname{tr}(\sigma_\mu A \sigma_\nu A^\dagger), \qquad \ker \rho = \{I, -I\}. \tag{202}$$

For every $A \in \mathrm{GL}(2, \mathbb{C})$, the same action instead satisfies

$$\eta(AXA^\dagger, AYA^\dagger) = |\det A|^2 \eta(X, Y). \tag{203}$$

Its image is the connected, time-orientation-preserving linear conformal Lorentz group $\mathbb{R}_{>0}\mathrm{SO}^+(1, 3)$ and its kernel is $\{e^{i\theta} I\}$. The metric-preserving subgroup in $\mathrm{GL}(2, \mathbb{C})$ is $|\det A| = 1$; the determinant-one subgroup gives the stated double cover.

Proof. Multiplicativity of the determinant gives

$$\det(AXA^\dagger) = |\det A|^2 \det X.$$

Polarization proves Eq. (203), and matrix multiplication proves the homomorphism law. Congruence preserves positive definiteness and positive semidefiniteness. The eigenvalues of X are $x^0 \pm |\boldsymbol{x}|$, so $X \succ 0$ is exactly the future timelike cone $x^0 > |\boldsymbol{x}|$. Thus the action preserves its time orientation. For $\det A = 1$, write $A = HU$ with $H = (AA^\dagger)^{1/2} \succ 0$ and $U \in \mathrm{SU}(2)$. The positive eigenvalues of H have product one, so H is joined to I by H^s, $0 \le s \le 1$. Every $U \in \mathrm{SU}(2)$ has the form

$\cos(\theta/2)I - i\sin(\theta/2)\boldsymbol{n}\cdot\boldsymbol{\sigma}$ and is also joined to I. Consequently $\rho(A)$ lies in the identity component and is proper.

For surjectivity, let $\Lambda \in \mathrm{SO}^+(1,3)$ and write its image of the unit future vector as $\Lambda e_0 = (u^0, \boldsymbol{u})$. The matrix $H_0 = u^0 I + \boldsymbol{u}\cdot\boldsymbol{\sigma}$ is positive definite with determinant one. Set $A = H_0^{1/2}$; then $\rho(A)e_0 = \Lambda e_0$. The transformation $\rho(A)^{-1}\Lambda$ fixes e_0, hence acts on its orthogonal complement as some $R \in \mathrm{SO}(3)$. Every such R is an axis-angle rotation. Direct multiplication with the Pauli identity $\sigma_a\sigma_b = \delta_{ab}I + i\epsilon_{abc}\sigma_c$ shows that $U = \cos(\theta/2)I - i\sin(\theta/2)\boldsymbol{n}\cdot\boldsymbol{\sigma}$ induces this rotation. Thus $\Lambda = \rho(AU)$.

If $AXA^\dagger = X$ for every Hermitian X, take $X = I$ to obtain $AA^\dagger = I$. The remaining identities say that A commutes with all Pauli matrices, which forces $A = \lambda I$. Unitarity gives $|\lambda| = 1$ and determinant one gives $\lambda = \pm 1$. The differential of ρ at the identity sends a traceless complex matrix a to the map $X \mapsto aX + Xa^\dagger$. If this map vanishes, $X = I$ first makes a anti-Hermitian and the other Hermitian X force a to be scalar; tracelessness then gives $a = 0$. The differential is therefore an isomorphism between two six-dimensional real tangent spaces. Local invertibility, translated around each group element, and the two-element kernel give the two-sheeted covering. The same scalar argument proves the stated kernel for the general linear action. Finally, choose a complex square root d of $\det A$ and put $S = A/d \in \mathrm{SL}(2,\mathbb{C})$. Then $AXA^\dagger = |\det A| SXS^\dagger$. Both choices of d give the same action. Arbitrary positive scalar dilations and all proper orthochronous Lorentz transformations are therefore obtained. □

For example, $A = \mathrm{diag}(e^{\chi/2}, e^{-\chi/2})$ gives the ordinary boost

$$\begin{pmatrix} x^{0\prime} \\ x^{3\prime} \end{pmatrix} = \begin{pmatrix} \cosh\chi & \sinh\chi \\ \sinh\chi & \cosh\chi \end{pmatrix} \begin{pmatrix} x^0 \\ x^3 \end{pmatrix}, \qquad x^{1\prime} = x^1, \quad x^{2\prime} = x^2. \tag{204}$$

Together with the rotations in the proof, these are all six Lorentz generators, not only one $\mathrm{SO}^+(1,1)$ subgroup. No imaginary base coordinate or real-part projection was introduced.

There is also a limited uniqueness statement. Every real quadratic form invariant under this $\mathrm{SL}(2,\mathbb{C})$ action is a scalar multiple of $\det X$. Rotation invariance first restricts it to $a(x^0)^2 + b|\boldsymbol{x}|^2$: a nonzero temporal-spatial cross vector and an anisotropic spatial quadratic form would break some rotation. Differentiating invariance under Eq. (204) at $\chi = 0$ then gives $2(a+b)x^0x^3 = 0$, hence $b = -a$. The invariant quadratic form is therefore fixed up to scale once this representation has been selected. This is representation-theoretic uniqueness, not a principle selecting the representation from the general VFE theory.

There is a related operational characterization with stronger hypotheses than positivity preservation alone. Let T be a real-linear bijection of $\mathcal{H}(\mathbb{C}^2)$ that maps the whole positive semidefinite cone onto itself. Its inverse then preserves that cone as well, and T maps its boundary onto itself. In Pauli coordinates write $Q(x) = \det T(x^\mu\sigma_\mu) = a(x^0)^2 + 2x^0\boldsymbol{b}\cdot\boldsymbol{x} + \boldsymbol{x}^\top D\boldsymbol{x}$, with D symmetric. Every $(1,\boldsymbol{n})$ with $|\boldsymbol{n}| = 1$ is a boundary point. Applying $Q = 0$ to $\boldsymbol{n}$ and $-\boldsymbol{n}$ gives $\boldsymbol{b} = 0$ and $\boldsymbol{n}^\top D\boldsymbol{n} = -a$ on the unit sphere, hence $D = -aI$. Since $T(I)$ lies in the positive interior, $a = \det T(I) > 0$. Therefore

$$\det T(X) = a\det X, \qquad a > 0. \tag{205}$$

After division by $\sqrt{a}$, T is a time-orientation-preserving Lorentz transformation. The identity component of this cone-automorphism group is exactly $\mathbb{R}_{>0}\mathrm{SO}^+(1,3)$; the other component also permits spatial reflection. This gives an observable route to the same conformal group if the entire cone is accessible, observables transform linearly, and transformations are reversible and preserve order in both directions. A generic lossy channel meets none of the reversibility conclusion merely by preserving positive moments. These hypotheses characterize a conditional symmetry class; they are not a derivation of that class from VFE.

20.3 Selective links and the positive cone

The relation to selective communication can now be stated without identifying every interaction with a Lorentz transformation. A message z contributes the rank-one tensor $zz^\dagger$, which lies on the future null cone. For nonnegative weights w_r and messages $z_r \in \mathbb{C}^2$, define

$$M = \sum_r w_r z_r z_r^\dagger, \qquad \det M = \sum_{r<s} w_r w_s |z_{r1} z_{s2} - z_{r2} z_{s1}|^2. \tag{206}$$

To verify the identity, expand $M_{11}M_{22} - |M_{12}|^2$. Terms with the same index cancel, and pairing indices r, s gives the squared modulus displayed above. Thus M is future timelike exactly when the messages with positive weight span two complex directions; it is nonzero future null exactly when they span one. The determinant measures a precise failure of the messages to share one complex ray. No physical mass or particle law follows from assigning this scalar a suggestive name.

For a received message $L_{ij} z_j$, its second moment is $L_{ij} M_j L_{ij}^\dagger$. If L_{ij} is invertible this is a conformal Lorentz map, and it is Lorentz-isometric exactly when $|\det L_{ij}| = 1$. If L_{ij} has rank one, write $L_{ij} = uv^\dagger$ in coordinates. Then

$$L_{ij} X L_{ij}^\dagger = (v^\dagger X v) u u^\dagger. \tag{207}$$

Its real-linear image is at most one-dimensional; every positive input is sent to the same null ray or zero. A singular selective link is therefore a cone-preserving communication map, not a Lorentz frame change. A weighted sum of such channels need not preserve the null cone either: the positive map $\Phi(X) = \frac{1}{2}\operatorname{tr}(X) I$ sends $\operatorname{diag}(1, 0)$ to $I/2$, changing determinant zero to $1/4$. It is a sum of the four rank-one maps with matrices $e_a e_b^\dagger/\sqrt{2}$, $a, b \in \{1, 2\}$. General selective communication thus supplies a larger family of operations than Lorentz symmetry.

The distinction between a classical second moment and a normalized density matrix also matters. In a chosen frame, fixing $\operatorname{tr} X = 1$ fixes $x^0 = 1/2$; its tangent vectors have $dx^0 = 0$, and the restriction of η to that tangent space is positive definite. The fourth, radial coordinate has been removed. The nonlinear normalization $AXA^\dagger / \operatorname{tr}(AXA^\dagger)$ retains the projective Lorentz action on cone rays, represented in this three-dimensional slice. It does not retain the fourth coordinate or a linear four-dimensional representation. Retaining the amplitude of a moment requires no unnormalized probability law. Likewise the determinant-one orbit of a fixed positive moment has fixed determinant and only three dimensions; one needs a family of determinant values to cover the cone interior.

Ordinary probability remains available throughout. On $\mathbb{C}^2 \cong \mathbb{R}^4$, an invertible complex matrix has real Jacobian $|\det A|^2$. The pushforward density is $q'(z') = q(A^{-1}z')/|\det A|^2$; the same change for a prior cancels the Jacobians in their likelihood ratio and preserves KL. This does not turn the Fisher metric into η, nor does it make the second moment a sufficient statistic for an arbitrary law. A quantum interpretation would add further structure beyond the classical random messages used here.

A passive detector comparison also differs from a physical filtering operation. For a detector covector represented by w, the mean recorded intensity is $w^\dagger M w$. Re-expressing both $M' = AMA^\dagger$ and $w' = A^{-\dagger} w$ leaves this number unchanged; transforming the message while holding the detector fixed generally changes it. Moreover, choosing a particular positive reference moment M_0 reduces its determinant-one stabilizer to

$$\{A \in \mathrm{SL}(2, \mathbb{C}) : AM_0A^\dagger = M_0\} = M_0^{1/2} \mathrm{SU}(2) M_0^{-1/2}. \tag{208}$$

Indeed $M_0^{-1/2} A M_0^{1/2}$ is unitary exactly when the displayed congruence fixes M_0, and it has determinant one. A selected positive moment can therefore define a rest direction even while the

representation laws remain Lorentz covariant. It is not a Lorentz-invariant vacuum merely because its law is written covariantly.

20.4 From an internal cone to a tensor on the base

A positive moment is not a causal update. For example, $M(t) = \mathrm{diag}(e^t, e^{-t})$ is positive definite for every t, while $\dot{M}(0) = \mathrm{diag}(1, -1)$ has $\eta(\dot{M}(0), \dot{M}(0)) = 1$ and is spacelike. The cone of moment values therefore supplies no signal-speed or causal-trajectory theorem by itself.

The four coefficients x^μ above are fiber coordinates. They are not automatically coordinates on the contextual base $\mathcal{C}$, which may have another dimension and an unknown geometry. To obtain a Lorentzian tensor on a four-dimensional base region, one needs a smooth fiberwise isomorphism

$$e : T\mathcal{C} \longrightarrow \mathcal{H}(W), \qquad \gamma(v, w) = \eta(e(v), e(w)). \tag{209}$$

The map e is a soldering form: it identifies base tangent vectors with the internal Lorentz vector space. In local coordinates it is a Hermitian-matrix-valued one-form. Under a determinant-one re-expression it obeys $e' = geg^\dagger$, so Theorem 1 makes γ gauge invariant. In a chosen component frame its matrix is $E^\top \eta E$. If E is invertible, changing variables $u = Ev$ shows that γ has exactly one negative and three positive directions. This proves the conditional construction, including its nondegeneracy hypothesis.

If $\dim \mathcal{C} > 4$, a surjective e instead gives a degenerate base tensor with radical $\ker e$ and a Lorentzian metric on the rank-four quotient bundle $T\mathcal{C}/\ker e$. This quotient bundle is not automatically the tangent bundle of a four-dimensional observable space. A local quotient by base leaves requires an integrable kernel and a tensor constant along those leaves. Neither condition follows from the rank statement. If $\dim \mathcal{C} < 4$, no such surjective e exists.

There is a simple exact local example. On the open cone $c^0 > |\boldsymbol{c}|$ in $\mathbb{R}^4$, take the trivial bundle, the classical second-moment section $M(c) = c^\mu \sigma_\mu \succ 0$, and $e = dM$. Every $M(c)$ can be realized by a centered complex Gaussian or by a suitable finite classical message law. Then $\gamma = -(dc^0)^2 + \sum_a (dc^a)^2$. This is an existence example with chosen readout, not a VFE selection result. The equality $e = dM$ uses the declared reference connection $B = 0$; it is not preserved as an ordinary-derivative formula under c-dependent frame changes. Its covariant version is $e = D^B M = dM + BM + MB^\dagger$, with B transformed as in Eq. (211). That derivative obeys $D^{B'} M' = g(D^B M)g^\dagger$. Moreover, whenever $e = dX$ in a fixed trivialization and dX is invertible, X itself gives local coordinates and the metric is flat. Curvature cannot be inferred from this exact-differential example.

Conversely, an arbitrary nonexact soldering can carry curvature, but that curvature was supplied with the soldering. For example, $e = dt\sigma_0 + a(t) \sum_a dx^a \sigma_a$ gives $\gamma = -dt^2 + a(t)^2 \sum_a (dx^a)^2$. Its nonzero Christoffel components are $\Gamma^0_{ab} = a\dot{a}\delta_{ab}$ and $\Gamma^a_{0b} = \Gamma^a_{b0} = (\dot{a}/a)\delta^a_b$. With $R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$, contraction gives $R = 6[\ddot{a}/a + (\dot{a}/a)^2]$. The choice $a = e^{Ht}$ has $R = 12H^2$. No field equation or variational mechanism in this example selects a or H. The construction distinguishes the ability to represent curved Lorentzian geometry from its dynamical emergence.

For a general $\mathrm{GL}(2, \mathbb{C})$ bundle, the determinant takes values in the real positive-scale line inside $\det W \otimes \overline{\det W}$. Its local coefficients change by $|\det g|^2$. Equation (209) therefore first supplies a conformal Lorentzian tensor. A positive scale field κ with $\kappa' = |\det g|^2 \kappa$ gives the invariant representative $\gamma = \kappa^{-1}\eta(e, e)$. Choosing this scale, or reducing to $|\det g| = 1$, is additional geometric information; it is not the trace normalization of a probability law. A complex volume form reduces further to $\mathrm{SL}(2, \mathbb{C})$. Physical length and time units require their own calibration.

On a good trivializing cover, with contractible nonempty finite intersections, transition matrices must satisfy $g_{ab} g_{bc} = g_{ac}$ and $e_a = g_{ab} e_b g^\dagger_{ab}$. Independent network links need satisfy neither identity.

Starting with a Lorentz-frame cocycle and trying to reconstruct its spinor lift can encounter a further global obstruction: local determinant-one lifts satisfy $\widetilde{g}_{ab}\widetilde{g}_{bc}\widetilde{g}_{ca} = \epsilon_{abc}I$ with $\epsilon_{abc} \in \{\pm 1\}$. Changing signs of the lifts modifies these signs by a coboundary. A global lift exists exactly when signs can be chosen to make every triple product $+I$. Starting instead with an actual $\mathrm{SL}(2,\mathbb{C})$ bundle supplies this compatibility, but still supplies no isomorphism with $T\mathcal{C}$. A soldering must be established globally as well as pointwise.

20.5 Why frame derivatives do not repair the missing soldering

The earlier constructions remain useful diagnostics, but no longer supply the primary route to Lorentz geometry.

Proposition 11 (The earlier single-generator rank obstruction)**.** *If $A = T(id\psi_0 + d\psi_1)$ and $\mathrm{tr}(T^2) = 2$, then $\mathrm{Re}\,\mathrm{tr}(A \otimes A) = 2(d\psi_1 \otimes d\psi_1 - d\psi_0 \otimes d\psi_0)$. Its rank is at most two on any base. Where $d\psi_0, d\psi_1$ are independent, its nondegenerate quotient has signature $(-,+)$.*

Proof. Expand the two factors and take their real part. Every vector annihilated by both differentials lies in the radical. On a complement, the two independent differentials are coordinates for the displayed diagonal form. □

Proposition 12 (The earlier commuting all-real construction)**.** *Let S, T_1, T_2, T_3 be commuting real matrices with $\mathrm{tr}(S^2) = -2$, $\mathrm{tr}(ST_a) = 0$, and $\mathrm{tr}(T_aT_b) = 2\delta_{ab}$. Set $U = \exp(\psi_0 S + \sum_a \psi_a T_a)$. On a four-dimensional region where the four $d\psi_\mu$ are independent, $\mathrm{tr}[(U^{-1}dU) \otimes (U^{-1}dU)]$ has signature $(-,+,+,+)$ and is flat.*

Proof. Commutation gives $U^{-1}dU = S d\psi_0 + \sum_a T_a d\psi_a$ exactly, so the tensor is $2[-d\psi_0^2 + \sum_a d\psi_a^2]$. The ψ_μ are local coordinates, where the metric is constant. The hypotheses have the earlier disjoint-support realization for real dimension at least six: put one standard skew 2×2 block in the first two coordinates and take an orthonormal basis of traceless diagonal matrices on the remaining four coordinates, scaled to squared trace norm two. □

These calculations establish available signatures for specified frame fields. They do not establish physical invariance or selection.

There is a separate error to avoid in replacing a single frame derivative by an agent-pair difference. Put $A_i = U_i^{-1}dU_i$ and $\Omega_{ij} = U_iU_j^{-1}$. The identity

$$\mathrm{tr}[(A_i - A_j)_\mu (A_i - A_j)_\nu] = \mathrm{tr}[(\Omega_{ij}^{-1}\partial_\mu \Omega_{ij})(\Omega_{ij}^{-1}\partial_\nu \Omega_{ij})] \tag{210}$$

is correct, since $\Omega_{ij}^{-1}d\Omega_{ij} = U_j(A_i - A_j)U_j^{-1}$. The expression is invariant under a common right change $U_i \mapsto U_i h(c)$, because the inhomogeneous terms in $A_i - A_j$ cancel. It is not invariant under independent local re-expressions $U_i \mapsto g_i(c)U_i$. On a one-dimensional interval, take $U_i = U_j = I_2$, $g_i = e^c I_2$, and $g_j = I_2$. The expression changes from zero to $2dc^2$. Thus common-reference invariance does not establish the required passive gauge invariance.

If base connections are separately chosen, there is a covariant replacement on invertible links. With coordinate convention $D_i = d + B_i$,

$$B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}, \qquad \nabla L_{ij} = dL_{ij} + B_i L_{ij} - L_{ij}B_j, \tag{211}$$

so $\nabla L_{ij}' = g_i(\nabla L_{ij})g_j^{-1}$. Consequently $K_{ij} = (\nabla L_{ij})L_{ij}^{-1}$ transforms by conjugation at i, and $\mathrm{Re}\,\mathrm{tr}(K_{ij,\mu}K_{ij,\nu})$ is a real gauge-invariant tensor. It is connection relative and has no guaranteed Lorentzian signature. For the pure frame links $L_{ij} = U_iU_j^{-1}$ with their induced flat connections

$B_i = -dU_i U_i^{-1}$, direct substitution gives $\nabla L_{ij} = 0$. The apparent frame-gradient geometry disappears. Singular selective links do not admit L_{ij}^{-1}, so this formula does not cover the general communication model. None of these corrections imposes new base-connection dynamics on the pointwise formation theory.

20.6 What remains to obtain physical spacetime

The determinant construction removes the need for the former imaginary-time, real-part-projection, and commuting-generator postulates. It replaces them with one explicit representation hypothesis and, only when base geometry is claimed, a soldering and scale hypothesis. It provides the full proper Lorentz group and a positive cone directly connected to selective messages. The physical claim remains conditional because arbitrary selective updates need not preserve this cone boundary, no VFE result here selects a complex two-component retained sector, and no result constructs the required collective soldering from those updates.

A finite-speed alternative retains a different conditional status. Given an independently specified shared time coordinate t, a positive spatial tensor h_{ab}, and an isotropic signal bound $h_{ab}dx^a dx^b \leq c_{\mathrm{sig}}^2 dt^2$, its boundary is the null cone of $-c_{\mathrm{sig}}^2 dt^2 + h_{ab}dx^a dx^b$. This identifies a Lorentzian conformal class under those hypotheses; it does not derive the time coordinate, three spatial dimensions, isotropy, exact finite propagation, or boost covariance of the underlying dynamics. Exponentially small influence outside an effective graph cone is not zero influence outside a physical null cone. A gradient-flow history also remains a history of beliefs: away from stationary points its velocity singles out a line in a positive Fisher tangent space but supplies no negative norm. That line vanishes at a stationary point.

The next discriminating task is therefore concrete: derive, within a specified normalized generative model and its VFE dynamics, a covariantly selected complex rank-two communication statistic together with a collective, nondegenerate base readout whose evolution has the derived null cone as its characteristic cone. Failure of rank, gluing, cone preservation, or characteristic compatibility would refute that proposed mechanism. Success would still leave physical units and field dynamics to establish, but it would connect the Lorentz representation proved here to the theory's actual updating law.

21 One World out of Many: Collective Geometry and Operational Agreement

Wheeler's question of how one world arises out of many participants motivates a collective description [1]. The mathematical problem is to say what becomes shared when agents communicate selectively. Agreement on a public channel is weaker than equality of cognitive states; a stable parent law is stronger than an instantaneous agreement; and agreement between statistical predictions is different from a common geometry of how those predictions vary with context. These distinctions make it possible to retain observer dependence without attributing physical consequences to a passive frame change.

21.1 Collective information geometry

For a regular collective recognition section $Q(c)$ in the normalized population model, define its covariant Fisher tensor by the same construction as in Section 19.1. It reads the joint law, including

correlations. A convenient descriptive alternative is a weighted sum of marginal tensors,

$$\bar{G}_{\alpha\beta}(c) = \sum_i w_i(c) G^{\omega}_{i,\alpha\beta}(c), \qquad w_i \geq 0, \quad \sum_i w_i = 1. \tag{212}$$

This tensor is gauge invariant if the weights are scalar observables, but it is not selected by gauge invariance and generally is not the Fisher tensor of Q. Different joint laws can share all the same marginals.

An exact example makes the missing correlation geometry visible. Let Q_ρ be a centered bivariate Gaussian with covariance

$$\Sigma_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad |\rho| < 1. \tag{213}$$

Each marginal is $\mathcal{N}(0,1)$ independently of ρ, so every marginal tensor in the ρ direction is zero. The joint tensor is instead

$$G_{Q_\rho} = \tfrac{1}{2} \operatorname{tr}\big[(\Sigma_\rho^{-1} \partial_\rho \Sigma_\rho)^2\big] d\rho^2 = \frac{1+\rho^2}{(1-\rho^2)^2} d\rho^2. \tag{214}$$

Indeed, $\Sigma_\rho^{-1}\partial_\rho\Sigma_\rho = (1-\rho^2)^{-1} \left(\begin{smallmatrix} -\rho & 1 \\ 1 & -\rho \end{smallmatrix}\right)$, and squaring it gives the displayed trace. A collective geometry can therefore carry information absent from every individual marginal geometry.

There is a specific probabilistic meaning for the weighted sum when it is appropriate. Let $r_i(o \mid c)$ be normalized laws on a common record space and retain a source label I with joint law $R(i, do \mid c) = w_i(c) r_i(do \mid c)$, where all active weights are positive. Its score is the sum of the weight score and conditional record score, and the conditional score has zero expectation. Therefore

$$G_R = \sum_i w_i G_{r_i} + \sum_i \frac{dw_i \otimes dw_i}{w_i}. \tag{215}$$

Constant weights remove the second term. Context-dependent attention weights do not. Forgetting the source label gives the mixture record law, whose score is the conditional expectation of the joint score given the record. Conditional variance then gives $G_{\text{mixture}} \preceq G_R$. These statements require regular scores and a compatible connection; they do not identify a source mixture with the correlated population law.

No orbit average over cognitive frames is needed for any of these tensors. Their invariance follows from their typed construction. The older frame-twist proposal can be disposed of by a smaller counterexample than an analysis of regulated Haar measures.

Proposition 13 (A removable frame does not define perceived length)**.** *Suppose only* $\Omega_{ij} = U_i U_j^{-1}$ *and frame-independent data are read. Then* $\operatorname{tr}(U_i^{-1} dU_i \otimes U_i^{-1} dU_i)$ *is not an invariant of those data.*

Proof. The common right change $U_j \mapsto U_j h(c)$ leaves every Ω_{ij} unchanged. Choosing $h = U_i^{-1}$ makes the selected frame U_i constant and its derivative tensor zero. For example, $U_i = e^c I$ gives a nonzero tensor before that change. Thus two presentations of the same specified data assign different lengths. □

This proposition concerns the pure reference-frame expression. It does not remove the content of independent matrix links, a chosen base connection, or the Hermitian determinant construction of Section 20.

21.2 Public agreement and private variation

For receiver i and sender j, the interface model compares laws on a declared common output space. Write these as $r_{i|ij}$ and $r_{j|ij}$, obtained from the endpoint laws by their respective receiving and sending kernels. The edge disagreement is

$$K_{ij} = D_{\mathrm{KL}}(r_{i|ij} \| r_{j|ij}). \tag{216}$$

It vanishes exactly when the compared laws agree, up to the standard almost-everywhere qualification. It need not compare every direction of either agent. For instance, the projection $(x, y) \mapsto x$ maps both $\mathcal{N}((0, a), I_2)$ and $\mathcal{N}((0, b), I_2)$ to $\mathcal{N}(0, 1)$ for arbitrary a, b. Communication gives perfect public agreement while private means differ.

This is a mechanism for public agreement without requiring uniform internal descriptions. Stability and whether such a record becomes an autonomous parent are separate closure questions of Sections 10–12. Hidden variables can remain relevant to later interactions despite being invisible to the current public comparison. Retained conditional memory is then part of the parent's actual law rather than an optional interpretation.

Even full pointwise belief agreement is insufficient for geometric agreement. In a fixed reference chart, $q_1(c) = \mathcal{N}(c, 1)$ and $q_2(c) = \mathcal{N}(2c, 1)$ agree at $c = 0$, but their pullbacks are dc^2 and $4dc^2$. Nor does uniformly small KL control their derivatives. For $\epsilon > 0$, let

$$q_\epsilon(c) = \mathcal{N}\big(\epsilon \sin(c/\epsilon^2), 1\big), \qquad q_0(c) = \mathcal{N}(0, 1). \tag{217}$$

Then

$$D_{\mathrm{KL}}(q_\epsilon(c) \| q_0(c)) \le \epsilon^2/2, \qquad G_{q_\epsilon}(c) = \epsilon^{-2} \cos^2(c/\epsilon^2) dc^2, \qquad G_{q_0} = 0. \tag{218}$$

The discrepancy is unbounded at $c = 0$ as $\epsilon \downarrow 0$. A theorem of common perceived geometry therefore needs derivative control, not only small disagreement values. Exact equality of smooth compared laws throughout an open region does imply equal Fisher tensors there when their comparison maps and connections are compatible; a merely pointwise or approximate equality does not.

21.3 Observable directions from records

A record law $r(o \mid c)$ gives its own Fisher tensor J, computed with the declared comparison rule. Its radical consists of directions invisible to first order in that record family. This is an operational distinction: invisibility is relative to the available record, and does not assert that the underlying direction is absent. If a fixed, or covariantly constant, Markov observation kernel maps a regular latent law to the record law, the record score is a conditional expectation of the latent score. The conditional-variance identity proves $J \preceq G$ and identifies the lost information. A context-dependent kernel carries its own score and must be differentiated as part of the full model; the same inequality cannot simply be assumed after omitting that dependence.

Finite sensitivity can be described by the generalized eigenproblem

$$Jv = \lambda H v, \tag{219}$$

where H is an explicitly chosen positive reference tensor on the comparison space. When $J \preceq H$, all its generalized eigenvalues lie in $[0, 1]$, because $v^\top J v / v^\top H v$ lies in that interval. A positive collective Fisher tensor can supply H on a regular observable quotient; otherwise a further reference must be declared. If H is singular, the pencil first requires a well-defined quotient or complement rather than an unqualified ordinary eigendecomposition. Smooth sector bundles require constant rank and a spectral gap around any chosen sensitivity threshold.

Nothing in this construction forces four observable directions or assigns one a temporal sign. The dimensions and thresholds are properties to determine from a specified record model. The Lorentz section names the separate rank-four and soldering conditions needed for a spacetime interpretation. Passive coordinate changes preserve these tensor statements; changing an agent's records, statistical law, interface or comparison connection can change them.

21.4 Physical quantities and calibration

A dimension is not a numerical choice of units. Bits and nats are dimensionless conventions related by $I_{\mathrm{bit}} = I_{\mathrm{nat}}/\log 2$. A physical action, length or temperature carries another type. An equation $S_{\mathrm{phys}} = a_0 I_{\mathrm{nat}}$ requires a_0 to have action units; changing the logarithm convention rescales its numerical coefficient. Choosing $a_0 = \hbar/\log 2$ so that one bit corresponds to $\hbar$ calibrates a conversion and does not derive $\hbar$ or a smallest possible update.

The most direct tests concern dimensionless predictions: ratios of response times, probabilities of records, relative coarse-graining errors and symmetry violations across calibrated observers. If a physical identification succeeds, its dimensions and conversion factors must survive composition across agents and scales. The same word should not stand for a statistic, a physical observable and a fitted conversion between them.

Physical target	Available mathematical object	Required connection to experiment
Duration	Fisher length or expected record information	Calibration and compatibility of histories and records
Spacetime	A conditional Lorentz cone and soldering	Observable rank, causal characteristics and physical scale
Inertia	Retained response and memory	An independently identified kinetic response law
Energy	Dimensionless VFE or effective action	A physical ensemble and an energy or action conversion
Entropy	Relative entropy or ensemble entropy	A specified reference measure and thermodynamic identification
Objectivity	Agreement on communicated records	Persistence and successful predictions under intervention

Table 3: Physical targets and their current mathematical counterparts. Each final column is a research obligation, not an additional universal postulate.

22 Physical Targets: Gravity, Persistence and Measurement

22.1 What would turn a geometric readout into gravity?

A curved statistical tensor is not yet a gravitational field. The covariant Fisher pullback is positive semidefinite; the separate Lorentz construction supplies an indefinite tensor only after a complex channel, soldering and scale have been chosen. To interpret a resulting Lorentz metric γ gravitationally, the same collective construction must also specify the matter or record variables that couple to it and an evolution law for that coupling. Otherwise one can fit a geometry to observations without explaining how geometry affects the participating systems.

A conditional route is available in the normalized population model. Suppose a proposed collective geometry changes a prior, a record kernel or an effective action. Its dependence then enters the evidence and all variational derivatives through that normalized law. For example, if a conditional density is written as $p_\gamma(y) = e^{-S_\gamma(y)}/Z_\gamma$ against a fixed reference measure and differentiation under the integral is justified, then

$$\delta \log Z_\gamma = -\mathbb{E}_{p_\gamma} \delta S_\gamma, \qquad \delta \log p_\gamma = -\delta S_\gamma + \mathbb{E}_{p_\gamma} \delta S_\gamma. \tag{220}$$

If the measure itself depends on γ, its logarithmic variation belongs in the same calculation. A proposed geometric equation that varies only S_γ and discards this term is not the equation of the declared probability model.

The physical target is a common causal geometry with consistent source coupling, conservation identities, and an observational limit. In a weak and slowly varying regime, a candidate must predict the appropriate acceleration and clock comparisons from its retained dynamics; naming a curvature scalar or a precision parameter does not supply those predictions. No derivation of Einstein dynamics or its constants is asserted here. The exact coarse effective action of Section 12 provides a place to investigate induced interactions and their symmetries, while the Lorentz construction specifies the geometric type such a limit would have to support.

22.2 Persistence and the meaning of an object

The formation theory distinguishes coherence, persistence and autonomy. The proposed physical reading should keep the distinction. A cluster can agree on a communicated statistic while its private state changes; a long-lived memory can retain that statistic while failing to support a second formation episode; and an exact instantaneous marginal can fail to have closed future dynamics. Calling the cluster a macroscopic object requires evidence for its retained state and response across those changes, rather than only a small divergence at one time.

Precision by itself does not solve this problem. In the scalar composite consensus specialization, take a receiver $q_i = \mathcal{N}(a, v)$ with fixed $v > 0$ and a source $q_j = \mathcal{N}(0, \delta)$, $\delta > 0$. Their cost is

$$K_{ij} = \tfrac{1}{2}\left[\frac{v + a^2}{\delta} - 1 + \log\frac{\delta}{v}\right]. \tag{221}$$

At fixed nonzero coupling, the mean gradient $\partial_a K_{ij} = a/\delta$ becomes strong as δ decreases. With Gibbs source selection, however, a competing source of bounded cost gives

$$\beta_{ij} \le C e^{-K_{ij}/\tau}, \qquad \lim_{\delta \downarrow 0} \beta_{ij} = 0 \tag{222}$$

for fixed $\tau > 0$ and fixed positive prior weights. The bound follows by retaining the competing source's positive term in the softmax denominator. Its exponential suppresses the $1/\delta$ growth of the gradient as well. A highly precise but incompatible source can therefore be ignored instead of becoming a universal anchor. A noisy interface or an adaptive receiver changes the calculation and must be modeled explicitly.

Appendix B develops the response and memory calculations relevant to inertia and persistence. Their physical use requires a retained observable, an external perturbation, and an independently identified response law. Curvature of a free-energy landscape supplies stiffness near a stationary state; a kinetic coefficient, dissipation and a memory kernel have different roles. Assigning all of them the name mass would obscure the mechanism. The useful question is whether eliminating internal degrees of freedom produces a reproducible effective response that remains valid over the claimed time window and through another coarse-graining step.

22.3 Measurement as a recorded interaction

In the normalized theory a measurement is an interaction with a specified record kernel. The record may be endogenous to the agent network. Its status as a recorded value is distinct from the agents' later beliefs about it, and several agents reading the same record does not multiply the likelihood by the number of readers. This distinction prevents agreement from being manufactured by counting one observation repeatedly.

A classical account of public measurement can seek three concrete properties: the record is stable on the readout timescale; different access channels recover compatible predictions from it; and interventions distinguish the inferred system from alternative explanations. The source of stability could be a metastable collective law or retained memory, but it must be established in the chosen dynamics. Merely increasing a Gaussian precision, or obtaining identical posterior means, proves none of these properties in general.

The present probability model has no noncommuting measurement algebra, Born rule or quantum instrument. Hermitian classical moments in Section 20 do not add those structures. Consequently neither consensus nor the Lorentz representation resolves quantum measurement. A quantum extension would need to specify its state space, composition rule and operational probabilities before asking whether its classical, coarse-grained records reproduce the present theory. This preserves a physical research target while avoiding a claim of quantum emergence from notation alone.

23 Consciousness: An Interpretive Stance

The theory developed here concerns statistical agents, communication and retained laws. It does not attach experience to a probability distribution by mathematical necessity. This section records a possible cognitive interpretation and the additional empirical question it would create. No theorem in the monograph depends on that interpretation.

A cognitive frame can represent how a system presents and communicates its variables. Its invariant content lies in the laws, records, interfaces and responses that survive a passive change of presentation. A relabeling of internal coordinates changes none of that content. It therefore cannot, by itself, model an altered experience, a perceptual intervention or a change of cognitive architecture. Such a proposal must change some operational part of the model, for example the record-access channel, the represented latent variables, a prior or the learned interface. It must then explain how that change would be detected.

The covariant Fisher tensor is a candidate descriptor of statistical sensitivity, and the collective law can describe integration of information unavailable in individual marginals. Neither object supplies a correspondence between a geometric value and a particular phenomenal quality. Two systems can share a Fisher tensor while differing in higher statistical structure, history and observation mechanisms. Conversely, a coordinate change can alter the matrix entries of a representation without altering any experience-related observable the model actually defines. These distinctions rule out a formal solution to the inverted-spectrum problem by coordinate relabeling alone.

Meta-agent formation supplies no criterion of consciousness either. A retained collective may be coherent, persistent and predictively closed, with or without explicitly retained memory. Those are properties of its law and dynamics. To propose them as necessary or sufficient for conscious experience would require an independent operational criterion and cases that distinguish the proposal from rival explanations. Recursion or cross-scale feedback does not establish that criterion merely because it occurs at several levels.

The participatory interpretation is consequently compatible with several positions about the relation between inference and experience. It does not derive panpsychism, eliminate an observer-

independent reality, or settle why information processing should be accompanied by experience. The narrower research opportunity is to construct competing models of report, discrimination and access, and test which collective variables improve their predictions under intervention. Such results could constrain a cognitive interpretation while leaving the broader philosophical question open.

24 Gauge Covariance, Empirical Symmetry and Scientific Agreement

Scientific statements must distinguish a change in the system from a change in its description. Gauge covariance addresses the latter distinction. Consensus addresses whether different participants make compatible statements about records. An empirical symmetry addresses whether specified transformations of a physical preparation and its measurements preserve predictive relations. These questions can interact, but they are not equivalent.

24.1 Coordinate consistency does not select a physical symmetry

For a bimeasurable invertible change of representation g, pushing both a law and its reference forward preserves their relative entropy,

$$D_{\mathrm{KL}}(g_{\#}q \| g_{\#}p) = D_{\mathrm{KL}}(q\|p). \tag{223}$$

The same consistency requires links, record kernels, noise and priors to transform with their declared types. It holds for one agent before any agreement with another agent occurs. Consensus therefore does not create this invariance. Changing a coordinate frame while holding a coordinate-dependent noise covariance or likelihood fixed generally changes the model as well.

Remark 1 (Covariance and selection)**.** *An invertible re-expression of one model cannot select a physical law over its re-expression, because it changes no predictive probabilities. A claim that variational dynamics select a physical symmetry must instead specify distinct candidate models or states and a gauge-invariant criterion distinguishing their predictions or stability.*

The Lorentz construction illustrates the distinction precisely. Once a complex rank-two communication sector is chosen, its determinant-preserving complex congruences have the proper, orthochronous Lorentz group as their image. This theorem supplies a representation and its invariant. It does not make every selective interaction a Lorentz transformation, select that channel from an arbitrary cognitive space, or establish that observed signals propagate along its null cone. Likewise, embedding a familiar compact group in a sufficiently large general linear group would show representational availability, not derive the interaction law of a physical gauge field.

Empirical symmetry needs an operational relation. If T acts on latent preparations and R is a bimeasurable bijection on records, a proposed covariant record mechanism must satisfy

$$K(RA \mid Ty) = K(A \mid y) \tag{224}$$

for every measurable record event A and every preparation y in its declared domain, together with compatible laws for preparation and evolution. The relation becomes testable when T and R correspond to specified interventions and calibrated comparisons. If they merely rename every symbol, Eq. (224) is a consistency identity. If the physically realizable interventions obey it approximately, their measured residual and regime of validity are part of the physical claim.

24.2 Relational invariants of the communication network

Independent links carry content that pure coordinate frames do not. For a closed directed path at one fixed context, form the composable operator

$$\mathcal{L}_\gamma = L_{i_0 i_1} L_{i_1 i_2} \cdots L_{i_{r-1} i_0} : E_{i_0} \longrightarrow E_{i_0}. \tag{225}$$

Endpoint re-expressions cancel at intermediate agents, giving $\mathcal{L}'_\gamma = g_{i_0} \mathcal{L}_\gamma g_{i_0}^{-1}$. Its rank, characteristic polynomial and traces of powers are therefore invariant. This calculation does not require each link to be invertible. For singular links the product is a loop operator, not an element of a transport group; for invertible group-valued links it can be called network holonomy. Neither construction alone is curvature of a connection along paths in the base.

These invariants suggest targeted relational observations: compare responses to different composable communication routes, controlling the endpoint law and output channel. A nontrivial internal loop operator is observable only to the extent that those records distinguish its action; a law invariant under that action can hide it. This is why zero transported disagreement does not force identity holonomy and why changing a graph loop need not change a particular loss. The required claim is about a specified observable response, not a universal relation between an invariant's magnitude and performance.

24.3 Agreement, evidence and model comparison

The normalized population law separates shared evidence from shared conclusions. A record appears once in its generative factorization even when several agents can access it. Selective communication can promote agreement on a low-dimensional public statistic while retaining different private states. A community that agrees is therefore not, on that fact alone, calibrated or predictively successful. Priors, source preferences and restrictions on record access can sustain agreement about an incorrect model.

The evidential comparison is predictive. A candidate should specify record probabilities before the relevant outcome, compare them with those of an alternative under the same access constraints, and assess interventions that were not used to fit the retained description. In the monograph's mathematical examples, the frozen-source contrast, interface-rank changes and removal of retained memory isolate different mechanisms. Their theoretical predictions are not empirical results about nature until a corresponding system and record protocol are supplied.

This also limits the cognitive reading of scientific change. A passive change of representational frame is scientifically empty. A change in latent variables, conditional independence assumptions, accessible records or predictive kernels can alter what a model explains. Describing both changes as a new gauge would conceal the substantive one. The general framework permits a rigorous comparison only after those altered model components have been specified.

24.4 A falsifiable participatory program

The claim that every possible observation can be redescribed as a consequence of cognitive architecture has no discriminating prediction in that unrestricted form. The present program instead asks for mechanisms with a restricted domain: an actual normalized model, independently specified observations, a retained variable and a forecast about its response or composition. Conditional mathematical results identify which hypothesis can fail. A parent-link square may not exist, a retained law may require memory, a proposed geometry may be degenerate, or its causal characteristics may disagree with its determinant cone. These are informative failures of specified constructions.

The underlying contextual base remains part of the model, and observer access remains mediated by records and inference. Neither fact proves that an observer-independent reality is absent or that scientific success is exhausted by agreement. The participatory perspective directs attention to how stable shared descriptions are formed and tested. Its physical content will be whatever quantitative relations survive that testing, with their assumptions and limits retained.

25 Open Problems with Decisive Next Steps

The unresolved program is narrower than deriving everything from a scalar functional. Each problem asks whether a specified operation can be justified by the actual agent dynamics, with a result or counterexample that would change the theory.

Predictive parent state. For the coupled selective-link dynamics, construct an equivariant statistic whose future record law depends only on its retained state and a controlled memory. Test the link square and generator descent, then compute the projected memory kernel. If a finite family is proposed, prove closure or quantify its error on the formation and handoff horizons. Failure of one finite family does not exclude a different family or a measure-valued parent.

Recursive formation. Extend the established fixed-temperature selection mechanism to an inherited parent dynamics without resetting its laws or supplying a preferred second partition. Separate the asymmetry in the initial preparation from the learning that amplifies it. A second episode must follow from the actual first parent, including retained attention events and hidden conditional information. The current four-cycle theorem does not settle this.

Emergent model variables. One joint bundle is already sufficient for state and model laws. The stronger problem is to derive a stable predictive coordinate and its slow evolution from an unsplit state. Accumulating sufficient statistics and exact projected memory provide examples. Local Fisher descent at an exactly represented posterior supplies no intrinsic spectral hierarchy, so a claim of slow model emergence must identify the mechanism that avoids that special case.

Learned selective interfaces. A likelihood involving low-rank maps can be unidentifiable or favor uninformative maps. Determine which records, conservation constraints, structural priors or kinetic mechanisms select a nontrivial interface without encoding the desired hierarchy in advance. Track the term caused by motion of the retention itself. Rank alone is neither a capacity nor an agency criterion.

Dynamic links and their coarse laws. The latent-generator prior and the local Proca-type history equation of Section 6 establish conditional link inference. The open question is a predictive coarse law that retains the resulting generator–matter dependence. Product Gaussian generator laws fail to close under nonabelian series composition, and on $\mathrm{GL}^{+}(K)$ a coarse product can have no real logarithm (Section 12, Eq. (165)). Construct an admitted family of group-valued link laws closed under the required products, retain their exact conditional law, or quantify a chosen approximation. A weak-current linear solution does not settle this closure or global nonlinear recognition convergence. Learning the interaction graph and deriving a slow link sector also remain separate obligations.

Operational Lorentz geometry. The determinant construction gives exact $1 + 3$ kinematics. The missing implication is selection of a complex two-component channel, a nondegenerate collective soldering to a four-dimensional base region, and a scale convention compatible across overlaps. A physical causal cone must then be the characteristic cone of a derived propagation law. A cone of positive moment matrices alone is not a cone of allowed trajectory velocities. Einstein dynamics, physical units and coupling to matter remain additional targets.

Physical inertia. Derive position and momentum observables, a calibrated forced response and the corresponding conservation or balance law. A second derivative obtained by eliminating hidden first-order variables does not suffice. Nor does the positive episode-history Proca operator: its Euclidean sign and inference index must be related to an operational physical propagation law before it represents mechanical inertia. Memory and metastable lifetimes give concrete persistence mechanisms; a claim that they become inertial mass must survive frequency-dependent response tests and distinguish friction from kinetic storage.

Registration and the quantum question. Binary record kernels provide the information-gain bound. They do not derive irreversible registration, incompatibility of measurements, the Born rule or quantum amplitudes. Classical complex messages already support the Hermitian determinant construction, so complex notation is not evidence of quantum physics. A quantum extension requires its own state space, update law and empirical tests.

Empirical identification. Choose a target with accessible records, a noise model and observables sensitive to the proposed links or hierarchy. Infer fine-scale parameters with uncertainty, then predict retained-scale records without independent refitting. Specify what observation would reject the added physical identification. Internal agreement of derivations or a successful model simulation is not that comparison.

26 Conclusion

A participatory theory needs a disciplined account of partial access. Local statistical sections over a continuous base, connected through selective matrix-valued channels, supply such an account. One normalized population law yields a collective evidence bound and source-selection attention. One joint statistical bundle accommodates state beliefs and model uncertainty. Exact coarse-graining produces parent laws, induced interactions and conditional cross-scale operators; predictive updating supplies later priors without a same-time shadow postulate.

Latent Lie-algebra links add a normalized prior and an expected link-field equation to that collective inference. Their Gaussian recognition follows Fisher descent; their separately declared episode law supplies Euclidean history inertia and a locally controlled Proca-type operator. Its physical interpretation and the closure of nonabelian coarse link laws remain open.

The framework also gives concrete limits. Coarse-graining does not select its own retained variables, a raw VFE decrease does not certify formation, and slow memory does not establish recursive coalescence. Fisher length is an information clock, not yet physical time. Hermitian determinant geometry carries the physical Lorentz group's kinematics under explicit assumptions, but its realization as spacetime remains to be derived. Persistent dissipative structures and memory can emerge without a kinetic postulate, while physical mass requires further dynamical structure.

These distinctions preserve Wheeler's ambition while making progress assessable. The next results should identify which collective variables the actual dynamics retain, how their laws compose,

and what physical observations those laws predict.

27 Derivational and Source Methods

The manuscript uses analytic probability identities, typed geometric constructions and explicit finite-dimensional examples. New algebraic steps were checked independently, with symbolic calculations used for finite identities rather than as substitutes for general proofs. The formation-selection result is stated with its fixed-temperature and almost-everywhere hypotheses, and its complete proof is given in Appendix J. No new empirical experiment or physical validation is reported.

27.1 Source use

Primary articles were consulted for the information-geometric, filtering, coarse-graining and Lorentz constructions. Literature-assisted research was used to identify candidate extensions and contrary cases. A suggested citation was not treated as proof that its hypotheses apply to the present model.

27.2 Language-model assistance

Language models assisted with literature discovery, mathematical reconstruction, symbolic checking and manuscript organization and drafting under author direction. Exa Agent was used at ultra effort for the three focused research questions on inference and formation, Lorentz structure, and persistence and inertia. Claims retained in the manuscript rest on their stated derivations and primary sources, not on agreement among models.

28 Statements and Declarations

Competing Interests

The author declares no competing interests, either financial or non-financial, related to this work.

Author Contributions

R.C.D. conceived the research program and directed its theoretical development and manuscript preparation. Language models assisted with literature discovery, derivation and counterexample development, symbolic checking, and drafting and revision, as described in Section 27. The manuscript states the assumptions and evidence for each result independently of that assistance.

Data and Supporting Material

This is a theoretical monograph. It reports analytic constructions and limited symbolic checks, with no new empirical dataset, simulation study or experimental validation of the proposed physical interpretation. The derivations cited in the text supply the mathematical context. The local research archive preserves source records, analytical reconstruction notes and the checks accompanying this revision.

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Part III: Technical Appendices

A Explicit Gradient Expressions for $\mathrm{GL}(K_q)$ Gaussian Agents

This appendix collects the gradients and natural gradients of the explicitly chosen consensus potential $\mathcal{E}$ of Section 5 for Gaussian beliefs with $\mathrm{GL}^+(K_q)$ transport, the connected component containing the transports $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ of Eq. (4). The compact restriction $\mathrm{SO}(K_q)$ is the special case in which the frame coefficients ϕ_i are skew-symmetric, so that each factor $\exp(\phi_i)$ and hence every transport Ω_{ij} is a rotation. The Gaussian-sector operations are covariant under the congruence action of $\mathrm{GL}(K_q)$ on the covariance fiber; the frame sector is discussed with the update equations.

The inverse-link formulas below assume nonsingular Gaussian laws, fixed priors and records, distinct endpoints, and invertible noiseless links. Frame-factorized derivatives additionally assume $\Omega_{ij} = U_i U_j^{-1}$. They are a specialization of the general channel model, not gradients of an arbitrary collective $\mathcal{F}_V$. A separate formula below covers rank-deficient noisy communication.

Setup and Notation

Use the optional product restriction $\zeta_i = q_i \otimes s_i$ of the joint statistical section, with Gaussian state belief $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and Gaussian model-parameter belief $s_i = \mathcal{N}(\mu_{s,i}, \Sigma_{s,i})$. A model presentation becomes predictive through its normalized evaluator; s_i alone is not a generative model. The state prior $p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i})$ and parameter prior $r_i = \mathcal{N}(\mu_{r,i}, \Sigma_{r,i})$ are fixed during this update. A lagged parent prediction is one possible source of these priors, not a requirement. In a regular local chart write $U_i = \exp \phi_i$. The state representation is the defining matrix action and $\Omega_{ij} = U_i U_j^{-1}$. A different model representation gives $\widetilde{\Omega}_{ij} = \rho_{\mathrm{model}}(U_i U_j^{-1})$. Model-sector derivatives have the analogous probabilistic form, but their contribution to a shared frame gradient must be pulled back through $D\rho_{\mathrm{model}}$; matrices of different representation dimensions cannot simply be added.

The attention temperatures are τ on the belief fiber and τ_γ on the model fiber, as in Eq. (35); with the parameterization $\tau = \kappa\sqrt{K_q}$ and $\tau_\gamma = \kappa\sqrt{K_m}$, the dimensionless $\kappa > 0$ is a declared constant, not a variational parameter of $\mathcal{E}$. The computation is made at the base point c_* with every agent present, so the receiver-presence weight of each row equals one. Condition numbers are written $\mathrm{cond}(\cdot)$ to avoid a collision with κ. The pair energies and the attention rows are

$$E_{ij} := D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j]), \qquad \beta_{ij} = \frac{\pi_{ij} \exp\left(-E_{ij}/\tau\right)}{\sum_m \pi_{im} \exp\left(-E_{im}/\tau\right)}, \tag{226}$$

so that $\beta_{ij} = \beta_{ij}(\{q\}, \{U\})$ depends on the beliefs and frames of the receiver i and of every source m in its row. The scalar α_i is the per-agent prior precision of Eq. (52), with optimal value $\alpha_i^* = c_0/(b_0 + D_{\mathrm{KL}}(q_i \| p_i))$ from Eq. (53); the constant-precision limit $\alpha_i = 1$ recovers the unit-weight self term of Eq. (39). On the model fiber a fixed hyper-prior weight λ_h plays the analogous role. A diagonal extension would replace the scalar α_i coordinate-wise by Eq. (58); it is not developed here.

The Canonical Envelope Gradient

The consensus specialization considered here descends the reduced functional $\mathcal{E}_{\mathrm{red}}$, in which every attention row has been replaced by its Gibbs value $-\tau \log Z_i$ (Section 7.2). At the Gibbs row the derivative of the row block vanishes on the simplex tangent space, so the gradient of the row value is the gradient taken at frozen weights. The reduced slow descent is then the natural-gradient flow of the reduced functional by this envelope identity. The adaptive-precision block has the same

cancellation (see the discussion following Eq. (59)), so α_i enters with its optimal value α_i^*. A state variable x_i of agent i appears in the row of agent i and, as the sender, in the row of every agent k that attends to i. For $x_i \in \{\mu_i, \Sigma_i, U_i\}$ the complete gradient is therefore

$$\nabla_{x_i}\mathcal{E}_{\mathrm{red}} = \alpha_i^* \nabla_{x_i} D_{\mathrm{KL}}(q_i\|p_i) + \sum_j \beta_{ij}\nabla_{x_i}E_{ij} + \sum_k \beta_{ki}\nabla_{x_i}E_{ki} + \nabla_{x_i}\mathcal{E}_{\mathrm{obs},i}, \tag{227}$$

with every weight at its Gibbs value, $\mathcal{E}_{\mathrm{obs},i}$ the observation term of agent i, and the model channel contributing terms of the same form, with $(\gamma_{ij}, \tilde{\Omega}_{ij}, s_i, r_i, \lambda_h)$ in place of $(\beta_{ij}, \Omega_{ij}, q_i, p_i, \alpha_i^*)$, to the gradients with respect to the model variables and, when the model-fiber transport is generated by the same frames, with respect to U_i. The observation and self terms carry no frame dependence in this specialization.

Belief mean and covariance. Write $A_{ij} := \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ for the transported covariance of the source. Differentiating the Gaussian relative entropy of Appendix C.2, the self and receiver pieces are

$$\begin{aligned} \frac{\partial D_{\mathrm{KL}}(q_i\|p_i)}{\partial \mu_i} &= \Sigma_{p,i}^{-1}(\mu_i - \mu_{p,i}), & \frac{\partial D_{\mathrm{KL}}(q_i\|p_i)}{\partial \Sigma_i} &= \frac{1}{2}\left(\Sigma_{p,i}^{-1} - \Sigma_i^{-1}\right), \\ \frac{\partial E_{ij}}{\partial \mu_i} &= A_{ij}^{-1}(\mu_i - \Omega_{ij}\mu_j), & \frac{\partial E_{ij}}{\partial \Sigma_i} &= \frac{1}{2}\left(A_{ij}^{-1} - \Sigma_i^{-1}\right). \end{aligned} \tag{228}$$

Differentiating $E_{ki} = D_{\mathrm{KL}}(q_k\|\Omega_{ki}[q_i])$ with respect to the sender's mean and covariance, using the precision transport of Eq. (110), gives

$$\frac{\partial E_{ki}}{\partial \mu_i} = \Lambda_{q_i}\Omega_{ki}^{-1}\left(\Omega_{ki}\mu_i - \mu_k\right), \tag{229}$$

$$\frac{\partial E_{ki}}{\partial \Sigma_i} = \frac{1}{2}\Omega_{ki}^\top\left[\tilde{\Lambda}_{q_i}^{(k)} - \tilde{\Lambda}_{q_i}^{(k)}\left(\Sigma_k + d_{ki}d_{ki}^\top\right)\tilde{\Lambda}_{q_i}^{(k)}\right]\Omega_{ki}, \tag{230}$$

where $\Lambda_{q_i} = \Sigma_i^{-1}$, $\tilde{\Lambda}_{q_i}^{(k)} := (\Omega_{ki}\Sigma_i\Omega_{ki}^\top)^{-1} = \Omega_{ki}^{-\top}\Lambda_{q_i}\Omega_{ki}^{-1}$ is agent i's precision transported into agent k's frame, and $d_{ki} := \Omega_{ki}\mu_i - \mu_k$ is the mean residual in agent k's frame. The rank-one piece $-\tilde{\Lambda}_{q_i}^{(k)}d_{ki}d_{ki}^\top\tilde{\Lambda}_{q_i}^{(k)}$ vanishes at consensus and contributes a stiffness aligned with d_{ki} away from it. The inverse-transpose form of Eq. (110) is what makes these expressions valid for general $\Omega_{ki} \in \mathrm{GL}^+(K_q)$, where Ω_{ki}^{-1} does not reduce to $\Omega_{ki}^\top$. Substituting into Eq. (227) gives the complete mean and covariance gradients,

$$\begin{aligned} \nabla_{\mu_i}\mathcal{E}_{\mathrm{red}} = \alpha_i^*\Sigma_{p,i}^{-1}(\mu_i - \mu_{p,i}) &+ \sum_j \beta_{ij}A_{ij}^{-1}(\mu_i - \Omega_{ij}\mu_j) \\ &+ \sum_k \beta_{ki}\Lambda_{q_i}\Omega_{ki}^{-1}\left(\Omega_{ki}\mu_i - \mu_k\right) + \nabla_{\mu_i}\mathcal{E}_{\mathrm{obs},i}, \end{aligned} \tag{231}$$

$$\begin{aligned} \nabla_{\Sigma_i}\mathcal{E}_{\mathrm{red}} = \frac{\alpha_i^*}{2}\left(\Sigma_{p,i}^{-1} - \Sigma_i^{-1}\right) &+ \frac{1}{2}\sum_j \beta_{ij}\left(A_{ij}^{-1} - \Sigma_i^{-1}\right) \\ &+ \frac{1}{2}\sum_k \beta_{ki}\Omega_{ki}^\top\left[\tilde{\Lambda}_{q_i}^{(k)} - \tilde{\Lambda}_{q_i}^{(k)}\left(\Sigma_k + d_{ki}d_{ki}^\top\right)\tilde{\Lambda}_{q_i}^{(k)}\right]\Omega_{ki} + \nabla_{\Sigma_i}\mathcal{E}_{\mathrm{obs},i}, \end{aligned} \tag{232}$$

plus the model-channel terms. Eq. (231) is Eq. (121) with the sender index relabeled $k \to j$, and Eq. (121) gives the observation terms for a Gaussian identity readout.

Frame sector: the transported relative entropy. The frames enter only through the links. For Gaussians write $A := \Omega_{ij}\Sigma_j\Omega_{ij}^\top$, $r := \mu_i - \Omega_{ij}\mu_j$, and $\Lambda := A^{-1}$. The Gaussian relative entropy has three Ω_{ij}-dependent pieces ($\log|A|$, $\mathrm{tr}(\Lambda\Sigma_i)$, $r^\top\Lambda r$), and the group is $\mathrm{GL}^+(K_q)$, so the directional derivative lives in the full Lie algebra $\mathfrak{gl}(K_q)$ with no skew-symmetry constraint. Along $X \in \mathfrak{gl}(K_q)$ acting by $\Omega_{ij} \to e^{tX}\Omega_{ij}$, with $\partial_X A = XA + AX^\top$ and $\partial_X r = -X\Omega_{ij}\mu_j$, the derivative is

$$\partial_X E_{ij} = \mathrm{tr}\left[X^\top\Big(I - \Lambda\Sigma_i - \Lambda r(\Omega_{ij}\mu_j)^\top - \Lambda r r^\top\Big)\right], \tag{233}$$

where the leading identity is the log det contribution $\partial_X \log|A| = 2\,\mathrm{tr}(X)$ carried at the half weight of the Gaussian relative entropy, and the quadratic term uses $\Lambda A = I$ to reduce $\Lambda r r^\top \Lambda A$ to $\Lambda r r^\top$. Writing $N_{ij} := I - \Lambda\Sigma_i - \Lambda r(\Omega_{ij}\mu_j)^\top - \Lambda r r^\top$ for the bracket of Eq. (233), the unconstrained matrix derivative is

$$\frac{\partial E_{ij}}{\partial \Omega_{ij}} = N_{ij}\Omega_{ij}^{-\top} = \Omega_{ij}^{-\top} - \Lambda\Sigma_i\Omega_{ij}^{-\top} - \Lambda(\mu_i - \Omega_{ij}\mu_j)\mu_j^\top - \Lambda r r^\top\Lambda\Omega_{ij}\Sigma_j, \tag{234}$$

where the mean term carries a negative sign because r depends on Ω_{ij} through $-\Omega_{ij}\mu_j$, and the simplifications $(\Omega_{ij}\mu_j)^\top\Omega_{ij}^{-\top} = \mu_j^\top$ and $\Omega_{ij}^{-\top} = \Lambda\Omega_{ij}\Sigma_j$ reduce the remaining factors. At consensus, $q_i = \Omega_{ij}[q_j]$, the bracket N_{ij} vanishes. The compact subgroup $\mathrm{SO}(K_q)$ is the special case $X \in \mathfrak{so}(K_q) \subset \mathfrak{gl}(K_q)$ ($X^\top = -X$), in which the log det piece vanishes because $\mathrm{tr}(X) = 0$ and the gradient is obtained by projecting onto $\mathfrak{so}(K_q)$ through antisymmetrization.

Frame sector: gradient in the group variable. Since $\Omega_{ij} = U_i U_j^{-1}$, a variation of U_i moves the receiver link by $\delta\Omega_{ij} = (\delta U_i)U_j^{-1}$ and the sender link by $\delta\Omega_{ki} = -\Omega_{ki}(\delta U_i)U_i^{-1}$. Combining with Eq. (234) and the identities $\Omega_{ij}^{-\top}U_j^{-\top} = U_i^{-\top}$ and $\Omega_{ki}^{-\top}U_i^{-\top} = U_k^{-\top}$ gives

$$\nabla_{U_i} E_{ij} = N_{ij}U_i^{-\top}, \qquad \nabla_{U_i} E_{ki} = -\Omega_{ki}^\top N_{ki}U_k^{-\top}, \tag{235}$$

with N_{ki} the bracket of Eq. (233) for the pair (k, i). The complete frame gradient is therefore

$$\nabla_{U_i}\mathcal{E}_{\mathrm{red}} = \sum_j \beta_{ij}N_{ij}U_i^{-\top} - \sum_k \beta_{ki}\Omega_{ki}^\top N_{ki}U_k^{-\top}, \tag{236}$$

plus the model-channel terms of the same form when the model-fiber transport $\tilde{\Omega}_{ij}$ is generated by the same frames through ρ_{model} (tied frames); with independent model-fiber frames the model-channel gradient is taken with respect to those frames instead. Every term vanishes at consensus. In the frame of the receiving agent of each row, the body covector $U_i^\top\nabla_{U_i}\mathcal{E}_{\mathrm{red}}$ reads $\sum_j \beta_{ij}U_i^\top N_{ij}U_i^{-\top} - \sum_k \beta_{ki}U_k^\top N_{ki}U_k^{-\top}$, so each pair contributes its bracket conjugated into the frame of the agent whose row contains it.

Frame sector: chart gradients. In the chart $U_i = \exp(\phi_i)$ with generators G_a, the chain rule gives $\partial\mathcal{E}_{\mathrm{red}}/\partial\phi_i^a = \mathrm{tr}[(D\exp_{\phi_i}[G_a])^\top\nabla_{U_i}\mathcal{E}_{\mathrm{red}}]$. For the receiver link the exact Fréchet derivative of the matrix exponential is

$$\frac{\partial\Omega_{ij}}{\partial\phi_i^a} = \left(\int_0^1 e^{(1-s)\phi_i}G_a e^{s\phi_i}\mathrm{d}s\right)e^{-\phi_j}, \tag{237}$$

and for the sender link, where ϕ_i enters through the right factor $\exp(-\phi_i)$,

$$\frac{\partial\Omega_{ki}}{\partial\phi_i^a} = -\exp(\phi_k)J_a, \qquad J_a := \int_0^1 e^{-(1-s)\phi_i}G_a e^{-s\phi_i}\mathrm{d}s, \tag{238}$$

so that

$$\frac{\partial E_{ij}}{\partial \phi_i^a} = \operatorname{tr}\left[\left(\frac{\partial \Omega_{ij}}{\partial \phi_i^a}\right)^\top N_{ij}\Omega_{ij}^{-\top}\right], \qquad \frac{\partial E_{ki}}{\partial \phi_i^a} = -\operatorname{tr}\left[J_a^\top (e^{\phi_k})^\top N_{ki}\Omega_{ki}^{-\top}\right]. \tag{239}$$

When G_a commutes with ϕ_i the integrals collapse, $\partial\Omega_{ij}/\partial\phi_i^a = G_a\Omega_{ij}$ and $\partial\Omega_{ki}/\partial\phi_i^a = -\Omega_{ki}G_a$, and the sender derivative becomes

$$\frac{\partial E_{ki}}{\partial \phi_i^a} = -\operatorname{tr}\left[G_a^\top\left(\Omega_{ki}^\top N_{ki}\Omega_{ki}^{-\top}\right)\right] \qquad \text{(abelian approximation)}, \tag{240}$$

while the receiver derivative becomes the projection of Eq. (233) along $X = G_a$. For noncommuting frames the first omitted term is a commutator: $\operatorname{dexp}_\phi^L(G) = G - \frac{1}{2}[\phi, G] + O(\|\phi\|^2\|G\|)$ in a bounded matrix norm as $\phi \to 0$. Its size depends on both the frame and the generator, so no universal empirical percentage is assigned. The group-variable formula (236) remains exact without this truncation.

General selective-link derivative

For a fixed Gaussian source and receiver and a fixed positive noise covariance, let

$$C_{ij} = L_{ij}\Sigma_j L_{ij}^\top + N_{ij}, \quad W_{ij} = C_{ij}^{-1}, \quad d_{ij} = \mu_i - L_{ij}\mu_j, \quad N_{ij} \succ 0.$$

Differentiating the same Gaussian channel cost gives

$$\nabla_{L_{ij}} E_{ij} = \left[W_{ij} - W_{ij}(\Sigma_i + d_{ij}d_{ij}^\top)W_{ij}\right]L_{ij}\Sigma_j - W_{ij}d_{ij}\mu_j^\top. \tag{241}$$

Indeed, $\delta C = \delta L \Sigma_j L^\top + L\Sigma_j \delta L^\top$ and $\delta d = -\delta L \mu_j$ give the two terms by the matrix trace pairing. No inverse of L_{ij} is needed. For $L_{ij} = R_i S_j$, the endpoint derivatives are $(\nabla_L E)S_j^\top$ and $R_i^\top(\nabla_L E)$, summed over incident edges when an interface is shared. These are covectors; a natural descent still requires a declared interface metric. If noise or source laws also depend on the interface, their chain-rule terms must be included.

Natural Gradients and Frame Covariance

The natural gradient raises the index of the gradient with the Fisher metric of Appendix C.3: $\tilde\nabla_{\mu_i}\mathcal{E} = \Sigma_i \nabla_{\mu_i}\mathcal{E}$ on the mean and $\tilde\nabla_{\Sigma_i}\mathcal{E} = 2\Sigma_i(\nabla_{\Sigma_i}\mathcal{E})\Sigma_i$ on $\mathcal{S}_{++}^{K_q}$.

Frame covariance. Under the congruence action of $g \in \mathrm{GL}(K_q)$, $\Sigma_i \mapsto g\Sigma_i g^\top$, which preserves the cone $\mathcal{S}_{++}^{K_q}$ for any invertible g, the Euclidean gradient $\nabla_{\Sigma_i}\mathcal{E}$ is a covector on the symmetric matrices and transforms by the inverse-transpose congruence, whereas the natural gradient transforms like Σ_i itself:

$$\Sigma_i \mapsto g\Sigma_i g^\top, \tag{242}$$

$$\nabla_{\Sigma_i}\mathcal{E} \mapsto g^{-\top}(\nabla_{\Sigma_i}\mathcal{E})g^{-1}, \tag{243}$$

$$\tilde\nabla_{\Sigma_i}\mathcal{E} \mapsto g(\tilde\nabla_{\Sigma_i}\mathcal{E})g^\top. \tag{244}$$

The two transformation laws agree for every covector if and only if g is orthogonal, $g^{-\top} = g$, that is $g \in \mathrm{O}(K_q)$, so for general g only the natural gradient is covariant on $\mathcal{S}_{++}^{K_q}$. Riemannian metrics on $\mathcal{S}_{++}^{K_q}$ invariant under the full congruence action form the two-parameter family $a\operatorname{tr}(\Sigma^{-1}U\Sigma^{-1}V) + b\operatorname{tr}(\Sigma^{-1}U)\operatorname{tr}(\Sigma^{-1}V)$ with $a > 0$ and $a + K_q b > 0$ [33]. The Gaussian Fisher metric on the covariance block is the member $a = \frac{1}{2}$, $b = 0$ (Proposition 37), and $2\Sigma(\cdot)\Sigma$ is its gradient structure; the member

$a = 1$, $b = 0$ would give $\Sigma(\cdot)\Sigma$. On the unit-determinant directions $\mathrm{tr}(\Sigma^{-1}U) = 0$, which form the $\mathfrak{sl}(K_q)$ sector after whitening, the second term vanishes and all members coincide up to scale. The gradient structure $2\Sigma(\cdot)\Sigma$ is therefore one invariant choice among many, singled out by the Fisher metric.

Exponential-Map Update on the SPD Manifold

After computing $\tilde{\nabla}_{\Sigma_i}\mathcal{E}$, the covariance is updated through a retraction, a smooth map from the tangent bundle to the manifold whose differential at the zero vector is the identity [34], so that Σ_i remains positive definite:

$$\Sigma_i^{\mathrm{new}} = \mathcal{R}_{\Sigma_i}(-\eta_\Sigma \tilde{\nabla}_{\Sigma_i}\mathcal{E}). \tag{245}$$

The retraction used is the affine-invariant exponential map,

$$\mathcal{R}_\Sigma(\Delta) = \Sigma^{1/2} \exp\left(\Sigma^{-1/2}\Delta\Sigma^{-1/2}\right)\Sigma^{1/2}, \tag{246}$$

where $\exp(\cdot)$ is the matrix exponential. It preserves positive-definiteness, since the exponential of a symmetric matrix is SPD and congruence preserves the cone, and it is equivariant under congruence, $\mathcal{R}_{A\Sigma A^\top}(A\Delta A^\top) = A\mathcal{R}_\Sigma(\Delta)A^\top$ for any invertible A. With the eigendecomposition $\Sigma = VDV^\top$,

$$\mathcal{R}_\Sigma(\Delta) = VD^{1/2} \exp\left(D^{-1/2}V^\top \Delta V D^{-1/2}\right) D^{1/2}V^\top. \tag{247}$$

This is the Riemannian exponential map of the affine-invariant metric, so the update follows the geodesic from Σ in direction Δ for unit parameter.

The Entropy-Suppressed Surrogate

Replacing each row block by $S_k := \sum_m \beta_{km}E_{km}$, with the Gibbs weights attached but the attention entropy removed, defines a different scalar S_{surr}. Applying the covariance identity of Section 5.6 to every row in which x_i appears gives

$$\nabla_{x_i} S_{\mathrm{surr}} = \nabla_{x_i}\mathcal{E}_{\mathrm{red}} - \frac{1}{\tau}\sum_k \mathrm{Cov}_{\beta_k}\left(E_{k\cdot}, \nabla_{x_i}E_{k\cdot}\right), \tag{248}$$

where the sum runs over the receiver row $k = i$ and every sender row containing i. In a sender row only E_{ki} depends on x_i, and the covariance reduces to $\beta_{ki}(E_{ki} - \langle E\rangle_{\beta_k})\nabla_{x_i}E_{ki}$. In the receiver row the surrogate contribution is $\sum_j \beta_{ij}[1 - (E_{ij} - \langle E\rangle_{\beta_i})/\tau]\nabla_{x_i}E_{ij}$. It reduces to the frozen-weight gradient whenever $\mathrm{Cov}_{\beta_i}(E, \nabla E) = 0$, for example when all E_{ij} are equal; it tends to the π-weighted mean gradient as $\tau \to \infty$; and it tends to ∇E_{ij^*} as $\tau \to 0$ when the minimizer j^* is unique, since for an energy gap $\Delta > 0$ the response term is bounded by a multiple of $(\Delta/\tau)e^{-\Delta/\tau}$. The response is largest near ties, where energy differences are of order τ. The split of the same quantity into a direct piece weighted by $(1 - E_{ij}/\tau)$ and a reweighting piece $\tau^{-1}\langle E\rangle_{\beta_i}\sum_m \beta_{im}\nabla E_{im}$ is algebraically equal but not centered: a common shift of all energies moves weight between the two pieces, each grows like E/τ as $\tau \to 0$ while their sum stays bounded, and evaluating them separately cancels catastrophically at small τ. The centered form above avoids this. On the model fiber the same statements hold with $(\gamma, \tilde{\Omega}, s, \tau_\gamma)$. For the precision weight, differentiating $\alpha_i^*(\mu_i, \Sigma_i)D_{\mathrm{KL}}(q_i\|p_i)$ without the regularizer of Eq. (52) gives the prefactor $(\alpha_i^*)^2 b_0/c_0$ in place of α_i^* (Eq. (59)).

Discarding the sender rows defines a further object, the receiver-only field $V_i^{\mathrm{rec}}(x) := \partial_{x_i}S_i(x)$, obtained by dropping $\sum_{k\neq i}\partial_{x_i}S_k$. It is not the gradient of S_{surr} and is generically nonintegrable; the receiver and sender terms together form ∇S_{surr}, whose Hessian is symmetric. Neither the surrogate nor the receiver-only field is the consensus envelope update, which descends $\mathcal{E}_{\mathrm{red}}$ with the gradients of Eqs. (227)–(236).

Complete Update Equations

The update uses a Fisher-Rao natural-gradient step on the Gaussian sector and a left-trivialized group retraction on the frame sector (cf. Eqs. (103)–(106)). Every derivative below is the complete receiver-plus-sender envelope gradient of $\mathcal{E}_{\mathrm{red}}$:

$$\mu_i^{t+1} = \mu_i^t - \eta_\mu \Sigma_i^t \nabla_{\mu_i} \mathcal{E}_{\mathrm{red}}, \tag{249}$$

$$\Sigma_i^{t+1} = \left(\Sigma_i^t\right)^{1/2} \exp\left[\left(\Sigma_i^t\right)^{-1/2} \left(-\eta_\Sigma \tilde{\nabla}_{\Sigma_i} \mathcal{E}_{\mathrm{red}}\right) \left(\Sigma_i^t\right)^{-1/2}\right] \left(\Sigma_i^t\right)^{1/2}, \tag{250}$$

$$\xi_i^t = \left(U_i^t\right)^{-1} \nabla_{U_i} \mathcal{E}_{\mathrm{red}}, \tag{251}$$

$$U_i^{t+1} = U_i^t \exp\left(-\eta_\phi \xi_i^t\right). \tag{252}$$

Here ξ_i is the left-trivialized frame velocity of Eq. (104): the chart natural-gradient vector $\mathcal{G}_{\mathrm{pull}}^{-1} \nabla_{\phi_i} \mathcal{E}$ pushes forward to $\nabla_{U_i} \mathcal{E}$, so its left-trivialized velocity is exactly $U_i^{-1} \nabla_{U_i} \mathcal{E}$ when the generators span $\mathfrak{gl}(K_q)$, while on a subalgebra the ambient-metric velocity is instead defined by $\langle U_i \xi_i, U_i \zeta \rangle_F = \langle \nabla_{U_i} \mathcal{E}, U_i \zeta \rangle_F$ for every tangent generator ζ, with $\nabla_{U_i} \mathcal{E}_{\mathrm{red}}$ given by Eq. (236). Evaluated in this form the velocity costs one $K_q \times K_q$ solve per agent; forming and inverting the $K_q^2 \times K_q^2$ pullback metric explicitly would cost $O(K_q^6)$ per agent and square its condition number. The alternative left-invariant Frobenius preconditioner replaces ξ_i by the body covector $U_i^\top \nabla_{U_i} \mathcal{E}_{\mathrm{red}}$. The body-covector update is equivariant under per-agent left translations with the beliefs co-transformed, and under right translations only by orthogonal g; the ambient-pullback update above is equivariant only under orthogonal translations. Neither descends to the quotient by the global right redundancy described below. A right-trivialized update would descend to that quotient at the cost of left equivariance, and for $K_q \geq 2$ no Riemannian metric on $\mathrm{GL}^+(K_q)$ is invariant under both translations [35]. The exponential map in the last line keeps $U_i \in G$ in exact arithmetic. Positivity and group membership do not ensure decrease of the potential for an arbitrary finite step; a step-size or descent condition is separate. The learning rates follow the separation declared in Section 8, with $\eta_\mu \equiv \eta_q$ the belief-fiber rate written for the separately updated mean block.

Computational Complexity

Per gradient step with N agents and dimension K_q, forming the links costs $O(NK_q^3)$ for the frames and their inverses plus $O(N^2 K_q^3)$ for the pairwise products; the transported relative entropies require factorizations of the $O(N^2)$ transported covariances, $O(N^2 K_q^3)$ in total; and the natural-gradient steps and exponential-map updates, with the frame velocity evaluated as $U_i^{-1} \nabla_{U_i} \mathcal{E}_{\mathrm{red}}$, cost $O(K_q^3)$ per agent, $O(NK_q^3)$ in total. The overall complexity is $O(N^2 K_q^3)$ per step, dominated by the pairwise coupling.

Numerical Stability and Verification

Positive-definiteness in finite precision. In exact arithmetic the exponential-map update maps into $\mathcal{S}_{++}^{K_q}$ for any symmetric Δ, so $\Sigma_i \succ 0$ is preserved by construction. In single precision, with unit roundoff $u = 2^{-24} \approx 5.96 \times 10^{-8}$, a symmetric matrix whose condition number exceeds about $u^{-1} \approx 1.68 \times 10^7$ can already be indefinite as stored, before any eigendecomposition is computed; an eigenvalue floor on the stored matrices, or storage of a factor, is what restores definiteness, and a higher-precision eigendecomposition does not. The same bound applies to the transported covariances $A_{ij} = \Omega_{ij} \Sigma_j \Omega_{ij}^\top$ as well as to Σ_i, and the bound

$$\mathrm{cond}(A_{ij}) \leq \mathrm{cond}(\Omega_{ij})^2 \, \mathrm{cond}(\Sigma_j) \tag{253}$$

holds, where $\mathrm{cond}_2(\Omega_{ij})$ can grow exponentially: $\mathrm{cond}_2(\Omega_{ij}) \le \exp(2\|\phi_i\|_2 + 2\|\phi_j\|_2)$. Large frame norm need not cause ill-conditioning, as skew-symmetric frames show; the displayed bound is an upper bound. Cholesky factorization of an $n \times n$ matrix is backward stable independently of conditioning: the computed factor satisfies $R^\top R = \Sigma + \delta\Sigma$ with $|\delta\Sigma| \le \gamma_{n+1}|R^\top||R|$ componentwise, $\gamma_{n+1} = (n+1)u/(1-(n+1)u)$ [36, Thm. 10.3]. What depends on conditioning is completion and forward error. The factorization runs to completion if $\lambda_{\min}(H) > n\gamma_{n+1}/(1-\gamma_{n+1})$, where $H = D^{-1}\Sigma D^{-1}$ with $D = \mathrm{diag}(\Sigma)^{1/2}$ is the unit-diagonal scaling of Σ [36, Thm. 10.7]; since $\lambda_{\max}(H) \ge 1$, this holds in particular when $\mathrm{cond}_2(H)$ is below about $1/(n(n+1)u)$, which for $n = K_q = 10$ in single precision is about 1.5×10^5.

Right-translation redundancy. Under the global right translation $U_i \mapsto U_i g$ by a single $g \in G$ shared by all agents (Section 4.3), every link $\Omega_{ij} = U_i U_j^{-1} \mapsto U_i g g^{-1} U_j^{-1} = \Omega_{ij}$ is unchanged. The free energy and the belief updates, which depend on the frames only through the links, are therefore unchanged. The frame update is not: since $\nabla_{U_i g}\mathcal{E} = (\nabla_{U_i}\mathcal{E})g^{-\top}$, the velocity becomes $g^{-1}\xi_i g^{-\top}$, and $U_i g \exp(-\eta_\phi g^{-1}\xi_i g^{-\top})$ agrees with $U_i \exp(-\eta_\phi \xi_i) g$ for generic ξ_i only when $g^{-\top} = g$, that is for orthogonal g; the body-covector update behaves in the same way. Initial frames related by a nonorthogonal right translation, which describe the same links, therefore evolve to different links. Where the required local logarithms exist, $U_i g$ corresponds to $\log(e^{\phi_i} e^{\zeta}) = \phi_i + \psi(\mathrm{ad}_{\phi_i})(\zeta) + O(\|\zeta\|^2)$ with $\zeta = \log g$ and $\psi(z) = z/(1-e^{-z}) = 1 + z/2 + z^2/12 + \cdots$. The additive shift $\phi_i \mapsto \phi_i + \zeta$ is therefore correct only to first order in both ϕ_i and ζ, or exactly in the abelian sector, and the group-level statement $U_i \mapsto U_i g$ is the one to use for the nonabelian redundancy.

Status of the formulas. The expressions above follow from the displayed matrix differentials, the Gaussian relative-entropy identity, and the declared metric. Historical numerical agreement is not used as evidence here. The regularity and finite-step restrictions are independent of the exact differential identities.

B Persistence, Hidden Memory, and the Limits of an Inertial Interpretation

All small-covariance response estimates below hold for uniformly bounded current mean discrepancy from the compared message. A positive-definite quadratic linearization requires a nondegenerate minimum. Metric whitening must preserve the declared retained–hidden split, or the retained variables must be redefined explicitly.

A persistent object is recognizable through changes in its surroundings. The statistical framework can express several different reasons for such recognition: small response to a disturbance, slow relaxation of an internal coordinate, a long residence time in a basin, or a record channel insensitive to changes elsewhere. These mechanisms have different mathematics. None is, by itself, the acceleration resistance measured by mechanical mass. This appendix derives the first three mechanisms in explicit models and identifies what an inertial extension would still have to supply. The construction is pointwise over a fixed c_* in an overlap of agent domains. It makes no identification of atoms, photons, or macroscopic matter with agents.

B.1 A sharp belief: exact response and its limitations

Let $q = \mathcal{N}(\mu, \Sigma)$ and $p = \mathcal{N}(\bar{\mu}, P^{-1})$ on $\mathbb{R}^d$, with $P \succ 0$. A sender supplies a fixed law through the channel

$$z = Lz_j + \epsilon, \qquad L = R_i S_j, \qquad \epsilon \sim \mathcal{N}(0, D), \qquad D \succ 0, \tag{254}$$

independently of $z_j \sim \mathcal{N}(\mu_j, \Sigma_j)$. Here j sends to i; L may be rectangular or rank deficient. Its received law is $r = \mathcal{N}(m, C)$, where $m = L\mu_j$, $C = L\Sigma_j L^\top + D \succ 0$, and $W = C^{-1}$. The sender, channel, prior, and weight $\beta \geq 0$ are held fixed. This is an open single-agent response calculation, not the complete two-agent variational update, which also differentiates sender contributions.

For the explicitly chosen consensus objective

$$\mathcal{E}_{\mathrm{KL}}(q) = D_{\mathrm{KL}}(q\|p) + \beta D_{\mathrm{KL}}(q\|r), \qquad A = P + \beta W, \tag{255}$$

the Gaussian derivatives are

$$\nabla_\mu \mathcal{E}_{\mathrm{KL}} = A\mu - P\bar{\mu} - \beta W m, \qquad \nabla_\Sigma \mathcal{E}_{\mathrm{KL}} = \frac{1}{2}\big[A - (1+\beta)\Sigma^{-1}\big]. \tag{256}$$

The unique minimizer over $\mathbb{R}^d \times \mathbb{S}^d_{++}$ is therefore

$$\mu_* = A^{-1}(P\bar{\mu} + \beta W m), \qquad \Sigma_* = (1+\beta)A^{-1}. \tag{257}$$

Uniqueness follows directly: the mean term has Hessian $A \succ 0$ and the covariance term is $\frac{1}{2}\operatorname{tr}(A\Sigma) - \frac{1}{2}(1+\beta)\log\det\Sigma$ plus a constant, strictly convex on the positive cone.

With the Gaussian Fisher metric and common rate $\eta > 0$, the continuous dynamics are

$$\dot{\mu} = -\eta\Sigma A(\mu - \mu_*), \qquad \dot{\Sigma} = -\eta\Sigma A\Sigma + \eta(1+\beta)\Sigma. \tag{258}$$

Their covariance evolution is particularly transparent in precision coordinates:

$$\dot{\Lambda} = \eta[A - (1+\beta)\Lambda], \qquad \Lambda(t) = \Lambda_* + e^{-\eta(1+\beta)t}(\Lambda(0) - \Lambda_*), \quad \Lambda_* = A/(1+\beta). \tag{259}$$

For every positive initial covariance this convex combination remains positive at every finite time. Its eigenvalues remain bounded above and away from zero, so the mean equation also has a unique global solution. For $V = \frac{1}{2}(\mu - \mu_*)^\top A(\mu - \mu_*)$,

$$\dot{V} = -\eta(\mu - \mu_*)^\top A\Sigma A(\mu - \mu_*) < 0 \quad \text{unless } \mu = \mu_*. \tag{260}$$

The uniform positive lower bound on Σ gives exponential convergence. At equilibrium the linearized mean rate is $\eta(1+\beta)$ in every direction; with $\beta = 0$, both mean and precision relax at rate η, independently of prior precision. These are continuous-time results. A covariance Euler step need not preserve positivity; the exact precision update in Eq. (259) does.

Set $P = \delta^{-1}I$ and let $\delta \downarrow 0$, holding β, W, and $m - \bar{\mu}$ bounded. Then

$$\mu_* - \bar{\mu} = \delta\beta(I + \delta\beta W)^{-1}W(m - \bar{\mu}) = O(\delta), \qquad \delta^{-1}\Sigma_* = (1+\beta)(I + \delta\beta W)^{-1}. \tag{261}$$

The mean is resistant to this specified disturbance, whereas the covariance changes by a finite relative factor. At $\Sigma = \delta I$, the coupling contributions to the velocities are

$$\dot{\mu}_{\mathrm{coup}} = -\eta\beta\delta W(\mu - m), \tag{262}$$

$$\dot{\Sigma}_{\mathrm{coup}} = \eta\beta\delta(I - \delta W). \tag{263}$$

The first tends to zero in the mean Fisher norm, but the covariance Fisher speed tends to $\eta\beta\sqrt{d/2}$ when $\beta > 0$. A sharply localized belief need not have a slowly advancing full Fisher clock. Small mean motion is insufficient when covariance evolution contributes to the clock.

Why the objective matters. The covariance broadening is caused by the $(1+\beta)$ copies of the receiver's negative entropy in Eq. (255). A different, probabilistically specified interaction gives a different prediction. Suppose a registered message m is an observation with Gaussian likelihood $p(m \mid z) = \mathcal{N}(m; z, (\beta W)^{-1})$, for $\beta > 0$, and hold its precision fixed. The ordinary variational free energy for this prior and likelihood is, up to a constant,

$$\mathcal{F}_V(q) = D_{\mathrm{KL}}(q\|p) + \frac{\beta}{2}\mathbb{E}_q[(z-m)^\top W(z-m)]. \tag{264}$$

It has the same optimal mean but $\Sigma_* = A^{-1}$. Consequently

$$\delta^{-1}\Sigma_* - I = -\delta\beta W(I + \delta\beta W)^{-1} = O(\delta). \tag{265}$$

Both relative covariance change and mean displacement are now small. The two objectives are legitimate different models. Replacing one by the other requires specifying whether the message is a distribution to imitate or a registered datum in a likelihood. A sum of consensus divergences is not automatically the variational free energy of that likelihood model. This distinction is essential before relating back-action to a physical observation.

B.2 What becomes of hidden internal coordinates?

Internal coordinates can store information about earlier perturbations even when the full state follows first-order descent. Eliminating them produces memory, a mechanism with a long history in statistical mechanics [21, 22]. The following finite-dimensional calculation is exact and requires no projection approximation.

Let $x \in \mathbb{R}^n$ be retained and $y \in \mathbb{R}^m$ be hidden. Assume the quadratic potential and continuous dynamics

$$F(x,y) = \frac{1}{2}\begin{pmatrix}x\\y\end{pmatrix}^\top H \begin{pmatrix}x\\y\end{pmatrix}, \qquad H = \begin{pmatrix}A & B\\ B^\top & C\end{pmatrix} \succ 0, \qquad \dot{x} = -Ax - By + f(t), \quad \dot{y} = -B^\top x - Cy, \tag{266}$$

where f is continuous, A, C are symmetric, and t is the declared flow parameter. The state space is all of $\mathbb{R}^{n+m}$ and no boundary conditions are required. Constant positive block mobilities can be reduced to this form by whitening the two coordinate blocks. Thus Eq. (266) also describes the linearization of a smooth Fisher gradient flow at a nondegenerate strict local minimum, in a chart with its equilibrium metric fixed. For a nonquadratic energy this interpretation is local, not an exact global equivalence.

Variation of constants gives

$$y(t) = e^{-Ct}y_0 - \int_0^t e^{-C(t-s)}B^\top x(s)ds. \tag{267}$$

Substitution and integration by parts yield the exact retained equation

$$\dot{x}(t) + \int_0^t K(t-s)\dot{x}(s)ds = -Sx(t) + f(t) - Be^{-Ct}r_0, \tag{268}$$

where

$$S = A - BC^{-1}B^\top \succ 0, \qquad K(u) = Be^{-Cu}C^{-1}B^\top, \qquad r_0 = y_0 + C^{-1}B^\top x_0. \tag{269}$$

Indeed, differentiating $e^{-C(t-s)}C^{-1}B^\top x(s)$ with respect to s produces both the original convolution and the velocity convolution. The endpoint terms give $BC^{-1}B^\top x(t)$ and the stated initial-condition

term. Positivity of S follows by evaluating the positive quadratic form of H at $(x, -C^{-1}B^\top x)$. For every $u \geq 0$, $K(u)$ is positive semidefinite; diagonalizing C shows that it is a sum of positive rank-one matrices multiplied by decaying exponentials.

The same algebra also displays the energy retained in the hidden coordinates:

$$F(x,y) = \frac{1}{2}x^\top S x + \frac{1}{2}(y + C^{-1}B^\top x)^\top C(y + C^{-1}B^\top x). \tag{270}$$

For $f = 0$ the full energy obeys $\dot{F} = -\|\nabla F\|^2$, and the full state converges exponentially to zero. The reduced energy $x^\top Sx/2$ need not decrease: it can increase while stored hidden energy is released. Thus persistent or delayed response after a disturbance does not contradict descent of the complete free energy. It indicates that the retained description has discarded part of the state.

When an instantaneous approximation is controlled. If $C \succeq \gamma I$ with $\gamma > 0$, $x \in C^2([0,t])$, and matrix norms are operator norms, then

$$\left\| \int_0^t K(u)\dot{x}(t-u)du - BC^{-2}B^\top \dot{x}(t) \right\| \leq \frac{\|B\|^2}{\gamma^3} \sup_{0\leq s\leq t} \|\ddot{x}(s)\| + \frac{\|B\|^2}{\gamma^2} e^{-\gamma t}\|\dot{x}(t)\|. \tag{271}$$

To prove this, subtract $\dot{x}(t)$ inside the integral, use $\|\dot{x}(t-u) - \dot{x}(t)\| \leq u \sup \|\ddot{x}\|$, and bound the missing tail from t to infinity. The identities $\int_0^\infty e^{-\gamma u}du = \gamma^{-1}$ and $\int_0^\infty ue^{-\gamma u}du = \gamma^{-2}$ give the two constants. After the initial transient, when this error is small, the leading local equation is

$$(I + BC^{-2}B^\top)\dot{x} \simeq -Sx + f. \tag{272}$$

Hidden relaxation therefore contributes positive friction. Small eigenvalues of C invalidate the short-memory approximation and retain a long internal record. The rate separation must be checked; it is not selected by the existence of a free energy.

B.3 A second-order equation that does not establish mass

Take $n = m = 1$, $A = a > 0$, $C = c > 0$, $B = b \neq 0$, and $ac > b^2$. For $f \in C^1$, direct differentiation of Eq. (266) gives

$$\ddot{x} + (a+c)\dot{x} + (ac - b^2)x = \dot{f} + cf, \qquad \dot{x}(0) = -ax_0 - by_0 + f(0). \tag{273}$$

This is exactly equivalent to the two-dimensional first-order model: reconstruct $y = (-\dot{x} - ax + f)/b$ and substitute into its equation. The apparent initial velocity is a hidden-state coordinate. The characteristic discriminant is

$$(a+c)^2 - 4(ac - b^2) = (a-c)^2 + 4b^2 > 0, \tag{274}$$

so both characteristic roots are real and negative. No underdamped mode has been created. Moreover, with zero initial conditions, the response to the original force is

$$\frac{X(s)}{F(s)} = \frac{s+c}{(s+a)(s+c) - b^2}. \tag{275}$$

The numerator is part of the observable response. Replacing $\dot{f} + cf$ by an unfiltered force gives a different system. At high frequency the response decays as $1/s$, as expected from the original first-order law, rather than as the $1/s^2$ of a directly forced mechanical position with nonzero mass.

There is also a sign warning in a local expansion. The exact inverse response in the matrix case is

$$sI + A - B(sI + C)^{-1}B^\top = S + s(I + BC^{-2}B^\top) - s^2BC^{-3}B^\top + O(s^3), \tag{276}$$

valid for $|s| < \lambda_{\min}(C)$. Its first acceleration correction is negative semidefinite. Treating this truncated expansion as a globally valid equation with a positive mass would reverse its sign and lose the original response. Equation (273) and this expansion describe different local presentations of the same memory, not independent derivations of an inertial coefficient.

For a general positive constant mobility M and symmetric positive Hessian H, the full drift $-MH$ is similar to $-M^{1/2}HM^{1/2}$. Its eigenvalues are real and negative. Any linear projection has poles only among these full-system poles. Signed sums of relaxations can overshoot, so monotonicity of every observable is not claimed. What is excluded is an underdamped linear mode generated solely by projection of this equilibrium gradient system. Nonreciprocal dynamics, explicit driving, an antisymmetric flow, or a nonequilibrium reference trajectory lie outside that statement.

B.4 Noise preserves the need for a history law

Add independent standard Brownian motions to the full model, with a declared noise scale $T > 0$:

$$dx = -(Ax + By)dt + \sqrt{2T}dW_x, \qquad dy = -(B^\top x + Cy)dt + \sqrt{2T}dW_y. \tag{277}$$

This linear stochastic system has a unique global solution. Its stationary covariance is TH^{-1}, as follows by substituting it into $H\Sigma + \Sigma H = 2TI$, and its explicit exponential solution converges in law to that centered Gaussian. This noise is an additional specification; deterministic free-energy descent alone does not supply it.

Exact elimination gives the semimartingale identity

$$dx_t = \left[-Sx_t - \int_0^t K(t-s)dx_s + \zeta_t\right]dt + \sqrt{2T}dW_x(t), \tag{278}$$

where the integral has its usual deterministic-kernel stochastic meaning and

$$\zeta_t = -Be^{-Ct}r_0 - \sqrt{2T}B\int_0^t e^{-C(t-s)}dW_y(s). \tag{279}$$

If y_0 is prepared from its equilibrium conditional law given x_0, then $r_0 \sim \mathcal{N}(0, TC^{-1})$ independently of x_0 and the future noises, and

$$\mathbb{E}[\zeta_t\zeta_s^\top] = TK(|t-s|). \tag{280}$$

For $t \geq s$, this follows by adding the initial covariance $TBe^{-Ct}C^{-1}e^{-Cs}B^\top$ to the stochastic-integral covariance $2TB\int_0^s e^{-C(t-u)}e^{-C(s-u)}duB^\top$. Their sum is $TBe^{-C(t-s)}C^{-1}B^\top$. The effective noise and memory are consequently linked; deleting the colored term or replacing it by arbitrary white noise changes the path law. An unprepared initial hidden law also changes the transient noise law.

For a concrete failure of a Markov replacement, choose $a = c = 2$ and $b = 1$. At equilibrium the scalar retained covariance is

$$C_x(t) = \frac{T}{2}\left(e^{-|t|} + \frac{1}{3}e^{-3|t|}\right). \tag{281}$$

For a stationary Gaussian scalar Markov process, conditional independence of the future and past given the present requires $C_x(t+s)C_x(0) = C_x(t)C_x(s)$ for $s, t \geq 0$. This follows by computing the Gaussian conditional covariance. At $s = t > 0$, the difference for the displayed law is $T^2(e^{-t} - e^{-3t})^2/12 > 0$. Thus the retained process is not Markov even though its one-time marginal is a perfectly ordinary Gaussian. Enlarging the retained state by y restores the Markov property exactly.

B.5 Persistence by an energy barrier

A second mechanism is long residence in a potential well. To separate this from precision, consider an explicitly supplied one-dimensional potential $U(x) = (x^2 - 1)^2/4$ and the diffusion

$$dX_t = -U'(X_t)dt + \sqrt{2T}dW_t + d\ell_t, \qquad -2 \le X_t \le 0, \tag{282}$$

reflected at -2 by the boundary local time ℓ_t and stopped at the first hit τ_0 of 0. Smoothness of the drift on this compact interval gives a unique reflected process up to stopping. Its mean exit time $u_T(x) = \mathbb{E}_x\tau_0$ solves $Tu_T'' - U'u_T' = -1$, $u_T'(-2) = 0$, $u_T(0) = 0$. Multiplication by $e^{-U/T}$ and two integrations give

$$u_T(x) = \frac{1}{T}\int_x^0 e^{U(y)/T}\left(\int_{-2}^y e^{-U(z)/T}dz\right)dy. \tag{283}$$

This positive solution is the mean exit time by the stopped Itô identity; bounded drift and nonzero diffusion give a positive probability, uniform over starting points, to exit within a fixed time interval, and iteration gives a geometric tail and integrability. At the left well $x = -1$, the inner integral concentrates at $z = -1$ and the outer integral at the boundary maximum $y = 0$. Continuity bounds above and below on neighborhoods of these extrema give

$$\lim_{T\downarrow 0} T\log u_T(-1) = U(0) - U(-1) = \frac{1}{4}. \tag{284}$$

The polynomial factor T^{-1} does not affect the limit. For an upper bound use $U(y) \le 1/4$ and $U(z) \ge 0$; for a lower bound restrict y near 0 and z near -1, where the exponent difference is at least $1/4 - \epsilon$, then let $\epsilon \downarrow 0$. This supplies the logarithmic statement without invoking an asymptotic prefactor. Sharp exit-time laws for more general reversible diffusions require additional landscape assumptions [37].

A well with a finite deterministic local relaxation rate can therefore have an exponentially long stochastic lifetime. Persistence is governed by a barrier relative to the noise scale, not by a large Fisher precision alone. The potential in this example is supplied. For the agent theory to explain persistence intrinsically, its actual normalized free energy and dynamics must produce the relevant basins, barriers, and disturbance law. A successful example of slow supplied model coordinates establishes storage or delayed relaxation; it does not establish recursive formation of another agent level.

B.6 The remaining physical question

These calculations support a modest but constructive reading of objecthood: a retained law or record can be stable because perturbations produce little change, because hidden coordinates preserve a history, or because escape from a basin is rare. A fixed communication kernel may also conceal changes in its null directions. Stability of a record therefore does not alone prove stability of the underlying state. Agreement between several records additionally requires compatible channels and identifiable shared content; products of independent selective links need not telescope to a direct link.

Mechanical inertia requires more. An explicit reversible sector illustrates the missing structure. On coordinates (x, v), choose

$$E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2, \qquad J = \begin{pmatrix} 0 & 1/m \\ -1/m & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & 0 \\ 0 & \gamma/m^2 \end{pmatrix}, \qquad k, m > 0, \quad \gamma \ge 0. \tag{285}$$

The law $\dot{z} = (J - D)\nabla E$ gives $\dot{x} = v$, $m\dot{v} = -kx - \gamma v$, and $\dot{E} = -\gamma v^2$. The antisymmetric operator generates the reversible position–momentum exchange, while the positive operator dissipates energy. This is a consistent mechanical extension, not a result of specifying E alone. The same scalar energy also permits purely relaxational dynamics with $J = 0$.

The minimal unresolved physical obligation is consequently to identify, from the agent dynamics and an operational observation map, a retained position and momentum, the reversible coupling between them, and their calibrated force and time units. One must then show that the reduced joint law reproduces the relevant forced trajectories, not merely a static displacement or a fitted second derivative. Until those steps are supplied, statistical stiffness, internal memory, metastable lifetime, and mechanical mass remain distinct quantities. The next discriminating calculation is the retained response kernel for a concrete coherently formed cluster under a specified perturbation; its poles, forcing map, and noise correlations can decide whether the memory mechanism above suffices or an additional reversible sector is required.

C Mathematical Details

C.1 Variational Free Energy Preliminaries

For a normalized generative law with observed evidence $Z_o \in (0, \infty)$ and posterior P_o, the variational free energy is

$$\mathcal{F}_V[Q] = D_{\mathrm{KL}}(Q\|P_o) - \log Z_o \geq -\log Z_o. \tag{286}$$

When the density and integrability assumptions of Section 5 hold, this is also

$$\mathcal{F}_V[Q] = D_{\mathrm{KL}}(Q\|P_0) - \mathbb{E}_Q \log \ell_o. \tag{287}$$

The posterior-relative-entropy formula defines the extended value when the split expression would subtract infinities. Exact inference can require an intractable integral, but tractable models also exist. Minimization over all probability laws attains the bound exactly at $Q = P_o$, while a restricted family can have a positive irreducible approximation gap (Propositions 32 and 38). An update scheme need not reach even its restricted optimum. The notation $\mathcal{E}$ remains reserved for the separate consensus potential.

C.2 Gaussian KL Divergence

For multivariate Gaussian distributions $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$ in $\mathbb{R}^{K_q}$ with $\Sigma_q, \Sigma_p \succ 0$, the Kullback-Leibler divergence is

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left[\log\frac{|\Sigma_p|}{|\Sigma_q|} + \mathrm{tr}(\Sigma_p^{-1}\Sigma_q) + (\mu_p - \mu_q)^\top \Sigma_p^{-1}(\mu_p - \mu_q) - K_q\right], \tag{288}$$

the Gaussian reverse relative entropy (Proposition 37). This formula is used throughout for the self and alignment terms of the consensus functional.

C.3 Fisher Information Matrix for Gaussians

The Fisher information matrix for Gaussian distributions parametrized by $\theta = (\mu, \Sigma)$ has the block structure

$$G = \begin{pmatrix} \Sigma^{-1} & 0 \\ 0 & \frac{1}{2}(\Sigma^{-1} \otimes \Sigma^{-1}) \end{pmatrix}, \tag{289}$$

and every affine map $(\mu, \Sigma) \mapsto (Q\mu + v, Q\Sigma Q^\top)$ with $Q \in \mathrm{GL}(K_q)$ is an isometry of it [38] (Proposition 37). The covariance block is to be read as the Fisher quadratic form restricted to symmetric perturbations, $g(d\Sigma, d\Sigma) = \frac{1}{2}\,\mathrm{tr}(\Sigma^{-1}d\Sigma\Sigma^{-1}d\Sigma)$. As an operator on symmetric matrices, $X \mapsto \frac{1}{2}\Sigma^{-1}X\Sigma^{-1}$ has inverse $X \mapsto 2\Sigma X\Sigma$, so the natural gradient is

$$\tilde{\nabla}_\mu \mathcal{E} = \Sigma(\nabla_\mu \mathcal{E}), \tag{290}$$

$$\tilde{\nabla}_\Sigma \mathcal{E} = 2\Sigma(\nabla_\Sigma \mathcal{E})\Sigma, \tag{291}$$

where $\nabla_\mu \mathcal{E}$ and $\nabla_\Sigma \mathcal{E}$ are the standard Euclidean gradients, with $\nabla_\Sigma \mathcal{E}$ the symmetric matrix gradient. For the non-redundant $\mathrm{vech}(\Sigma)$ parameterization the matrix representation of this block is $\frac{1}{2}\mathcal{D}^\top(\Sigma^{-1} \otimes \Sigma^{-1})\mathcal{D}$ with $\mathcal{D}$ the duplication matrix, and its matrix inverse is not simply $2(\Sigma \otimes \Sigma)$; the inverse as an operator on symmetric matrices, and hence the displayed natural gradient, is unchanged.

C.4 SO(3) Generators and Transport Operators

For the compact group $G = \mathrm{SO}(3)$ acting through a real irreducible representation of integer spin $j \in \{0, 1, 2, \ldots\}$, the belief dimension is $K_q = 2j + 1$, and the generators $\{G_x, G_y, G_z\}$ are real skew-symmetric $K_q \times K_q$ matrices satisfying

$$[G_a, G_b] = \epsilon_{abc} G_c. \tag{292}$$

Half-integer spin representations are representations of the double cover SU(2) and are not included in this real SO(3) statement. Flat reference identifications are

$$\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j), \tag{293}$$

where $\phi_i = \phi_i^x G_x + \phi_i^y G_y + \phi_i^z G_z \in \mathfrak{so}(3)$ in the spin-j representation.

The product $\exp(\phi_i)\exp(-\phi_j)$ may be replaced by the first-order Baker-Campbell-Hausdorff truncation $\exp(\phi_i - \phi_j)$, with error controlled by the leading commutator, $\Omega_{ij} = \exp\big(\phi_i - \phi_j - \frac{1}{2}[\phi_i, \phi_j] + \mathcal{O}(\|\phi\|^3)\big)$. The truncated transport is not of the form $U_i U_j^{-1}$: it breaks the cocycle identity, so its holonomy around a triangle of agents is of the order of $\|[\phi_i, \phi_j]\|$, and it is not invariant under the right translation $U_i \mapsto U_i g$. The vanishing field strength of Regime I (Lemma 1) holds for the exact two-exponential product only. The truncation is acceptable only when the frames are small enough that this holonomy is negligible for the purpose at hand; otherwise the exact product is used.

C.5 Numerical Stability and Cholesky Parametrization

Covariance matrices Σ_i must remain positive definite throughout optimization. Two parameterizations of the covariance sector achieve this, and they differ in their dynamics. The first is the Cholesky parametrization

$$\Sigma_i = L_i L_i^\top, \tag{294}$$

where L_i is lower triangular with positive diagonal entries and gradients are taken with respect to L_i. Any lower-triangular L_i gives a positive-semidefinite Σ_i, but Σ_i is positive definite only when every diagonal entry of L_i is nonzero, and a Euclidean gradient step on L_i can drive a diagonal entry through zero. Strict positive-definiteness is therefore enforced by writing the diagonal through a diffeomorphism of the real line onto the positive half-line, such as softplus or the exponential, so that the stored degrees of freedom are unconstrained while the resulting diagonal entries cannot

cross zero in exact arithmetic. The absolute value is not admissible: it vanishes at zero, is not differentiable there, and is two-to-one, so the map from stored parameters to $\mathcal{S}_{++}^{K_q}$ would not be a diffeomorphism. The second parameterization is the Fisher-Rao natural-gradient step with the affine-invariant exponential retraction of Eq. (103) and Appendix C.3.

With a diffeomorphic positive diagonal map, the unconstrained Cholesky coordinates give a chart of $\mathcal{S}_{++}^{K_q}$. The chain-rule gradient vanishes exactly when the symmetric covariance gradient vanishes, so the parameterizations share stationary points. Euclidean descent in those coordinates does not share the Fisher trajectories; its local rates depend on the chart Jacobian.

For example, the covariance objective $\frac{1}{2}\operatorname{tr}(M\Sigma) - \log\det\Sigma$, with $M \succ 0$, has $\Sigma_* = 2M^{-1}$. Its covariance Hessian is $V \mapsto \Sigma_*^{-1} V \Sigma_*^{-1}$. The Fisher exponential update therefore has linearization $(1-2\eta_\Sigma)V$ and is locally stable for $0 < \eta_\Sigma < 1$. Direct Euclidean covariance descent is locally stable for $0 < \eta_\Sigma < 2\lambda_{\min}(\Sigma_*)^2$. An arbitrary positive-diagonal reparameterization supplies no universal Euclidean learning-rate bound. A retraction preserves positive definiteness in exact arithmetic; continuous natural-gradient dynamics and a finite retraction step remain different objects.

D Covariance Dynamics and Equilibrium Analysis

Scope of this appendix. The analysis below is a local, fixed-attention, receiver-only surrogate. We treat the attention weights β_{ij} as exogenous constants when differentiating with respect to Σ_i, vary only the receiver covariance, and never differentiate the sender covariances Σ_j or the weights β_{ij} themselves. The full reduced functional, obtained by envelope-eliminating β_{ij} via its entropy-regularized softmax optimum and including the complete sender-row contributions, is not derived here; no explicit first-derivative $\partial\beta_{ij}$ response survives in that canonical functional. The fixed point characterized below is therefore a partial Euler-Lagrange stationarity condition, not a stationary point of the fully reduced free energy.

The covariance fixed point is also a property of the consensus potential of Section 5.4, which weights the belief entropy of agent i by $1 + B_i$ with $B_i := \sum_j \beta_{ij}$. Minimizing $\mathbb{E}_q[U] - (1+B_i)H(q)$ over laws q gives $q \propto e^{-U/(1+B_i)}$, so each belief is a Gibbs state at temperature $1 + B_i = 2$ in units in which Bayesian updating has temperature one. In a normalized Gaussian joint the entropy enters once, and the mean-field covariance is the inverse diagonal precision block, fixed in one step and independent of the neighbors' covariances (Proposition 38). The factor $\frac{1}{2}$ in Eq. (300) and the homogeneous limit $\Sigma_\infty = \Sigma_0$ below both come from this doubled entropy weight.

D.1 Covariance Gradient of the Per-Agent Block

All laws in this appendix are nonsingular Gaussians, priors and invertible noiseless links are fixed, and self edges are excluded. These are receiver-response equations for the consensus specialization; the normalized collective VFE and general noisy selective channels have different equations. Here we derive the gradient of agent i's block $\mathcal{E}_i$ of the consensus functional with respect to its covariance Σ_i, a standard computation for Gaussian families [29, 39].

The block is

$$\mathcal{E}_i = D_{\mathrm{KL}}(q_i \| p_i) + \sum_{j \neq i} \beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)], \tag{295}$$

where $q_i = \mathcal{N}(\mu_i, \Sigma_i)$, $p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i})$, and $\sum_j \beta_{ij} = 1$ by construction.

D.1.1 Gaussian relative entropy and its derivative

Differentiating the Gaussian relative entropy of Appendix C.2, with first argument $\mathcal{N}(\mu_1, \Sigma_1)$ and second argument $\mathcal{N}(\mu_2, \Sigma_2)$, with respect to Σ_1 at fixed μ_1, μ_2, Σ_2 gives

$$\frac{\partial D_{\mathrm{KL}}}{\partial \Sigma_1} = \frac{1}{2}\left[-\Sigma_1^{-1} + \Sigma_2^{-1}\right]. \tag{296}$$

Applying this to each term in $\mathcal{E}_i$:

$$\frac{\partial}{\partial \Sigma_i} D_{\mathrm{KL}}(q_i \| p_i) = \frac{1}{2}\left[-\Sigma_i^{-1} + \Sigma_{p,i}^{-1}\right], \tag{297}$$

$$\frac{\partial}{\partial \Sigma_i} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) = \frac{1}{2}\left[-\Sigma_i^{-1} + (\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}\right]. \tag{298}$$

The observation term, for the Gaussian likelihood $-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)] = \frac{1}{2}(o_i - \mu_i)^\top \Lambda_{o_i}(o_i - \mu_i) + \frac{1}{2}\mathrm{tr}(\Lambda_{o_i}\Sigma_i) + \text{const}$, contributes a constant precision correction $\partial[-\mathbb{E}_{q_i}\log p(o_i \mid k_i)]/\partial\Sigma_i = \frac{1}{2}\Lambda_{o_i}$, independent of Σ_i. Including it,

$$\boxed{\frac{\partial \mathcal{E}_i}{\partial \Sigma_i} = \frac{1}{2}\left[-2\Sigma_i^{-1} + \Sigma_{p,i}^{-1} + \sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} + \Lambda_{o_i}\right].} \tag{299}$$

Because $\sum_j \beta_{ij} = 1$, the prefactor $(1 + \sum_j \beta_{ij})$ in front of Σ_i^{-1} reduces to the constant 2, as displayed in the boxed gradient.

D.2 Fixed-Point Equation and Symmetric Solution

At equilibrium, we set $\partial\mathcal{E}_i/\partial\Sigma_i = 0$, giving

$$\Sigma_i^{-1} = \frac{1}{2}\left[\Sigma_{p,i}^{-1} + \sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} + \Lambda_{o_i}\right]. \tag{300}$$

The sensory precision Λ_{o_i} enters with the same prefactor as the other precision contributions; setting $\Lambda_{o_i} = 0$ gives the observation-free case used in the homogeneous limit below. Equation (300) couples all agents: each agent's precision is half the sum of its prior and sensory precision and the β-weighted transported neighbor precisions.

Precision coordinates. Write $\Lambda_i := \Sigma_i^{-1}$ for the belief precision, $A_i := \Sigma_{p,i}^{-1} + \Lambda_{o_i} \succ 0$ for the prior-and-sensory anchor, and define the linear map

$$\mathcal{L}(\Lambda)_i := \sum_j \beta_{ij}\Omega_{ij}^{-\top}\Lambda_j\Omega_{ij}^{-1} \tag{301}$$

on the space $\mathrm{Sym}(K)^N$ of N-tuples of symmetric matrices. Since $(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} = \Omega_{ij}^{-\top}\Lambda_j\Omega_{ij}^{-1}$, Eq. (300) is linear in precision coordinates,

$$2\Lambda = A + \mathcal{L}(\Lambda), \tag{302}$$

and the covariance fixed-point map

$$T(\Sigma)_i = 2\big(\Sigma_{p,i}^{-1} + \sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} + \Lambda_{o_i}\big)^{-1} \tag{303}$$

is, in precision coordinates, the affine map $\Lambda \mapsto \frac{1}{2}(A + \mathcal{L}(\Lambda))$. Each term of $\mathcal{L}$ is a congruence with a nonnegative weight, so $\mathcal{L}$ carries the product cone of positive semidefinite tuples into itself. Under a change of frames, $\Lambda_i \mapsto g_i^{-\top}\Lambda_i g_i^{-1}$ and $\Omega_{ij} \mapsto g_i\Omega_{ij}g_j^{-1}$, the map $\mathcal{L}$ transforms by similarity, so its spectral radius $\rho(\mathcal{L})$ is frame independent.

Proposition 14 (Frozen-attention covariance fixed point)**.** *Fix $\beta_{ij} \geq 0$ with $\sum_j \beta_{ij} = 1$ and $\Omega_{ij} \in \mathrm{GL}(K)$, and assume $\Sigma_{p,i} \succ 0$ and $\Lambda_{o_i} \succeq 0$.*

(i) Equation (300) *has a solution with every $\Sigma_i \succ 0$ if and only if $\rho(\mathcal{L}) < 2$. The solution is then unique, $\Lambda^* = \frac{1}{2}\sum_{k\geq 0}(\mathcal{L}/2)^k A$, and the iteration of T converges to it. For every $r \in (\rho(\mathcal{L})/2, 1)$ its precision error is bounded by $C_r r^n$; the spectral radius is the worst-case asymptotic root factor, not necessarily the ratio of successive errors on every trajectory.*

(ii) If $\rho(\mathcal{L}) < 2$, the simultaneous receiver-only natural-gradient flow of all agents, $d\Sigma_i/dt = -2\eta_\Sigma\Sigma_i(\partial\mathcal{E}_i/\partial\Sigma_i)\Sigma_i$ with $\eta_\Sigma > 0$, remains in the SPD cone and converges to Λ^* from every SPD initial condition, at every exponential rate strictly smaller than $\eta_\Sigma(2-\rho(\mathcal{L}))$. A polynomial factor may accompany the spectral decay rate when $\mathcal{L}$ has a nontrivial Jordan block.

(iii) If $\rho(\mathcal{L}) \geq 2$, no SPD fixed point exists, and along the same flow the precision grows without bound, so that some covariance direction collapses.

(iv) $\rho(\mathcal{L}) \leq \max_i \lambda_{\max}(C_i)$ with $C_i := \sum_j \beta_{ij}(\Omega_{ij}\Omega_{ij}^\top)^{-1} = \mathcal{L}(I)_i$. If every row sums to one and a change of frames makes every link orthogonal, in particular if they are graph-flat, $\Omega_{ij} = U_iU_j^{-1}$, then $\rho(\mathcal{L}) = 1$.

Proof. With the boxed gradient, $\partial\mathcal{E}_i/\partial\Sigma_i = \frac{1}{2}(A_i + \mathcal{L}(\Lambda)_i - 2\Lambda_i)$, and $d\Lambda_i/dt = -\Lambda_i(d\Sigma_i/dt)\Lambda_i$, so the flow of (ii) is the linear system

$$\frac{d\Lambda}{dt} = \eta_\Sigma\big(A + \mathcal{L}(\Lambda) - 2\Lambda\big). \tag{304}$$

For (i), if $\rho(\mathcal{L}) < 2$ the Neumann series converges, $2I - \mathcal{L}$ is invertible, and $\Lambda^* \succeq A/2 \succ 0$ because every term of the series is positive semidefinite; the iteration error is $(\mathcal{L}/2)^n(\Lambda_0 - \Lambda^*)$. Conversely, let $\Lambda \succ 0$ solve Eq. (302), and choose $\epsilon > 0$ with $A_i \succeq \epsilon\Lambda_i$ for every i. Then $\mathcal{L}(\Lambda) = 2\Lambda - A \preceq (2-\epsilon)\Lambda$. In the order-unit norm $\|X\|_\Lambda := \min\{s \geq 0 : -s\Lambda \preceq X \preceq s\Lambda\}$, which is a norm because Λ is interior to the cone, cone preservation gives $\|\mathcal{L}(X)\|_\Lambda \leq (2-\epsilon)\|X\|_\Lambda$, hence $\rho(\mathcal{L}) \leq 2 - \epsilon < 2$. The same argument with $\Lambda = I$ and $\mathcal{L}(I) \preceq \max_i \lambda_{\max}(C_i)I$ proves the bound in (iv). For (ii), variation of constants gives

$$\Lambda(t) = e^{-2\eta_\Sigma t}e^{\eta_\Sigma t\mathcal{L}}\Lambda(0) + \eta_\Sigma\int_0^t e^{-2\eta_\Sigma s}e^{\eta_\Sigma s\mathcal{L}}(A)\mathrm{d}s.$$

The exponential $e^{s\mathcal{L}} = \sum_k (s\mathcal{L})^k/k!$ preserves the cone and satisfies $e^{s\mathcal{L}}A \succeq A$, so $\Lambda(t) \succeq (1 - e^{-2\eta_\Sigma t})A/2$ plus a positive semidefinite term, and $\Lambda(t)$ stays SPD. The deviation is $\Lambda(t) - \Lambda^* = e^{-\eta_\Sigma(2I-\mathcal{L})t}(\Lambda(0) - \Lambda^*)$. In a finite-dimensional Jordan form, each block contributes an exponential multiplied by a polynomial; this proves the stated bound at any strictly smaller rate, without assuming diagonalizability. Every eigenvalue of $2I - \mathcal{L}$ has real part at least $2 - \rho(\mathcal{L})$, with equality at $2 - \rho(\mathcal{L})$ itself, because the spectral radius of a cone-preserving map is an eigenvalue [40]. For (iii),

the same theorem applied to the adjoint $\mathcal{L}^*(W)_j = \sum_i \beta_{ij}\Omega_{ij}^{-1}W_i\Omega_{ij}^{-\top}$, which preserves the positive semidefinite cone (self-dual under $\langle X, Y\rangle = \sum_i \operatorname{tr}(X_iY_i)$), supplies $W \succeq 0$, $W \neq 0$, with $\mathcal{L}^*W = \rho(\mathcal{L})W$. Pairing a fixed point with W gives $(2-\rho(\mathcal{L}))\langle W, \Lambda\rangle = \langle W, A\rangle > 0$, which is impossible for $\rho(\mathcal{L}) \geq 2$ and $\Lambda \succ 0$. Along Eq. (304), $d\langle W, \Lambda\rangle/dt = \eta_\Sigma(\langle W, A\rangle + (\rho(\mathcal{L})-2)\langle W, \Lambda\rangle) \geq \eta_\Sigma\langle W, A\rangle > 0$, since $\Lambda(t)$ stays positive semidefinite, so $\langle W, \Lambda(t)\rangle$ grows at least linearly in t. For the second part of (iv), if frames g_i make every link orthogonal, then in those frames $\mathcal{L}(I)_i = \sum_j \beta_{ij}I = I$, so I is an interior eigenvector with eigenvalue one and the bound gives $\rho(\mathcal{L}) = 1$; similarity carries this back to the original frames. For graph-flat links the frames $g_i = U_i^{-1}$ make every link the identity. □

Frame dependence and holonomy. The directly computable hypothesis $\lambda_{\max}(C_i) < 2$ for every i is sufficient by part (iv), but it is not frame independent, because C_i picks up the factor $g_j^{-1}g_j^{-\top}$ under a change of frames. For graph-flat $\mathrm{GL}(K)$ links it can fail by a large margin while $\rho(\mathcal{L}) = 1$: for two scalar agents with $U_1 = a$, $U_2 = 1$, $a > 1$, and $\beta_{12} = \beta_{21} = 1$, one has $C_1 = a^{-2}$ and $C_2 = a^2$, while $\rho(\mathcal{L}) = 1$ for every a. The value of $\rho(\mathcal{L})$ departs from one only through loop holonomy whose holonomy group is not conjugate into $\mathrm{O}(K)$, and such holonomy can push it past two. For $K = 1$ and two agents with $\beta_{12} = \beta_{21} = 1$ and independent links $w = \Omega_{12}$, $v = \Omega_{21}$, the map is $\mathcal{L} = \left(\begin{smallmatrix} 0 & w^{-2} \\ v^{-2} & 0 \end{smallmatrix}\right)$ with $\rho(\mathcal{L}) = 1/|h|$, where $h = wv$ is the backtrack holonomy; no SPD fixed point exists for $|h| \leq 1/2$, and the iterated covariances decay to zero. With orientation reversal $\Omega_{ji} = \Omega_{ij}^{-1}$ imposed, as for the independent links of Regime II, a two-cycle carries no holonomy, but the obstruction reappears on a triangle: with rows $\beta_{ij} = 1/2$ and holonomy h around the loop, the characteristic polynomial of $\mathcal{L}$ is $\lambda^3 - \frac{3}{4}\lambda - (h^2 + h^{-2})/8$, and $\rho(\mathcal{L}) \geq 2$ exactly when $h^2 + h^{-2} \geq 52$.

D.2.1 Homogeneous limit

In the homogeneous limit we assume that (i) all agents are identical, so $\Sigma_i = \Sigma_\infty$ for all i; (ii) every link preserves the shared prior covariance, $\Omega_{ij}\Sigma_0\Omega_{ij}^\top = \Sigma_0$ (for instance $\Omega_{ij} = I$); (iii) $\Sigma_{p,i} = \Sigma_0$; and (iv) $\Lambda_{o_i} = 0$, the observation-free case. Under (i)–(iv), the candidate $\Sigma_\infty = \Sigma_0$ has every transported covariance equal to Σ_0, so it satisfies Eq. (300):

$$\Sigma_0^{-1} = \frac{1}{2}\left[\Sigma_0^{-1} + \Sigma_0^{-1}\right], \qquad \Sigma_\infty = \Sigma_0. \tag{305}$$

and the solution is unique by Proposition 14, since links preserving Σ_0 are orthogonal in the common frame $\Sigma_0^{-1/2}$. With a common sensory precision $\Lambda_{o_i} = \Lambda_o > 0$ and links preserving the combined precision $A = \Sigma_0^{-1} + \Lambda_o$ by $\Omega_{ij}^{-\top}A\Omega_{ij}^{-1} = A$, the same computation gives $\Sigma_\infty = (\Sigma_0^{-1} + \Lambda_o)^{-1}$.

Hence, in a symmetric observation-free population, the equilibrium covariance reproduces the shared prior. This reflects the doubled entropy weight of the consensus potential: a normalized Gaussian joint with prior precision Σ_0^{-1} and edge precision Σ_0^{-1} gives the mean-field covariance $\Sigma_0/2$, which is precision addition (Proposition 38(ii)).

D.2.2 Alignment-dominated regime

Although $\sum_j \beta_{ij} = 1$, one might expect the effective strength of alignment to be controlled by the attention temperature τ in

$$\beta_{ij} = \frac{\exp\left[-\frac{1}{\tau}D_{\mathrm{KL}}\left(q_i \| \Omega_{ij}q_j\right)\right]}{\sum_k \exp\left[-\frac{1}{\tau}D_{\mathrm{KL}}\left(q_i \| \Omega_{ik}q_k\right)\right]}. \tag{306}$$

As $\tau \to 0$, β_{ij} becomes sharply peaked on $j^\star = \arg\min_j D_{\mathrm{KL}}(q_i \| \Omega_{ij}q_j)$. Since $\sum_j \beta_{ij} = 1$ for every $\tau > 0$, sharpening β by itself does not amplify the coefficient on the alignment block: in the

limit, $\sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} \to (\Omega_{ij^\star}\Sigma_{j^\star}\Omega_{ij^\star}^\top)^{-1}$, which is bounded. To make the alignment-dominated regime precise we place an explicit coupling strength $a > 0$ on the alignment block,

$$\mathcal{E}_i^{(a)} = D_{\mathrm{KL}}(q_i\|p_i) + a\sum_{j\neq i}\beta_{ij}D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)], \tag{307}$$

of which the derivation above is the case $a = 1$. Repeating the differentiation term by term, the stationarity condition becomes

$$(1+a)\Sigma_i^{-1} = \Sigma_{p,i}^{-1} + a\sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} + \Lambda_{o_i}, \tag{308}$$

which reduces to (300) at $a = 1$. In precision coordinates it reads $(1+a)\Lambda = A + a\mathcal{L}(\Lambda)$, and the argument of Proposition 14(i), applied to $a\mathcal{L}/(1+a)$, shows that an SPD solution exists if and only if $\rho(\mathcal{L}) < 1 + 1/a$. The high-coupling limit $a \to \infty$ therefore exists only if $\rho(\mathcal{L}) \leq 1$. Dividing by a gives formally

$$\boxed{\Sigma_i^{-1} \longrightarrow \sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} = \left\langle(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}\right\rangle_\beta \qquad \text{(alignment-dominated)},} \tag{309}$$

where $\langle\cdot\rangle_\beta$ denotes a β-weighted average over neighbors. This is the eigenvalue equation $\Lambda = \mathcal{L}(\Lambda)$. If $\rho(\mathcal{L}) < 1$, the finite-coupling solution is $\Lambda_a = (I + a(I - \mathcal{L}))^{-1}A$, which tends to zero like $1/a$, so the covariances diverge. The boxed relation has a nonzero SPD limit only when $\rho(\mathcal{L}) = 1$ with an SPD Perron eigenvector. Graph-flat links are the basic case. In the frames $g_i = U_i^{-1}$ the map acts as $\tilde{\Lambda}_i \mapsto \sum_j \beta_{ij}\tilde{\Lambda}_j$ with $\tilde{\Lambda}_i := U_i^\top\Lambda_i U_i$. For an irreducible row-stochastic β with stationary distribution π, contracting the finite-coupling equation with π gives $\sum_i \pi_i\tilde{\Lambda}_i = \sum_i \pi_i\tilde{A}_i$ for every a, with $\tilde{A}_i := U_i^\top A_i U_i$, while the components off the Perron direction decay like $1/a$. The limit is therefore the common precision $\tilde{\Lambda}_\infty = \sum_k \pi_k\tilde{A}_k$, and the limit covariances $\Sigma_i = U_i\tilde{\Lambda}_\infty^{-1}U_i^\top$ satisfy $\Sigma_i = \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ on every edge.

Per-edge covariance alignment is thus exact in the flat-link, high-coupling limit, and small τ alone does not produce it. At finite coupling, or with holonomy, the equilibrium satisfies only the β-averaged relation (300), and with several heterogeneous neighbors the per-edge mismatch is generically nonzero, since Σ_i cannot equal every transported neighbor covariance at once.

D.2.3 Stability

Proposition 14(ii) gives global stability of the simultaneous receiver-only natural-gradient flow whenever $\rho(\mathcal{L}) < 2$. The own-block Hessian is also exact,

$$\frac{\partial^2\mathcal{E}_i}{\partial\Sigma_i\partial\Sigma_i}[X] = \Sigma_i^{-1}X\Sigma_i^{-1}, \tag{310}$$

since each KL term contributes $\frac{1}{2}\Sigma_i^{-1}X\Sigma_i^{-1}$ from its log-determinant, the self and alignment weights sum to $2 \times \frac{1}{2} = 1$, and the trace and observation terms are linear in Σ_i. It is positive definite for every $\Sigma_i \succ 0$, so each agent's problem with its neighbors frozen is strictly convex in Σ_i. Neither statement covers the complete canonical objective, whose Hessian carries sender cross-blocks and the reduced-Hessian softmax covariance correction; stability of that coupled equilibrium, and existence of an equilibrium of the live-softmax dynamics, are not established here. A Euclidean flow $d\Sigma_i/dt = -\eta_\Sigma\partial\mathcal{E}_i/\partial\Sigma_i$ has the same fixed point, but its linearization depends on the conditioning of the fixed point, and we have no general stability theorem for it.

Mean update where covariances align. The natural-gradient mean update $\Sigma_i \nabla_{\mu_i} \mathcal{E}_i$ carries the per-edge alignment term $\beta_{ij}\Sigma_i(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}(\mu_i - \Omega_{ij}\mu_j)$. Where per-edge alignment $\Sigma_i = \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ holds, as in the flat-link, high-coupling limit above, the precision factors cancel and

$$\beta_{ij}\Sigma_i(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}(\mu_i - \Omega_{ij}\mu_j) \longrightarrow \beta_{ij}(\mu_i - \Omega_{ij}\mu_j), \tag{311}$$

so the step is an attention-weighted sum of transported residuals with no covariance factor. This is again a property of the consensus potential: in the exact receiver update of a normalized comparison-record model the corresponding step keeps a Kalman-type gain set by the receiver's posterior covariance (Proposition 39).

E Relating the Quadratic Forms to Transported Relative Entropies

This appendix treats the invertible, noiseless, product-Gaussian consensus specialization at c_*. It relates quadratic expectations to transported relative entropies, including the determinant correction. The direction of the divergence follows from building the coupling precision from the transported sender covariance, as shown below; it is not supplied by the representation theorem of Appendix F.

E.1 Exact expansion for Gaussian beliefs

For independent Gaussians $q_i = \mathcal{N}(\mu_{q,i}, \Sigma_{q,i})$ and $q_j = \mathcal{N}(\mu_{q,j}, \Sigma_{q,j})$ and any symmetric matrix M, the expectation of a quadratic form is

$$\mathbb{E}_{q_i q_j}[\delta^\top M \delta] = \mathrm{tr}\big(M\mathrm{Cov}(\delta)\big) + \bar{\delta}^\top M \bar{\delta}, \tag{312}$$

where $\delta = k_i - \Omega_{ij}k_j$, $\bar{\delta} = \mathbb{E}[\delta] = \mu_{q,i} - \Omega_{ij}\mu_{q,j}$, and

$$\mathrm{Cov}(\delta) = \Sigma_{q,i} + \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top. \tag{313}$$

Applying (312) with $M = W_{ij}^{(q)}$:

$$\begin{aligned}\mathbb{E}_{q_i q_j}[(k_i - \Omega_{ij}k_j)^\top W_{ij}^{(q)}(k_i - \Omega_{ij}k_j)] = \mathrm{tr}\big[W_{ij}^{(q)}(\Sigma_{q,i} + \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)\big] \\ + (\mu_{q,i} - \Omega_{ij}\mu_{q,j})^\top W_{ij}^{(q)}(\mu_{q,i} - \Omega_{ij}\mu_{q,j}).\end{aligned} \tag{314}$$

An identical expansion holds for the model channel with $W_{ij}^{(s)}$ and $\tilde{\Omega}_{ij}$. For a fixed coupling $W_{ij} \succ 0$ the mean-dependent part is exactly a relative entropy at common covariance, by Eq. (15) with equal covariances: $\frac{1}{2}\mathbb{E}[\delta^\top W_{ij}\delta] = D_{\mathrm{KL}}\big(\mathcal{N}(\mu_{q,i}, W_{ij}^{-1}) \| \mathcal{N}(\Omega_{ij}\mu_{q,j}, W_{ij}^{-1})\big) + \frac{1}{2}\mathrm{tr}\big(W_{ij}(\Sigma_{q,i} + \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)\big)$.

To obtain the transported relative entropy of the beliefs themselves we choose state-dependent couplings,

$$W_{ij}^{(q)} := c_{ij}^{(q)}(\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)^{-1}, \qquad W_{ij}^{(s)} := c_{ij}^{(s)}(\tilde{\Omega}_{ij}\Sigma_{s,j}\tilde{\Omega}_{ij}^\top)^{-1}, \tag{315}$$

where $c_{ij}^{(q)}, c_{ij}^{(s)} > 0$ are dimensionless coupling strengths. Because $W_{ij}^{(q)}$ is built from the current variational covariance $\Sigma_{q,j}$ (and $W_{ij}^{(s)}$ from $\Sigma_{s,j}$), the identification below is a value-level matching at the current variational state, not a property of a fixed generative model, consistent with the status of the construction in Section 5.4. This identity does not license differentiating the quadratic form while holding its state-dependent $W_{ij}^{(q)}$ fixed. The normalized comparison-record model of Section 5 instead supplies a fixed likelihood precision and is a different objective.

Inserting the Gaussian KL closed form of Appendix C.2 for $q_i = \mathcal{N}(\mu_{q,i}, \Sigma_{q,i})$ and $\Omega_{ij} q_j = \mathcal{N}(\Omega_{ij}\mu_{q,j}, \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)$ into (314) gives the exact identity

$$\begin{aligned}\frac{1}{4}\mathbb{E}_{q_i q_j}[(k_i - \Omega_{ij}k_j)^\top W_{ij}^{(q)}(k_i - \Omega_{ij}k_j)] = \frac{c_{ij}^{(q)}}{2} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) + \frac{c_{ij}^{(q)} K_q}{2} \\ + \frac{c_{ij}^{(q)}}{4} \log\det\big((\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)^{-1}\Sigma_{q,i}\big).\end{aligned} \tag{316}$$

The identification with the transported relative entropy is thus exact up to the constant and the single log-determinant term. The direction comes from the transported sender covariance in $W_{ij}^{(q)}$: with $W = c\Sigma_{q,i}^{-1}$ the same expansion yields $D_{\mathrm{KL}}(\Omega_{ij} q_j \| q_i)$. The log-determinant term vanishes when $\det \Sigma_{q,i} = \det(\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)$. Expanded in the covariance mismatch $\Delta := \Sigma_{q,i} - \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top$, its leading contribution is $\frac{c_{ij}^{(q)}}{4}\mathrm{tr}\big((\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)^{-1}\Delta\big)$, so the correction is $O(\|\Delta\|)$ in general and is $O(\|\Delta\|^2)$ along first-order volume-preserving perturbations with $\mathrm{tr}(V^{-1}\Delta) = O(\|\Delta\|^2)$, where $V = \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top$. Along exactly determinant-preserving deviations it vanishes identically. With several heterogeneous neighbors the per-edge mismatch Δ_{ij} is generically nonzero at the stationary point of Eq. (300) (Appendix D), and the first-order correction persists there. The model channel satisfies the same identity with $(m, s, \tilde{\Omega}_{ij}, W_{ij}^{(s)}, c_{ij}^{(s)}, K_m)$ in place of $(k, q, \Omega_{ij}, W_{ij}^{(q)}, c_{ij}^{(q)}, K_q)$.

E.2 Defining raw alignment weights

To connect with the main text we define the raw alignment weights

$$\tilde{\beta}_{ij} := \frac{c_{ij}^{(q)}}{2}, \qquad \tilde{\gamma}_{ij} := \frac{c_{ij}^{(s)}}{2}, \tag{317}$$

the bare belief- and model-alignment strengths between agents i and j. For a sequence of minimizers, a diverging coefficient forces its nonnegative edge divergence to zero only if a feasible zero-cost comparison state gives a uniform upper bound and the remaining objective is bounded below. Incompatible cycle constraints can defeat that hypothesis. Under these conditions communicated belief agreement need not imply model agreement; $\tilde{\gamma}_{ij} = 0$ decouples model alignment entirely. The normalized weights β_{ij}, γ_{ij} of the main text, which sum to one over each row, are not identified with the raw couplings. They arise by composition: at a uniform coupling $c_{ij}^{(q)} \equiv c^{(q)}$, the identity (316) supplies the per-edge energies $E_{ij} = D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ up to the common prefactor, and minimizing the entropy-regularized row objective of Section 5.4 (Eqs. (33) and (33)) over the row simplex returns the softmax weights. The quadratic form reproduces the canonical functional only through this composition, and only up to the log-determinant term of Eq. (316).

F Conditional Representation Theorem for the Transported Relative Entropy

This appendix characterizes the belief coupling $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$, with the receiver law in the first slot (the variational or I-projection direction), within a class of divergences and for one problem: agent i's best response with its neighbors' beliefs and its attention weights held fixed. This transported divergence is a declared consensus choice rather than a derived quantity, and nothing below fixes the divergence of the joint scalar functional; the scope paragraph at the end of this appendix shows why the closure criterion cannot. Within the class of local f-divergences that enter the

free energy linearly in β_{ij}, the relative entropy is the only divergence whose best response has the geometric-mean Boltzmann form for all priors and neighbor beliefs; richness of the attainable likelihood ratios, which the converse needs, holds automatically in the Gaussian fiber (Lemma 3). We work at fixed c_* and use x for a sample coordinate, not a base coordinate. Put $r_j = (\Omega_{ij})_{\#} q_j$ and use its density in the receiver's sample space. More general normalized channels can supply the fixed template r_j when it has a density. The universal characterization quantifies over the stated rich template family; it is not automatically a theorem about every restricted channel image.

The best-response problem

With the neighbor beliefs q_j and the weights $\beta_{ij} \geq 0$ held fixed, agent i solves

$$\mathcal{E}_i^{\mathrm{br}}[\beta_i] = \min_{q_i} \Big\{ D_{\mathrm{KL}}(q_i \| p_i) + \sum_{j \neq i} \beta_{ij} \mathcal{D}(q_i, \Omega_{ij} q_j) \Big\}, \qquad \mathcal{D}(q, r) = \int r(x) f\left(\frac{q(x)}{r(x)}\right) dx, \tag{318}$$

with the second-argument density as the integration measure. For a fixed candidate law q_i, optimizing an entropy-regularized row uses the fixed costs $\mathcal{D}(q_i, r_j)$ and gives the Gibbs formula of Section 5.4. If q_i has already been minimized out, the reduced row objective is $\mathcal{E}_i^{\mathrm{br}}(\beta_i) + \tau D_{\mathrm{KL}}(\beta_i \| \pi_i)$, not $\sum_j \beta_{ij} (\partial \mathcal{E}_i^{\mathrm{br}} / \partial \beta_{ij}) + \tau D_{\mathrm{KL}}(\beta_i \| \pi_i)$. Where the value function is differentiable, its marginal costs enter an implicit row stationarity equation.

Lemma 2 (KL-sum identity)**.** *Let p and $r_1, \ldots, r_n$ be probability densities such that the weighted common-support normalizer below satisfies $Z_g > 0$, let $\beta_j \geq 0$ with $B := \sum_j \beta_j$, and define*

$$g := Z_g^{-1} p^{1/(1+B)} \prod_j r_j^{\beta_j/(1+B)}, \qquad Z_g := \int p^{1/(1+B)} \prod_j r_j^{\beta_j/(1+B)} dx. \tag{319}$$

Then for every density q with $D_{\mathrm{KL}}(q\|p)$ and every $D_{\mathrm{KL}}(q\|r_j)$ finite,

$$D_{\mathrm{KL}}(q\|p) + \sum_j \beta_j D_{\mathrm{KL}}(q\|r_j) = (1+B) D_{\mathrm{KL}}(q\|g) - (1+B) \log Z_g. \tag{320}$$

Proof. The left-hand side is $\int q[(1+B)\log q - \log p - \sum_j \beta_j \log r_j] dx = (1+B) \int q[\log q - \log g - \log Z_g] dx$, which is the right-hand side. □

With $r_j = \Omega_{ij} q_j$ and $\mathcal{D} = D_{\mathrm{KL}}$, the identity solves the best-response problem (318) at once: the unique minimizer is $q_i^* = g$, the geometric-mean form (321) below, and the minimum value is $\mathcal{E}_i^{\mathrm{br}} = -(1+B)\log Z_g \geq 0$, nonnegative because $Z_g \leq 1$ by Hölder's inequality (the exponents sum to one). The exponents $1/(1+B)$ are the reciprocal of the entropy weight discussed in Appendix D: the best response is a Gibbs state at temperature $1+B$ relative to Bayesian updating. In a normalized joint the analogous coordinate update is the prior times the exponential of the expected log factors, and its stationarity includes each agent's appearance in its neighbors' factors [41, §10.1], [42].

Conditional representation theorem

Theorem 2 (Conditional representation for the frozen-neighbor best response)**.** *Let $\mathcal{D}(q, r) = \int r(x) f(q(x)/r(x)) dx$ with f real-analytic on $(0, \infty)$, convex, $f(1) = 0$, and, without loss of generality*[1]*, $f'(1) = 0$. Fix agent i's neighbor beliefs $\{q_j\}$ and weights $\beta_{ij} \geq 0$ with $B := \sum_j \beta_{ij} > 0$,*

[1] Any f-divergence is invariant under $f(t) \mapsto f(t) + c(t-1)$ when q and r are normalized, so we may always choose the representative with $f'(1) = 0$.

and let $\mathcal{D}$ enter the best-response problem (318) *linearly. Then the stationary best response has the geometric-mean Boltzmann form*

$$q_i^*(x) \propto p_i(x)^{1/(1+B)} \prod_j \big[r_j(x)\big]^{\beta_{ij}/(1+B)} \tag{321}$$

for every prior and template in a family containing all nonsingular Gaussian laws, with the asserted law required to be a finite stationary solution, if and only if, $\mathcal{D}$ is the relative entropy (up to the affine equivalence above):

$$\mathcal{D}(q, r) = D_{\mathrm{KL}}(q\|r).$$

For $B = 1$, as when β_i is a probability vector, the exponents are $1/2$.

Corollary 1 (Closure forces the KL generator among nondegenerate divergences)**.** *Within the same class, with real-analytic generator and nondegeneracy $f''(1) > 0$, exponential-family closure of the best response for all Gaussian priors and neighbor beliefs already forces $f'(t) = \gamma \log t$ with $\gamma > 0$, that is, $\mathcal{D} = \gamma D_{\mathrm{KL}}$, where the positive multiple γ is absorbable into the coupling weights β_{ij}; the geometric-mean exponents of Eq.* (321) *normalize $\gamma = 1$. The nondegeneracy hypothesis cannot be dropped: $f \equiv 0$ is real-analytic and convex, satisfies the normalization, and is closed for every Gaussian prior and neighbor set, since stationarity then reduces to $q_i^* = p_i$; it yields $\gamma = 0$ and the identically zero divergence. Because $f'(t) = \gamma \log t$ gives $f''(1) = \gamma$, the hypothesis $f''(1) > 0$ is exactly the statement $\gamma > 0$ and excludes that degenerate solution alone.*

Proof. Interpret closure as existence of a finite unrestricted minimizer that is Gaussian for every Gaussian prior and template choice. The prior KL is strictly convex in the law and the other terms are convex, so such a minimizer is unique. Choose all templates equal to $r = \mathcal{N}(0, I)$ and the prior $p = \mathcal{N}(0, aI)$ with $a > 0$, $a \neq 1$. Rotational invariance and uniqueness force a Gaussian minimizer to be $\mathcal{N}(0, bI)$. It cannot equal r: if $b = 1$, stationarity would make $\log(r/p)$ constant, which it is not. Thus

$$\log(q/r) = u + v\|x\|^2, \qquad v \neq 0.$$

The stationarity equation makes $h(u + v\|x\|^2)$ affine in $\|x\|^2$, where $h(z) = f'(e^z)$. Hence h is affine on an open half-line, and real analyticity extends this identity to all real z. The normalization $f'(1) = 0$ gives $f'(t) = \gamma \log t$; convexity gives $\gamma \geq 0$, and $f''(1) > 0$ gives $\gamma > 0$. Integrating and using $f(1) = 0$ proves the claim. Conversely the KL-sum identity gives a Gaussian geometric pool for every positive multiple, with rescaled weights. □

Proof of Theorem 2. Step 1. Stationarity for a general f-divergence. With ratio $r(x) = q_i(x)/\big[r_j(x)\big]$, the first-slot functional derivative is $\delta\mathcal{D}/\delta q_i(x) = f'(r(x))$, so the stationarity condition $\delta\mathcal{E}_i^{\mathrm{br}}/\delta q_i(x) = \lambda$, with Lagrange multiplier λ for normalization, is

$$\log \frac{q_i(x)}{p_i(x)} + 1 + \sum_j \beta_{ij} f'\left(\frac{q_i(x)}{r_j(x)}\right) = \lambda. \tag{322}$$

Step 2. Sufficiency. For $\mathcal{D} = D_{\mathrm{KL}}$ the generator is $f(t) = t \log t - t + 1$ with $f'(t) = \log t$, and Lemma 2 gives the minimizer (321) directly.

Step 3. Necessity. Assume the best response has the form (321) for all priors p_i and neighbor beliefs $\{q_j\}$, with general B. Taking logarithms, $(1 + B) \log q_i^* = \log p_i + \sum_j \beta_{ij} \log(\Omega_{ij} q_j) + C$, and splitting $(1 + B) \log q_i^* = \log q_i^* + \sum_j \beta_{ij} \log q_i^*$ gives

$$\log \frac{q_i^*}{p_i} + \sum_j \beta_{ij} \log \frac{q_i^*}{\Omega_{ij} q_j} = C', \tag{323}$$

with $C' = C$. It remains to pass from Eqs. (322) and (323) to the per-term identity

$$f'\left(\frac{q_i}{\Omega_{ij}q_j}\right) = \log\frac{q_i}{\Omega_{ij}q_j} + k, \tag{324}$$

with a single constant k, and from there to $f'(t) = \log t + k$ on all of $(0,\infty)$. The Lagrange multiplier and the normalizer vary from configuration to configuration, and the geometric-mean form constrains the minimizer, not the neighbor beliefs. Both points are handled by fixing one configuration and varying the sample coordinate x at fixed base point c_*, where the multiplier and the normalizer are constants.

Lemma 3 (Per-term separation and richness)**.** *Let f be real-analytic on $(0,\infty)$ with $f'(1) = 0$, and suppose the stationary solution of* (322) *has the geometric-mean form* (321) *for all priors and neighbor beliefs. Then $f'(t) = \log t$ as a functional identity on $(0,\infty)$.*

Proof. Subtracting the rearranged target from stationarity gives

$$\sum_j \beta_{ij}\left[f'(q_i^*(x)/r_j(x)) - \log(q_i^*(x)/r_j(x))\right] = \text{constant}. \tag{325}$$

Respect the fixed weights by choosing all their positive-weight templates equal to $r = \mathcal{N}(0, I)$, with prior $p = \mathcal{N}(a, I)$ for $a \neq 0$. The asserted geometric pool is $q^* = \mathcal{N}(a/(1+B), I)$. Its ratio to r is

$$q^*(x)/r(x) = \exp\left(\frac{a^\top x}{1+B} - \frac{\|a\|^2}{2(1+B)^2}\right),$$

which ranges over all of $(0,\infty)$. The displayed equation is therefore $B[f'(t) - \log t] = \text{constant}$ for every $t > 0$. Evaluating at $t = 1$ gives that constant as zero. This necessity argument uses the full Gaussian ratio range and does not require analytic continuation; the corollary's weaker closure hypothesis still uses analyticity. □

Integrating $f'(t) = \log t$ gives $f(t) = t\log t - t + C_0$, and $f(1) = 0$ gives $C_0 = 1$. Therefore $f(t) = t\log t - t + 1$, the generator of $D_{\mathrm{KL}}(q\|r)$. □

Scope: the joint scalar functional

The theorem concerns a frozen-neighbor receiver response. The complete consensus scalar also contains every term in which the varied law is a sender. Its second-slot functional derivative is $f(t) - tf'(t)$, equal to $1 - t$ for KL. Under the additional *slotwise* requirement that both slot derivatives be affine in $\log t$, write $f'(t) = a\log t + b$ and $f(t) - tf'(t) = c\log t + d$. Differentiating gives $f''(t) = a/t$ and $-tf''(t) = c/t$, which force $a = c = 0$. This proves incompatibility of that strong slotwise criterion with a nonzero divergence. It is not a general nonexistence theorem for coupled equilibria or for restricted Gaussian optimization.

An exact witness distinguishes receiver response from joint stationarity. Take two scalar agents, identity links, unit weights, $p_1 = \mathcal{N}(0,1)$ and $p_2 = \mathcal{N}(3, 1/4)$. The laws $q_1 = \mathcal{N}(2, 1/2)$ and $q_2 = \mathcal{N}(8/3, 1/3)$ are mutual geometric-pool responses: their precisions and precision-weighted means satisfy the two pool equations. Their receiver mean derivatives vanish, but the omitted sender derivatives in the joint scalar are

$$\partial_{\mu_1}\mathcal{E} = -\frac{4}{3}, \qquad \partial_{\mu_2}\mathcal{E} = 2.$$

Thus this mutual response is not a stationary point of the full scalar. A Gaussian variational restriction supplies Gaussian closure by definition and cannot itself select the divergence.

Dual cost identification

By the envelope identity for minimizing over q_i at fixed β_i, where the value function is differentiable, the marginal attention cost is

$$C_{ij} \equiv \frac{\partial \mathcal{E}_i^{\mathrm{br}}}{\partial \beta_{ij}}\bigg|_{q_i^*} = \mathcal{D}(q_i^*, \Omega_{ij} q_j), \tag{326}$$

which for the relative entropy identifies the cost of attention with the divergence between the agent's best response and the neighbor's transported law. The identity holds for every divergence that enters linearly in β_{ij}, so it is a consistency property and does not select the divergence.

Relation to probabilistic opinion pooling

The best response (321) carries exponents $\beta_{ij}/(1+B)$ together with the prior exponent $1/(1+B)$, which sum to one over the prior-and-neighbor set, so it is the normalized weighted geometric (logarithmic) opinion pool, whose Gaussian case gives the precision-weighted aggregate $\Sigma_*^{-1} = \sum_k \lambda_k \Sigma_k^{-1}$. Dietrich and List [43] characterize this geometric pool axiomatically as externally Bayesian and unanimity/zero-preserving; Bordley [44] and Genest and Zidek [45] supply the multiplicative, log-linear precursor, which coincides with the normalized geometric pool in the continuous Gaussian fiber used here. Theorem 2 re-derives this operator, for the frozen-neighbor best response, by an f-divergence route within its hypothesis class. The meta-agent barycenter of Section 10.1 is the Gaussian M-projection of the linear pool that Dietrich and List characterize from eventwise independence. The axiomatic properties belong to the pool, not to its projection, and a form that drops the between-mean dispersion approximates the projection only in the high-coherence regime.

Interpretation and summary

Gauge covariance fixes what is compared, the law q_i and the transported law $\Omega_{ij} q_j$ in one frame, while the theorem above constrains how it is compared, and only for the best response. On the information-geometric side, the relative entropy is the Bregman divergence of the negative-entropy potential in affine density coordinates, and it is the canonical divergence of the dually flat e/m structure, whence the Pythagorean relation [46]. The Fisher metric, by contrast, is the local Hessian of every smooth f-divergence, up to the factor $f''(1)$.

The characterization concerns the exact unrestricted receiver response, not the whole joint dynamics. Equal-covariance Gaussians with different means already make the likelihood ratio range over every positive value, as the proof shows. The separate Gaussian-closure corollary uses analyticity to extend an identity from a half-line. Coordinate invariance of KL follows from the common-pushforward theorem and holds for real general linear changes, including realifications of complex unitary actions. No physical symmetry or preferred gauge group follows from this invariance alone.

G Augmented Hierarchical Model, the Rigid-Link Predictive Density, and Mean-Field Fixed Points

The transported-copy specialization of the main text reads the cross-scale shadow $p_i^{(\ell)} = \Omega_{i,I}[q_I^{(\ell+1)}]$ of Section 11 as the one-episode-lagged predictive prior of an episode-indexed joint. When a parent forms at episode n, the prior of its children at episode $n+1$ is the filtered parent law pushed through

a declared transition, and for a transported-copy transition this is the shadow formula exactly (Eq. (157)). This appendix records what a same-time hierarchical joint does and does not supply. We define an augmented hierarchical generative model with Gaussian link kernels on the tree of agent latents, prove that it is a proper joint, state its mean-field fixed-point equations, and compute the variational predictive density whose rigid-link limit reproduces the shadow formula. That density is not generally an evidence-consistent downward message: evaluated against the full parent posterior it can reuse a constituent's own evidence, and the functional built from it is not an evidence bound (Remark 2). The general statements behind this are the typing prohibition of Section 5, under which no generative kernel reads a recognition law, and the fact that an energy appended to the free energy of a joint is an evidence bound only after its normalizer is restored (Proposition 41). Downward kernels and the conditional cross-scale divergence give the evidence-consistent form of top-down influence (Proposition 42).

G.1 The Augmented Joint

Let the hierarchy be a finite tree with one root, depth $\ell_{\max}$ and finite breadth, with I_ℓ the set of agents at level ℓ and $\pi(i)$ the parent of agent i. Each agent $i \in I_\ell$ carries a latent variable $k_i^{(\ell)} \in \mathbb{R}^{K_q}$. The construction is made at a single base point, so it is a finite-dimensional probability law on the product space $\mathbb{K} := \prod_{\ell, i \in I_\ell} \mathbb{R}^{K_q}$. The root prior is proper. In this appendix the records are conditionally independent leaf records, $p(o \mid k^{(0)}) = \prod_{i \in I_0} \ell_i(o_i \mid k_i^{(0)})$, with each record kernel normalized. This leaf-local assumption is what keeps the factor graph a tree and licenses the cavity-message formulas; the general record hypergraph of Section 5 need not have this property.

Definition 3 (Augmented hierarchical joint)**.** *For each parent-child edge $(i, \pi(i))$ with $i \in I_\ell$, define the Gaussian link kernel*

$$\mathcal{K}_{i,\pi(i)}^{(\ell)}\big(k_i^{(\ell)} \mid k_{\pi(i)}^{(\ell+1)}\big) = (2\pi\sigma^2)^{-K_q/2} \exp\left[-\frac{1}{2\sigma^2}\big\|k_i^{(\ell)} - \Omega_{i,\pi(i)} k_{\pi(i)}^{(\ell+1)}\big\|^2\right], \tag{327}$$

where $\Omega_{i,\pi(i)} = \exp(\phi_i^{(\ell)}) \exp(-\phi_{\pi(i)}^{(\ell+1)}) \in G$ is the inter-level transport of Section 4.7 and $\sigma^2 > 0$ is a link noise variance. The kernel is a normalized transition density, a propagator from parent to child, and not a Boltzmann weight. Its isotropic covariance is specified in this chart. Under a general receiver change g_i, the noise covariance becomes $\sigma^2 g_i g_i^\top$; isotropy is not preserved by every general linear frame change. The top-level prior $p_{\text{top}}(k^{(\ell_{\max})})$ is a declared root, fixed as a boundary condition (Section 13). The augmented joint over all agent latents is

$$p_\sigma\big(\{k_i^{(\ell)}\}, o\big) = p(o \mid \{k_i^{(0)}\}) \prod_{\ell=0}^{\ell_{\max}-1} \prod_{i \in I_\ell} \mathcal{K}_{i,\pi(i)}^{(\ell)}\big(k_i^{(\ell)} \mid k_{\pi(i)}^{(\ell+1)}\big) p_{\text{top}}\big(k^{(\ell_{\max})}\big). \tag{328}$$

Lemma 4 (Well-definedness)**.** *Assume the observation likelihood $p(o \mid \cdot)$ is bounded above by a finite constant (unity in the discrete-observation case, the normalization constant in the Gaussian case). Then for every finite $\sigma^2 > 0$ and every finite hierarchical tree, the augmented joint* (328) *is integrable over $\mathbb{K}$ for every fixed o, with normalization equal to the evidence $p(o)$; the prior portion, obtained by omitting the likelihood factor, is a proper probability density on $\mathbb{K}$.*

Proof. Each kernel $\mathcal{K}_{i,\pi(i)}^{(\ell)}$ is a Gaussian density on $k_i^{(\ell)}$ with mean $\Omega_{i,\pi(i)} k_{\pi(i)}^{(\ell+1)}$ and covariance $\sigma^2 I$, and it integrates to unity for every value of $k_{\pi(i)}^{(\ell+1)}$. The factor graph of (328) is the hierarchical tree itself. For tree-structured products of proper conditional densities, the integral over the leaves factorizes through the tree [47, §3.2] and leaves the top-level prior p_{top}, which is normalized, so the

prior portion is a proper probability density on $\mathbb{K}$. The likelihood enters as a multiplicative factor at the leaves and is bounded above by hypothesis, so it does not affect integrability, and integrating it against the prior portion yields $p(o)$. □

G.2 Mean-Field Fixed-Point Equations

At a record with finite positive evidence, mean-field inference factorizes recognition across agent latents. The displayed density variations assume finite terms and normalizable coordinate exponentials; they assert no update when a coordinate normalizer fails to exist.

Lemma 5 (Mean-field fixed-point equations)**.** *The variational free energy*

$$\mathcal{F}_{V,\sigma}[q] = \sum_{\ell,i} \mathbb{E}_{q_i^{(\ell)}}\big[\log q_i^{(\ell)}\big] - \sum_{\ell<\ell_{\max},i} \mathbb{E}_{q_i^{(\ell)} q_{\pi(i)}^{(\ell+1)}}\big[\log \mathcal{K}_{i,\pi(i)}^{(\ell)}\big] - \mathbb{E}_{q^{(\ell_{\max})}}\big[\log p_{\text{top}}\big] - \mathbb{E}_{q^{(0)}}\big[\log p(o \mid k^{(0)})\big] \tag{329}$$

is stationary under the mean-field factorization $q(\{k_i^{(\ell)}\}) = \prod_{\ell,i} q_i^{(\ell)}(k_i^{(\ell)})$ when each factor satisfies

$$\begin{aligned} \log q_i^{(\ell)}(k_i) = \mathit{const} &+ \mathbb{E}_{q_{\pi(i)}^{(\ell+1)}}\big[\log \mathcal{K}_{i,\pi(i)}^{(\ell)}(k_i \mid k_{\pi(i)})\big] + \sum_{j:\pi(j)=i} \mathbb{E}_{q_j^{(\ell-1)}}\big[\log \mathcal{K}_{j,i}^{(\ell-1)}(k_j \mid k_i)\big] \\ &+ \delta_{\ell,0}\mathbb{E}_{q_{\setminus i}^{(0)}}\big[\log p(o \mid k^{(0)})\big] + \delta_{\ell,\ell_{\max}} \log p_{\text{top}}(k_i), \end{aligned} \tag{330}$$

where $q_{\setminus i}^{(0)}$ is the product of the other leaf factors. The first expectation is the top-down message from the parent of agent i (absent at $\ell = \ell_{\max}$), the sum collects the bottom-up messages from its children (absent at $\ell = 0$), and the data likelihood enters only at the leaves.

Proof. Equation (330) is the standard coordinate-ascent mean-field update [48, §5.3], [41, §10.1]. It follows by varying (329) with respect to $q_i^{(\ell)}$ at fixed values of the other factors, imposing $\int q_i^{(\ell)} = 1$ by a Lagrange multiplier, and collecting the terms that depend on $k_i^{(\ell)}$. □

G.3 The Rigid-Link Variational Predictive Density

Lemma 6 (Rigid-link variational predictive density)**.** *Let the variational posterior of the parent be Gaussian, $q_{\pi(i)}^{(\ell+1)} = \mathcal{N}(\mu_{\pi(i)}^{(\ell+1)}, \Sigma_{\pi(i)}^{(\ell+1)})$. The variational predictive density of agent i, the marginal of the link kernel against the parent's variational posterior, is*

$$\hat{p}_i^{(\ell),\sigma}(k_i) := \int \mathcal{K}_{i,\pi(i)}^{(\ell)}(k_i \mid k_{\pi(i)}) q_{\pi(i)}^{(\ell+1)}(k_{\pi(i)}) dk_{\pi(i)} = \mathcal{N}\big(k_i; \Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}, \Omega_{i,\pi(i)}\Sigma_{\pi(i)}^{(\ell+1)}\Omega_{i,\pi(i)}^{\top} + \sigma^2 I\big), \tag{331}$$

and as $\sigma^2 \to 0$ it converges to the pushforward $(\Omega_{i,\pi(i)})_* q_{\pi(i)}^{(\ell+1)}$, which is the shadow formula of Eq. (158)*. The lemma computes a density. That density is generally different from the mean-field factor of Eq.* (330) *and from an evidence-consistent downward message of the joint* (328) *(Remark 2).*

Proof. Conditional on $k_{\pi(i)}$, the kernel (327) is Gaussian with mean $\Omega_{i,\pi(i)}k_{\pi(i)}$, linear in the parent, and covariance $\sigma^2 I$. The linear-Gaussian marginalization rule [41, §2.3.3] gives a Gaussian marginal with mean $\Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}$ and covariance $\Omega_{i,\pi(i)}\Sigma_{\pi(i)}^{(\ell+1)}\Omega_{i,\pi(i)}^{\top} + \sigma^2 I$. As $\sigma^2 \to 0$ the kernel converges weakly to $\delta(k_i - \Omega_{i,\pi(i)}k_{\pi(i)})$, and the marginal converges to

$$\int \delta\big(k_i - \Omega_{i,\pi(i)}k_{\pi(i)}\big) q_{\pi(i)}^{(\ell+1)}(k_{\pi(i)}) dk_{\pi(i)} = \big(\Omega_{i,\pi(i)}\big)_* q_{\pi(i)}^{(\ell+1)}. \tag{332}$$

□

General selective hierarchical channels. The normalization and coordinate-variation arguments extend to $k_i \mid k_{\pi(i)} \sim \mathcal{N}(L_{i,\pi(i)}k_{\pi(i)}, N_i)$ with $N_i \succ 0$, including rectangular or rank-deficient $L_{i,\pi(i)}$ and different latent dimensions. Their predictive law is $\mathcal{N}(L\mu, L\Sigma L^\top + N_i)$. With $N_i = \epsilon N_{i,0}$ and $\epsilon \downarrow 0$, it converges weakly to $L_{\#}q_{\pi(i)}$; this limit can be singular and is then a measure, not an ambient density. The isolated child mean-field coordinate is instead $\mathcal{N}(L\mu, N_i)$, because the parent covariance enters its expected log kernel only as a constant. No inverse link is used by these conclusions. The noiseless invertible Ω case is retained below only to identify the earlier transported-copy formula.

Remark 2 (Predictive marginals and cavity messages)**.** *Four facts fix the status of Lemma 6. First, the strict mean-field factor is a different density. Substituting the kernel into the top-down term of Eq.* (330) *gives*

$$\mathbb{E}_{q_{\pi(i)}^{(\ell+1)}}\big[\log \mathcal{K}_{i,\pi(i)}^{(\ell)}\big] = const - \frac{1}{2\sigma^2}\Big[\big\|k_i - \Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}\big\|^2 + \operatorname{tr}\big(\Omega_{i,\pi(i)}\Sigma_{\pi(i)}^{(\ell+1)}\Omega_{i,\pi(i)}^\top\big)\Big],$$

so in isolation from children and data the mean-field factor is $\mathcal{N}(\Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}, \sigma^2 I)$: the parent covariance enters only the constant, and in the rigid limit the factor collapses to a point mass. Second, the predictive density (331) *equals the prior marginal of k_i under p_σ when $q_{\pi(i)}^{(\ell+1)}$ is the prior marginal of the parent. A parent posterior, by contrast, has absorbed the evidence of every leaf below the parent, including the leaves below agent i.*

Third, evaluated against the full parent posterior, the shadow reuses each constituent's own evidence; the evidence-consistent downward message uses the parent posterior with that constituent's contribution removed. On a tree, belief propagation is exact [41, §8.4]: the exact posterior of k_i is proportional to the bottom-up evidence of agent i times the cavity predictive density $\int \mathcal{K}_{i,\pi(i)}^{(\ell)}(k_i \mid k_{\pi(i)}) q_{\pi(i)}^{\setminus i}(k_{\pi(i)})dk_{\pi(i)}$, where $q_{\pi(i)}^{\setminus i}$ is the parent posterior computed without the message from agent i. Multiplying the full-posterior predictive density by agent i's likelihood counts that likelihood twice. An exact scalar example uses $Z \sim \mathcal{N}(0,1)$, $K \mid Z \sim \mathcal{N}(Z,1)$, and observed $o = 1$ under $O \mid K \sim \mathcal{N}(K,1)$. The exact child posterior is $\mathcal{N}(2/3, 2/3)$. The full parent posterior is $\mathcal{N}(1/3, 2/3)$; predicting through the child kernel gives $\mathcal{N}(1/3, 5/3)$, and treating this as a new prior before multiplying the same likelihood gives $\mathcal{N}(3/4, 5/8)$. The difference is exact and exhibits the repeated evidence.

Fourth, the functional built from the predictive density is not an evidence bound. Replacing the mean-field cross term $\mathbb{E}_{q_i q_{\pi(i)}}[-\log \mathcal{K}_{i,\pi(i)}]$ by the predictive cross-entropy $\mathbb{E}_{q_i}[-\log \hat{p}_i]$ can only lower the objective, since $-\log$ is convex and Jensen's inequality gives $\mathbb{E}_{q_i}[-\log \int \mathcal{K}_{i,\pi(i)} q_{\pi(i)}] \le \mathbb{E}_{q_i q_{\pi(i)}}[-\log \mathcal{K}_{i,\pi(i)}]$. The resulting same-time shadow functional $D_{\mathrm{KL}}(q_{\mathrm{top}} \| p_{\mathrm{top}}) + \sum_i [D_{\mathrm{KL}}(q_i \| \hat{p}_i) - \mathbb{E}_{q_i} \log p(o_i \mid k_i)]$ can therefore fall below the surprisal $-\log p_\sigma(o)$, which no variational free energy of p_σ can do. A direct witness needs no numerical minimization. Take one standard-normal parent, two children with link variance $1/2$, independent record noise variance $1/2$, and records $(0,0)$. The record covariance is $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, so the true surprisal is $\log(2\pi) + \frac{1}{2}\log 3$. Choose $q_{\mathrm{top}} = \mathcal{N}(0, 1/2)$ and both child laws $\mathcal{N}(0, 1/3)$. Each predictive child variance is one. Substitution into the shadow functional gives $\log(2\pi) + \log 3 - \frac{1}{2}\log 2 - \frac{1}{4}$, whose difference from the surprisal is $\frac{1}{2}\log(3/2) - \frac{1}{4} < 0$, since $\log(3/2) < 1/2$. Thus this functional is not an evidence bound of the stated joint. In the rigid limit the child-parent term becomes $D_{\mathrm{KL}}(q_i \| (\Omega_{i,\pi(i)})_* q_{\pi(i)})$ with both factors varying. Its mixed third variation (338) *is nonzero, so Theorem 3 applies to it exactly as to the peer terms, and the same-time shadow edge has no fixed state-level evidence-bound representation.*

G.4 Evidence-Consistent Readings of Top-Down Influence

Lemmas 4–6 do not make the same-time shadow a derived prior. Four readings of top-down influence are consistent with one normalized joint. The first is the conditional coupling $\mathbb{E}_{q_{\pi(i)}} D_{\mathrm{KL}}(q_i \| \mathcal{K}_{i,\pi(i)}(\cdot \mid k_{\pi(i)}))$, the mean-field cross term of Eq. (329); it is the conditional cross-scale divergence, and its rigid limit diverges as $1/\sigma^2$ unless the factors degenerate. The second is the cavity predictive density of Remark 2, the exact downward message on a tree. The third is the random-effects, or empirical-Bayes, reading: the parent is a point hyperparameter $\theta = (m, S)$, the constituents are conditionally independent draws $k_i \sim \mathcal{N}(\Omega_{i,I} m, \Omega_{i,I} S \Omega_{i,I}^\top)$, and the shadow is then an honest prior. Fitting θ is type-II maximum likelihood, the barycenter of Eq. (151) with its dispersion term is the exact update of θ at fixed constituent factors [26], and the parent carries no belief of its own; Σ_I is then a population covariance and not a posterior uncertainty, so this is a different model from Definition 3. The fourth is the one-episode-lagged reading of the main text, stated at the opening of this appendix, with the transported-copy transition $k_i^{(n+1)} = \Omega_{i,I} k_I^{(n)}$; Eq. (158) is its formula with the variational parent standing in for the filtered law. The first three readings are same-time; the fourth carries the participatory reading of Section 11.

In hierarchical generative models of cortical inference [49, 50] the level-ℓ prior is the conditional $p(x^{(\ell)} \mid x^{(\ell+1)})$ of the generative model; the link kernel (327) plays that role here, with a gauge-equivariant mean and an isotropic variance. It is also a special case of the top-down Gaussian conditional of hierarchical variational autoencoders [51], whose conditional variances are learned per dimension. Their bound evaluates this conditional at parent samples, which is the conditional coupling above and not the predictive density of Lemma 6, and the rigid limit $\sigma^2 \to 0$ is singular for that variational problem.

The transported-copy specialization is the zero-noise limit. The finite-σ^2 density (331) adds $\sigma^2 I$ to the transported parent covariance, giving a nonsingular predictive law. Choosing that noisy channel changes the generative model; it is distinct from changing the variational update within a fixed model.

The lemmas concern the cross-scale terms alone. Reinstating the within-scale alignment terms gives a fixed-point system in which each level-ℓ posterior balances the transported parental pull against the within-scale alignment energy. Because the softmax weights β_{ij} are row-stochastic, a "small β_{ij}" is not a small parameter. The dimensionless coupling strengths are the raw alignment weights $\tilde{\beta}_{ij}, \tilde{\gamma}_{ij}$ of Section E.2, and the within-scale terms vanish from the functional in the limit $\tilde{\beta}_{ij}, \tilde{\gamma}_{ij} \to 0$. Perturbative control of the within-scale corrections is open (Section G.5).

G.5 Status of the Mean-Field Fixed-Point Question

At zero within-scale coupling and under the Gaussian hypotheses below, the mean-field optimum on the augmented joint is exactly computable and unique, although the true posterior generally remains correlated (Section G.5.2). Transported-copy prediction between episodes, Eq. (158), is a different operation from these same-time mean-field coordinate equations. Its dynamical guarantees require a separate analysis. The same-time substitution of a predictive marginal also differs from the coordinate equations (Remark 2). The within-scale question has three separate answers: a general obstruction for a fixed joint on the original state variables, a self-consistent auxiliary model that becomes fixed only after an equilibrium is selected, and an exact Gibbs lift whose random variable is an entire belief configuration (Theorem 6, Appendix I).

G.5.1 Fixed Representation on the State Variables: General Obstruction

Let (X_i, ν_i) be the state space of agent i, and let T_{ij} denote a fixed mass-preserving positive linear map on densities induced by a normalized communication channel, including $(\Omega_{ij})_\#$ as a special case. A family $\mathcal{Q}_i$ of strictly positive densities is called mixture-open here when every zero-mass tangent h_i under consideration with $h_i/q_i \in L^\infty(q_i)$ generates $q_i(t) = q_i + th_i \in \mathcal{Q}_i$ for all sufficiently small $|t|$. All differentiated integrals below are assumed finite.

Theorem 3 (State-level mean-field ELBO obstruction)**.** *Let the factors $q_i \in \mathcal{Q}_i$ vary independently, let at least one nonself coefficient $\beta_{ij} > 0$, and consider*

$$\mathcal{C}_{\mathrm{cons}}(q) = \mathcal{E}_{\mathrm{rest}}(q) + \sum_{a \neq b} \beta_{ab} D_{\mathrm{KL}}(q_a \| \mathsf{T}_{ab} q_b), \tag{333}$$

where $D^3_{q_i q_j q_j} \mathcal{E}_{\mathrm{rest}} = 0$ for $i \neq j$. Assume that for one active edge (i, j) there is an admissible tangent h_j such that

$$g_{ij} := \left(\frac{\mathsf{T}_{ij} h_j}{\mathsf{T}_{ij} q_j} \right)^2 \tag{334}$$

is bounded and nonconstant q_i-almost everywhere. Then there is no density p_θ on $\prod_i \mathsf{X}_i$, independent of q, and no constant C_0 such that

$$\mathcal{C}_{\mathrm{cons}}(q) = D_{\mathrm{KL}} \left(\bigotimes_i q_i \middle\| p_\theta \right) + C_0, \tag{335}$$

throughout the open product family.

Proof. Write $Q_q = \bigotimes_i q_i$. The standard fixed-model mean-field decomposition [48] gives

$$D_{\mathrm{KL}}(Q_q \| p_\theta) = \sum_i \int_{\mathsf{X}_i} q_i \log q_i d\nu_i - \int_{\prod_i \mathsf{X}_i} \left(\prod_i q_i \right) \log p_\theta d\nu_{1:N}. \tag{336}$$

The entropy is separable, while the fixed-model energy is affine in each factor. Therefore

$$D^3_{q_i q_j q_j} D_{\mathrm{KL}}(Q_q \| p_\theta)[a_i, h_j, h_j] = 0, \qquad i \neq j. \tag{337}$$

For the active consensus edge, take $q_i(s) = q_i + sa_i$ and $q_j(t) = q_j + th_j$. Direct differentiation gives

$$\frac{\partial^3}{\partial s \partial t^2} D_{\mathrm{KL}}(q_i(s) \| \mathsf{T}_{ij} q_j(t)) \Big|_{s=t=0} = \int_{\mathsf{X}_i} a_i g_{ij} d\nu_i. \tag{338}$$

Choose $a_i = q_i(g_{ij} - \mathbb{E}_{q_i}[g_{ij}])$, which has zero mass and satisfies $a_i/q_i \in L^\infty(q_i)$ because g_{ij} is bounded; it is therefore admissible for a sufficiently small mixture path. Equation (338) becomes $\mathrm{Var}_{q_i}(g_{ij}) > 0$, contradicting Eq. (337). The terms collected in $\mathcal{E}_{\mathrm{rest}}$ cannot cancel this mixed variation by hypothesis. Every other ordered-edge term also has zero q_i, q_j, q_j variation: an unrelated edge omits one varied factor, while the reverse edge is affine in q_j outside its entropy and that entropy's second variation is independent of q_i. □

Remark 3 (Scope of the obstruction)**.** *The theorem applies directly to the entropy-retaining functional by holding its independently varied attention rows fixed. The self terms, likelihood terms with fixed observation models, and categorical attention entropies lie in $\mathcal{E}_{\mathrm{rest}}$. The same argument*

applies to the child-parent edge of a same-time shadow (Remark 2). The result concerns an ordinary mean-field factorization on the original agent-state variables. It does not exclude frozen source templates, restricted quadratic families, compatible auxiliary variables, a model selected after a fixed point, or a probability law over belief configurations. If the attention optimum $\beta^*(q)$ has already been substituted, its response derivatives define a different reduced functional and require a separate representation test.

G.5.2 Exact Cross-Scale Gaussian Exception

At zero raw within-scale coupling ($\tilde{\beta}_{ij} = \tilde{\gamma}_{ij} = 0$, so the alignment sectors are absent), the augmented joint of Definition 3 is a closed variational object. Assume the top-level prior p_{top} is a nondegenerate Gaussian with full-rank precision Λ_{top}, and take the leaf observation term absent or Gaussian; this is needed both for the negative log joint to be a quadratic form in the stacked latents and for the positive-definiteness below. The precision matrix of that quadratic form is block-sparse along the hierarchical tree, with nonzero blocks only on the diagonal and on parent-child edges. The diagonal block is $\sigma^{-2}\sum_{j:\pi(j)=i}\Omega_{j,i}^{\top}\Omega_{j,i}$ plus $\sigma^{-2}I$ when i has a parent, plus Λ_{top} at the root and any local Gaussian likelihood precision; the off-diagonal block on each parent-child edge is $-\sigma^{-2}\Omega_{i,\pi(i)}$ together with its transpose. The link terms alone are only positive semidefinite, with kernel the consistent-propagation subspace on which $k_i = \Omega_{i,\pi(i)}k_{\pi(i)}$ along every edge. The full-rank prior precision at the top removes this kernel, so for finite $\sigma^2 > 0$ the stacked precision is symmetric positive definite, the joint is a proper Gaussian, and the coordinate-ascent mean-field updates of Lemma 5 are Gauss-Seidel sweeps on a symmetric positive-definite system, which converge to the unique global optimum. In this regime the evidence-bound interpretation is exact and the mean-field fixed point exists and is unique. The factorized optimum is not generally the exact correlated posterior; its covariance blocks are inverses of the diagonal precision blocks, rather than the diagonal blocks of the full inverse.

G.5.3 Self-Consistent Equilibrium Representation

Theorem 4 (Existence of a self-consistent source model)**.** *Let the latent alphabet X and the agent set be finite. Let $p_i(k) > 0$, let $\pi_{ij} > 0$ with $\sum_j \pi_{ij} = 1$, and let $a_i > 0$, $\lambda > 0$, and $\tau > 0$. Assume every stochastic transport has a uniform floor: there is $\epsilon > 0$ such that $(\mathsf{T}_{ij}q)(k) \geq \epsilon$ for every probability vector q, state k, and active edge. Define*

$$E_{ij}(q) := D_{\mathrm{KL}}(q_i \| \mathsf{T}_{ij}q_j), \qquad \beta_{ij}(q) := \frac{\pi_{ij}\exp[-E_{ij}(q)/\tau]}{\sum_m \pi_{im}\exp[-E_{im}(q)/\tau]}, \tag{339}$$

and $\bar{a}_i := a_i + \lambda$. The geometric-pooling response map

$$B_i(q)(k) := \frac{p_i(k)^{a_i/\bar{a}_i}\prod_j[\mathsf{T}_{ij}q_j(k)]^{\lambda\beta_{ij}(q)/\bar{a}_i}}{\sum_{u\in\mathsf{X}} p_i(u)^{a_i/\bar{a}_i}\prod_j[\mathsf{T}_{ij}q_j(u)]^{\lambda\beta_{ij}(q)/\bar{a}_i}} \tag{340}$$

has at least one receiver-response equilibrium $q^\star = B(q^\star)$. Write $\beta^\star := \beta(q^\star)$.

At any such fixed point, define normalized row distributions

$$P_i^\star(k,j) := \pi_{ij}[\mathsf{T}_{ij}q_j^\star](k), \qquad Q_i^\star(k,j) := \beta_{ij}^\star q_i^\star(k). \tag{341}$$

They satisfy the exact identity

$$D_{\mathrm{KL}}(Q_i^\star \| P_i^\star) = \sum_j \beta_{ij}^\star D_{\mathrm{KL}}(q_i^\star \| \mathsf{T}_{ij}q_j^\star) + D_{\mathrm{KL}}(\beta_i^\star \| \pi_i). \tag{342}$$

The identity is algebraic: it holds with $(q^\star, \beta^\star)$ replaced by any (q, β) in which each q_i and each row β_i is a probability vector, and in particular at the equilibrium.

Proof. The uniform floor makes every energy in Eq. (339) continuous on the closed product simplex. Hence $\beta(q)$ and $B(q)$ are continuous, and B maps that compact convex simplex into itself. Brouwer's theorem gives a receiver-response equilibrium. For fixed source templates and fixed β_i, the Euler equation for

$$a_i D_{\mathrm{KL}}(r\|p_i) + \lambda \sum_j \beta_{ij} D_{\mathrm{KL}}(r\|\mathsf{T}_{ij} q_j), \tag{343}$$

under $\sum_k r(k) = 1$ gives $r = B_i(q)$, so the fixed point is self-consistent with the frozen-row optimum. Expanding $\log[Q_i/P_i] = \log(\beta_{ij}/\pi_{ij}) + \log[q_i(k)/\mathsf{T}_{ij}q_j(k)]$ and summing against Q_i over (k, j), using $\sum_k q_i(k) = \sum_j \beta_{ij} = 1$, proves Eq. (342) for every such (q, β). □

Remark 4 (What the equilibrium construction proves)**.** *The distributions $P_i^\star$ are fixed and normalized only after $q^\star$ has been selected. Their product is a proper auxiliary model over independent row variables, but it duplicates the transported source templates. It is not automatically a joint over shared latent variables $(k_1, \ldots, k_N)$. Such a joint additionally requires compatible conditionals, an acyclic factorization, or an undirected normalized potential. Equation* (342) *matches the ordinary row KL at unit attention temperature; arbitrary τ gives the tempered loss-based update of Eq.* (339)*, a generalized-Bayesian construction and not one unscaled KL [52]. Because the response holds incoming source roles fixed, its equilibrium is not thereby a stationary point of the global smoothing scalar. The theorem proves existence, not uniqueness or convergence of a particular update schedule.*

G.5.4 Scalar Objectives and Update Fields

The within-scale construction requires a distinction between three objects. First, the canonical entropy-retaining row objective is $\mathcal{E}_{i,\mathrm{full}} = \sum_j \beta_{ij} E_{ij} + \tau \sum_j \beta_{ij} \log(\beta_{ij}/\pi_{ij})$. At its simplex optimum it is exactly $-\tau \log Z_i$, and the envelope theorem gives $\mathrm{d}\mathcal{E}_{i,\mathrm{red}} = \sum_j \beta^*_{ij} \mathrm{d}E_{ij}$. Summing over rows therefore gives one scalar potential with a symmetric Hessian. Second, the entropy-suppressed quantity $S = \sum_{ij} \beta^*_{ij}(x) E_{ij}(x)$ is a different scalar. Its response terms change its gradient but do not change the symmetry of its Hessian. Third, the deliberately asymmetric field

$$V_i^{\mathrm{rec}}(x) := \partial_{x_i} \left(\sum_j \beta^*_{ij}(x) E_{ij}(x) \right), \qquad \text{with all } \partial_{x_i} \left(\sum_m \beta^*_{\ell m} E_{\ell m} \right),\ \ell \neq i,\ \text{discarded}, \tag{344}$$

is not the gradient of either scalar and is generically nonintegrable because cross-row mixed derivatives need not match. This nonpotential field is an optional update approximation, not a property of the softmax, the noncompact group, or differentiation of a scalar. The consensus functional remains a composite scalar potential and not a fixed state-level generative-model evidence bound; that distinction does not require a nonsymmetric Hessian.

Existence for the full Gaussian system is more delicate than the compact finite-state result of Theorem 4. A restricted stationary point exists under a symmetry hypothesis: the mean-only consensus state $\mu_i \equiv 0$ is stationary for every prior strength $\alpha > 0$, temperature $\tau > 0$, and finite N when the prior means vanish and observation forcing is absent, because every pairwise alignment mean-gradient $\Lambda_{t,ij}(\mu_i - \Omega_{ij}\mu_j)$ and the prior pull vanish there. This does not prove a fixed point for the covariance and frame coordinates. For live state-dependent softmax weights on $\mathrm{GL}^+(K_q)$, local stability, global uniqueness, and global convergence remain open, and noncompact frame directions prevent a direct compactness argument unless one imposes a gauge slice, a

bounded invariant container, or a coercive frame prior. A Banach guarantee holds in a restricted setting. Freeze covariances and attention weights, restrict transports to the orthogonal group, and take isotropic couplings $\Lambda_{t,ij} = \lambda_{ij} I$ with an arbitrary symmetric positive-definite Λ_p. The mean sub-map $\mu_i \leftarrow (\alpha\Lambda_p + b_i I)^{-1}(\alpha\Lambda_p \mu_{p,i} + \sum_j \beta_{ij}\lambda_{ij}\Omega_{ij}\mu_j)$, with $b_i = \sum_j \beta_{ij}\lambda_{ij}$, then has row norm $\sum_j \|(\alpha\Lambda_p + b_i I)^{-1}\beta_{ij}\lambda_{ij}\Omega_{ij}\|_2 = b_i/[\alpha\lambda_{\min}(\Lambda_p) + b_i] < 1$, so it is a contraction in the norm $\max_i \|\mu_i\|_2$. This row-norm proof uses isotropy. For anisotropic positive matrices, dividing by the largest eigenvalue of their sum is not a valid upper bound. For example, take $\alpha\Lambda_p = I/10$, $B = \mathrm{diag}(1, 9/10)$, and $C = \mathrm{diag}(99, 1/10)$. Then

$$\|(\alpha\Lambda_p + B + C)^{-1}B\|_2 = 9/11, \qquad \lambda_{\max}(B)/\lambda_{\max}(B + C) = 1/100.$$

Even commuting anisotropic matrices invalidate that proposed estimate. A valid general bound uses $\lambda_{\min}(\alpha\Lambda_p + B + C)$ in the denominator. Failure of a sufficient contraction estimate is not proof of divergent dynamics. A contraction, monotonicity, or Lyapunov argument for any enlarged class must be supplied separately.

The remaining obligations are a quantitative comparison between a normalized cross-scale model and an explicitly supplied consensus perturbation, compatibility when cyclic latent conditionals are asserted, and existence and stability for the chosen fully coupled selective dynamics. Setting one consensus channel weight to zero does not by itself reduce the other channels to this Gaussian hierarchical model.

H Beyond Mean Field: Correlated Posteriors, the Nested Bound, and Fluctuation Corrections

Several distinct reductions enter the construction. The factorization $Q = \prod_i q_i$ suppresses state-posterior correlations. A deterministic saddle approximation suppresses fluctuations of whole configurations. An empirical measure is an exact representation of a finite unlabeled population, but replacing its random dynamics by a deterministic large-population law requires an additional limit. Projecting an exact coarse-grained law into a selected functional class is a further approximation. Improving one reduction does not establish the others.

The configuration-space Gibbs lift of Theorem 6 places a probability law on whole belief configurations and introduces the distinct configuration temperature T_{cfg}, which is a bookkeeping scale of that ensemble and not a physical temperature. The Laplace term in Eq. (364) also retains a local Gaussian fluctuation determinant when internal cluster variables are eliminated. Those results do not yet define a correlated posterior over the latent states of different agents, a generating functional for connected configuration correlations, or a self-consistent two-point theory. This appendix states the nested bound that organizes these objects, the correlated state families that preserve evidence semantics, and an outlook on fluctuation corrections, without reclassifying the peer-KL functional as an ordinary state-level evidence bound.

H.1 A Normalized Nested Variational Model

Let X denote a generative configuration of prior and transition parameters, supports, graph or source priors, precision fields, and frame or edge variables. It does not contain the variational posterior densities q_i or s_i. Let $Y = (Y_1, \ldots, Y_N)$ denote the state-level and model-level latent variables, and let o denote the observations. No slow evolution of X is implied by this probabilistic decomposition. The state model is a fixed conditional law normalized jointly over (Y, o), with

likelihood densities relative to one declared observation reference measure

$$p_\theta(dY, o \mid X) := p_\theta(dY \mid X) p_\theta(o \mid Y, X), \tag{345}$$

whose factors do not depend on the variational posterior used to approximate them. For a possibly correlated conditional posterior $Q_X(dY)$, define the state variational free energy in nats by

$$\mathcal{F}_{V,\mathrm{state}}[Q_X; X, o] := D_{\mathrm{KL}}\big(Q_X \| p_\theta(\cdot \mid X)\big) - \mathbb{E}_{Q_X} \log p_\theta(o \mid Y, X). \tag{346}$$

It obeys

$$\mathcal{F}_{V,\mathrm{state}}[Q_X; X, o] = -\log p_\theta(o \mid X) + D_{\mathrm{KL}}\big(Q_X \| p_\theta(\cdot \mid o, X)\big), \tag{347}$$

so correlated inference changes the approximation family without changing the evidence semantics.

Theorem 5 (Nested state and configuration VFE)**.** *Let $T_{\mathrm{cfg}} > 0$, let $P_0(dX)$ be a proper configuration prior, let Eq.* (345) *be normalized for P_0-almost every X, and define $p_\theta(o) := \int p_\theta(o \mid X) P_0(dX) \in (0, \infty)$. Let $R \ll P_0$ be a configuration posterior and let $X \mapsto Q_X$ be a measurable probability kernel with $Q_X \ll p_\theta(\cdot \mid X)$ for R-almost every X. Assume $0 < p_\theta(o \mid X) < \infty$ for R-almost every X. Assume the terms below are finite, or interpret the identity in the extended-real sense without an indeterminate $\infty - \infty$. Then*

$$\begin{aligned} \mathcal{J}[R, Q] &:= T_{\mathrm{cfg}} D_{\mathrm{KL}}(R \| P_0) + T_{\mathrm{cfg}} \int \mathcal{F}_{V,\mathrm{state}}[Q_X; X, o] R(dX) \\ &= -T_{\mathrm{cfg}} \log p_\theta(o) + T_{\mathrm{cfg}} D_{\mathrm{KL}}\big(R(dX) Q_X(dY) \| P_\theta(dX, dY \mid o)\big). \end{aligned} \tag{348}$$

Consequently, $-\mathcal{J}/T_{\mathrm{cfg}}$ is an evidence lower bound for one normalized hierarchical model, with equality if and only if the joint variational law equals the exact posterior almost everywhere.

Proof. Expand the joint KL divergence by the chain rule, use $P_\theta(dX, dY, o) = P_0(dX) p_\theta(dY, o \mid X)$, and collect the conditional terms into Eq. (346). □

The quantity $\mathcal{J}/T_{\mathrm{cfg}}$ is a two-level tower variational free energy with latent (X, Y). Its configuration temperature and the vacuum prior below are specific to this paper. If P_0 is itself defined by a vacuum functional,

$$P_0(dX) := Z_0^{-1} \exp[-\mathcal{E}_{\mathrm{vac}}(X)/T_{\mathrm{cfg}}] \nu_0(dX), \tag{349}$$

assume that ν_0 is a declared sigma-finite reference measure and $0 < Z_0 < \infty$. For finite terms, the configuration contribution is exactly

$$T_{\mathrm{cfg}} D_{\mathrm{KL}}(R \| P_0) = \mathbb{E}_R \mathcal{E}_{\mathrm{vac}} + T_{\mathrm{cfg}} \int \log \frac{dR}{d\nu_0} dR + T_{\mathrm{cfg}} \log Z_0.$$

The reference-entropy integral is a nonnegative KL divergence only when ν_0 is a probability law; for Lebesgue reference it is the negative differential entropy and may have either sign. This nests state-level evidence inside the configuration ensemble. The vacuum functional may contain only observation-free structural energies in the declared variables X; it cannot contain live q_i, s_i, or their peer-KL terms without promoting separate measure-valued fields to generative variables and specifying their relation to Q_X. The held-out likelihood is excluded from $\mathcal{E}_{\mathrm{vac}}$ and enters once through Eq. (346). Environmental variables may be represented as ordinary latent nodes or observed boundary conditions; the construction does not treat a variational belief of another agent as a fixed generative factor.

Both terms of $\mathcal{J}$ carry T_{cfg}, so at the optimal state posterior the optimal configuration posterior is

$$R^*(dX) \propto \exp[-\mathcal{E}_{\mathrm{vac}}(X)/T_{\mathrm{cfg}}]p_\theta(o \mid X)\nu_0(dX).$$

A precise concentration statement requires hypotheses on the vacuum energy, reference measure, and likelihood. For example, on a compact configuration space with continuous vacuum energy having a unique minimizer X_0, finite reference measure assigning positive mass to each neighborhood of X_0, and bounded continuous likelihood positive at X_0, the posterior concentrates at X_0 as $T_{\mathrm{cfg}} \downarrow 0$. To see this, compare the positive energy gap outside any neighborhood with a smaller gap inside a neighborhood on which the likelihood is bounded below. The resulting exterior-to-interior mass ratio decays exponentially. Multiple minima can retain data-dependent relative weights, and a likelihood vanishing on their neighborhoods can exclude vacuum minima entirely. For the separate noncompact Gaussian example $\nu_0(dX) = dX$, $\mathcal{E}_{\mathrm{vac}}(X) = X^2/2$, and $p_\theta(o \mid X) = \mathcal{N}(o; X, 1)$, completing the square gives

$$R^* = \mathcal{N}\left(\frac{oT_{\mathrm{cfg}}}{1+T_{\mathrm{cfg}}}, \frac{T_{\mathrm{cfg}}}{1+T_{\mathrm{cfg}}}\right).$$

Here the data move the mean at order T_{cfg}. This scaling does not in general recover minimization of a configuration action in which the data term has an order-one coefficient.

H.2 Correlated State Posteriors and Region Free Energies

The first extension beyond mean field preserves the normalized model above while enlarging Q_X. For the following marginal-polytope statement, take finite state alphabets and strictly positive finite potentials. Suppose the conditional state prior has a gauge-covariant pairwise factorization on a graph $\mathcal{G} = (V, E)$,

$$p_\theta(Y \mid X) := Z_X^{-1} \prod_{i \in V} \psi_i(Y_i; X) \prod_{(i,j) \in E} \psi_{ij}(Y_i, Y_j; X), \tag{350}$$

where each edge factor is a fixed model potential and not a function of live variational densities. Introduce node beliefs b_i and edge beliefs b_{ij} satisfying normalization and local marginal consistency. Their Bethe entropy is

$$H_{\mathrm{B}}[b] := \sum_{(i,j) \in E} H[b_{ij}] - \sum_{i \in V} (d_i - 1) H[b_i]. \tag{351}$$

The corresponding region free energy is the expectation of the fixed node and edge energies $-\log \psi_i$ and $-\log \psi_{ij}$ minus H_{B}. On a tree the local marginal polytope equals the marginal polytope [48, Prop. 4.1], and minimizing this functional over it returns the exact marginals and $-\log Z_X$. On a loopy graph it is the Bethe approximation. Larger regions give the cluster-variation method [53] and its region-graph generalizations, a family of more expressive approximations whose accuracy depends on the choice of regions and is not guaranteed to improve [54, Sec. VIII.C]. Generic loopy pseudomarginals need not be the marginals of any global Q_X, and their region objective is not a generic upper or lower evidence bound. Bound-preserving constructions require additional structure, such as a globally realizable variational family or an appropriate tree-reweighted relaxation. Ordinary mean field is recovered by the restriction $b_{ij} = b_i b_j$ on every edge when those beliefs arise from a global product law. It is therefore a lower-complexity variational subfamily of the correlated theory, not the definition of the theory.

Gauge covariance acts on the region beliefs as a joint pushforward. If $Y_i \mapsto G_i Y_i$ and $Y_j \mapsto G_j Y_j$, then b_{ij} is pushed forward by $G_i \times G_j$, while the edge potential is transformed contragrediently so that its scalar energy is unchanged. Entropies must be defined relative to a covariantly transformed or

quasi-invariant reference measure, including its Jacobian; a finite invariant reference is not generally available for noncompact $\mathrm{GL}^+(K_q)$. A Regime I link $\Omega_{ij} = U_i U_j^{-1}$ has trivial loop holonomy and is removed by a passive frame re-expression. A correlated theory with nontrivial holonomy must instead retain the independent links on the interaction graph of Regime II (Section 4.7), with inverse orientation reversal for the reversible group-valued identification sector. Singular selective channels instead form composable maps or Markov kernels, whose cycle compositions are not group holonomies. Integrating only flat vertex-frame identifications cannot generate nontrivial group holonomy.

For the Gaussian sector, a computationally direct correlated family is

$$Q_X(Y) = \mathcal{N}(\bar{Y}, C), \qquad C^{-1} = \Lambda_Q, \tag{352}$$

with a sparse block precision Λ_Q supported on the state interaction graph. The off-diagonal blocks C_{ij} carry inter-agent correlations, and if $Y_i = (k_i, m_i)$ contains fast state and slow model variables, the within-node block $C_i^{(km)}$ carries cross-channel correlations that the separate q_i and s_i factors discard. Under a local linear frame change, $C_{ij} \mapsto \rho_i(G_i) C_{ij} \rho_j(G_j)^\top$, where ρ_i is block diagonal if the state and model channels use distinct representations, so the covariance family is closed under the group action. The block-diagonal restriction $C_{ij} = 0$ for $i \neq j$ is the Gaussian mean-field limit. At a Gaussian target of precision Λ, its exact cost at the optimum is the determinant gap $\frac{1}{2}[\sum_b \log\det \Lambda_{bb} - \log\det \Lambda] \geq 0$, which vanishes exactly when all cross-block precisions vanish (Proposition 38). Sparse block-Gaussian inference is therefore a tractable extension that exposes a measurable connected two-point state function while retaining tractable linear algebra.

The live alignment term $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ does not become an exact instance of Eq. (350) merely because edge beliefs have been introduced: its target still moves with another variational factor, which is the obstruction proved by Theorem 3. It has two coherent placements. It may remain a configuration-level consensus potential, or it may be replaced by a fixed state-transition or pair potential from which the correlated posterior is inferred. Using both for the same interaction would count that edge twice.

Remark 5 (No-double-counting conditions)**.** *An observation likelihood enters the state free energy once and is not also placed in the vacuum action; a physical interaction is represented either by a fixed edge factor of the state model or by a consensus potential on configurations, not both; a categorical source variable J_i is marginalized exactly or assigned one categorical variational law with probabilities β_{ij}, and its entropy is counted once; introducing a random simplex-valued parameter is a separate model requiring its own prior and conditional law, not another description of that categorical entropy; and marginal Gaussian entropies and the correlation correction are components of one state-posterior entropy, while attention-row entropy and configuration meta-entropy belong to separate sample spaces. These are instances of two general statements: an energy appended to the free energy of a joint is an evidence bound only after its normalizer is restored (Proposition 41), and the complexity of a correlated recognition law relative to a product prior splits exactly into its total correlation and its marginal divergences (Proposition 40).*

H.3 Outlook: Fluctuation Corrections and the Effective Action

After the state family has been specified, define a state-reduced configuration action

$$\mathcal{A}_o[X] := \mathcal{E}_{\mathrm{vac}}[X] + T_{\mathrm{cfg}} \inf_{Q_X \in \mathcal{Q}_{\mathrm{corr}}} \mathcal{F}_{V,\mathrm{state}}[Q_X; X, o]. \tag{353}$$

With the unrestricted posterior family the second term is $-T_{\mathrm{cfg}} \log p_\theta(o \mid X)$; with a normalized family such as a sparse block-Gaussian Q_X it is a variational upper approximation to that conditional

surprisal, while a Bethe or Kikuchi pseudomarginal optimization does not inherit this evidence-bound statement. The live multi-agent functional cannot serve as $\mathcal{E}_{\mathrm{vac}}$ in Theorem 5, because it depends on the variational q_i, s_i and contains the observation term. The peer-KL sector is a gauge-covariant consensus energy and, when its belief-configuration partition function is finite, an exact generalized-Bayesian energy on that separate sample space; inside $\mathcal{A}_o$ it is only a candidate truncation pending an explicit summary map from Q_X to belief fields and quantitative energy and force residuals.

Let $\mathfrak{X}_{\mathrm{phys}}$ be a regular configuration quotient by transformations explicitly declared to be redundancies; the subscript denotes retained coordinates and does not establish a physical interpretation. Assume a free proper redundancy action on the region used, or work directly on a specified smooth quotient chart. Active transformations remain coordinates of X. Choose an injective chart $X = \mathrm{Exp}_{\bar{X}}\,\xi$ on a domain D in the tangent space. The following construction describes an ensemble supported in that chart; a global ensemble requires compatible charts and their measure factors, since an exponential map need not be globally injective. For $T = T_{\mathrm{cfg}} > 0$, define

$$Z_{\bar{X}}[J] := \int_D \exp\left[-\frac{\mathcal{A}_o[\mathrm{Exp}_{\bar{X}}\,\xi] - J_A\xi^A}{T}\right] d\nu_{\mathrm{phys}}(\xi), \qquad W_{\bar{X}}[J] := T\log Z_{\bar{X}}[J]. \tag{354}$$

Assume the integral and its first two source derivatives exist on an open neighborhood of the sources considered, with differentiation under the integral justified. Direct differentiation gives

$$\partial_{J_A} W = \mathbb{E}_J \xi^A, \qquad \partial_{J_A}\partial_{J_B} W = T^{-1}\,\mathrm{Cov}_J(\xi^A, \xi^B).$$

Where this covariance is positive definite, the local Legendre transform $\Gamma_{\bar{X}}(\varphi) = J_A\varphi^A - W_{\bar{X}}(J)$ therefore satisfies

$$\Gamma^{(2)}_{\bar{X}} = T\,\mathrm{Cov}_J(\xi)^{-1}.$$

At the source-free mean $\varphi_0 = \mathbb{E}_0\xi$ its value is exactly $\Gamma_{\bar{X}}(\varphi_0) = -T\log Z_{\bar{X}}[0]$. This configuration covariance differs from the state covariance C_{ij} in Eq. (352). The Legendre effective action also differs from the conditional coarse energy $\mathcal{E}^{\mathrm{exact}}_{\ell+1}$ in Appendix I, obtained by integrating eliminated variables at a fixed retained value.

A finite-dimensional Laplace statement can now be made precisely. Write $d\nu_{\mathrm{phys}} = \rho_{\mathrm{phys}}\,\mathrm{vol}_{g_{\mathrm{phys}}}$. Assume that $\mathcal{A}_o$ is C^6, has one dominant interior minimum $\bar{X}$ with positive Hessian H_{phys}, and has a positive energy gap outside every sufficiently small neighborhood of that minimum in a compact integration region. Assume ρ_{phys} is C^4 and positive near $\bar{X}$. If the action depends on T, these regularity, nondegeneracy, and domination hypotheses must hold uniformly. Taylor expansion after $\xi = \sqrt{T}v$, Gaussian integration, and the cancellation of odd Gaussian moments give

$$\Gamma_{\bar{X}}(\varphi_0) = \mathcal{A}_o(\bar{X}) + \frac{T}{2}\log\det\left(\frac{g^{-1}_{\mathrm{phys}}H_{\mathrm{phys}}}{2\pi T}\right) - T\log\rho_{\mathrm{phys}}(\bar{X}) + O(T^2). \tag{355}$$

The exterior gap makes the omitted exterior integral exponentially small, and the local integral has relative error $O(T)$; multiplication of its logarithm by T gives the stated remainder. In general the mean displacement φ_0 is $O(T)$ rather than exactly zero, so this equation is evaluated at the source-free mean, not at the classical minimum as an argument of Γ. An off-stationary effective action requires a source selecting that mean. The factor $2\pi T$ contributes at order $T\log T$ and cannot be suppressed into an $O(T^2)$ remainder. An infinite-dimensional functional determinant additionally needs a regulator, a reference normalization, and a controlled limiting procedure; Eq. (355) alone supplies none of these.

Gauge-fixed integrals require both a transverse Jacobian and the measure induced on the slice. In a finite-dimensional ambient Riemannian space, let a regular gauge condition be $\Phi = 0$, let V_a be the redundancy generators, and put

$$(M_{\mathrm{FP}})_{\alpha a} = d\Phi_\alpha(V_a), \qquad J_\Phi = \sqrt{\det(D\Phi g^{-1} D\Phi^\top)}.$$

The coarea formula gives the local identity

$$\delta(\Phi)|\det M_{\mathrm{FP}}|\,\mathrm{vol}_g = \frac{|\det M_{\mathrm{FP}}|}{J_\Phi}\,\mathrm{vol}_{\Phi=0}\,.$$

Any ambient density multiplies the right-hand side after restriction to the slice. This is the measure content of the local Faddeev–Popov construction [55]. Under $\Phi \mapsto A\Phi$ with nonsingular constant A, both $|\det M_{\mathrm{FP}}|$ and J_Φ acquire $|\det A|$, leaving their ratio unchanged. A formula retaining the former determinant alone with the intrinsic slice measure fails this elementary rescaling check. Direct quotient integration and a correctly normalized gauge-fixed representation are alternative descriptions of the same specified measure; neither a second determinant nor an orbit-volume factor can be appended without deriving the relevant disintegration. Stabilizers, multiple slice intersections, and noncompact orbit volume require separate treatment.

The Frobenius frame metric in Eq. (105) is not invariant under the right action $U_i \mapsto U_i g$ for general $g \in \mathrm{GL}^+(K_q)$, so a quotient metric cannot be inferred from that formula. It must be declared on the quotient or obtained from a suitable invariant ambient metric. Background covariance is a further condition on a redundant-field expansion [56–58]. Noncompact redundant directions require a quotient or a valid gauge-fixing construction; active noncompact directions require an integrable combined action and reference measure. Haar measure by itself supplies no such normalization.

A two-particle-irreducible extension would introduce a symmetric bilocal source and a second Legendre transform, producing an action $\Gamma_{2\mathrm{PI}}[\varphi, G]$ whose covariance stationarity condition is $\delta\Gamma_{2\mathrm{PI}}/\delta G = 0$ [59]. A gauge-consistent realization for the present configuration model remains an unresolved construction. It would become relevant if fluctuations change the saddle or if soft modes invalidate the nondegenerate Laplace approximation.

Exact elimination can generate interactions outside the original ansatz, including higher-order region terms, nonlocal kernels, and measure contributions; symmetry constrains the allowed terms but does not require every allowed coefficient to be nonzero [60]. Closure is therefore a conditional approximation to be assessed through the oscillation-to-KL control of Eq. (366) and the separate force discrepancy of Eq. (367). Correlated state inference, local configuration fluctuations, and a self-consistent two-point approximation address different errors and do not automatically form nested families or a monotone sequence of evidence bounds. A concrete comparison must specify each normalized model or approximation family and report its evidence gap, correlations, and closure residuals. No spacetime, gauge-field kinetic term, or physical mass follows merely from adopting one of these fluctuation constructions.

I Gibbs Ensembles, Exact Reduction, and Approximation Error

This appendix separates a probability model over belief configurations from the sample-level population model, then states what exact reduction and a controlled approximation preserve. All constructions are pointwise over the base.

I.1 A normalized configuration ensemble

Theorem 6 (Belief-configuration Gibbs lift)**.** *Let ν_0 be a proper reference probability on a measurable configuration space $\mathfrak{X}$. Let $\mathcal{E} : \mathfrak{X} \to (-\infty, \infty]$ be measurable, bounded below, and finite on a set of positive ν_0-measure. For $T_{\mathrm{cfg}} > 0$,*

$$Z_{\mathcal{E}} = \int e^{-\mathcal{E}(X)/T_{\mathrm{cfg}}} \nu_0(dX), \qquad 0 < Z_{\mathcal{E}} < \infty, \tag{356}$$

and

$$\mathbb{P}_{\mathcal{E}}(dX) = Z_{\mathcal{E}}^{-1} e^{-\mathcal{E}(X)/T_{\mathrm{cfg}}} \nu_0(dX) \tag{357}$$

defines a probability law. For R for which the terms are finite,

$$\mathbb{E}_R \mathcal{E} + T_{\mathrm{cfg}} D_{\mathrm{KL}}(R\|\nu_0) = T_{\mathrm{cfg}} D_{\mathrm{KL}}(R\|\mathbb{P}_{\mathcal{E}}) - T_{\mathrm{cfg}} \log Z_{\mathcal{E}}. \tag{358}$$

Consequently the left-hand side has unique minimizer $\mathbb{P}_{\mathcal{E}}$ and minimum

$$-T_{\mathrm{cfg}} \log Z_{\mathcal{E}}. \tag{359}$$

Proof. If $\mathcal{E} \geq m$, the partition function is at most $e^{-m/T_{\mathrm{cfg}}}$; the positive-measure finite-energy set gives strict positivity. On the common support,

$$\log \frac{dR}{d\mathbb{P}_{\mathcal{E}}} = \log \frac{dR}{d\nu_0} + \frac{\mathcal{E}}{T_{\mathrm{cfg}}} + \log Z_{\mathcal{E}}. \tag{360}$$

Integration proves the identity. Nonnegativity and the equality case of KL give uniqueness. The minimizing terms are finite: boundedness below controls the negative part, and $xe^{-x/T_{\mathrm{cfg}}}$ is bounded for $x \geq 0$. □

Remark 6 (The ensemble is an extra probability model)**.** *The random variable X contains laws, frames or other configuration parameters. Its Gibbs law does not turn the consensus objective into a sample-level ELBO, and deterministic gradient descent does not sample this ensemble. A stochastic kinetics with the required invariant law would be additional structure. The reference ν_0 also matters. An invariant energy integrated against unregulated Haar volume along a noncompact gauge orbit generally has infinite partition function. One must use a proper reference, an admissible quotient, or a specified gauge fixing with its measure factor.*

I.2 Exact channels and their composition

Proposition 15 (Preservation of retained laws)**.** *Let $C : X \rightsquigarrow Z$ be a normalized channel and put $\nu^c = \nu_0 C$. Pushing the unnormalized Gibbs measure through C gives*

$$e^{-\mathcal{E}^c(z)/T_{\mathrm{cfg}}} = \mathbb{E}_{\nu_0}[e^{-\mathcal{E}(X)/T_{\mathrm{cfg}}} \mid Z = z]. \tag{361}$$

Its partition function is $Z_{\mathcal{E}}$ and its normalized law is $\mathbb{P}_{\mathcal{E}} C$. Every bounded observable $f(Z)$ has the same expectation before and after reduction.

Proof. Integrate the proposed density against $f(z)\nu^c(dz)$ and use the definition of conditional expectation under $\nu_0(dX)C(X, dz)$. The result is the unnormalized fine expectation of the same retained observable. Set $f = 1$ for the partition function. □

With a different reference $\bar{\nu}$ satisfying $\nu^c \ll \bar{\nu}$, the energy changes to $\bar{\mathcal{E}}^c = \mathcal{E}^c - T_{\mathrm{cfg}} \log(d\nu^c/d\bar{\nu})$. An additive constant also changes the partition function. It cannot be discarded while claiming exact evidence agreement.

Proposition 16 (Composition and equivariance)**.** *Successive normalized channels $C_1 : X \rightsquigarrow Z$ and $C_2 : Z \rightsquigarrow W$ satisfy $(\mathbb{P}C_1)C_2 = \mathbb{P}(C_1C_2)$. If frame transformations g_X, g_Z intertwine C, $C(g_X x, g_Z A) = C(x, A)$, then $(g_{X\#}\mathbb{P})C = g_{Z\#}(\mathbb{P}C)$.*

Proof. The first statement is iterated integration of a bounded test function. For the second substitute the intertwining relation into its integral under the transformed fine law. □

This is composition of channels, not a renormalization semigroup on an identified coupling space. Such a semigroup additionally needs a closed family, rescaling and a meaning for comparison across levels. A finite tower can exhibit useful effective behavior without supplying a critical fixed point.

I.3 Gaussian elimination and the Laplace regime

For a globally positive-definite quadratic energy in hidden x and retained z,

$$E(x,z) = \frac{1}{2}x^\top A x + x^\top B z + \frac{1}{2}z^\top D z, \qquad \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix} \succ 0, \tag{362}$$

integration against Lebesgue measure gives

$$E^c(z) = \frac{1}{2}z^\top (D - B^\top A^{-1} B) z + \frac{T_{\mathrm{cfg}}}{2} \log\det A - \frac{d_x T_{\mathrm{cfg}}}{2} \log(2\pi T_{\mathrm{cfg}}). \tag{363}$$

Completing the square proves the formula. It is exact closure in the family of all such quadratic energies; it need not preserve a restricted graph, the absence of self terms, or a chosen shared $R_i S_j$ parameterization.

For a smooth nonquadratic $E(x,z)$, suppose that at each retained z in a compact region there is a unique isolated global minimum $x_*(z)$ in the integrated variable, with uniformly positive Hessian $H_x(z)$. Suppose also that a smooth positive density $a(x,z)$ and uniformly controlled derivatives and tails justify Laplace expansion. Then

$$\begin{aligned} -T_{\mathrm{cfg}} \log \int e^{-E(x,z)/T_{\mathrm{cfg}}} a(x,z)\,dx = E(x_*(z),z) + \frac{T_{\mathrm{cfg}}}{2} \log\det H_x(z) \\ - \frac{d_x T_{\mathrm{cfg}}}{2} \log(2\pi T_{\mathrm{cfg}}) - T_{\mathrm{cfg}} \log a(x_*(z),z) + O(T_{\mathrm{cfg}}^2). \end{aligned} \tag{364}$$

The Gaussian integral gives the displayed terms; the stated domination and regularity control the remainder [61]. Multiple equal minima require a sum of their contributions. A vanishing Hessian eigenvalue invalidates this expansion. Neither a small within-cluster KL nor a threshold on apparent coherence supplies these hypotheses.

The determinant term is a conditional fluctuation contribution. It can change the retained interactions. It does not universally renormalize the coefficient of a parent belief entropy: that identification requires an explicit parameterization in which this determinant has the proposed dependence. The zero-temperature limit is the conditional energy profile under these hypotheses,

$$E_{\mathrm{eff}}(z) = E(x_*(z), z). \tag{365}$$

The Hessian of this profile is the corresponding Schur complement. The reduced kinetic metric, however, depends on the original mobility and on the motion of $x_*(z)$; it is not fixed by that Hessian alone.

I.4 A quantitative closure criterion

Suppose P has dimensionless energy H against a fixed reference and $\delta \in L^\infty(P)$ is a dimensionless perturbation. Define $\widetilde{P}$ by the normalized density proportional to $e^{-\delta}dP$, so the two laws are mutually absolutely continuous. Their density ratio obeys

$$\log \frac{dP}{d\widetilde{P}} = \delta + \log \mathbb{E}_P e^{-\delta}, \qquad D_{\mathrm{KL}}(P\|\widetilde{P}) \le \operatorname{esssup} \delta - \operatorname{essinf} \delta. \tag{366}$$

Indeed, $\log \mathbb{E}_P e^{-\delta} \le -\operatorname{essinf} \delta$ and $\mathbb{E}_P \delta \le \operatorname{esssup} \delta$. Exchanging the laws gives the same upper bound for the reverse KL. Thus a uniformly small residual modulo constants controls distributional closure after normalization.

This is not a force bound. On a fixed interval, $\delta_\epsilon(z) = \epsilon \sin(z/\epsilon^2)$ has vanishing amplitude but derivative of order ϵ^{-1}. For reduced flows $\dot{z} = b(z)$ and $\dot{\tilde{z}} = \tilde{b}(\tilde{z})$ with the same initial state, if b is Lipschitz with constant L and $\|b - \tilde{b}\| \le r$ throughout a region containing both trajectories, direct integration and Gronwall give

$$\|z(t) - \tilde{z}(t)\| \le \begin{cases} r(e^{Lt} - 1)/L, & L > 0, \\ rt, & L = 0. \end{cases} \tag{367}$$

A gradient model therefore needs a derivative residual and control of its mobility, not only an energy fit. An exact retained process may also have memory, as Appendix B demonstrates. The appropriate reduction then retains that memory or proves an approximation bound for discarding it.

These results provide usable acceptance conditions for a proposed effective agent: an induced normalized law, compatible communication, quantified distributional error when approximated, and a retained or controlled predictive dynamics. They do not impose a universal detector or assume that every reduction yields a new physical object.

J Proof of Fixed-Temperature Pairing Selection

This appendix proves the selection theorem of Section 14. Subsection J.1 declares the model completely and Subsection J.2 restates the theorem. The proof has five parts. At frozen rows the dynamics is uniformly geometrically ergodic and contracts relative entropy (Lemmas 9 and 13). While the rows are close to uniform, learning integrates a small matching-mode imprint of the initial mean, and this imprint vanishes only on a Lebesgue-null set of means (Lemmas 10 and 11). An entropy balance controls how closely the recognition law follows the invariant law of its instantaneous rows (Lemma 12). The reduced row field has a saddle at uniform rows whose single expanding direction is the matching mode, and its diagonal flows into the two matching vertices (Lemma 14). Finally, Subsection J.10 amplifies the imprint through the saddle on a logarithmic slow horizon and carries the rows to the selected vertex.

J.1 The declared model

Definition 4 (Homogeneous four-cycle with learned source rows)**.** *Index four agents at c_* by $i \in \mathbb{Z}/4$ and orient the edge e_i from i to $i+1$. A configuration is $\mathbf{x} = (k, m, w) \in \mathbb{R}^{12}$: agent i has the state belief $\mathcal{N}(k_i, 1)$ and the model law $\mathcal{N}(m_i, \frac{1}{2})$, and w_i is the scalar translation on e_i. With the residuals $r_i^b = k_{i+1} - k_i - w_i$ and $r_i^m = m_{i+1} - m_i - w_i$, the comparison cost* (173) *of e_i is*

$$\begin{aligned} D_{e_i}(\mathbf{x}) &= \tfrac{1}{2}\big[(r_i^b)^2 + (r_i^m)^2\big] \\ &= \mathrm{KL}\big(\mathcal{N}(k_{i+1}, 1) \| \mathcal{N}(k_i + w_i, 1)\big) + \tfrac{1}{2}\,\mathrm{KL}\big(\mathcal{N}(m_{i+1}, \tfrac{1}{2}) \| \mathcal{N}(m_i + w_i, \tfrac{1}{2})\big). \end{aligned}$$

Reversing the orientation of an edge reverses the signs of its translation and of both residuals, so the cost does not depend on the orientation. Each agent i, acting as a receiver, has both neighbors as candidate sources. An assignment is $J = (J_i)_i \in \mathcal{J} = \prod_i \{i-1, i+1\}$, so $|\mathcal{J}| = 16$; $e(i, j)$ is the edge joining i and j, and $V_J(\mathbf{x}) = \sum_i D_{e(i,J_i)}(\mathbf{x})$. The occupied edge set of J is $S(J) = \{e(i, J_i) : i \in \mathbb{Z}/4\}$, and the perfect matchings of the cycle are $M_0 = \{e_0, e_2\}$ and $M_1 = \{e_1, e_3\}$.

The generative ingredients at the temperature $T > 0$ are the following. The configuration prior is $\mathcal{N}(0, I_{12})$. The source prior is a product $\pi_J = \prod_i \pi_{iJ_i}$ of receiver rows with $\pi_{i,i-1} + \pi_{i,i+1} = 1$. Given $(\mathbf{x}, J)$, binary records E_i are independent with $\Pr(E_i = 1 \mid \mathbf{x}, J) = \frac{1}{2}\exp[-D_{e(i,J_i)}(\mathbf{x})/T]$, and $E = (1, 1, 1, 1)$ is observed. Agent i's model presentation is a sample $\tilde{m}_i \sim \mathcal{N}(m_i, \frac{1}{2})$, and its evaluator sends $\tilde{m}_i$ to the normalized kernel $\mathcal{N}(\tilde{m}_i, \frac{1}{2})$ on a belief sample $\tilde{k}_i$. At a fixed configuration the recognition coupling of the two samples is

$$\tilde{m}_i = m_i + \xi_i, \qquad \tilde{k}_i = k_i + \xi_i + \xi_i', \qquad \xi_i, \xi_i' \sim \mathcal{N}(0, \tfrac{1}{2}) \quad \textit{independent},$$

which has the marginals $\mathcal{N}(m_i, \frac{1}{2})$ and $\mathcal{N}(k_i, 1)$. Put $\mathfrak{D}\mathbf{x} = k - m$, $R_{\mathrm{ev}} = 2\mathfrak{D}^\top\mathfrak{D}$, $\Omega = I_{12} + R_{\mathrm{ev}}$ and

$$U_J(\mathbf{x}) = V_J(\mathbf{x}) + \frac{T}{2}\mathbf{x}^\top\Omega\mathbf{x}, \qquad \beta_{ij}(\mathbf{x}; \pi) = \frac{\pi_{ij} e^{-D_{e(i,j)}(\mathbf{x})/T}}{\sum_{j'=i\pm1} \pi_{ij'} e^{-D_{e(i,j')}(\mathbf{x})/T}}, \tag{368}$$

with $\beta_J(\mathbf{x}; \pi) = \prod_i \beta_{iJ_i}(\mathbf{x}; \pi)$.

The kinetics at a continuous row path $t \mapsto \pi(t)$ form the following process on $(\mathbf{x}, J)$. Between rings of a rate-one Poisson clock the assignment is held and

$$d\mathbf{X}_t = -\nabla U_{J_t}(\mathbf{X}_t)dt + \sqrt{2T}dW_t,$$

with W a standard Brownian motion on $\mathbb{R}^{12}$. At a ring τ the complete assignment is redrawn from $\beta_J(\mathbf{X}_\tau; \pi(\tau))$, self-proposals allowed: receiver i takes source $i+1$ exactly when a uniform variable

$\mathsf{U}_{\tau,i}$ *on* $[0,1]$ *is at most* $\beta_{i,i+1}(\mathbf{X}_\tau;\pi(\tau))$*. The uniform variables are independent of each other and of* W*, the clock and the initial state. The recognition law* q_t *is the law of* $(\mathbf{X}_t, J_t)$*. In the learned arm the rows start uniform and evolve by* (174) *with mobility* $\epsilon_L > 0$*; in the frozen arm they stay at* $\frac{1}{2}$*.*

The following reading of the declaration explains its form; the proof uses only the formulas it produces. Under the recognition coupling the conditional law of $\tilde{k}_i$ given $\tilde{m}_i$ is $\mathcal{N}(\tilde{m}_i + k_i - m_i, \frac{1}{2})$, whereas the evaluator kernel is $\mathcal{N}(\tilde{m}_i, \frac{1}{2})$, and the $\tilde{m}_i$ marginals agree. The relative entropy of the coupling to the evaluator factor is therefore $(k_i - m_i)^2$, and summed over agents it is $\frac{1}{2}\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}$. Adding the prior term $\frac{1}{2}|\mathbf{x}|^2$ and the record terms $D_{e(i,J_i)}/T + \log 2$, the negative evidence bound of a recognition law q on $(\mathbf{x}, J)$, extended by the recognition coupling, equals $\mathrm{KL}(q\|q^\star_\pi)$ plus a constant, where

$$q^\star_\pi(\mathbf{x}, J) \propto \pi_J \exp[-U_J(\mathbf{x})/T].$$

The factor $\exp[-\mathbf{x}^\top\Omega\mathbf{x}/2]$ is common to all assignments and $\pi_J e^{-V_J/T}$ factorizes over receivers, so $\beta_J(\mathbf{x};\pi)$ is the conditional law of J given $\mathbf{x}$ under $q^\star_\pi$. The redraw therefore refreshes the assignment from its conditional law, and the configuration motion is the Langevin dynamics of U_J at temperature T.

J.2 Statement

Theorem 7 (Fixed-temperature pairing selection)**.** *In the model of Definition 4, let the initial law be* $\mathcal{N}(\mathbf{m}_0, I_{12}) \otimes \mathrm{Unif}(\mathcal{J})$ *with* $\mathbf{m}_0 \in \mathbb{R}^{12}$*. Write* $q_t^{\epsilon_L}$ *for the recognition law of the learned arm,* q_t^0 *for that of the frozen arm, and* $S = S(J)$*.*

(i) There is $T_0 > 0$ *such that for every* $T \in (0, T_0]$ *there is* $C_T < \infty$ *for which, for Lebesgue-almost every* $\mathbf{m}_0 \in \mathbb{R}^{12}$*, one perfect matching* $M \in \{M_0, M_1\}$ *satisfies*

$$\lim_{\epsilon_L \to 0} \sup_{t_L \le t \le 2t_L} q_t^{\epsilon_L}(S \neq M) = 0, \qquad t_L = C_T \epsilon_L^{-1} \log(1/\epsilon_L).$$

(ii) For every $T > 0$ *and every* $\mathbf{m}_0 \in \mathbb{R}^{12}$*,* $q_t^0(S = M_0) - q_t^0(S = M_1) \to 0$ *as* $t \to \infty$*.*

J.3 Reduced coordinates, symmetry and frozen invariant laws

Throughout, $T > 0$ is fixed and $\varepsilon = \epsilon_L$. Encode an assignment by $\sigma \in \{\pm 1\}^4$, with $\sigma_i = +1$ exactly when receiver i selects $i + 1$, and write the rows as $p_i = \pi_{i,i+1}$, so $\pi_{i,i-1} = 1 - p_i$ and

$$\pi_\sigma(p) = \prod_i p_i^{1_{\{\sigma_i = +1\}}} (1 - p_i)^{1_{\{\sigma_i = -1\}}}.$$

The number of receivers attending the edge e_i is its multiplicity $n_i(\sigma) = 1_{\{\sigma_i = +1\}} + 1_{\{\sigma_{i+1} = -1\}}$, so $V_\sigma = \sum_i n_i(\sigma) D_{e_i}$; write $C_\sigma = \mathrm{diag}(n_i(\sigma))$. Put $\mathbf{v} = (1, -1, 1, -1)$. The assignment $\sigma = \mathbf{v}$ is $J^{(0)} = (1, 0, 3, 2)$, which selects e_0 and e_2 from both ends, and $\sigma = -\mathbf{v}$ is $J^{(1)} = (3, 2, 1, 0)$. These are the only assignments whose occupied edge set is a perfect matching: if $S(J) = M_0$, receiver 0 must select e_0 rather than e_3, receiver 1 must select e_0 rather than e_1, and similarly receivers 2 and 3 must select e_2, which forces $J = J^{(0)}$; the case M_1 is the same after rotation. Hence $\{S = M_0\} = \{\sigma = \mathbf{v}\}$ and $\{S = M_1\} = \{\sigma = -\mathbf{v}\}$.

Write $H_\sigma = \nabla^2 V_\sigma$, $A_\sigma = H_\sigma + T\Omega$ and $B_\sigma = A_\sigma/T$. Then $\nabla U_\sigma(\mathbf{x}) = A_\sigma \mathbf{x}$ and $A_\sigma \succeq TI$, because H_σ and R_{ev} are positive semidefinite. Let

$$\theta_\sigma = \det(B_\sigma)^{-1/2}, \qquad \mathsf{Z}(p) = \sum_\sigma \pi_\sigma(p)\theta_\sigma, \qquad \mu_p(\mathbf{x}, \sigma) = \frac{\pi_\sigma(p)}{(2\pi)^6 \mathsf{Z}(p)} \exp\left[-\frac{\mathbf{x}^\top A_\sigma \mathbf{x}}{2T}\right],$$

so that $\mu_p = q^\star_\pi$, and let $b_\sigma(\mathbf{x};p) = \mu_p(\mathbf{x},\sigma)/\sum_\eta \mu_p(\mathbf{x},\eta) = \beta_\sigma(\mathbf{x};\pi)$ be its conditional assignment law. The reduced row field is

$$\bar\beta_i(p) = \mu_p(\sigma_i = +1), \qquad g_i(p) = \partial_{p_i} \log \mathsf{Z}(p), \qquad \Phi_i(p) = \bar\beta_i(p) - p_i = p_i(1-p_i)g_i(p). \tag{369}$$

The last identity follows from $\partial_{p_i}\pi_\sigma = \pi_\sigma(1_{\{\sigma_i=+1\}} - p_i)/(p_i(1-p_i))$ by summing against θ_σ/Z. Since Z is a multilinear polynomial that is positive on the closed cube $[0,1]^4$, the functions g and Φ extend analytically to a neighborhood of the cube and have bounded derivatives there.

For constant rows p write P^p_t for the transition semigroup of the kinetics. Let $\mathcal{G}_s$ denote the held motion for a time s, acting on functions of $(\mathbf{x},\sigma)$ with σ fixed, and let $\mathcal{R}_p\phi(\mathbf{x},\sigma) = \sum_\eta b_\eta(\mathbf{x};p)\phi(\mathbf{x},\eta)$ be the redraw. Because the clock has constant rate one and is independent of the other inputs,

$$P^p_t = \sum_{n\ge 0} e^{-t} \int_{0<t_1<\cdots<t_n<t} \mathcal{G}_{t_1}\mathcal{R}_p\mathcal{G}_{t_2-t_1}\cdots\mathcal{R}_p\mathcal{G}_{t-t_n}\,dt_1\cdots dt_n. \tag{370}$$

Lemma 7 (Frozen invariant laws and symmetry)**.** *For every $p \in [0,1]^4$, μ_p is invariant for P^p_t. The law $\mu_{1/2}$ is invariant under the automorphisms of the cycle; consequently $\bar\beta_i(\frac12) = \frac12$, $g(\frac12) = \Phi(\frac12) = 0$, $\mu_{1/2}f_{\mathbf{v}} = 0$ for $f_{\mathbf{v}}(\sigma) = \frac12\sum_i \mathbf{v}_i\sigma_i$, and $\mu_{1/2}(S = M_0) = \mu_{1/2}(S = M_1)$.*

Proof. On a held sheet σ the configuration is an Ornstein–Uhlenbeck process with drift $-A_\sigma\mathbf{x}$ and diffusion matrix $2TI$. Since A_σ is symmetric, its stationary covariance Σ solves $A_\sigma\Sigma + \Sigma A_\sigma = 2TI$, so $\Sigma = TA_\sigma^{-1}$ and each $\mathcal{G}_s$ preserves μ_p. The redraw $\mathcal{R}_p$ resamples σ from its conditional law given $\mathbf{x}$ under μ_p and also preserves it. By (370), $\mu_pP^p_t = \sum_n e^{-t}t^n\mu_p/n! = \mu_p$. When some $p_i \in \{0,1\}$ the same argument applies on the sheets of positive weight.

The rotation $i \mapsto i+1$ permutes k, m and w cyclically. The reflection $i \mapsto -i$ sends k_i and m_i to the coordinates of agent $-i$ and, since it carries e_i onto e_{-i-1} with reversed orientation, it sends w_i to the translation $-w_i$ on e_{-i-1}. Each automorphism g thus acts by an orthogonal signed permutation O_g with $D_{e(gi,gj)}(O_g\mathbf{x}) = D_{e(i,j)}(\mathbf{x})$, and $\mathfrak{D}O_g\mathbf{x}$ is a permutation of $\mathfrak{D}\mathbf{x}$, hence $U_{gJ}(O_g\mathbf{x}) = U_J(\mathbf{x})$, where $(gJ)_{gi} = gJ_i$. At uniform rows $\pi_\sigma = 1/16$ for every σ, so $\mu_{1/2}(O_g\mathbf{x}, gJ) = \mu_{1/2}(\mathbf{x},J)$. The reflection $j \mapsto 2i - j$ fixes receiver i and exchanges its two sources, which gives $\bar\beta_i(\frac12) = \frac12$ and, by (369), $\Phi(\frac12) = g(\frac12) = 0$. The rotation by one maps $J^{(0)}$ to $J^{(1)}$, that is $\mathbf{v}$ to $-\mathbf{v}$, and changes the sign of $f_{\mathbf{v}}$, which gives the last two assertions. □

J.4 Rows and geometric ergodicity

Lemma 8 (Well-posedness and row bounds)**.** *For every $\varepsilon > 0$ the kinetics and (174), started from uniform rows, have exactly one joint solution on $[0,\infty)$. Its rows are deterministic and continuously differentiable, and*

$$\min_i\{p_i(t), 1-p_i(t)\} \ge \tfrac12 e^{-\varepsilon t}, \qquad \|p(t) - \tfrac12\|_\infty \le \tfrac12(1-e^{-\varepsilon t}) \le \tfrac12\varepsilon t. \tag{371}$$

Proof. For a continuous row path π construct the kinetics pathwise from the inputs of Definition 4. Between rings the configuration solves a linear equation with one of sixteen matrices, and rings are finitely many on bounded intervals, so the process exists, is pathwise unique given the inputs, and does not explode. The configuration drift does not involve the rows. Write q^π_t for the law at time t. Suppose the entries of two row paths π, π' lie in $[\underline{p}, 1-\underline{p}]$ with $\underline{p} \in (0, \frac12]$, and drive both constructions by the same initial data, Brownian motion, clock and uniform variables. They coincide until the first ring at which the uniform variable of some receiver falls between its posterior rows under π and π'. Given agreement before a ring τ, this happens at τ with conditional probability

at most $\sum_i |\beta_{i,i+1}(\mathbf{x};\pi(\tau)) - \beta_{i,i+1}(\mathbf{x};\pi'(\tau))|$ at the common configuration. With $\mathrm{p} = \pi_{i,i+1}$, the derivative of a posterior row in p is $\beta_{i,i+1}(1-\beta_{i,i+1})/(\mathrm{p}(1-\mathrm{p})) \le 1/(4\underline{p}(1-\underline{p})) \le 1/(2\underline{p})$. The clock has rate one, so

$$\big|q_t^{\pi}(J_i = j) - q_t^{\pi'}(J_i = j)\big| \le \|q_t^{\pi} - q_t^{\pi'}\|_{\mathrm{TV}} \le \frac{2}{\underline{p}} \int_0^t \|\pi(s) - \pi'(s)\|_\infty ds, \tag{372}$$

where $\|\cdot\|_\infty$ is the maximum over receivers and sources and $\|q-\nu\|_{\mathrm{TV}} = \sup_A |q(A) - \nu(A)|$.

A solution of (174) that starts uniform satisfies

$$\pi_{ij}(t) - \tfrac{1}{2} = \varepsilon \int_0^t e^{-\varepsilon(t-s)} \big(q_s(J_i = j) - \tfrac{1}{2}\big) ds, \qquad \big|\pi_{ij}(t) - \tfrac{1}{2}\big| \le \tfrac{1}{2}(1 - e^{-\varepsilon t}),$$

because $0 \le q_s(J_i = j) \le 1$; each row sums to one, and on $[0, t_1]$ its entries lie in $[\underline{p}, 1-\underline{p}]$ with $\underline{p} = e^{-\varepsilon t_1}/2$. Fix $t_1 > 0$ and let Γ send a row path π to the right-hand side of the identity with $q_s = q_s^{\pi}$. It maps the closed set of continuous row paths on $[0, t_1]$ with $|\pi_{ij}(t) - \frac{1}{2}| \le \frac{1}{2}(1 - e^{-\varepsilon t})$ into itself, and by (372)

$$\|\Gamma\pi(t) - \Gamma\pi'(t)\|_\infty \le \frac{2\varepsilon t_1}{\underline{p}} \int_0^t \|\pi(s) - \pi'(s)\|_\infty ds, \qquad t \le t_1.$$

The nth iterate of Γ therefore contracts the supremum distance by the factor $(2\varepsilon t_1^2/\underline{p})^n/n!$, and Γ has exactly one fixed point in that set. Every solution lies in the set, and solutions on nested intervals agree. Since J changes only at rings, $|q_{s+h}^{\pi}(J_i = j) - q_s^{\pi}(J_i = j)| \le h$, so the fixed point is continuously differentiable and solves (174). The rows are a deterministic function of time and satisfy (371). □

Lemma 9 (Uniform geometric ergodicity at frozen rows)**.** *For every row path and every starting point,*

$$\mathbb{E}_{\mathbf{x}} |\mathbf{X}_t|^2 \le e^{-2Tt} |\mathbf{x}|^2 + 12(1 - e^{-2Tt}). \tag{373}$$

There are $K_T < \infty$ and $\lambda_T > 0$, depending only on T, such that for every $p \in [0,1]^4$, every bounded measurable ϕ, every $(\mathbf{x}, \sigma)$ and every $t \ge 0$,

$$|P_t^p \phi(\mathbf{x}, \sigma) - \mu_p \phi| \le K_T e^{-\lambda_T t} (1 + |\mathbf{x}|^2) \|\phi\|_\infty. \tag{374}$$

Proof. Redraws leave $\mathbf{X}$ unchanged, and on a held sheet Itô's formula gives $d|\mathbf{X}|^2 = (24T - 2\mathbf{X}^\top A_\sigma \mathbf{X})dt + dN_t$ with a martingale N. Since $A_\sigma \succeq TI$, the drift is at most $24T - 2T|\mathbf{X}|^2$, which gives (373). Let $P = P_1^p$ and $\mathcal{V}(\mathbf{x}, \sigma) = |\mathbf{x}|^2$; then $P\mathcal{V} \le \gamma_0 \mathcal{V} + K_0$ with $\gamma_0 = e^{-2T}$ and $K_0 = 12$. Choose $R > 2K_0/(1 - \gamma_0)$.

Minorization. For any p some assignment $\sigma_*(p)$ has $\pi_{\sigma_*} \ge 1/16$. Consider the clock event of exactly one ring in $[0, \frac{1}{2}]$ and none in $(\frac{1}{2}, 1]$, of probability $e^{-1}/2$. Starting with $|\mathbf{x}| \le \sqrt{R}$, the held motion up to the ring time τ is Gaussian with mean $e^{-A_\sigma \tau}\mathbf{x}$, of norm at most $|\mathbf{x}|$, and a fluctuation with expected squared norm at most $24T\tau \le 12T$; by Chebyshev's inequality it lies in the ball of radius $r' = \sqrt{R} + \sqrt{24T}$ at the ring with probability at least $\frac{1}{2}$. On that ball the redraw selects σ_* with probability $\pi_{\sigma_*} e^{-V_{\sigma_*}/T} / \sum_\eta \pi_\eta e^{-V_\eta/T} \ge 16^{-1} e^{-v(r')/T}$, where $v(r') = \max_{|\mathbf{z}| \le r', \sigma} V_\sigma(\mathbf{z}) < \infty$, because the denominator is at most one. The subsequent held Gaussian transition lasts between $\frac{1}{2}$ and 1, its means are bounded and its covariance matrices have common positive lower and finite upper bounds, so its density has a positive common lower bound on the unit ball B_1. Hence, for some $\alpha_T > 0$ independent of p,

$$P((\mathbf{x}, \sigma), \cdot) \ge \alpha_T \nu_p, \qquad \nu_p = \mathrm{Unif}(B_1) \otimes \delta_{\sigma_*(p)}, \qquad \text{whenever } |\mathbf{x}|^2 \le R.$$

Contraction. The weighted-norm argument below follows [62]. Fix $\gamma_1 \in (\gamma_0 + 2K_0/R, 1)$, put $\vartheta = \alpha_T/(2K_0)$, and for functions ϕ on the state space define

$$|||\phi||| = \sup_{y \neq y'} \frac{|\phi(y) - \phi(y')|}{2 + \vartheta\mathcal{V}(y) + \vartheta\mathcal{V}(y')}.$$

If ϕ is bounded and $|||\phi||| \le 1$, then $\sup_y[\phi(y) - 1 - \vartheta\mathcal{V}(y)] \le \inf_{y'}[\phi(y') + 1 + \vartheta\mathcal{V}(y')]$, and a constant c between the two satisfies $|\phi - c| \le 1 + \vartheta\mathcal{V}$. Since P reproduces constants, we may replace ϕ by $\phi - c$. Consider two points y, y'. If $\mathcal{V}(y) + \mathcal{V}(y') \ge R$, then $(\gamma_1 - \gamma_0)(\mathcal{V}(y) + \mathcal{V}(y')) \ge 2K_0$, and

$$\begin{aligned} |P\phi(y) - P\phi(y')| &\le 2 + \vartheta P\mathcal{V}(y) + \vartheta P\mathcal{V}(y') \le 2 + \vartheta\gamma_1(\mathcal{V}(y) + \mathcal{V}(y')) \\ &\le \bar\alpha_1(2 + \vartheta\mathcal{V}(y) + \vartheta\mathcal{V}(y')), \end{aligned}$$

with $\bar\alpha_1 = (2+\vartheta R\gamma_1)/(2+\vartheta R) < 1$, because $(2+\vartheta\gamma_1 W)/(2+\vartheta W)$ decreases in W. If $\mathcal{V}(y)+\mathcal{V}(y') < R$, both points lie in the minorization set, $P(y,\cdot) = \alpha_T\nu_p + (1-\alpha_T)\tilde P(y,\cdot)$ with a Markov kernel $\tilde P$, and the same holds for y'. Then $P\phi(y) - P\phi(y') = (1-\alpha_T)(\tilde P\phi(y) - \tilde P\phi(y'))$ and $(1-\alpha_T)\tilde P\mathcal{V} \le P\mathcal{V}$, so

$$\begin{aligned} |P\phi(y) - P\phi(y')| &\le 2(1-\alpha_T) + 2\vartheta K_0 + \vartheta\gamma_0(\mathcal{V}(y) + \mathcal{V}(y')) \\ &\le \bar\alpha_2(2 + \vartheta\mathcal{V}(y) + \vartheta\mathcal{V}(y')), \end{aligned}$$

with $\bar\alpha_2 = \max(1 - \alpha_T/2, \gamma_0)$, since $2\vartheta K_0 = \alpha_T$. Hence $|||P\phi||| \le \bar\alpha|||\phi|||$ for every bounded ϕ, with $\bar\alpha = \max(\bar\alpha_1, \bar\alpha_2) < 1$ depending only on T; this is the weighted-norm form of Harris's ergodic theorem. Since μ_p is invariant (Lemma 7), $|||\phi||| \le \|\phi\|_\infty$, and $\mu_p\mathcal{V} \le 12$ because $TA_\sigma^{-1} \preceq I$,

$$|P^n\phi(y) - \mu_p\phi| \le \int |P^n\phi(y) - P^n\phi(y')|\mu_p(dy') \le \bar\alpha^n\|\phi\|_\infty(2 + 12\vartheta + \vartheta\mathcal{V}(y)).$$

For $t = n + r$ with $0 \le r < 1$, apply this to $P_r^p\phi$, whose supremum norm is at most $\|\phi\|_\infty$. This gives (374) with $\lambda_T = -\log\bar\alpha$. □

J.5 The initial imprint

From here on K_T and c_T denote finite positive constants depending only on T, whose values may change between occurrences, and a subscript $\mathbf{m}_0$ indicates additional dependence on the initial mean.

Lemma 10 (Imprint)**.** *For every $\mathbf{m}_0$ the integrals*

$$\mathsf{I}_i(\mathbf{m}_0) = \int_0^\infty \big[q_t^0(\sigma_i = +1) - \tfrac12\big]dt, \qquad \mathsf{j}_T(\mathbf{m}_0) = \mathbf{v}\cdot\mathsf{I}(\mathbf{m}_0) = \int_0^\infty q_t^0 f_{\mathbf{v}}\,dt \tag{375}$$

converge absolutely, and j_T extends to an entire function of $\mathbf{m}_0 \in \mathbb{C}^{12}$. The Poisson integral

$$\mathcal{Q}_T(\mathbf{x},\sigma) = \int_0^\infty P_t^{1/2} f_{\mathbf{v}}(\mathbf{x},\sigma)dt \quad \textit{satisfies} \quad |\mathcal{Q}_T(\mathbf{x},\sigma)| \le \mathrm{K}_T(1 + |\mathbf{x}|^2). \tag{376}$$

Moreover, with $\underline{p} = 0.45$, for $t \le 1/(10\varepsilon)$,

$$\begin{aligned} \|q_t^\varepsilon - q_t^0\|_{\mathrm{TV}} &\le \frac{\varepsilon t^2}{2\underline{p}}, \\ \big|p(t) - \tfrac12 - \varepsilon\mathsf{I}(\mathbf{m}_0)\big|_\infty &\le \mathrm{K}_{T,\mathbf{m}_0}(\varepsilon e^{-\lambda_T t} + \varepsilon^2 t) + \frac{\varepsilon^2 t^3}{6\underline{p}}. \end{aligned} \tag{377}$$

In particular $p(t_1) - \frac12 = \varepsilon\mathsf{I}(\mathbf{m}_0) + o(\varepsilon)$ at $t_1 = \varepsilon^{-1/4}$.

Proof. Since $\mu_{1/2} f_{\mathbf{v}} = 0$ and $|f_{\mathbf{v}}| \le 2$, estimate (374) at $p = \frac{1}{2}$ gives $|P_t^{1/2} f_{\mathbf{v}}(\mathbf{x}, \sigma)| \le 2K_T e^{-\lambda_T t}(1 + |\mathbf{x}|^2)$, which proves (376). The same estimate applied to $1_{\{\sigma_i = +1\}} - \frac{1}{2}$ and integrated against the initial law proves absolute convergence of I_i, with $|q_t^0(\sigma_i = +1) - \frac{1}{2}| \le \mathrm{K}_{T,\mathbf{m}_0} e^{-\lambda_T t}$. By Fubini's theorem, $\mathsf{j}_T(\mathbf{m}_0) = \frac{1}{16} \sum_\sigma \int \mathcal{Q}_T(\mathbf{x}, \sigma) \mathcal{N}(\mathbf{m}_0, I_{12})(d\mathbf{x})$. Writing the Gaussian density as $(2\pi)^{-6} e^{-|\mathbf{x}|^2/2} e^{\mathbf{x} \cdot \mathbf{m}_0} e^{-|\mathbf{m}_0|^2/2}$ exhibits j_T as an entire function of the complexified mean: for every compact set of complex means, all differentiated integrands are bounded by an integrable polynomial times a Gaussian.

For $t \le 1/(10\varepsilon)$, (371) places every row entry in $[\underline{p}, 1 - \underline{p}]$. Coupling the learned and frozen arms by common inputs as in the proof of Lemma 8, estimate (372) and $\|p(s) - \frac{1}{2}\|_\infty \le \varepsilon s/2$ give the first inequality of (377). For the second, subtract $\varepsilon \mathsf{I}_i$ from the variation-of-constants identity in the proof of Lemma 8:

$$\begin{aligned} p_i(t) - \tfrac{1}{2} - \varepsilon \mathsf{I}_i = \varepsilon \int_0^t e^{-\varepsilon(t-s)} \big[q_s^\varepsilon(\sigma_i = +1) - q_s^0(\sigma_i = +1) \big] ds \\ + \varepsilon \int_0^t \big(e^{-\varepsilon(t-s)} - 1 \big) \big[q_s^0(\sigma_i = +1) - \tfrac{1}{2} \big] ds \\ - \varepsilon \int_t^\infty \big[q_s^0(\sigma_i = +1) - \tfrac{1}{2} \big] ds. \end{aligned}$$

The first term is at most $\varepsilon \int_0^t \varepsilon s^2/(2\underline{p}) ds = \varepsilon^2 t^3/(6\underline{p})$. With $|e^{-\varepsilon(t-s)} - 1| \le \varepsilon(t-s)$ and the exponential bound on $q_s^0(\sigma_i = +1) - \frac{1}{2}$, the second is at most $\mathrm{K}_{T,\mathbf{m}_0} \varepsilon^2 t$ and the third at most $\mathrm{K}_{T,\mathbf{m}_0} \varepsilon e^{-\lambda_T t}$. At $t_1 = \varepsilon^{-1/4}$ the three terms are $O(\varepsilon^{5/4})$, $O(\varepsilon^{7/4})$ and $o(\varepsilon^n)$ for every n. □

J.6 Nonvanishing of the matching imprint

Lemma 11 (Nonvanishing imprint)**.** *Let $\mathbf{u}_\circ = (0_4, 0_4, (0, 1, 0, -1))$. For every $T > 0$, as $s \to \infty$,*

$$\mathsf{j}_T(s\mathbf{u}_\circ) \ge \frac{\log s}{2T} - \frac{5}{T} \log\log s - O_T(1). \tag{378}$$

Consequently j_T is not identically zero, and its real zero set is Lebesgue-null.

Proof. The tied residual motion. Both residual vectors of $\mathbf{u}_\circ$ equal $(0, -1, 0, 1)$. Consider a switched motion $\dot{\mathbf{y}} = -A_{\sigma(t)} \mathbf{y}$ along a prescribed piecewise constant assignment path. On the tied subspace $\{k = m\}$ we have $R_{\mathrm{ev}} \mathbf{y} = 0$, the two residual vectors agree, and, writing ρ for the common one, $V_\sigma = \sum_i n_i(\sigma) \rho_i^2$. Let $D = \mathsf{P} - I$ be the cyclic difference matrix, $(\mathsf{P}k)_i = k_{i+1}$. Then $\partial V_\sigma / \partial k = \partial V_\sigma / \partial m = D^\top C_\sigma \rho$ and $\partial V_\sigma / \partial w = -2 C_\sigma \rho$ there, so the tied subspace is invariant. After removal of the scalar factor e^{-Tt} contributed by $T\Omega = TI$ on it, $\dot\rho = D\dot k - \dot w = -(DD^\top + 2I) C_\sigma \rho$, that is

$$\dot\rho = -G C_\sigma \rho, \qquad G = 4I - \mathsf{P} - \mathsf{P}^\top. \tag{379}$$

Of the sixteen assignments, six have $n_1 = n_3$: the two matchings, with multiplicity patterns $(2, 0, 2, 0)$ and $(0, 2, 0, 2)$, the two oriented cycles, with pattern $(1, 1, 1, 1)$, and the patterns $(2, 1, 0, 1)$ and $(0, 1, 2, 1)$. Since $G(0, -1, 0, 1)^\top = 4(0, -1, 0, 1)^\top$, their first held motion is $e^{-ct}(0, -1, 0, 1)$ with $c = 4n_1 \in \{0, 4, 8\}$. At their first clock ring the even residuals vanish and the odd ones do not, so every receiver prefers its even edge, that is the matching $J^{(0)}$. Holding $J^{(0)}$, with $C_\sigma = \mathrm{diag}(2, 0, 2, 0)$, then leaves the odd residuals constant and the even residuals zero.

Four further assignments have the multiplicity patterns $(1, 1, 2, 0)$, $(1, 0, 2, 1)$, $(2, 0, 1, 1)$ and $(2, 1, 1, 0)$. The exchange of the residual coordinates 1 and 3 and the rotation of coordinates by

two commute with G, permute these four patterns transitively, fix the pattern $(2,0,2,0)$ of $J^{(0)}$, and carry the initial residual to its negative; by linearity they reduce these four cases to $(1,1,2,0)$. Write its residual as $\rho = (-\mathsf{a}, -\mathsf{b}, -\mathsf{d}, \mathsf{h})$. Then

$$\begin{pmatrix} \dot{\mathsf{a}} \\ \dot{\mathsf{b}} \\ \dot{\mathsf{d}} \end{pmatrix} = \begin{pmatrix} -4 & 1 & 0 \\ 1 & -4 & 2 \\ 0 & 1 & -8 \end{pmatrix} \begin{pmatrix} \mathsf{a} \\ \mathsf{b} \\ \mathsf{d} \end{pmatrix}, \qquad \dot{\mathsf{h}} = -(\mathsf{a} + 2\mathsf{d}),$$

with $(\mathsf{a}, \mathsf{b}, \mathsf{d}, \mathsf{h})(0) = (0, 1, 0, 1)$. The off-diagonal entries are nonnegative, so the nonnegative cone is invariant, and the row sums are negative, so $\max(\mathsf{a}, \mathsf{b}, \mathsf{d}) \le 1$; hence $\mathsf{a} \le \frac{1}{4}$ and $\mathsf{d} \le \frac{1}{8}$. The half-spaces $\mathsf{a} - \frac{3}{4}\mathsf{b} - \frac{11}{25}\mathsf{d} \le 0$ and $\mathsf{d} - \frac{1}{4}\mathsf{b} \le 0$ are invariant: on the corresponding boundaries the outward derivatives are $-\mathsf{b}/400 - 7\mathsf{d}/100$ and $-\mathsf{a}/4 - \mathsf{b}/8$. Consequently $\mathsf{a} \le \frac{43}{50}\mathsf{b}$, $\mathsf{d} \le \frac{1}{4}\mathsf{b}$, and $\mathsf{b} \ge e^{-4t}$ because $\dot{\mathsf{b}} \ge -4\mathsf{b}$. The attended 3×3 block is similar to a negative definite symmetric matrix because $G \succ 0$, so $(\mathsf{a}, \mathsf{b}, \mathsf{d}) \to 0$. Integrating (379), $\rho(\infty) = \rho(0) - Gz$ with $z = \int_0^\infty C_\sigma \rho dt$ and $z_3 = 0$; the first three coordinates of $\rho(\infty)$ vanish, which gives $z = (-\frac{1}{14}, -\frac{4}{14}, -\frac{1}{14}, 0)$ and $\mathsf{h}(\infty) = \frac{6}{7}$. Since h decreases, $\mathsf{h} \ge \frac{6}{7}$.

If the first ring occurs at t_{r} and redraws $J^{(0)}$, the subsequent scalar-free held motion, in elapsed time r, is

$$\mathsf{a}_r = \mathsf{a}_0 e^{-8r}, \qquad \mathsf{d}_r = \mathsf{d}_0 e^{-8r},$$
$$\mathsf{b}_r = \mathsf{b}_0 + \tfrac{1}{4}(\mathsf{a}_0 + \mathsf{d}_0)(1 - e^{-8r}), \quad \mathsf{h}_r = \mathsf{h}_0 - \tfrac{1}{4}(\mathsf{a}_0 + \mathsf{d}_0)(1 - e^{-8r}) \ge \tfrac{171}{224}.$$

Each receiver has one odd and one even edge, and on every receiver the odd-edge absolute residual exceeds the even-edge one by at least $\frac{7}{50}e^{-4t_{\mathrm{r}}}$, both at the ring and afterward. On the tied subspace $D_{e_i} = \rho_i^2$ in scale-free coordinates, so $\sqrt{D_{e_i}}$ is the scale times $e^{-Tt}|\rho_i|$. Combining these four cases with the preceding six, the deterministic motion $\mathbf{Y}$ from $s\mathbf{u}_\circ$ satisfies, for every receiver i with odd edge $o(i)$ and even edge $\bar{o}(i)$, throughout the hypothesized hold of $J^{(0)}$,

$$\sqrt{D_{o(i)}(\mathbf{Y}_t)} - \sqrt{D_{\bar{o}(i)}(\mathbf{Y}_t)} \ge \frac{7}{50} s e^{-Tt - 8t_{\mathrm{r}}}, \qquad t \ge t_{\mathrm{r}}. \tag{380}$$

There are therefore ten good initial assignments. For the remaining six, with patterns $(1,2,0,1)$, $(1,2,1,0)$, $(0,1,1,2)$, $(0,2,1,1)$, $(1,0,1,2)$ and $(1,1,0,2)$, only $f_{\mathbf{v}} \ge -2$ will be used.

Noise. Set $h_s = (\log s)/T$ for $s > e$, and write the initial configuration as $s\mathbf{u}_\circ + \xi$ with $\xi \sim \mathcal{N}(0, I_{12})$. Along the actual assignment path of the frozen arm decompose $\mathbf{X} = \mathbf{Y} + \mathbf{Y}' + \mathbf{Z}$, where $\mathbf{Y}$ and $\mathbf{Y}'$ solve the switched deterministic equation from $s\mathbf{u}_\circ$ and from ξ, while

$$d\mathbf{Z} = -A_{\sigma_t}\mathbf{Z}dt + \sqrt{2T}dW_t, \qquad \mathbf{Z}_0 = 0.$$

All three are adapted, and $|\mathbf{Y}'_t| \le e^{-Tt}|\xi|$ because $A_\sigma \succeq TI$. For $\varphi(\mathbf{z}) = e^{|\mathbf{z}|^2/4}$ the generator of $\mathbf{Z}$ satisfies

$$\mathcal{L}\varphi = \big[T(6 + |\mathbf{z}|^2/4) - \tfrac{1}{2}\mathbf{z}^\top A_\sigma \mathbf{z}\big]\varphi \le T(6 - |\mathbf{z}|^2/4)\varphi \le Te^5.$$

Stopping at radius $\log s$ and using $\mathbb{E}e^{|\xi|^2/4} = 2^6 = 64$ bounds the probability that $|\xi| > \log s$ or $\sup_{t \le h_s} |\mathbf{Z}_t| > \log s$ by $(65 + Te^5 h_s)e^{-(\log s)^2/4}$. On the complement, each $\sqrt{D_e}$ differs from its value at $\mathbf{Y}$ by at most $2c_N \log s$, where c_N is the largest operator norm of a residual-pair map $\mathbf{x} \mapsto (r_i^b, r_i^m)$ divided by $\sqrt{2}$.

Lock. Define

$$L_s = h_s - \frac{8}{T}t_{\mathrm{r}} - \frac{2}{T}\log\log s, \qquad \tau_s = (L_s - t_{\mathrm{r}})_+.$$

For a good initial assignment, (380) gives a square-root gap at least $\frac{7}{50}(\log s)^2$ on $[t_{\mathrm{r}}, L_s]$, since there $se^{-Tt-8t_{\mathrm{r}}} \ge se^{-TL_s - 8t_{\mathrm{r}}} = (\log s)^2$. For sufficiently large s, while both noise bounds hold, the actual

square-root gap is at least $\frac{7}{100}(\log s)^2$, and since $a-b \geq (\sqrt{a}-\sqrt{b})^2$ for $a \geq b \geq 0$ the energy gap is at least $c_\star(\log s)^4$ with $c_\star = (7/50)^2/4$. At uniform rows a receiver's posterior then chooses its odd edge with probability at most $e^{-c_\star(\log s)^4/T}$. Apply this bound at the first ring, then use the rate-one compensator up to the first wrong redraw or exit from the noise bounds, stopping at $L_s \wedge h_s$. The endpoint L_s is known after the first ring, and the gap estimate holds until the first failure. Thus $J^{(0)}$ is held throughout $[t_{\mathrm{r}}, L_s]$ except on an event of probability at most

$$\delta_s = (65 + Te^5 h_s)e^{-(\log s)^2/4} + 4(1+h_s)e^{-c_\star(\log s)^4/T}, \qquad h_s\delta_s \to 0. \tag{381}$$

Since $f_{\mathbf{v}}(\pm\mathbf{v}) = \pm 2$ and $\mathbb{E}t_{\mathrm{r}} = 1$, a good initial assignment contributes at least

$$-2h_s + 4\mathbb{E}\tau_s - 4h_s\delta_s \geq 2h_s - 4(1+8/T) - (8/T)\log\log s - 4h_s\delta_s$$

to $\mathbb{E}\int_0^{h_s} f_{\mathbf{v}}(\sigma_t)dt$, and each of the other six contributes at least $-2h_s$. Averaging over the sixteen equally likely initial assignments,

$$\mathbb{E}\int_0^{h_s} f_{\mathbf{v}}(\sigma_t)dt \geq \tfrac{1}{2}h_s - \frac{5}{T}\log\log s - O_T(1).$$

By (373), $\mathbb{E}|\mathbf{X}_{h_s}|^2 \leq s^{-2}(2s^2+12) + 12(1-s^{-2}) = 14$. The Markov property and (376) bound the integral after h_s in absolute value by $15\mathrm{K}_T$, independently of s. This proves (378).

Zero set. A real-analytic function $f \not\equiv 0$ on $\mathbb{R}^d$ has a Lebesgue-null zero set. In one variable its zeros are isolated. In d variables some coefficient $\partial_d^\ell f(\mathbf{x}', 0)$ is a real-analytic function of $\mathbf{x}' \in \mathbb{R}^{d-1}$ that is not identically zero, for otherwise f would vanish identically. Outside the zero set of that coefficient, null by induction, each one-dimensional slice $x_d \mapsto f(\mathbf{x}', x_d)$ is a nonzero analytic function with isolated zeros, and Fubini's theorem concludes. Apply this to j_T. □

J.7 Entropy balance for the evolving recognition law

Write $q_\sigma(t, \mathbf{x})$ for the density of the learned recognition law on sheet σ, $Q = \sum_\sigma q_\sigma$, $b_\sigma(t, \mathbf{x}) = b_\sigma(\mathbf{x}; p(t))$, $f_\sigma = q_\sigma/\mu_{p(t)}(\cdot, \sigma)$ and $\mathsf{H}(t) = \mathrm{KL}(q_t \| \mu_{p(t)})$.

Lemma 12 (Entropy balance)**.** *On every finite interval $[0, h]$, H is finite and absolutely continuous, and almost everywhere*

$$\begin{aligned} \mathsf{H}'(t) &= -\mathcal{P}_{p(t)}(f) - \sum_\sigma \int q_\sigma \partial_t \log \mu_{p(t)}(\mathbf{x}, \sigma) d\mathbf{x}, \\ \mathcal{P}_p(f) &= T\sum_\sigma \int \mu_p(\mathbf{x}, \sigma)\frac{|\nabla f_\sigma|^2}{f_\sigma} d\mathbf{x} \\ &\quad + \frac{1}{2}\sum_{\sigma,\eta}\int \mu_p(\mathbf{x}, \sigma) b_\eta(\mathbf{x}; p)(f_\eta - f_\sigma)(\log f_\eta - \log f_\sigma) d\mathbf{x}, \end{aligned} \tag{382}$$

with both production integrals finite and locally integrable in time.

Proof. All rows are bounded away from zero on $[0, h]$ by (371). For the forward Ornstein–Uhlenbeck semigroup $\mathcal{K}_\sigma$ of sheet σ, acting on densities, conditioning on the last ring before t gives the last-proposal formula

$$q_\sigma(t) = e^{-t}\mathcal{K}_\sigma(t)q_\sigma(0) + \int_0^t e^{-(t-r)}\mathcal{K}_\sigma(t-r)[b_\sigma(r)Q(r)]dr. \tag{383}$$

Iterating this positive formula, and bounding each proposal factor by one, reduces each term to a Gaussian motion along a deterministic sequence of sheets. Put $\bar{a} = \max_\sigma \|A_\sigma\|$. The means of these motions have norm at most $|\mathbf{m}_0|$, and their covariances lie between $e^{-2\bar{a}h} I$ and $(1 + 2Th)I$. Summing the at most 16^n sequences on each time simplex, of volume $t^n/n!$, gives a finite upper Gaussian envelope, and the no-proposal term gives a lower one. Consequently, uniformly on $[0, h]$,

$$0 < \mathrm{c}_h e^{-\mathrm{C}_h(1+|\mathbf{x}|^2)} \le q_\sigma(t, \mathbf{x}) \le \mathrm{C}_h e^{-\mathrm{c}_h|\mathbf{x}|^2}. \tag{384}$$

This bound does not assert Gaussianity conditional on a state-dependent assignment history.

Differentiating one Ornstein–Uhlenbeck kernel against a function bounded by a polynomial times a Gaussian gives a bound of the same type with the factor $(t - r)^{-1/2}$: the kernel covariance is bounded above and below by multiples of $t - r$, its linear derivative is absorbed into a Gaussian with twice the covariance, and completing the square in the convolution concludes. This singularity is integrable in (383), which proves a first-derivative Gaussian bound. Moreover

$$\nabla b_\sigma = -\frac{b_\sigma}{T}\Big(H_\sigma \mathbf{x} - \sum_\eta b_\eta H_\eta \mathbf{x}\Big), \qquad |\nabla b_\sigma| \le \mathrm{K}_T |\mathbf{x}|,$$

so the source $b_\sigma Q$ has a first derivative with the same type of bound. For the second derivative, transfer one derivative to this source using $\nabla \mathcal{K}_\sigma(r) g = e^{rA_\sigma} \mathcal{K}_\sigma(r) \nabla g$, which follows by integration by parts in the Gaussian transition density, whose inverse mean map is uniformly bounded on this finite interval, and differentiate the other kernel factor; the singularity is again only $r^{-1/2}$. The explicit initial Gaussian term supplies the bounds near zero. The forward equation obtained from a time difference quotient in (383) then bounds the time derivative, and

$$|\nabla q_\sigma| + |\nabla^2 q_\sigma| + |\partial_t q_\sigma| \le \mathrm{C}_h (1 + |\mathbf{x}|)^\ell e^{-\mathrm{c}_h|\mathbf{x}|^2} \tag{385}$$

for some integer ℓ. These estimates justify the differentiations below without assuming finite Fisher information.

By (384) the logarithms of q_σ and $\mu_{p(t)}(\cdot, \sigma)$ grow at most quadratically, and $\partial_t \log \mu_p$ is bounded in $\mathbf{x}$, since only its prior weight and normalizer vary. Together with (385), this proves that H is finite, in particular $\mathsf{H}(0) < \infty$, and absolutely continuous on $[0, h]$, and permits differentiation under the integral. To integrate the diffusion term by parts, multiply first by a smooth radial cutoff χ_ϱ, equal to one on B_ϱ and zero outside $B_{2\varrho}$, with $|\nabla \chi_\varrho| \le \mathrm{C}/\varrho$, increasing to one as $\varrho \to \infty$. On each compact cylinder the densities have positive lower bounds, so $\log f_\sigma$ is a legitimate test function. The boundary term is bounded by

$$\frac{\mathrm{C}_h}{\varrho} \int_{\varrho \le |\mathbf{x}| \le 2\varrho} (1 + |\mathbf{x}|)^{\ell+3} e^{-\mathrm{c}_h|\mathbf{x}|^2} d\mathbf{x} \longrightarrow 0.$$

Detailed balance, $\mu_p(\mathbf{x}, \sigma) b_\eta(\mathbf{x}; p) = \mu_p(\mathbf{x}, \eta) b_\sigma(\mathbf{x}; p)$, symmetrizes the finite jump sum. Both resulting production integrands are nonnegative, so monotone convergence removes the cutoff and proves finiteness and local time integrability of the production. The jump term is integrable already by the Gaussian envelopes and the quadratic logarithmic bounds, so the cutoff argument does not presuppose a bound on the diffusion production. The result is (382), almost everywhere and in integrated form on every finite interval. □

J.8 Entropy contraction at frozen rows

For p in the open cube put

$$\underline{\kappa} = \min_\sigma \lambda_{\min}(B_\sigma) > 0, \qquad \bar{\kappa} = \max_\sigma \|B_\sigma\|, \qquad q^*_\sigma = \mu_p(\sigma) = \frac{\pi_\sigma \theta_\sigma}{\mathsf{Z}}, \qquad q^*_{\min} = \min_\sigma q^*_\sigma,$$

let $\gamma_\sigma = \mathcal{N}(0, B_\sigma^{-1})$, so that $\mu_p(\mathbf{x},\sigma)d\mathbf{x} = q_\sigma^* \gamma_\sigma(d\mathbf{x})$, and for a real function $u = (u_\sigma)_\sigma$ define

$$\mathcal{E}_c(u) = T\sum_\sigma \int \mu_p(\mathbf{x},\sigma)|\nabla u_\sigma|^2 d\mathbf{x}, \qquad \mathcal{E}_j(u) = \frac{1}{2}\sum_{\sigma,\eta}\int \mu_p(\mathbf{x},\sigma) b_\eta(\mathbf{x};p)(u_\sigma - u_\eta)^2 d\mathbf{x}.$$

Lemma 13 (Entropy contraction)**.** *With* $p_{\min} = \min_i\{p_i, 1-p_i\} > 0$ *and* $\mathrm{Ent}_\nu(f) = \int f \log(f/\int f d\nu) d\nu$ *for* $f \geq 0$*, every probability density* f *with respect to* μ_p *in the form domain satisfies*

$$\mathcal{P}_p(f) \geq \lambda(T,p)\,\mathrm{Ent}_{\mu_p}(f), \qquad \lambda(T,p) \geq \frac{\mathrm{c}_T}{\mathrm{K}_T + 8\log(1/p_{\min})}. \tag{386}$$

Proof. Gaussian inequalities. The Ornstein–Uhlenbeck semigroup P_t with generator $\Delta - B_\sigma \mathbf{x}\cdot\nabla$ has invariant law γ_σ, Dirichlet form $\int |\nabla g|^2 d\gamma_\sigma$, and gradient commutation $\nabla P_t g = e^{-tB_\sigma} P_t \nabla g$. Differentiating entropy and applying the Cauchy–Schwarz inequality inside the kernel, $(P_t|\nabla g|)^2 \leq P_t(|\nabla g|^2/g)P_t g$, gives

$$-\frac{d}{dt}\,\mathrm{Ent}_{\gamma_\sigma}(P_t g) = \int \frac{|\nabla P_t g|^2}{P_t g} d\gamma_\sigma \leq e^{-2\underline{\kappa} t}\int \frac{|\nabla g|^2}{g} d\gamma_\sigma.$$

For bounded positive smooth g the explicit Gaussian formula and dominated convergence make the entropy tend to zero. Integrating in t, with $g = u^2$, and differentiating the variance along the same semigroup,

$$\mathrm{Ent}_{\gamma_\sigma}(u^2) \leq \frac{2}{\underline{\kappa}}\int |\nabla u|^2 d\gamma_\sigma, \qquad \mathrm{Var}_{\gamma_\sigma}(u) \leq \frac{1}{\underline{\kappa}}\int |\nabla u|^2 d\gamma_\sigma. \tag{387}$$

Positive regularization, truncation and smooth cutoffs extend both estimates to their form domains: the cutoff errors tend to zero by integrability of the squared function and its gradient, and the entropy passes to the limit by its variational formula.

Mixture decomposition. Replace u by $|u|$, which decreases both energies, and normalize $\int u^2 d\mu_p = 1$. Put $z_\sigma = (\int u_\sigma^2 d\gamma_\sigma)^{1/2}$. The chain rule for entropy gives

$$\mathrm{Ent}_{\mu_p}(u^2) = \sum_\sigma q_\sigma^*\,\mathrm{Ent}_{\gamma_\sigma}(u_\sigma^2) + \mathrm{Ent}_{q^*}(z^2),$$

with $0 \leq z_\sigma \leq R_* = (q_{\min}^*)^{-1/2}$. The function $\psi(z) = z^2\log z^2 - z^2 + 1$ satisfies $\psi(1) = \psi'(1) = 0$ and $\psi''(z) = 4 + 4\log z$, so $\psi(z) \leq 2(1+\log R_*)(z-1)^2$ on $[0, R_*]$. Since $\sum_\sigma q_\sigma^* z_\sigma^2 = 1$, $\mathrm{Ent}_{q^*}(z^2) = \sum_\sigma q_\sigma^* \psi(z_\sigma)$ and $\sum_\sigma q_\sigma^*(z_\sigma - 1)^2 = 2(1 - \mathbb{E}_{q^*} z) \leq 2\,\mathrm{Var}_{q^*}(z)$, whence

$$\mathrm{Ent}_{q^*}(z^2) \leq L_p\,\mathrm{Var}_{q^*}(z), \qquad L_p = 4 + 2\log(1/q_{\min}^*).$$

Variance of the sheet masses. Fix a ball B_ϱ in $\mathbb{R}^{12}$ and define

$$\bar{u}_\sigma = |B_\varrho|^{-1}\int_{B_\varrho} u_\sigma, \quad \bar{u}_* = \sum_\sigma \pi_\sigma \bar{u}_\sigma, \quad \chi_\sigma = \left[\frac{(2\pi)^6 \theta_\sigma e^{\varrho^2\|B_\sigma\|/2}}{|B_\varrho|} - 1\right]_+, \quad \mathsf{B}_T = \max_\sigma (1+\sqrt{\chi_\sigma})^2.$$

Variance about any center bounds the variance, so

$$\mathrm{Var}_{q^*}(z) \leq 2\sum_\sigma q_\sigma^*(z_\sigma - \bar{u}_\sigma)^2 + 2\sum_\sigma q_\sigma^*(\bar{u}_\sigma - \bar{u}_*)^2.$$

For the first sum, $|z_\sigma - \gamma_\sigma u_\sigma| \leq \sqrt{\mathrm{Var}_{\gamma_\sigma}(u_\sigma)}$, and the Cauchy–Schwarz inequality bounds $|\bar{u}_\sigma - \gamma_\sigma u_\sigma|$ by the same standard deviation times $\sqrt{\chi^2(\mathrm{Unif}(B_\varrho)\|\gamma_\sigma)} \leq \sqrt{\chi_\sigma}$, because the density of γ_σ is at

least $(2\pi)^{-6}\theta_\sigma^{-1}e^{-\varrho^2\|B_\sigma\|/2}$ on the ball. With (387), the first sum is at most $(\mathsf{B}_T/(T\underline{\kappa}))\mathcal{E}_c(u)$. For the second sum, Jensen's inequality and $q_\sigma^* \le \pi_\sigma\theta_{\max}/\mathsf{Z}$ give

$$\sum_\sigma q_\sigma^*(\bar u_\sigma - \bar u_*)^2 \le \frac{\theta_{\max}}{\mathsf{Z}|B_\varrho|}\sum_{\sigma,\eta}\pi_\sigma\pi_\eta\int_{B_\varrho}(u_\sigma-u_\eta)^2.$$

On B_ϱ the explicit Gaussian densities give

$$\mu_p(\mathbf{x},\sigma)b_\eta(\mathbf{x};p) \ge \frac{\pi_\sigma\pi_\eta e^{-\varrho^2\bar\kappa}}{(2\pi)^6\mathsf{Z}}.$$

The normalizer therefore cancels, and the preceding sum is at most $2(2\pi)^6\theta_{\max}e^{\varrho^2\bar\kappa}\mathcal{E}_j(u)/|B_\varrho|$. Combining these bounds proves

$$\mathrm{Ent}_{\mu_p}(u^2) \le \frac{2}{T\underline{\kappa}}(1+L_p\mathsf{B}_T)\mathcal{E}_c(u) + \frac{4L_p(2\pi)^6\theta_{\max}e^{\varrho^2\bar\kappa}}{|B_\varrho|}\mathcal{E}_j(u).$$

Production. For $u=\sqrt{f}$ the production is at least $4(\mathcal{E}_c(u)+\mathcal{E}_j(u))$: the diffusion part equals $4\mathcal{E}_c(u)$, and the jump part uses $(a-b)(\log a-\log b)\ge 4(\sqrt{a}-\sqrt{b})^2$, which follows from the Cauchy–Schwarz inequality applied to the integral of $r^{-1/2}$ between a and b. The inequality extends to the form domain by the preceding approximations. Finally $q_{\min}^* \ge p_{\min}^4\theta_{\min}/\theta_{\max}$, because $\pi_\sigma \ge p_{\min}^4$ and $\mathsf{Z}\le\theta_{\max}$, so $L_p \le 4+2\log(\theta_{\max}/\theta_{\min})+8\log(1/p_{\min})$, and (386) follows with constants depending only on T. The factor T in $\mathcal{E}_c$ is the diffusion coefficient of (382). □

J.9 The matching branch of the small-temperature landscape

Lemma 14 (Landscape)**.** (a) Sheet weights. *The sheet weight θ_σ is constant on orbits of the cycle automorphisms. With $\mathsf{d}_\sigma=\det A_\sigma$, the two matchings, eight end-path assignments, four middle-path assignments and two oriented cycles, with representative multiplicity patterns $(2,0,2,0)$, $(2,1,1,0)$, $(1,2,1,0)$ and $(1,1,1,1)$, have*

$$\begin{aligned}
\mathsf{d}_M &= 25T^8(T+8)^2(5T+4)^2,\\
\mathsf{d}_E &= 5T^6(T^3+16T^2+77T+112)(125T^3+200T^2+85T+8),\\
\mathsf{d}_N &= 5T^6(T+4)(5T+2)(T^2+12T+28)(25T^2+30T+4),\\
\mathsf{d}_C &= 5T^5(T+2)(T+4)^2(T+6)(5T+2)^2(5T+4).
\end{aligned}$$

The ratios $\mathsf{r}_E=\sqrt{\mathsf{d}_M/\mathsf{d}_E}$, $\mathsf{r}_N=\sqrt{\mathsf{d}_M/\mathsf{d}_N}$ and $\mathsf{r}_C=\sqrt{\mathsf{d}_M/\mathsf{d}_C}$ of the corresponding sheet weights to the matching weight satisfy $\mathsf{r}_E,\mathsf{r}_N=\sqrt{40/7}T+O(T^2)$ and $\mathsf{r}_C=\sqrt{5/3}T^{3/2}(1+O(T))$. Choose $T_0>0$ with $0<\mathsf{r}_E,\mathsf{r}_N,\mathsf{r}_C\le\frac18$ for $0<T\le T_0$.

(b) Saddle. *For $T\le T_0$, the Jacobian of Φ at $p=\frac12$ is symmetric, with eigenvalues*

$$\begin{aligned}
\alpha = \frac{3-2\mathsf{r}_N-\mathsf{r}_C}{\mathsf{D}_0} \ge \frac75, \qquad \alpha_{\mathrm{alt}} = \frac{-1-2\mathsf{r}_N+3\mathsf{r}_C}{\mathsf{D}_0} \le -\frac13,\\
\alpha_{\mathrm{opp}} = \frac{-1+2\mathsf{r}_N-\mathsf{r}_C}{\mathsf{D}_0} \le -\frac25 \quad \textit{(twice)},
\end{aligned} \tag{388}$$

where $\mathsf{D}_0=1+4\mathsf{r}_E+2\mathsf{r}_N+\mathsf{r}_C$; the eigenvector of α is $\mathbf{v}$. The centered field $y\mapsto\Phi(\frac12+y)$ is odd, and the line $\frac12+\mathbb{R}\mathbf{v}$ is invariant.

(c) Diagonal. *On that line the reduced flow $\dot p = \Phi(p)$ moves from $\frac12 + a\mathbf{v}/2$ with $0 < a < 1$ into every prescribed neighborhood of the vertex $\sigma = \mathbf{v}$ in finite time, and from $-1 < a < 0$ into every neighborhood of the vertex $\sigma = -\mathbf{v}$.*

(d) Vertices. *Both matching vertices are sinks. At each, $|g_i| = \nu = 1 - \mathsf{r}_E > 0$ with g_i directed toward the vertex, and the Jacobian of Φ is $-\nu I$.*

Proof. (a) An automorphism g gives $A_{g\sigma} = O_g A_\sigma O_g^\top$, so θ_σ is constant on orbits; the sixteen assignments fall into the four listed classes of sizes 2, 8, 4 and 2. In the orthonormal coordinates $u = (k+m)/\sqrt2$, $v = (k-m)/\sqrt2$ and w, the residuals are $r^b = (Du + Dv)/\sqrt2 - w$ and $r^m = (Du - Dv)/\sqrt2 - w$, so

$$V_\sigma = \tfrac12 (Du - \sqrt2 w)^\top C_\sigma (Du - \sqrt2 w) + \tfrac12 (Dv)^\top C_\sigma Dv,$$

while $\mathbf{x}^\top R_{\mathrm{ev}} \mathbf{x} = 2|k-m|^2 = 4|v|^2$, so the discrepancy coordinate v has reference precision $5I$. The (u,w) block of A_σ is $TI_8 + \mathsf{F}^\top C_\sigma \mathsf{F}$ with $\mathsf{F} = (D, -\sqrt2 I_4)$, and Sylvester's determinant identity gives $\det(TI_8 + \mathsf{F}^\top C_\sigma \mathsf{F}) = T^4 \det(TI_4 + C_\sigma \mathsf{F}\mathsf{F}^\top)$ with $\mathsf{F}\mathsf{F}^\top = DD^\top + 2I$. Hence

$$\det A_\sigma = T^4 \det\big[TI + C_\sigma (DD^\top + 2I)\big] \det\big[5TI + D^\top C_\sigma D\big]. \tag{389}$$

Evaluating (389) on the four representative patterns gives the displayed determinants, and $\theta_\sigma/\theta_M = \sqrt{\mathsf{d}_M/\mathsf{d}_\sigma}$. Expanding the ratios at small T gives their leading terms. The ratios are continuous on $(0,\infty)$ and tend to zero with T, so T_0 exists.

(b) Use the matching coordinates $x = (p_0, 1-p_1, p_2, 1-p_3)$, in which x_i is the prior probability that receiver i makes its $J^{(0)}$ choice. A vertex of the cube in these coordinates is the set $\mathcal{A}$ of receivers agreeing with $J^{(0)}$. Normalizing the matching weight to one, the evidence Z is the multilinear polynomial with vertex weights 1 at $\mathcal{A} = \emptyset$ and $\mathcal{A} = \{0,1,2,3\}$, r_E for $|\mathcal{A}| \in \{1,3\}$, r_N for the four adjacent pairs and r_C for the two opposite pairs. For example, $\mathcal{A} = \{0\}$ is $\sigma = (1,1,-1,1)$ with pattern $(1,2,0,1)$, an end path, and $\mathcal{A} = \{0,2\}$ is $\sigma = (1,1,1,1)$, an oriented cycle. In these coordinates the field is $x_i(1-x_i)\partial_{x_i} \log \mathsf{Z}$, because $x \mapsto p$ is affine with diagonal signs. Complementation $\mathcal{A} \mapsto \mathcal{A}^c$ and the cyclic rotation of receivers preserve these weights, so the centered field is odd and commutes with the cyclic rotation of coordinates, which makes the diagonal $x = z\mathbf{1}$, that is the line $\frac12 + \mathbb{R}\mathbf{v}$, invariant. At $x = \frac12\mathbf{1}$ the gradient of $\log \mathsf{Z}$ vanishes, the row mobility is $\frac14$, the uniform value of Z is $\mathsf{D}_0/8$, the adjacent Hessian entry of Z is $(1-\mathsf{r}_C)/2$ and the opposite one is $(1 - 2\mathsf{r}_N + \mathsf{r}_C)/2$. The Jacobian is therefore the circulant

$$\frac{1}{\mathsf{D}_0} \begin{pmatrix} 0 & 1-\mathsf{r}_C & 1-2\mathsf{r}_N+\mathsf{r}_C & 1-\mathsf{r}_C \\ 1-\mathsf{r}_C & 0 & 1-\mathsf{r}_C & 1-2\mathsf{r}_N+\mathsf{r}_C \\ 1-2\mathsf{r}_N+\mathsf{r}_C & 1-\mathsf{r}_C & 0 & 1-\mathsf{r}_C \\ 1-\mathsf{r}_C & 1-2\mathsf{r}_N+\mathsf{r}_C & 1-\mathsf{r}_C & 0 \end{pmatrix},$$

with eigenvectors $\mathbf{1}$, $(1,-1,1,-1)$, $(1,0,-1,0)$ and $(0,1,0,-1)$ and the eigenvalues (388). The direction $\mathbf{1}$ in matching coordinates is $\mathbf{v}$ in the original rows, and the change of coordinates is a signed diagonal map, so the Jacobian is symmetric in the original rows too. With $\mathsf{r}_E, \mathsf{r}_N, \mathsf{r}_C \in (0, \frac18]$ we have $\mathsf{D}_0 \le \frac{15}{8}$, which gives the three numerical bounds.

(c) On the diagonal write $\varpi = z(1-z)$. The normalized evidence and the reduced motion are

$$\mathsf{Z}_{\mathrm{diag}}(z) = 1 + 4(\mathsf{r}_E - 1)\varpi + (2 - 8\mathsf{r}_E + 4\mathsf{r}_N + 2\mathsf{r}_C)\varpi^2, \qquad \dot z = \frac{z(1-z)}{4\mathsf{Z}_{\mathrm{diag}}(z)} \mathsf{Z}'_{\mathrm{diag}}(z).$$

The derivative is $(1-2z)[4(\mathsf{r}_E - 1) + 2(2 - 8\mathsf{r}_E + 4\mathsf{r}_N + 2\mathsf{r}_C)\varpi]$. The bracket is affine in ϖ and negative at both endpoints $\varpi = 0$ and $\varpi = \frac14$, where it equals $4(\mathsf{r}_E - 1)$ and $-3 + 2\mathsf{r}_N + \mathsf{r}_C$. It is

therefore negative throughout, so the diagonal motion tends to $z = 1$ from $z > \frac{1}{2}$ and to $z = 0$ from $z < \frac{1}{2}$, reaching every prescribed endpoint neighborhood in finite time.

(d) At the vertex $x = \mathbf{1}$ a single receiver flip changes the weight one to r_E, so $\partial_{x_i} \log \mathsf{Z} = 1 - \mathsf{r}_E = \nu$ there, directed toward the vertex. The mobility $x_i(1 - x_i)$ vanishes at the vertex with derivative -1, so the Jacobian is $-\nu I$. The vertex $x = 0$ follows by complementation. □

J.10 Tracking, escape and selection

Tracking. Let $\hat{q}_i(t) = q_t(\sigma_i = +1)$ and $\Delta_i(t) = \hat{q}_i(t) - \bar{\beta}_i(p(t))$, so that the learning equation reads $\dot{p}_i = \varepsilon(\Delta_i + \Phi_i(p))$. Differentiating the normalized product prior gives

$$\partial_{p_i} \log \mu_p(\mathbf{x}, \sigma) = \frac{1_{\{\sigma_i = +1\}} - \bar{\beta}_i(p)}{p_i(1 - p_i)}.$$

Inserting the learning equation and (369) in (382) yields

$$\begin{aligned} \mathsf{H}' &= -\mathcal{P}_p(f) - \varepsilon \sum_i g_i(p)\Delta_i - \varepsilon \sum_i \frac{\Delta_i^2}{p_i(1 - p_i)} \\ &\le -\lambda(T, p)\mathsf{H} + \varepsilon \|g(p)\|_1 \sqrt{\mathsf{H}/2}. \end{aligned} \tag{390}$$

The last bound uses Lemma 13 and $|\Delta_i| \le \|q_t - \mu_{p(t)}\|_{\mathrm{TV}} \le \sqrt{\mathsf{H}/2}$ by Pinsker's inequality. The nonpositive square is kept until this step, so no inverse row factor enters the forcing. Writing $\Lambda(t) = \int_0^t \lambda(T, p(r))dr$, comparison for $\sqrt{\mathsf{H} + \delta^2}$, followed by $\delta \downarrow 0$, gives

$$\sqrt{\mathsf{H}(t)} \le e^{-\Lambda(t)/2} \sqrt{\mathsf{H}(0)} + \frac{\varepsilon}{2\sqrt{2}} \int_0^t e^{-(\Lambda(t) - \Lambda(r))/2} \|g(p(r))\|_1 dr. \tag{391}$$

Let C_T be the window constant fixed below. On $t \le 2C_T \varepsilon^{-1} \log(1/\varepsilon)$, (371) gives $p_{\min} \ge \frac{1}{2}\varepsilon^{2C_T}$, so by (386) λ is bounded below by a positive constant divided by $1 + \log(1/\varepsilon)$. Since g is bounded, on this horizon

$$\|q_t - \mu_{p(t)}\|_{\mathrm{TV}} \le e^{-\Lambda(t)/2} \sqrt{\mathsf{H}(0)/2} + O_T(\varepsilon \log(1/\varepsilon)). \tag{392}$$

Near the saddle a finer bound is needed. Let $\|g(p) - g(p')\|_1 \le L_T \|p - p'\|_\infty$ on the cube, and suppose $\lambda \ge \lambda_\flat > 0$ on $[s_0, t]$. Inserting $\|g(p(r))\|_1 \le \|g(p(t))\|_1 + L_T \varepsilon (t - r)$, which follows from $|\dot{p}_i| \le \varepsilon$, in (391) restarted at s_0, and integrating 1 and $t - r$ against $e^{-\lambda_\flat (t-r)/2}$, gives

$$\|q_t - \mu_{p(t)}\|_{\mathrm{TV}} \le e^{-\lambda_\flat (t - s_0)/2} \sqrt{\mathsf{H}(s_0)/2} + \frac{\varepsilon}{2\lambda_\flat} \|g(p(t))\|_1 + \frac{L_T \varepsilon^2}{\lambda_\flat^2}. \tag{393}$$

Escape from the saddle. Fix $0 < T \le T_0$ and a mean with $\mathrm{j}_T(\mathbf{m}_0) \ne 0$; by Lemma 11 this excludes only a Lebesgue-null set. Put $\mathbf{e} = \mathbf{v}/2$, a unit vector. In slow time $\mathfrak{s} = \varepsilon t$, write $y = p - \frac{1}{2} = a\mathbf{e} + \mathbf{z}$ with $\mathbf{z} \perp \mathbf{e}$, so that $dy/d\mathfrak{s} = \Phi(\frac{1}{2} + y) + \Delta$. By (377), at $t_1 = \varepsilon^{-1/4}$,

$$a(t_1) = \varepsilon\big(\mathrm{j}_T(\mathbf{m}_0)/2 + o(1)\big), \qquad |\mathbf{z}(t_1)| = O(\varepsilon).$$

On $[0, t_1]$, $g(\frac{1}{2}) = 0$ (Lemma 7) gives $\|g(p(t))\|_1 \le L_T \varepsilon t/2$, and λ is bounded below by a positive constant, so (391) gives $\sqrt{\mathsf{H}(t_1)} = O(\varepsilon^2 t_1) = O(\varepsilon^{7/4})$. Consequently, until exit from any sufficiently small fixed ball about $\frac{1}{2}$, (393) with $s_0 = t_1$ and $\|g(p)\|_1 \le L_T |y|$ implies

$$|\Delta|_2 \le \kappa_\varepsilon |y| + b_\varepsilon, \qquad \kappa_\varepsilon = O(\varepsilon), \qquad b_\varepsilon = O(\varepsilon^{7/4}) = o(|a(t_1)|), \tag{394}$$

with constants that may depend on the fixed mean.

Let $\alpha_\perp = \max(\alpha_{\mathrm{alt}}, \alpha_{\mathrm{opp}}) < 0$ and $\gamma = \alpha - \alpha_\perp > 0$. Oddness and diagonal invariance of the analytic reduced field imply, for $|y| \le r$,

$$\big|\langle \Phi(\tfrac12 + y), \mathbf{e}\rangle - \alpha a\big| \le \mathrm{C}|y|^3, \qquad \big|P_\perp \Phi(\tfrac12 + y) - D\Phi(\tfrac12)\mathbf{z}\big| \le \mathrm{C}|y|^2|\mathbf{z}|,$$

where $P_\perp$ is the orthogonal projection onto $\mathbf{e}^\perp$. The first follows because the field has no quadratic term and $\mathbf{e}$ is an eigenvector of the symmetric Jacobian. For the second, subtract the perpendicular field on $a\mathbf{e}$, which vanishes by diagonal invariance, and integrate its derivative along the transverse segment; the derivative differs from its value at the center by $O(|y|^2)$ because the field is odd, and $D\Phi(\frac12)$ preserves $\mathbf{e}^\perp$. Writing $\omega = |\mathbf{z}|/|a|$, the upper right derivatives of norms give

$$\begin{aligned} \frac{d|a|}{d\mathfrak{s}} &\ge |a|\big[\alpha - (\mathrm{C}r^2 + \kappa_\varepsilon)(1+\omega)\big] - b_\varepsilon, \\ D^+\omega &\le -\big[\gamma - \mathrm{C}r^2 - (\mathrm{C}r^2 + \kappa_\varepsilon)(1+\omega)\big]\omega + (1+\omega)(\kappa_\varepsilon + b_\varepsilon/|a|). \end{aligned} \tag{395}$$

Choose ω_* larger than twice the limiting initial ratio $\lim_{\varepsilon\to 0}\omega(t_1)$, which is finite, then choose $r > 0$ so small that the terms $\mathrm{C}r^2(1+\omega_*)$ consume less than one quarter of both rates α and γ. For small enough ε, (394) controls the remaining terms. A bootstrap then keeps $\omega \le \omega_*$, makes $|a|$ increase without changing sign, and gives

$$\omega(\mathfrak{s}) \le \omega(\mathfrak{s}_1)e^{-\gamma(\mathfrak{s}-\mathfrak{s}_1)/2} + \frac{2(1+\omega_*)}{\gamma}\big(\kappa_\varepsilon + b_\varepsilon/|a(t_1)|\big), \qquad \mathfrak{s}_1 = \varepsilon t_1.$$

The lower growth bound for $|a|$ yields an exit time from the ball of radius r no larger than

$$\frac{1 + O(r^2) + o(1)}{\alpha}\log(1/\varepsilon)$$

in slow time, and the corresponding upper growth bound for $|y|$ forces the exit time to tend to infinity. Hence $\omega \to 0$ at exit, and the exit point tends to $\mathrm{sign}(\mathrm{j}_T(\mathbf{m}_0))r\mathbf{e}$.

Transit and capture. Let $M = M_0$ if $\mathrm{j}_T(\mathbf{m}_0) > 0$ and $M = M_1$ otherwise, and let p^M be the corresponding vertex, $\sigma = \mathbf{v}$ or $\sigma = -\mathbf{v}$. By Lemma 14(c) the reduced diagonal orbit from $\frac12 \pm r\mathbf{e}$ enters any fixed neighborhood of p^M in finite slow time. After t_1 the tracking error in (392) tends to zero, because $\Lambda(t)$ grows at least as a positive multiple of ε^{-1} once t is of order ε^{-1}; finite-time Grönwall comparison therefore carries the evolving rows into the same neighborhood. At p^M let $\zeta_i = |p_i - p_i^M|$. By Lemma 14(d) and continuity, a sufficiently small neighborhood has $(1-\zeta_i)|g_i(p)| \ge \nu/2$ with g_i directed toward the vertex. Since $\Phi_i = p_i(1-p_i)g_i$ and $p_i(1-p_i) = \zeta_i(1-\zeta_i)$ there, in slow time

$$D^+\zeta_i \le -\tfrac{\nu}{2}\zeta_i + \|q_t - \mu_{p(t)}\|_{\mathrm{TV}}.$$

The neighborhood remains invariant once the tracking floor is small enough, and ζ_i decays exponentially to a vanishing floor.

Selection window. Choose any $C_T > 1/\alpha$, independently of the mean; by (388) every $C_T > \frac57$ is admissible. Then choose the local radius r so that the coefficient of $\log(1/\varepsilon)$ in the exit time is less than C_T. The slow time remaining after the finite transit tends to infinity uniformly on the slow-time window $[C_T\log(1/\varepsilon), 2C_T\log(1/\varepsilon)]$, which is the window $t_L \le t \le 2t_L$ of the theorem. The row distances therefore vanish uniformly on that window. Every normalized sheet weight is at most the matching weight one, so

$$\mu_p(S = M) \ge \pi_M \ge 1 - \sum_i \zeta_i, \qquad q_t(S \ne M) \le \sum_i \zeta_i + \|q_t - \mu_{p(t)}\|_{\mathrm{TV}}.$$

The total variation tends to zero uniformly on the window by (392), since the integrated contraction exponent at the beginning of the window grows at least as a positive constant times ε^{-1}. This proves part (i) of Theorem 7 for every mean outside the null zero set of j_T, with T_0 from Lemma 14(a).

Frozen arm. For part (ii), fix any $T > 0$ and $\mathbf{m}_0$. Integrating (374) at $p = \frac{1}{2}$ against the initial law, which has a finite second moment, gives $\|q_t^0 - \mu_{1/2}\|_{\mathrm{TV}} \le \mathrm{K}_{T,\mathbf{m}_0} e^{-\lambda_T t}$, and $\mu_{1/2}(S = M_0) = \mu_{1/2}(S = M_1)$ by Lemma 7. Hence $q_t^0(S = M_0) - q_t^0(S = M_1) \to 0$.

The convergence in part (i) is pointwise in the fixed mean; no bound uniform near the zero set of the imprint has been used. The asymmetry is supplied by the initial mean through $\mathrm{j}_T(\mathbf{m}_0)$, and slow learning amplifies it along the expanding saddle direction $\mathbf{v}$, whereas at frozen rows the same initial asymmetry decays at the rate λ_T. □

K Proofs for the Collective Evidence Identity, Latent Lie-Algebra Links and Coarse Link Laws

This appendix proves the statements of Sections 5, 6 and 12 that concern the collective evidence identity, latent Lie-algebra links and coarse link laws. It uses the following standard facts with references: the chain rule for relative entropy [24] and the disintegration of laws on standard Borel spaces [23], the elementary theory of exponential families [63] and of convex conjugates [64], the naturality of the exponential map and the Baker–Campbell–Hausdorff series [65], the matrix exponential and logarithm [66], and Culver's criterion for real matrix logarithms [67].

Conventions. All laws live on standard Borel spaces. For laws Q and P, $D_{\mathrm{KL}}(Q\|P) = \int \log(dQ/dP)dQ$ when $Q \ll P$, and $D_{\mathrm{KL}}(Q\|P) = +\infty$ otherwise. In the first case the negative part of the integrand has Q-integral at most $1/e$, because $\int_{\{dQ/dP<1\}} -\log(dQ/dP)dQ = \int_{\{dQ/dP<1\}} -(dQ/dP)\log(dQ/dP)dP$ and $-u\log u \le 1/e$ on $[0,1]$; hence $D_{\mathrm{KL}}(Q\|P) \in [0,+\infty]$, with value zero exactly when $Q = P$. If Q and P have densities q and p with respect to a σ-finite reference, then $Q \ll P$ exactly when $Q(\{p=0\}) = 0$, since a P-null set A has $A \cap \{p>0\}$ null for the reference. The chain rule states that for laws Q and P on a product $\mathsf{Y}_1 \times \mathsf{Y}_2$ with first marginals Q_1, P_1 and conditional kernels $Q_{2|1}, P_{2|1}$, the conditional divergence $y_1 \mapsto D_{\mathrm{KL}}(Q_{2|1}(y_1,\cdot)\|P_{2|1}(y_1,\cdot))$ is measurable and

$$D_{\mathrm{KL}}(Q\|P) = D_{\mathrm{KL}}(Q_1\|P_1) + \int D_{\mathrm{KL}}\big(Q_{2|1}(y_1,\cdot)\big\|P_{2|1}(y_1,\cdot)\big)Q_1(dy_1) \quad \text{in } [0,+\infty]. \tag{396}$$

Matter coordinates, including source labels when present, are written y. For a link family Θ the held-link population law is $\mathbb{P}_\Theta(dy,do) = P_\Theta^Y(dy)K_\Theta(y,do)$, with the proper matter prior P_Θ^Y, which is $P_0\Pi$ of Eq. (21) at the links Θ, and a normalized record kernel with density $\ell_o(y;\Theta)$ against a reference ν^O on records, jointly measurable in (Θ,y,o). At a record o with $0 < p(o \mid \Theta) < \infty$ its evidence, posterior and free energy are

$$\begin{gathered} p(o \mid \Theta) = \int \ell_o(y;\Theta)P_\Theta^Y(dy), \qquad \Pi_\Theta = \frac{\ell_o(\cdot;\Theta)P_\Theta^Y}{p(o \mid \Theta)}, \\ \mathcal{F}_V[Q';X,\Theta,o] = D_{\mathrm{KL}}(Q'\|\Pi_\Theta) - \log p(o \mid \Theta). \end{gathered} \tag{397}$$

At $\Theta(\mathring{\omega})$ we write $p(o \mid \mathring{\omega})$, $\Pi_{\mathring{\omega}}$ and $P_{\mathring{\omega}}^Y$. On $\mathcal{T} = \mathfrak{g}^{E_+}$, gradients, Hessians, adjoints, traces and determinants refer to the product of the inner products chosen on its summands, its norm $\|\cdot\|$ and the Lebesgue measure normalized by it; on matrices $\|\cdot\|$ is the operator norm unless a subscript says otherwise. Constants $c, c_1, c_2, \dots$ are local to one statement.

K.1 The collective evidence identity

Let P^Y be a proper prior on the matter space, $\ell_o \ge 0$ a measurable record likelihood with evidence $Z_o = \int \ell_o dP^Y \in (0,\infty)$, and $\Pi_o = Z_o^{-1}\ell_o P^Y$ the posterior. For every law Q on the matter space put

$$\mathcal{F}_V[Q;X,o] = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o \in [-\log Z_o, +\infty]. \tag{398}$$

For Eq. (22), $P^Y = P_0\Pi$ and $\Pi_o = \mathbb{P}(\cdot \mid o, X)$.

Proposition 17 (Collective evidence identity)**.** *(a) $\mathcal{F}_V[Q;X,o] \ge -\log Z_o$, with equality exactly when $Q = \Pi_o$. (b) If $D_{\mathrm{KL}}(Q\|P^Y) < \infty$ and $\mathbb{E}_Q \max(\log \ell_o, 0) < \infty$, then $\mathcal{F}_V[Q;X,o] = D_{\mathrm{KL}}(Q\|P^Y) - \mathbb{E}_Q \log \ell_o$ in $(-\infty,+\infty]$, where $\log 0 = -\infty$ and $\mathbb{E}_Q \log \ell_o \in [-\infty,+\infty)$. (c) If Q and P^Y have densities q and p^Y with respect to a σ-finite reference, and $\mathbb{E}_Q|\log q|$ and $\mathbb{E}_Q|\log(\ell_o p^Y)|$ are finite, then $\mathcal{F}_V[Q;X,o] = \mathbb{E}_Q[\log q - \log(\ell_o p^Y)]$.*

Proof. Part (a) is the nonnegativity of relative entropy and its zero case. For (b), let $M = \ell_o P^Y$, a measure of total mass Z_o with $\Pi_o = M/Z_o$. Suppose first that $Q \ll M$, and let $r = dQ/dM$. Then $dQ/d\Pi_o = Z_o r$, and $\int_{\{r<1\}} -\log r dQ = \int_{\{r<1\}} -r \log r dM \le Z_o/e$, so $D_{\mathrm{KL}}(Q\|\Pi_o) = \log Z_o + \int \log r dQ$ with an integral in $(-\infty, +\infty]$. Since $M \ll P^Y$, $dQ/dP^Y = r\ell_o$, and $Q(\{\ell_o = 0\}) = \int_{\{\ell_o=0\}} r dM = 0$, so $\log r = \log(dQ/dP^Y) - \log \ell_o$ holds Q-almost surely. The first term is Q-integrable because $D_{\mathrm{KL}}(Q\|P^Y)$ is finite and its negative part is integrable, and the second has a Q-integrable negative part by hypothesis, so the integral splits and gives the claim. Suppose next that $Q \not\ll M$, and choose S with $M(S) = 0 < Q(S)$. Then $\Pi_o(S) = 0$, so the left side is $+\infty$. Since $\int_S \ell_o dP^Y = 0$, ℓ_o vanishes P^Y-almost everywhere on S, hence Q-almost everywhere on S because $Q \ll P^Y$; thus $\mathbb{E}_Q \log \ell_o = -\infty$ and the right side is $+\infty$ as well. For (c), Π_o has the density $\ell_o p^Y/Z_o$, which is positive Q-almost surely because $\log(\ell_o p^Y)$ is Q-integrable, so $Q \ll \Pi_o$ with $dQ/d\Pi_o = Z_o q/(\ell_o p^Y)$ Q-almost surely. Its logarithm is a sum of three Q-integrable terms, and integration gives $D_{\mathrm{KL}}(Q\|\Pi_o) = \log Z_o + \mathbb{E}_Q[\log q - \log(\ell_o p^Y)]$. □

K.2 The loop defect and the generator prior

Let $\rho : G \to \mathrm{GL}(\mathbb{V})$ be a smooth representation on $\mathbb{V} = \mathbb{R}^{n_V}$ with differential $\mathrm{d}\rho$, and write $\mathsf{G} = \mathsf{G}^{\mathrm{fib}}$ for the positive-definite fiber matrix of Eq. (24). For matrices on $\mathbb{V}$ put $\langle \mathbf{H}, \mathbf{H}' \rangle_{\mathsf{G}} = \mathrm{tr}(\mathsf{G}^{-1}\mathbf{H}^\top \mathsf{G}\mathbf{H}')$ and $\mathbf{H}^* = \mathsf{G}^{-1}\mathbf{H}^\top\mathsf{G}$, so that $\langle \mathbf{H}, \mathbf{H}' \rangle_{\mathsf{G}} = \mathrm{tr}(\mathbf{H}^*\mathbf{H}')$; a different subscript on the norm names the matrix used in place of G. Two identities of Lie theory are used repeatedly [65]: $\rho(\exp \xi) = \exp(\mathrm{d}\rho\xi)$, and $h \exp(\xi) h^{-1} = \exp(\mathrm{Ad}_h \xi)$ with $\mathrm{d}\rho(\mathrm{Ad}_h \xi) = \rho(h)\mathrm{d}\rho(\xi)\rho(h)^{-1}$. For a closed walk $p = (e_0, \dots, e_{r-1})$ the reference holonomy $\mathring{\Theta}_p$ is the ordered product of Eq. (24) with the reversal convention $\mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e$, and the cycle map is $\mathbf{C}_p \mathring{\omega} = \sum_{a=0}^{r-1} \mathring{\omega}_{e_a}$, which counts each traversal with its orientation.

Lemma 15 (Trace form)**.** *(a) $\langle \cdot, \cdot \rangle_{\mathsf{G}}$ is an inner product with $\|\mathbf{H}\mathbf{H}'\|_{\mathsf{G}} \le \|\mathbf{H}\|_{\mathsf{G}} \|\mathbf{H}'\|_{\mathsf{G}}$ and $\|I\|_{\mathsf{G}} = n_V^{1/2}$, and left or right multiplication by $\mathbf{H}$ has as adjoint the left or right multiplication by $\mathbf{H}^*$. (b) For positive-definite G' and G'', the norms $\|\cdot\|_{\mathsf{G}'}$ and $\|\cdot\|_{\mathsf{G}''}$ agree on every rank-one matrix exactly when G'' is a positive multiple of G', and then they agree on all matrices. (c) For invertible $\mathbf{R}$ and positive-definite G', $\|\mathbf{R}^{-1}\Delta\mathbf{R}\|_{\mathsf{G}'} = \|\Delta\|_{\mathbf{R}^{-\top}\mathsf{G}'\mathbf{R}^{-1}}$.*

Proof. (a) Cyclicity of the trace gives $\langle \mathbf{H}, \mathbf{H}' \rangle_{\mathsf{G}} = \mathrm{tr}((\mathsf{G}^{1/2}\mathbf{H}\mathsf{G}^{-1/2})^\top \mathsf{G}^{1/2}\mathbf{H}'\mathsf{G}^{-1/2})$, the Frobenius inner product of the images under the algebra isomorphism $\mathbf{H} \mapsto \mathsf{G}^{1/2}\mathbf{H}\mathsf{G}^{-1/2}$, which yields the inner product, submultiplicativity and the norm of I. Since $(\mathbf{A}\mathbf{B})^* = \mathbf{B}^*\mathbf{A}^*$ and $\mathbf{H}^{**} = \mathbf{H}$, $\mathrm{tr}(\mathbf{X}^*\mathbf{H}\mathbf{Y}) = \mathrm{tr}((\mathbf{H}^*\mathbf{X})^*\mathbf{Y})$ and $\mathrm{tr}(\mathbf{X}^*\mathbf{Y}\mathbf{H}) = \mathrm{tr}((\mathbf{X}\mathbf{H}^*)^*\mathbf{Y})$. (b) For $\mathbf{H} = \mathbf{b}\mathbf{c}^\top$, $\|\mathbf{H}\|^2_{\mathsf{G}'} = (\mathbf{b}^\top\mathsf{G}'\mathbf{b})(\mathbf{c}^\top\mathsf{G}'^{-1}\mathbf{c})$. If $\mathsf{G}'' = s\mathsf{G}'$ with $s > 0$, the factors s and s^{-1} cancel in the trace form for every matrix. Conversely, fixing $\mathbf{c} \neq 0$ gives $\mathbf{b}^\top\mathsf{G}''\mathbf{b} = s\mathbf{b}^\top\mathsf{G}'\mathbf{b}$ for every $\mathbf{b}$, with $s = (\mathbf{c}^\top\mathsf{G}'^{-1}\mathbf{c})/(\mathbf{c}^\top\mathsf{G}''^{-1}\mathbf{c}) > 0$, and symmetric matrices with proportional quadratic forms are proportional. (c) By cyclicity, $\|\mathbf{R}^{-1}\Delta\mathbf{R}\|^2_{\mathsf{G}'} = \mathrm{tr}(\mathbf{R}\mathsf{G}'^{-1}\mathbf{R}^\top\Delta^\top\mathbf{R}^{-\top}\mathsf{G}'\mathbf{R}^{-1}\Delta)$, and $(\mathbf{R}^{-\top}\mathsf{G}'\mathbf{R}^{-1})^{-1} = \mathbf{R}\mathsf{G}'^{-1}\mathbf{R}^\top$. □

Proposition 18 (Loop defect)**.** *Let $\mathfrak{Q}_p$ be the loop defect of Eq. (24). (a) $\mathfrak{Q}_p$ is continuous and nonnegative on $\mathcal{T}$ and vanishes exactly when $\rho(\mathring{\Theta}_p) = I$. (b) A passive frame change leaves $\mathfrak{Q}_p$ unchanged. Let the reference change by $h \in G$, so that the frames become $U_i h$ and the generators $\mathrm{Ad}_{h^{-1}} \mathring{\omega}$, and put $\mathbf{R} = \rho(h)$. The defect computed at the new reference with the matrix $\mathbf{R}^\top\mathsf{G}\mathbf{R}$, which represents the same inner product in the new coordinates, equals $\mathfrak{Q}_p$ for every generator family. The defect computed with G held fixed equals the defect at the old reference computed with $\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1}$; it equals $\mathfrak{Q}_p$ for every generator family whenever $\mathbf{R}$ is a nonzero multiple of a G-isometry, and, when $\mathrm{d}\rho(\mathfrak{g}) = \mathfrak{gl}(\mathbb{V})$ and p traverses some reversal pair exactly once, only then. (c) Put $\mathring{\Omega}_a = \mathrm{d}\rho\mathring{\omega}_{e_a}$ and*

$\varsigma_p = \sum_a \|\mathring{\Omega}_a\|_{\mathsf{G}}$. *Then* $\sum_a \mathring{\Omega}_a = \mathrm{d}\rho(\mathbf{C}_p\mathring{\omega})$ *and*

$$\left|\mathfrak{Omega}_p - \tfrac{1}{2}\|\mathrm{d}\rho(\mathbf{C}_p\mathring{\omega})\|_{\mathsf{G}}^2\right| \le \varsigma_p(e^{\varsigma_p} - 1 - \varsigma_p) + \tfrac{1}{2}(e^{\varsigma_p} - 1 - \varsigma_p)^2 \le \tfrac{23}{10}\varsigma_p^3 \quad \text{if } \varsigma_p \le 1. \tag{399}$$

If $\mathrm{d}\rho$ *is injective and* $\mathfrak{g}$ *carries the inner product pulled back from* $\langle\cdot,\cdot\rangle_{\mathsf{G}}$*, then* $\varsigma_p \le r\|\mathring{\omega}\|$*, so* $\mathfrak{Omega}_p = \frac{1}{2}\|\mathbf{C}_p\mathring{\omega}\|^2 + O(\|\mathring{\omega}\|^3)$.

Proof. Write $\Delta = I - \rho(\mathring{\Theta}_p)$. (a) The exponential, products and ρ are continuous, and the zero set of $\frac{1}{2}\|\Delta\|_{\mathsf{G}}^2$ is that of a norm. (b) A passive change alters neither the generators (Section 4.7) nor the reference and G. After the reference change each factor is $\exp(\mathrm{Ad}_{h^{-1}}\mathring{\omega}_{e_a}) = h^{-1}\exp(\mathring{\omega}_{e_a})h$, so the new reference holonomy is $h^{-1}\mathring{\Theta}_p h$ and its defect matrix is $\mathbf{R}^{-1}\Delta\mathbf{R}$. This is the old matrix written in the coordinates $v' = \mathbf{R}^{-1}v$, in which the inner product with matrix G has the matrix $\mathbf{R}^{\top}\mathsf{G}\mathbf{R}$. Lemma 15(c) with $\mathsf{G}' = \mathbf{R}^{\top}\mathsf{G}\mathbf{R}$ returns $\|\Delta\|_{\mathsf{G}}$, and with $\mathsf{G}' = \mathsf{G}$ gives $\|\Delta\|_{\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1}}$. If $\mathbf{R} = c\mathbf{O}$ with $c \neq 0$ and $\mathbf{O}^{\top}\mathsf{G}\mathbf{O} = \mathsf{G}$, then $\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1} = c^{-2}\mathsf{G}$, and Lemma 15(b) gives equality. Conversely, suppose that equality holds for every generator family, let p traverse the pair of e exactly once, let $\mathbf{H}$ be any matrix, choose $\xi \in \mathfrak{g}$ with $\mathrm{d}\rho\xi = \mathbf{H}$, and give the pair of e the generator $t\xi$ and every other pair the generator zero. Then $\rho(\mathring{\Theta}_p) = \exp(\pm t\mathbf{H})$, so $\Delta = \mp t\mathbf{H} + O(t^2)$; dividing the assumed equality by t^2 and letting $t \to 0$ gives $\|\mathbf{H}\|_{\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1}} = \|\mathbf{H}\|_{\mathsf{G}}$ for every $\mathbf{H}$. Lemma 15(b) gives $\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1} = s\mathsf{G}$ with $s > 0$, so $s^{1/2}\mathbf{R}$ is a G-isometry. (c) The first identity is the linearity of $\mathrm{d}\rho$ with the reversal convention. Multiplying the exponential series, $\mathbf{T}_p = \rho(\mathring{\Theta}_p) - I - \sum_a \mathring{\Omega}_a$ is the absolutely convergent sum of the ordered products $\mathring{\Omega}_0^{l_0}\cdots\mathring{\Omega}_{r-1}^{l_{r-1}}/(l_0!\cdots l_{r-1}!)$ over the multi-indices with $l_0 + \cdots + l_{r-1} \ge 2$, so Lemma 15(a) gives $\|\mathbf{T}_p\|_{\mathsf{G}} \le e^{\varsigma_p} - 1 - \varsigma_p$. Since $2\mathfrak{Omega}_p = \|\sum_a \mathring{\Omega}_a + \mathbf{T}_p\|_{\mathsf{G}}^2$ and $\|\sum_a \mathring{\Omega}_a\|_{\mathsf{G}} \le \varsigma_p$, expanding the square gives the first inequality. With $e^{\varsigma} - 1 - \varsigma \le \frac{1}{2}\varsigma^2 e^{\varsigma}$, the right side is at most $\frac{1}{2}\varsigma_p^3 e^{\varsigma_p} + \frac{1}{8}\varsigma_p^4 e^{2\varsigma_p} \le (\frac{e}{2} + \frac{e^2}{8})\varsigma_p^3$ for $\varsigma_p \le 1$, and $\frac{e}{2} + \frac{e^2}{8} < 2.29$. With the pulled-back inner product, $\|\mathring{\Omega}_a\|_{\mathsf{G}} = \|\mathring{\omega}_{e_a}\| \le \|\mathring{\omega}\|$. □

Proposition 19 (Continuous class functions are blind to unipotent holonomy)**.** *Let* $K \ge 2$*, let* $\mathbb{H}$ *be* $\mathrm{GL}(K,\mathbb{R})$ *or* $\mathrm{GL}^+(K)$*, and let* $f : \mathbb{H} \to \mathbb{R}$ *be continuous and constant on the conjugacy classes of* $\mathbb{H}$*. Then* $f(I + \mathbf{T}) = f(I)$ *for every nilpotent* $\mathbf{T}$*. In particular no such function vanishes exactly at* I*, since the shear* $I + E_{12} \neq I$ *is unipotent. Consequently, for* $G = \mathrm{GL}(K,\mathbb{R})$ *or* $\mathrm{GL}^+(K)$ *in its defining representation and a closed walk that traverses some reversal pair exactly once, a loop penalty that is a continuous function of the holonomy and is unchanged by every passive frame change takes at every unipotent holonomy the value it takes at the identity.*

Proof. Since $\mathbf{T}^K = 0$, the kernels $\ker\mathbf{T} \subseteq \ker\mathbf{T}^2 \subseteq \cdots \subseteq \ker\mathbf{T}^K = \mathbb{R}^K$ increase and $\mathbf{T}$ maps $\ker\mathbf{T}^{k+1}$ into $\ker\mathbf{T}^k$. A basis of $\ker\mathbf{T}$, extended successively to bases of $\ker\mathbf{T}^2, \ldots, \ker\mathbf{T}^K$, gives the columns of an invertible $\mathbf{B}$ with $\mathbf{T}_0 = \mathbf{B}^{-1}\mathbf{T}\mathbf{B}$ strictly upper triangular; replacing the first column by its negative preserves this property, so $\det\mathbf{B} > 0$ may be assumed. For $t > 0$ put $\mathbf{S}_t = \mathrm{diag}(1, t, \ldots, t^{K-1})$. Then $\mathbf{B}\mathbf{S}_t \in \mathrm{GL}^+(K) \subseteq \mathbb{H}$, and $(\mathbf{S}_t^{-1}\mathbf{T}_0\mathbf{S}_t)_{kl} = t^{l-k}(\mathbf{T}_0)_{kl} \to 0$ as $t \to 0$, because $(\mathbf{T}_0)_{kl} = 0$ unless $l > k$. Hence $f(I + \mathbf{T}) = f(I + \mathbf{S}_t^{-1}\mathbf{T}_0\mathbf{S}_t)$ for every $t > 0$, and continuity gives $f(I + \mathbf{T}) = f(I)$. For the last claim, give the traversed pair the link g' or g'^{-1}, according to the direction of traversal, and every other link the identity. The holonomy is then g', and by Eq. (18) a passive change at the base vertex replaces it by $gg'g^{-1}$, so the penalty is a continuous class function on G. □

Lemma 16 (Gaussian moments and growth)**.** *(a) For a Gaussian law* $\mathcal{N}(\mu,\Sigma)$ *on a Euclidean space of dimension* d *and every* $c \ge 0$*,* $\mathbb{E}e^{c\|\xi\|} \le e^{c\|\mu\|}(2e^{c^2\|\Sigma\|/2})^d$*. (b) The matrix exponential and its first two derivatives at* $\mathbf{B}$*, in directions* $\mathbf{T}$ *and* $\mathbf{T}'$*, have norms at most* $e^{\|\mathbf{B}\|}$*,* $e^{\|\mathbf{B}\|}\|\mathbf{T}\|$ *and* $e^{\|\mathbf{B}\|}\|\mathbf{T}\|\|\mathbf{T}'\|$.

(c) Every real polynomial f in the entries of the matrices $\exp(\pm \mathrm{d}\rho \mathring{\omega}_e)$, $e \in E_+$, is smooth on $\mathcal{T}$ with $|f| + \|\nabla f\| + \|\nabla^2 f\| \le c_1 e^{c_2 \|\mathring{\omega}\|}$. Each loop defect, and hence $\mathcal{A}_{\mathrm{loop}}$, is such a polynomial.

Proof. (a) Write $\xi = \mu + \Sigma^{1/2}\mathbf{b}$ with $\mathbf{b}$ standard Gaussian in an orthonormal basis. Then $\|\xi\| \le \|\mu\| + \|\Sigma\|^{1/2} \sum_l |b_l|$, the coordinates b_l are independent, and $\mathbb{E} e^{c'|b_l|} \le \mathbb{E} e^{c' b_l} + \mathbb{E} e^{-c' b_l} = 2e^{c'^2/2}$. (b) The first derivative of $\mathbf{B} \mapsto \mathbf{B}^l$ in direction $\mathbf{T}$ is a sum of l products of $l-1$ factors $\mathbf{B}$ and one factor $\mathbf{T}$, and the second derivative is a sum of $l(l-1)$ products of $l-2$ factors $\mathbf{B}$ and one factor each of $\mathbf{T}$ and $\mathbf{T}'$; summing the exponential series gives the bounds. (c) By (b) and the linearity of $\mathrm{d}\rho$, every entry of $\exp(\pm \mathrm{d}\rho \mathring{\omega}_e)$ and its first two derivatives in $\mathring{\omega}$ are bounded by $c e^{\|\mathrm{d}\rho\| \|\mathring{\omega}\|}$, and the Leibniz rule bounds a monomial of degree k and its first two derivatives by $c' e^{k \|\mathrm{d}\rho\| \|\mathring{\omega}\|}$. A loop defect is a quadratic polynomial in the entries of $\rho(\mathring{\Theta}_p)$, which is a product of the matrices $\exp(\pm \mathrm{d}\rho \mathring{\omega}_e)$. □

Proposition 20 (Normalization and covariance of the generator prior)**.** *Let G be a Lie group, compact or not, and let the data of Eq.* (23) *be given. (a) $\mathcal{A}_{\mathrm{loop}}$ is finite, continuous and nonnegative, has finite $\nu^{\mathcal{T}}$-mean, and $0 < e^{-\mathbb{E}_{\nu^{\mathcal{T}}} \mathcal{A}_{\mathrm{loop}}} \le Z_{\mathrm{loop}} \le 1$. If $\rho(G)$ consists of G-isometries, also $Z_{\mathrm{loop}} \ge \exp(-2 n_V \sum_p c_p)$. The prior P_{gen} is a probability law with a strictly positive continuous density with respect to Lebesgue measure on $\mathcal{T}$. (b) Passive frame changes leave P_{gen} unchanged. (c) Under the reference change by h, with $\mathbf{R} = \rho(h)$ and $\mathrm{Ad}_{h^{-1}}$ acting on every summand, the image of P_{gen} is the generator prior at the new reference with anchors $\mathrm{Ad}_{h^{-1}} \mathring{\alpha}_e$, covariances $\mathrm{Ad}_{h^{-1}} \mathring{\Sigma}_e \mathrm{Ad}_h$, fiber matrix $\mathbf{R}^\top \mathsf{G} \mathbf{R}$, inner product $\langle \mathrm{Ad}_h \cdot, \mathrm{Ad}_h \cdot \rangle$ on $\mathfrak{g}$, and the same walks and coefficients; both priors induce the same law of links. (d) If Ad_h is an isometry of the inner product on $\mathfrak{g}$, fixes every anchor and commutes with every covariance, and $\mathbf{R}$ is a nonzero multiple of a G-isometry, then P_{gen} is invariant under $\mathrm{Ad}_{h^{-1}}$.*

Proof. (a) Finiteness, continuity and nonnegativity follow from Proposition 18(a). The exponential series and Lemma 15(a) give $\|\exp \mathbf{H}\|_{\mathsf{G}} \le n_V^{1/2} + e^{\|\mathbf{H}\|_{\mathsf{G}}} - 1 \le n_V^{1/2} e^{\|\mathbf{H}\|_{\mathsf{G}}}$, so a walk of length r has $\mathfrak{Q}_p \le \frac{1}{2}(n_V^{1/2} + n_V^{r/2} e^{\varsigma_p})^2 \le 2 n_V^r e^{2\varsigma_p}$. Here ς_p is at most the operator norm of $\mathrm{d}\rho$ times the sum of the norms of the generators on the traversed pairs, each counted with its number of traversals. Under $\nu^{\mathcal{T}}$ the generators of distinct pairs are independent Gaussian vectors, so Lemma 16(a) gives $\mathbb{E}_{\nu^{\mathcal{T}}} e^{2\varsigma_p} < \infty$, and $\mathcal{A}_{\mathrm{loop}}$ has a finite mean. Since $0 < e^{-\mathcal{A}_{\mathrm{loop}}} \le 1$, $0 < Z_{\mathrm{loop}} \le 1$, and Jensen's inequality for the convex function $u \mapsto e^{-u}$ gives $Z_{\mathrm{loop}} \ge e^{-\mathbb{E}\mathcal{A}_{\mathrm{loop}}}$. If $\mathbf{B} = \rho(\mathring{\Theta}_p)$ satisfies $\mathbf{B}^\top \mathsf{G} \mathbf{B} = \mathsf{G}$, then $\Delta^\top \mathsf{G} \Delta = 2\mathsf{G} - \mathbf{B}^\top \mathsf{G} - \mathsf{G}\mathbf{B}$ and $\mathrm{tr}(\mathsf{G}^{-1}\mathbf{B}^\top \mathsf{G}) = \mathrm{tr}\, \mathbf{B}$, so $\mathfrak{Q}_p = n_V - \mathrm{tr}\, \mathbf{B} \le 2 n_V$, because $\mathsf{G}^{1/2} \mathbf{B} \mathsf{G}^{-1/2}$ is orthogonal and its eigenvalues have modulus one; hence $\mathcal{A}_{\mathrm{loop}} \le 2 n_V \sum_p c_p$. The density of P_{gen} is $Z_{\mathrm{loop}}^{-1} e^{-\mathcal{A}_{\mathrm{loop}}}$ times a product of nondegenerate Gaussian densities. (b) Passive changes alter neither the generators nor the anchors, covariances, inner product, walks and fiber matrix. (c) By construction $\mathrm{Ad}_{h^{-1}}$ is a linear isometry from $(\mathfrak{g}, \langle \cdot, \cdot \rangle)$ onto $(\mathfrak{g}, \langle \mathrm{Ad}_h \cdot, \mathrm{Ad}_h \cdot \rangle)$. It carries the normalized Lebesgue measure to the normalized Lebesgue measure and $\mathcal{N}(\mathring{\alpha}_e, \mathring{\Sigma}_e)$ to the Gaussian law with mean $\mathrm{Ad}_{h^{-1}} \mathring{\alpha}_e$ and covariance $\mathrm{Ad}_{h^{-1}} \mathring{\Sigma}_e \mathrm{Ad}_h$, since for every β the new pairing of β with $\mathrm{Ad}_{h^{-1}} \mathring{\omega}_e$ is $\langle \mathrm{Ad}_h \beta, \mathring{\omega}_e \rangle$, of variance $\langle \mathrm{Ad}_h \beta, \mathring{\Sigma}_e \mathrm{Ad}_h \beta \rangle$. When $\mathfrak{g}$ carries the pulled-back inner product, $\mathrm{d}\rho(\mathrm{Ad}_h \xi) = \mathbf{R} \mathrm{d}\rho(\xi) \mathbf{R}^{-1}$ and Lemma 15(c) identify the new inner product with the pullback for $\mathbf{R}^\top \mathsf{G} \mathbf{R}$. By Proposition 18(b) the new defect of $\mathrm{Ad}_{h^{-1}} \mathring{\omega}$ equals $\mathfrak{Q}_p(\mathring{\omega})$, so the transformed action composed with $\mathrm{Ad}_{h^{-1}}$ is $\mathcal{A}_{\mathrm{loop}}$ and the normalizer is unchanged. The links agree because $U_i h \exp(\mathrm{Ad}_{h^{-1}} \mathring{\omega}_e) h^{-1} U_j^{-1} = U_i \exp(\mathring{\omega}_e) U_j^{-1}$. (d) Under these hypotheses the data in (c) are the original anchors, covariances, inner product, walks and coefficients, with the fiber matrix multiplied by a positive scalar, which changes neither the trace form nor the defect by Lemma 15(b). □

K.3 Latent generators in the evidence bound

Assume the setting of Eq. (25): the map $\mathring{\omega} \mapsto \Theta(\mathring{\omega})$ of Eq. (19) is continuous, and the held-link prior and record kernels are measurable in the link values, so that $\mathring{\omega} \mapsto \mathbb{P}_{\Theta(\mathring{\omega})}$ is a probability kernel. Recognition laws are disintegrated as $Q(d\mathring{\omega}, dy) = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$.

Proposition 21 (Normalization, evidence and conditional decomposition)**.** *(a) For every finite record family, cyclic or not, the joint law* (25) *is a probability law. (b) The conditional evidence* $\mathring{\omega} \mapsto p(o \mid \mathring{\omega})$ *is measurable, and the evidence of the joint is* $p(o) = \int p(o \mid \mathring{\omega})P_{\mathrm{gen}}(d\mathring{\omega})$*. It is positive exactly when* $p(o \mid \cdot)$ *is positive on a set of positive Lebesgue measure. If* $\ell_o \le c_{\mathfrak{A}}$*, then* $p(o \mid \mathring{\omega}) \le c_{\mathfrak{A}}$ *for every* $\mathring{\omega}$ *and* $p(o) \le c_{\mathfrak{A}}$*. (c) Suppose that* $\ell_o \le c_{\mathfrak{A}}$ *and* $p(o \mid \mathring{\omega}) > 0$ *for every* $\mathring{\omega}$*. Then the posterior disintegrates as* $\Pi^{\mathcal{T}}(d\mathring{\omega})\Pi_{\mathring{\omega}}(dy)$ *with* $\Pi^{\mathcal{T}} = p(o)^{-1}p(o \mid \cdot)P_{\mathrm{gen}}$*, and every law* $Q = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$ *satisfies*

$$\mathcal{F}_V[Q; X, o] = D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) + \int \mathcal{F}_V[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o]R(d\mathring{\omega}) \quad \text{in } (-\infty, +\infty], \tag{400}$$

with a measurable integrand bounded below by $-\log c_{\mathfrak{A}}$*. If* $D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) < \infty$ *and* $\mathbb{E}_R\mathcal{A}_{\mathrm{loop}} < \infty$*, then* $D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) = D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) + \mathbb{E}_R\mathcal{A}_{\mathrm{loop}} + \log Z_{\mathrm{loop}}$.

Proof. (a) P_{gen} is a probability law by Proposition 20(a), and for every link family the held-link law has unit mass, by integrating the records and then the prior as in Eq. (21). Tonelli's theorem, integrating records, then matter, then generators, gives unit mass; cycles in the record hypergraph do not enter, because every record kernel is normalized conditional on the whole latent state. (b) Joint measurability of ℓ_o and Tonelli's theorem give the measurability of $p(o \mid \cdot)$ and the evidence formula. Since P_{gen} has a strictly positive Lebesgue density, the integral of the nonnegative function $p(o \mid \cdot)$ is positive exactly when that function is positive on a set of positive Lebesgue measure. The bounds follow by integrating $\ell_o \le c_{\mathfrak{A}}$ against probability laws. (c) The measure $\ell_o(y; \Theta(\mathring{\omega}))P^Y_{\mathring{\omega}}(dy)P_{\mathrm{gen}}(d\mathring{\omega})$ equals $p(o \mid \mathring{\omega})\Pi_{\mathring{\omega}}(dy)P_{\mathrm{gen}}(d\mathring{\omega})$, and $\mathring{\omega} \mapsto \Pi_{\mathring{\omega}}$ is a kernel by Tonelli's theorem; dividing by $p(o) \in (0, c_{\mathfrak{A}}]$ gives the disintegration. The chain rule (396) gives $D_{\mathrm{KL}}(Q\|\Pi) = D_{\mathrm{KL}}(R\|\Pi^{\mathcal{T}}) + \int D_{\mathrm{KL}}(Q_{\mathring{\omega}}\|\Pi_{\mathring{\omega}})R(d\mathring{\omega})$ with a measurable integrand. If $D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) < \infty$, then $R \ll P_{\mathrm{gen}}$, and since $d\Pi^{\mathcal{T}}/dP_{\mathrm{gen}} = p(o \mid \cdot)/p(o)$ is positive and finite, $R \ll \Pi^{\mathcal{T}}$ with $\log(dR/d\Pi^{\mathcal{T}}) = \log(dR/dP_{\mathrm{gen}}) - \log p(o \mid \mathring{\omega}) + \log p(o)$. The first term has an R-integrable negative part and the second is bounded below by $-\log c_{\mathfrak{A}}$, so the integral splits; adding the conditional term and $-\log p(o)$ gives Eq. (400), because $\mathcal{F}_V[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o] = -\log p(o \mid \mathring{\omega}) + D_{\mathrm{KL}}(Q_{\mathring{\omega}}\|\Pi_{\mathring{\omega}}) \ge -\log c_{\mathfrak{A}}$. If $D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) = +\infty$, then either $R \not\ll P_{\mathrm{gen}}$, hence $R \not\ll \Pi^{\mathcal{T}}$ because the two laws have the same null sets, or $R \ll P_{\mathrm{gen}}$ and $\Pi^{\mathcal{T}} \le (c_{\mathfrak{A}}/p(o))P_{\mathrm{gen}}$ gives $dR/d\Pi^{\mathcal{T}} \ge (p(o)/c_{\mathfrak{A}})dR/dP_{\mathrm{gen}}$. In both cases $D_{\mathrm{KL}}(R\|\Pi^{\mathcal{T}}) = +\infty$, and both sides of Eq. (400) are $+\infty$. The last claim integrates $\log(dR/dP_{\mathrm{gen}}) = \log(dR/d\nu^{\mathcal{T}}) + \mathcal{A}_{\mathrm{loop}} + \log Z_{\mathrm{loop}}$ against R. □

Proposition 22 (Gaussian recognition of the generators)**.** *Let* $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ *with* $\mathring{\Lambda} \succ 0$*, and let* $\mathring{\Sigma}$ *be the block-diagonal prior covariance. Then*

$$D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) = \tfrac{1}{2}\big[\mathrm{tr}(\mathring{\Sigma}^{-1}\mathring{\Lambda}^{-1}) + \langle \mathring{\mu} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha})\rangle - \dim \mathcal{T} + \log\det \mathring{\Sigma} + \log\det \mathring{\Lambda}\big], \tag{401}$$

$\mathbb{E}_R e^{c\|\mathring{\omega}\|} < \infty$ *for every* $c \ge 0$*, and* $\mathbb{E}_R\mathcal{A}_{\mathrm{loop}} < \infty$*. Hence, under the hypotheses of Proposition 21(c), the link sector of Eq.* (26) *is finite and equals* $D_{\mathrm{KL}}(R\|P_{\mathrm{gen}})$.

Proof. For a self-adjoint $\mathbf{B}$, $\mathbb{E}_R\langle \mathring{\omega} - \mathring{\alpha}, \mathbf{B}(\mathring{\omega} - \mathring{\alpha})\rangle = \mathrm{tr}(\mathbf{B}\mathring{\Lambda}^{-1}) + \langle \mathring{\mu} - \mathring{\alpha}, \mathbf{B}(\mathring{\mu} - \mathring{\alpha})\rangle$, and the R-expectation of the difference of the two Gaussian log densities gives Eq. (401). The moment bound is Lemma 16(a), and Lemma 16(c) gives $\mathbb{E}_R\mathcal{A}_{\mathrm{loop}} \le c_1\mathbb{E}_R e^{c_2\|\mathring{\omega}\|}$. The last claim is Proposition 21(c). □

K.4 Held links as a limit, point masses and conditional marginals

Hold the anchor $\mathring{\alpha}$, the record and the structural data, and let the prior covariance be $\mathring{\Sigma} = \mathring{\sigma}^2 \mathring{\Sigma}_0$ with $\mathring{\sigma} > 0$ and a block-diagonal $\mathring{\Sigma}_0 \succ 0$. Write $\nu_{\mathring{\sigma}}$, $P_{\mathring{\sigma}}$, $Z_{\mathring{\sigma}}$ and $\mathcal{F}_{\mathring{\sigma}}$ for the base law, generator prior, normalizer and latent-link free energy at width $\mathring{\sigma}$. For a class $\mathcal{Q}'$ of laws on the matter space put $\mathbf{F}_{Q'}(\mathring{\omega}) = \mathcal{F}_V[Q'; X, \Theta(\mathring{\omega}), o]$ and $\mathbf{F}(\mathring{\omega}) = \inf_{Q' \in \mathcal{Q}'} \mathbf{F}_{Q'}(\mathring{\omega})$.

Theorem 8 (Held links are the zero-width limit)**.** *Assume (i) $\ell_o \le c_{\mathfrak{A}}$ and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$; (ii) $\mathbf{F}$ is finite and lower semicontinuous at $\mathring{\alpha}$; and (iii) a subclass $\mathcal{Q}'_0 \subseteq \mathcal{Q}'$ with $\inf_{Q' \in \mathcal{Q}'_0} \mathbf{F}_{Q'}(\mathring{\alpha}) = \mathbf{F}(\mathring{\alpha})$ consists of laws for which $\mathbf{F}_{Q'}$ is continuous at $\mathring{\alpha}$ and bounded above by $c_{Q'} e^{c_{Q'} \|\mathring{\omega}\|}$. Then, as $\mathring{\sigma} \to 0$, the infimum of $\mathcal{F}_{\mathring{\sigma}}$ over the product laws $R \otimes Q'$ with $Q' \in \mathcal{Q}'$ and R in any class of laws containing $\nu_{\mathring{\sigma}}$, and its infimum over all laws $R(d\mathring{\omega}) Q_{\mathring{\omega}}(dy)$ whose kernel takes values in $\mathcal{Q}'$, both converge to $\mathbf{F}(\mathring{\alpha})$, the infimum over $\mathcal{Q}'$ of the held-link free energy at the anchor links $U_i \exp(\mathring{\alpha}_e) U_j^{-1}$.*

Proof. Write $\mathring{\omega} = \mathring{\alpha} + \mathring{\sigma} \mathring{\Sigma}_0^{1/2} \mathbf{b}$ under $\nu_{\mathring{\sigma}}$, with $\mathbf{b}$ standard Gaussian. Since $e^{-\mathcal{A}_{\mathrm{loop}}} \in (0, 1]$ is continuous, bounded convergence gives $Z_{\mathring{\sigma}} \to e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\alpha})} > 0$, so $P_{\mathring{\sigma}}\{\|\mathring{\omega} - \mathring{\alpha}\| \ge \epsilon\} \le \nu_{\mathring{\sigma}}\{\|\mathring{\omega} - \mathring{\alpha}\| \ge \epsilon\} / Z_{\mathring{\sigma}} \to 0$ for every $\epsilon > 0$. By Proposition 21(c), every law $Q = R(d\mathring{\omega}) Q_{\mathring{\omega}}(dy)$ has $\mathcal{F}_{\mathring{\sigma}}[Q] = D_{\mathrm{KL}}(R \| P_{\mathring{\sigma}}) + \int \mathbf{F}_{Q_{\mathring{\omega}}}(\mathring{\omega}) R(d\mathring{\omega})$ with $\mathbf{F}_{Q_{\mathring{\omega}}} \ge -\log c_{\mathfrak{A}}$.

For the upper bound, let $Q' \in \mathcal{Q}'_0$. The law $\nu_{\mathring{\sigma}} \otimes Q'$ is admissible, and $D_{\mathrm{KL}}(\nu_{\mathring{\sigma}} \| P_{\mathring{\sigma}}) = \mathbb{E}_{\nu_{\mathring{\sigma}}} \mathcal{A}_{\mathrm{loop}} + \log Z_{\mathring{\sigma}}$. For $\mathring{\sigma} \le 1$, Lemma 16(c) dominates $\mathcal{A}_{\mathrm{loop}}(\mathring{\alpha} + \mathring{\sigma} \mathring{\Sigma}_0^{1/2} \mathbf{b})$ by $c_1 e^{c_2(\|\mathring{\alpha}\| + \|\mathring{\Sigma}_0\|^{1/2} \|\mathbf{b}\|)}$, which is integrable by Lemma 16(a), so $\mathbb{E}_{\nu_{\mathring{\sigma}}} \mathcal{A}_{\mathrm{loop}} \to \mathcal{A}_{\mathrm{loop}}(\mathring{\alpha})$ and the divergence tends to zero. Hypothesis (iii), the lower bound $-\log c_{\mathfrak{A}}$ and the same domination give $\mathbb{E}_{\nu_{\mathring{\sigma}}} \mathbf{F}_{Q'} \to \mathbf{F}_{Q'}(\mathring{\alpha})$. Hence the lim sup of the product infimum is at most $\mathbf{F}_{Q'}(\mathring{\alpha})$, and the infimum over $\mathcal{Q}'_0$ gives at most $\mathbf{F}(\mathring{\alpha})$.

For the lower bound, let $c_0 > 0$. By (ii) there is $\epsilon > 0$ with $\mathbf{F} > \mathbf{F}(\mathring{\alpha}) - c_0$ on the ball $B_\epsilon = \{\|\mathring{\omega} - \mathring{\alpha}\| < \epsilon\}$. The bounded function $\mathbf{q} = (\mathbf{F}(\mathring{\alpha}) - c_0) \mathbf{1}_{B_\epsilon} - (\log c_{\mathfrak{A}}) \mathbf{1}_{B_\epsilon^c}$ satisfies $\mathbf{q}(\mathring{\omega}) \le \mathbf{F}_{Q_{\mathring{\omega}}}(\mathring{\omega})$ for every kernel with values in $\mathcal{Q}'$. For every R with $D_{\mathrm{KL}}(R \| P_{\mathring{\sigma}}) < \infty$, the law $P^{\mathbf{q}} \propto e^{-\mathbf{q}} P_{\mathring{\sigma}}$ gives $0 \le D_{\mathrm{KL}}(R \| P^{\mathbf{q}}) = D_{\mathrm{KL}}(R \| P_{\mathring{\sigma}}) + \mathbb{E}_R \mathbf{q} + \log \mathbb{E}_{P_{\mathring{\sigma}}} e^{-\mathbf{q}}$, hence $\mathcal{F}_{\mathring{\sigma}}[Q] \ge D_{\mathrm{KL}}(R \| P_{\mathring{\sigma}}) + \mathbb{E}_R \mathbf{q} \ge -\log \mathbb{E}_{P_{\mathring{\sigma}}} e^{-\mathbf{q}}$; the bound is trivial when the divergence is infinite. Since $P_{\mathring{\sigma}}(B_\epsilon^c) \to 0$, the right side tends to $\mathbf{F}(\mathring{\alpha}) - c_0$. The kernel infimum is at most the product infimum, so both converge to $\mathbf{F}(\mathring{\alpha})$. □

Proposition 23 (Point-mass recognition)**.** *For every $\mathring{\mu}$, $D_{\mathrm{KL}}(\delta_{\mathring{\mu}} \| P_{\mathrm{gen}}) = +\infty$. For $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$, $D_{\mathrm{KL}}(R \| P_{\mathrm{gen}}) \ge \frac{1}{2}(\log\det \mathring{\Lambda} - \log\det \mathring{\Sigma}^{-1} - \dim \mathcal{T}) + \log Z_{\mathrm{loop}}$, which tends to $+\infty$ as $\mathring{\Lambda}^{-1} \to 0$ with the prior held. Under the hypotheses of Proposition 21(c), the free energy of $\delta_{\mathring{\mu}} \otimes Q_Y$ is $+\infty$ and that of $R \otimes Q_Y$ tends to $+\infty$.*

Proof. P_{gen} has a Lebesgue density, so $\{\mathring{\mu}\}$ is P_{gen}-null and $\delta_{\mathring{\mu}} \not\ll P_{\mathrm{gen}}$. For Gaussian R, Propositions 21(c) and 22 with $\mathcal{A}_{\mathrm{loop}} \ge 0$ give $D_{\mathrm{KL}}(R \| P_{\mathrm{gen}}) \ge D_{\mathrm{KL}}(R \| \nu^{\mathcal{T}}) + \log Z_{\mathrm{loop}}$; in Eq. (401) the trace and quadratic terms are nonnegative, while $\log\det \mathring{\Lambda} \ge \dim \mathcal{T} \log \lambda_{\min}(\mathring{\Lambda}) \to +\infty$. The free-energy statements follow from Eq. (400), whose matter integrand is bounded below by $-\log c_{\mathfrak{A}}$. □

Proposition 24 (Gaussian conditionals with a non-Gaussian marginal)**.** *Let $G = \mathrm{GL}^+(1)$ act on $\mathbb{R}$ by multiplication, with trivial frames, $\mathcal{A}_{\mathrm{loop}} = 0$ and $\nu^{\mathcal{T}} = \mathcal{N}(0, 1)$ on $\mathfrak{g} = \mathbb{R}$. Let two scalar agents have $k_j \sim \mathcal{N}(0, 1)$ and $k_i \mid k_j, \mathring{\omega} \sim \mathcal{N}(e^{\mathring{\omega}} k_j, 1)$ for the copy $e : j \to i$, and let there be no record. Every held-link law of (k_i, k_j) is Gaussian, while the k_i-marginal of the latent-link law satisfies $\mathbb{E} k_i^4 = 3(1 + 2e^2 + e^8) > 3(1 + e^2)^2 = 3(\mathbb{E} k_i^2)^2$ and is therefore not Gaussian.*

Proof. Given $\mathring{\omega}$, (k_i, k_j) is centered Gaussian with $\mathbb{E}[k_i^2 \mid \mathring{\omega}] = 1 + e^{2\mathring{\omega}}$, and a centered Gaussian variable has $\mathbb{E} k^4 = 3(\mathbb{E} k^2)^2$. With $\mathbb{E} e^{t\mathring{\omega}} = e^{t^2/2}$, $\mathbb{E} k_i^2 = 1 + e^2$ and $\mathbb{E} k_i^4 = 3\mathbb{E}(1 + e^{2\mathring{\omega}})^2 = 3(1 + 2e^2 + e^8)$. The inequality is strict Jensen, since $e^{2\mathring{\omega}}$ is not almost surely constant. The marginal is centered, and a centered Gaussian law would give equality. □

K.5 The link field equation

For square matrices write $\langle \mathbf{S}, \mathbf{T} \rangle = \operatorname{tr}(\mathbf{S}^\top \mathbf{T})$, and let $(\mathrm{d}\rho)^\top$ be the adjoint of $\mathrm{d}\rho$ for the inner product on $\mathfrak{g}$ and this pairing.

Lemma 17 (Derivative of the exponential chart)**.** *Let* $\mathbf{L}(\mathbf{B}, \mathbf{T}) = \int_0^1 e^{u\mathbf{B}} \mathbf{T} e^{(1-u)\mathbf{B}} du$*. (a)* $\|e^{\mathbf{B}+\mathbf{T}} - e^{\mathbf{B}} - \mathbf{L}(\mathbf{B}, \mathbf{T})\| \le \frac{1}{2}\|\mathbf{T}\|^2 e^{\|\mathbf{B}\|+\|\mathbf{T}\|}$ *and* $\|\mathbf{L}(\mathbf{B}, \mathbf{T})\| \le e^{\|\mathbf{B}\|}\|\mathbf{T}\|$*, so* $\mathbf{T} \mapsto \mathbf{L}(\mathbf{B}, \mathbf{T})$ *is the Fréchet derivative of the exponential at* $\mathbf{B}$ *[66]. (b)* $\langle \mathbf{S}, \mathbf{L}(\mathbf{B}, \mathbf{T}) \rangle = \langle \mathbf{L}(\mathbf{B}^\top, \mathbf{S}), \mathbf{T} \rangle$ *and* $\langle \mathbf{S}, \mathbf{L}(\mathbf{B}, \mathbf{T}) \rangle_{\mathsf{G}} = \langle \mathbf{L}(\mathbf{B}^*, \mathbf{S}), \mathbf{T} \rangle_{\mathsf{G}}$*. (c) If* f *is differentiable at* $e^{\mathbf{B}}$*, then* $\mathbf{B} \mapsto f(e^{\mathbf{B}})$ *has gradient* $\mathbf{L}(\mathbf{B}^\top, \nabla f(e^{\mathbf{B}}))$ *at* $\mathbf{B}$*, and* $\xi \mapsto f(\exp(\mathrm{d}\rho\xi))$ *has gradient* $(\mathrm{d}\rho)^\top \mathbf{L}((\mathrm{d}\rho\xi)^\top, \nabla f(\exp(\mathrm{d}\rho\xi)))$ *at* ξ*.*

Proof. For square $\mathbf{B}, \mathbf{B}'$, $e^{\mathbf{B}'} - e^{\mathbf{B}} = \int_0^1 e^{u\mathbf{B}'}(\mathbf{B}' - \mathbf{B}) e^{(1-u)\mathbf{B}} du$, because the integrand is the u-derivative of $e^{u\mathbf{B}'} e^{(1-u)\mathbf{B}}$; with $\|e^{u\mathbf{B}}\| \le e^{u\|\mathbf{B}\|}$ this gives $\|e^{\mathbf{B}'} - e^{\mathbf{B}}\| \le \|\mathbf{B}' - \mathbf{B}\| e^{\max(\|\mathbf{B}'\|, \|\mathbf{B}\|)}$. With $\mathbf{B}' = \mathbf{B} + \mathbf{T}$ the remainder in (a) is $\int_0^1 (e^{u(\mathbf{B}+\mathbf{T})} - e^{u\mathbf{B}}) \mathbf{T} e^{(1-u)\mathbf{B}} du$, whose integrand is bounded by $u\|\mathbf{T}\|^2 e^{\|\mathbf{B}\|+\|\mathbf{T}\|}$; the bound on $\mathbf{L}$ is immediate. For (b), cyclicity and $(e^{u\mathbf{B}})^\top = e^{u\mathbf{B}^\top}$ give $\operatorname{tr}(\mathbf{S}^\top e^{u\mathbf{B}} \mathbf{T} e^{(1-u)\mathbf{B}}) = \operatorname{tr}((e^{u\mathbf{B}^\top} \mathbf{S} e^{(1-u)\mathbf{B}^\top})^\top \mathbf{T})$, and integration over u gives the first identity; the second follows in the same way from Lemma 15(a) and $(e^{u\mathbf{B}})^* = e^{u\mathbf{B}^*}$. Part (c) is the chain rule with (a) and (b), the derivative of $\xi \mapsto \exp(\mathrm{d}\rho\xi)$ in direction η being $\mathbf{L}(\mathrm{d}\rho\xi, \mathrm{d}\rho\eta)$. □

Lemma 18 (Gaussian derivative identities)**.** *Let* f *be twice continuously differentiable on a Euclidean space with* $|f| + \|\nabla f\| + \|\nabla^2 f\| \le c_1 e^{c_2\|\xi\|}$*. Then* $(\mu, S) \mapsto \mathbb{E}_{\mathcal{N}(\mu,S)} f$ *is continuously differentiable on the product of the space and the open cone of positive-definite operators, and for self-adjoint* Δ

$$\nabla_\mu \mathbb{E}_{\mathcal{N}(\mu,S)} f = \mathbb{E}_{\mathcal{N}(\mu,S)} \nabla f, \qquad \frac{d}{dt} \mathbb{E}_{\mathcal{N}(\mu, S+t\Delta)} f \Big|_{t=0} = \tfrac{1}{2} \operatorname{tr}\big(\Delta \mathbb{E}_{\mathcal{N}(\mu,S)} \nabla^2 f\big). \tag{402}$$

These are the identities used for variational Gaussian approximations [68].

Proof. All expectations below are finite by Lemma 16(a). For a continuously differentiable φ with $|\varphi| + \|\nabla\varphi\| \le c_1 e^{c_2\|\xi\|}$, Gaussian integration by parts gives

$$\mathbb{E}_{\mathcal{N}(\mu,S)}[(\xi - \mu)\varphi(\xi)] = S \mathbb{E}_{\mathcal{N}(\mu,S)} \nabla\varphi. \tag{403}$$

Indeed, for a standard Gaussian vector $\mathbf{b}$ in an orthonormal basis, integration by parts in the l-th coordinate against the standard normal density, whose derivative is $-b_l$ times the density, has vanishing boundary terms and gives $\mathbb{E}[b_l \varphi'(\mathbf{b})] = \mathbb{E}[\partial_l \varphi'(\mathbf{b})]$, and $\varphi'(\mathbf{b}) = \varphi(\mu + S^{1/2}\mathbf{b})$ has gradient $S^{1/2}\nabla\varphi$. The mean derivative follows by differentiating $\mathbb{E} f(\mu + S^{1/2}\mathbf{b})$ under the expectation, with domination by $c_1 e^{c_2(\|\mu\|+\|\eta\|+\|S\|^{1/2}\|\mathbf{b}\|)}$ along $\mu + t\eta$, $|t| \le 1$. For the covariance, $S_t = S + t\Delta$ stays in a compact subset of the cone for small $|t|$, so the t-derivative of the Gaussian density, $\frac{1}{2}[\langle \xi - \mu, S_t^{-1}\Delta S_t^{-1}(\xi - \mu)\rangle - \operatorname{tr}(S_t^{-1}\Delta)]$ times the density, is dominated uniformly by a Gaussian density times a quadratic polynomial, and the derivative passes under the integral:

$$\frac{d}{dt} \mathbb{E} f \Big|_{t=0} = \tfrac{1}{2} \mathbb{E}[f \langle \xi - \mu, S^{-1}\Delta S^{-1}(\xi - \mu)\rangle] - \tfrac{1}{2} \operatorname{tr}(S^{-1}\Delta) \mathbb{E} f.$$

Applying Eq. (403) to $\varphi = f\langle \eta, \xi - \mu \rangle$ for fixed η, and then to the partial derivatives of f, gives $\mathbb{E}[f(\xi - \mu) \otimes (\xi - \mu)] = S(\mathbb{E}\nabla^2 f)S + S\mathbb{E} f$, so the first term equals $\frac{1}{2}\operatorname{tr}(\Delta \mathbb{E}\nabla^2 f) + \frac{1}{2}\operatorname{tr}(S^{-1}\Delta)\mathbb{E} f$, which is Eq. (402). Both derivatives are continuous in (μ, S) by dominated convergence in the representation $\xi = \mu + S^{1/2}\mathbf{b}$, since $S \mapsto S^{1/2}$ is continuous on the cone. □

Let $\mathring{\Xi}$ be the space of self-adjoint operators on $\mathcal{T}$ that preserve each summand of a fixed orthogonal decomposition of $\mathcal{T}$: a single summand gives the full family, the edge summands the edge-block family, and the lines of an orthonormal basis the diagonal family. Let $\mathrm{pr}_{\mathring{\Xi}}$ be the orthogonal projection onto $\mathring{\Xi}$ in the trace inner product; it keeps the diagonal blocks of the decomposition. The inverse of a positive-definite element of $\mathring{\Xi}$ lies in $\mathring{\Xi}$.

Proposition 25 (Link field equation)**.** *Assume $\ell_o \le c_{\mathfrak{A}}$ and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$, a σ-finite matter reference ν^Y with respect to which every $P^Y_{\mathring{\omega}}$ has a density, and a matter recognition law Q_Y with density q_Y and $\mathbb{E}_{Q_Y}|\log q_Y| < \infty$. Let $H_o(\mathring{\omega}, y) = -\log p(o, y \mid X, \Theta(\mathring{\omega}))$, the negative logarithm of the held-link joint density with respect to $\nu^Y \otimes \nu^O$, be twice continuously differentiable in $\mathring{\omega}$ for every y, with*

$$\mathbb{E}_{Q_Y} \sup_{\|\mathring{\omega}\| \le r} \big(|H_o| + \|\nabla H_o\| + \|\nabla^2 H_o\|\big)(\mathring{\omega}, Y) \le c_1 e^{c_2 r} \qquad \text{for every } r \ge 0. \tag{404}$$

Restrict recognition to $R \otimes Q_Y$ with $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ and $\mathring{\Lambda}$ positive definite in $\mathring{\Xi}$, and put $\mathbf{U} = \mathcal{A}_{\mathrm{loop}} + \mathbb{E}_{Q_Y} H_o(\cdot, Y)$ and $\mathbf{V}(\mathring{\omega}) = \frac{1}{2}\langle \mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha})\rangle + \mathbf{U}(\mathring{\omega})$. (a) $\mathbf{U}$ is twice continuously differentiable with $|\mathbf{U}| + \|\nabla \mathbf{U}\| + \|\nabla^2 \mathbf{U}\| \le c_3 e^{c_4 \|\mathring{\omega}\|}$, and

$$\mathcal{F}_V[R \otimes Q_Y; X, o] = D_{\mathrm{KL}}(R \| \nu^{\mathcal{T}}) + \mathbb{E}_R \mathbf{U} + \log Z_{\mathrm{loop}} + \mathbb{E}_{Q_Y} \log q_Y \tag{405}$$

is finite and continuously differentiable in $(\mathring{\mu}, \mathring{\Lambda})$. (b) $(\mathring{\mu}, \mathring{\Lambda})$ is stationary exactly when $\mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha}) + \mathbb{E}_R \nabla \mathcal{A}_{\mathrm{loop}} = -\mathbb{E}_{R \otimes Q_Y} \nabla H_o$ and $\mathring{\Lambda} = \mathrm{pr}_{\mathring{\Xi}}(\mathring{\Sigma}^{-1} + \mathbb{E}_R \nabla^2 \mathcal{A}_{\mathrm{loop}} + \mathbb{E}_{R \otimes Q_Y} \nabla^2 H_o)$. For the full family these are Eq. (42). (c) If H_o reads $\mathring{\omega}_e$ only through $M_e = \exp(\mathrm{d}\rho \mathring{\omega}_e)$, as a differentiable function on an open set of matrices containing the represented twists, and $\partial H_o / \partial M_e$ is Q_Y-integrable at every $\mathring{\omega}$, then with $\mathbf{J}_e = -\mathbb{E}_{Q_Y} \partial H_o / \partial M_e$ the e component of the mean equation is Eq. (43). (d) The mean equation is $\mathbb{E}_R \nabla \mathbf{V} = 0$, and $\mathbb{E}_{\mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})} \nabla \mathbf{V} \to \nabla \mathbf{V}(\mathring{\mu})$ as $\mathring{\Lambda}^{-1} \to 0$, uniformly for $\mathring{\mu}$ in compact sets; every limit point of solutions as $\mathring{\Lambda}^{-1} \to 0$ satisfies $\nabla \mathbf{V} = 0$. If $\mathbf{U}$ is bounded below, $R^\star \propto e^{-\mathbf{U}} \nu^{\mathcal{T}}$ is the unique minimizer of $\mathcal{F}_V[R \otimes Q_Y; X, o]$ over all laws R on $\mathcal{T}$, and its log density is $-\mathbf{V}$ up to a constant.

Proof. (a) By Lemma 16(c), $\mathcal{A}_{\mathrm{loop}}$ has the stated growth. On each ball $\|\mathring{\omega}\| \le r$, Eq. (404) dominates H_o and its first two derivatives by a Q_Y-integrable function, so differentiation passes under $\mathbb{E}_{Q_Y}$, and $\mathbb{E}_{Q_Y} H_o$ is twice continuously differentiable with derivatives bounded by $c_1 e^{c_2 r}$ there. At each $\mathring{\omega}$ the held-link posterior has the density $e^{-H_o(\mathring{\omega}, \cdot)} / p(o \mid \mathring{\omega})$ with respect to ν^Y and $\mathbb{E}_{Q_Y}|H_o(\mathring{\omega}, Y)| < \infty$, so Proposition 17(c) gives $\mathcal{F}_V[Q_Y; X, \Theta(\mathring{\omega}), o] = \mathbb{E}_{Q_Y}[\log q_Y + H_o(\mathring{\omega}, Y)]$. Proposition 21(c) with $Q_{\mathring{\omega}} = Q_Y$, Proposition 22 and Fubini's theorem, justified by $\mathbb{E}_R \mathbb{E}_{Q_Y}|H_o| \le c_1 \mathbb{E}_R e^{c_2 \|\mathring{\omega}\|} < \infty$, give Eq. (405), which is finite by Lemma 16(a). The closed form (401) is smooth, $\mathbb{E}_R \mathbf{U}$ is continuously differentiable in the mean and the covariance $\mathring{\Lambda}^{-1}$ by Lemma 18, and inversion is a diffeomorphism of the positive-definite cone of $\mathring{\Xi}$. (b) By Eq. (401) and Lemma 18, the derivative of Eq. (405) in the mean along η is $\langle \mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha}) + \mathbb{E}_R \nabla \mathbf{U}, \eta \rangle$, and, because $d \log\det S = \mathrm{tr}(S^{-1}\Delta)$, its derivative in the covariance $S = \mathring{\Lambda}^{-1}$ along $\Delta \in \mathring{\Xi}$ is $\frac{1}{2}\mathrm{tr}(\Delta(\mathring{\Sigma}^{-1} - \mathring{\Lambda} + \mathbb{E}_R \nabla^2 \mathbf{U}))$. Stationarity in $(\mathring{\mu}, \mathring{\Lambda})$ is stationarity in $(\mathring{\mu}, S)$. The covariance condition holds for every $\Delta \in \mathring{\Xi}$ exactly when $\mathrm{pr}_{\mathring{\Xi}}(\mathring{\Sigma}^{-1} - \mathring{\Lambda} + \mathbb{E}_R \nabla^2 \mathbf{U}) = 0$, that is, since $\mathring{\Lambda} \in \mathring{\Xi}$, when $\mathring{\Lambda} = \mathrm{pr}_{\mathring{\Xi}}(\mathring{\Sigma}^{-1} + \mathbb{E}_R \nabla^2 \mathbf{U})$. Differentiation under $\mathbb{E}_{Q_Y}$ gives $\mathbb{E}_R \nabla \mathbf{U} = \mathbb{E}_R \nabla \mathcal{A}_{\mathrm{loop}} + \mathbb{E}_{R \otimes Q_Y} \nabla H_o$, and likewise for Hessians. (c) For fixed y and remaining generators, Lemma 17(c) gives $\nabla_e H_o = (\mathrm{d}\rho)^\top \mathbf{L}((\mathrm{d}\rho \mathring{\omega}_e)^\top, \partial H_o / \partial M_e)$. The map $\mathbf{S} \mapsto (\mathrm{d}\rho)^\top \mathbf{L}(\mathbf{B}, \mathbf{S})$ is linear and continuous, so it commutes with $\mathbb{E}_{Q_Y}$, and the e component of the mean equation becomes Eq. (43). (d) Since $\mathbb{E}_R(\mathring{\omega} - \mathring{\alpha}) = \mathring{\mu} - \mathring{\alpha}$, the mean equation is $\mathbb{E}_R \nabla \mathbf{V} = 0$. Write $\mathring{\omega} = \mathring{\mu} + \mathring{\Lambda}^{-1/2} \mathbf{b}$. For $\|\mathring{\Lambda}^{-1}\| \le 1$, $\|\nabla \mathbf{V}(\mathring{\omega})\|$ is dominated by $c(1 + \|\mathring{\mu}\| + \|\mathbf{b}\|) e^{c(\|\mathring{\mu}\| + \|\mathbf{b}\|)}$, so for $\mathring{\mu}$ in a compact set the contribution

of $\{\|\mathbf{b}\| > c'\}$ is uniformly small for large c', while on $\{\|\mathbf{b}\| \le c'\}$ the uniform continuity of $\nabla\mathbf{V}$ on bounded sets makes $\|\nabla\mathbf{V}(\mathring{\omega}) - \nabla\mathbf{V}(\mathring{\mu})\|$ uniformly small; this is the uniform convergence. Along solutions with $\mathring{\mu} \to \mathring{\mu}_*$, $\|\nabla\mathbf{V}(\mathring{\mu}_*)\| \le \|\nabla\mathbf{V}(\mathring{\mu}_*) - \nabla\mathbf{V}(\mathring{\mu})\| + \|\nabla\mathbf{V}(\mathring{\mu}) - \mathbb{E}_R\nabla\mathbf{V}\| \to 0$. For the last claim, $P_{\mathrm{gen}} \le Z_{\mathrm{loop}}^{-1}\nu^{\mathcal{T}}$ gives $D_{\mathrm{KL}}(R\|P_{\mathrm{gen}}) \ge D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) + \log Z_{\mathrm{loop}}$, so every R with $D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) = +\infty$ has infinite free energy by Eq. (400). For the others, Eq. (400) and the argument of (a) give Eq. (405) in $(-\infty, +\infty]$, because $\mathcal{A}_{\mathrm{loop}} \ge 0$ and the matter integrand is bounded below. Since $\mathbf{U}$ is finite and bounded below, $0 < \int e^{-\mathbf{U}} d\nu^{\mathcal{T}} < \infty$ and $D_{\mathrm{KL}}(R\|\nu^{\mathcal{T}}) + \mathbb{E}_R\mathbf{U} = D_{\mathrm{KL}}(R\|R^\star) - \log\int e^{-\mathbf{U}}d\nu^{\mathcal{T}}$, which is minimal exactly at $R = R^\star$; the bound $R^\star \le c\nu^{\mathcal{T}}$ makes its free energy finite. The density of $\nu^{\mathcal{T}}$ is proportional to $e^{-\frac{1}{2}\langle\mathring{\omega}-\mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega}-\mathring{\alpha})\rangle}$. □

Corollary 2 (The current of a comparison record)**.** *In the setting of Proposition 25(c), write samples and links in the reference gauge, in which every frame is the identity; a passive change alters no generator. Let the belief marginals of Q_Y have finite second moments, with means μ_i, μ_j, covariance Σ_j of k_j and cross covariance $\Sigma_{ij} = \mathbb{E}_{Q_Y}[(k_i - \mu_i)(k_j - \mu_j)^\top]$. (a) If the only term of H_o that reads $\mathring{\omega}_e$, $e : j \to i$, is $\frac{1}{2}(k_i - M_e k_j)^\top W_a (k_i - M_e k_j)$ with $W_a \succ 0$, then $\mathbf{J}_e$ is given by Eq.* (44)*. (b) Consider the selective record* (27) *of row i at the zero comparison record, with the source label J_i among the matter coordinates and the sending map of the source j reached through e equal to $B_{ij} = B_i^{\mathrm{ref}} M_e$. If Q_Y makes J_i independent of the remaining matter coordinates with a row β_i that does not depend on $(\mathring{\omega}, y)$, and the only terms of H_o that read $\mathring{\omega}_e$ are those of row i with $J_i = j$, then*

$$\mathbf{J}_e = \frac{\beta_{ij}}{\tau_i}(B_i^{\mathrm{ref}})^\top \Gamma_i^{-1}\big[(A_i\mu_i - B_i^{\mathrm{ref}} M_e \mu_j)\mu_j^\top + A_i\Sigma_{ij} - B_i^{\mathrm{ref}} M_e \Sigma_j\big]. \tag{406}$$

In both cases Eq. (43) *holds with this current.*

Proof. For a matrix $\mathbf{M}$ in place of M_e, the trace-pairing gradient of $\frac{1}{2}(k_i - \mathbf{M}k_j)^\top W_a(k_i - \mathbf{M}k_j)$ in $\mathbf{M}$ is $-W_a(k_i - \mathbf{M}k_j)k_j^\top$, which is Q_Y-integrable by the second moments, and $\mathbb{E}_{Q_Y}[(k_i - \mathbf{M}k_j)k_j^\top] = \mu_i\mu_j^\top + \Sigma_{ij} - \mathbf{M}(\Sigma_j + \mu_j\mu_j^\top)$, which gives (a). In (b), by Eq. (28) the row-i part of H_o is $D_{iJ_i}(y)/\tau_i - \log c_i - \log \pi_{iJ_i}$, and c_i does not depend on the links because Γ_i is common to the candidate sources. Only D_{ij} reads M_e, with gradient $-(B_i^{\mathrm{ref}})^\top\Gamma_i^{-1}(A_i k_i - B_i^{\mathrm{ref}}\mathbf{M}k_j)k_j^\top$, so $\partial H_o/\partial M_e$ is $\mathbf{1}\{J_i = j\}/\tau_i$ times this matrix; the independence of J_i gives the factor β_{ij}, and the expectation is computed as in (a). The last sentence is Proposition 25(c). □

K.6 Fisher recognition flow and the learning step

In this subsection $\mathsf{G} = g^F \oplus \mathsf{G}_Y$ is the product metric of Section 6.2. Let the agents' recognition coordinates λ range over a finite-dimensional C^2 manifold $\mathcal{Q}_Y$ with a C^1 positive-definite metric G_Y, let $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ with $\mathring{\Lambda}$ positive definite in the space $\mathring{\Xi}$ of Section K.5, and assume $\ell_o \le c_{\mathfrak{A}}$ and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$. With $\mathfrak{i}_\lambda$ the generator integrand of Eq. (45), taken without an additive constant, Proposition 21(c) with $Q_{\mathring{\omega}} = Q_\lambda$, the Gaussian density of $\nu^{\mathcal{T}}$ and the entropy of R give

$$F(\mathring{\mu}, \mathring{\Lambda}, \lambda) = \mathcal{F}_V[R \otimes Q_\lambda; X, o] = \mathbb{E}_R\mathfrak{i}_\lambda + \tfrac{1}{2}\log\det\mathring{\Lambda} + c_{\mathrm{gen}}, \tag{407}$$

where $c_{\mathrm{gen}} = \log Z_{\mathrm{loop}} + \frac{1}{2}\log\det(2\pi\mathring{\Sigma}) - \frac{1}{2}\dim\mathcal{T}\log(2\pi e)$ depends only on the fixed prior. Assume that F is C^2 on $\mathcal{T} \times \mathring{\Xi}_{\succ 0} \times \mathcal{Q}_Y$, where $\mathring{\Xi}_{\succ 0}$ is the positive-definite cone of $\mathring{\Xi}$, and that each $\mathfrak{i}_\lambda$ is C^2 with $|\mathfrak{i}_\lambda| + \|\nabla\mathfrak{i}_\lambda\| + \|\nabla^2\mathfrak{i}_\lambda\| \le c_1 e^{c_2\|\mathring{\omega}\|}$. For $\mathbf{a} \in \mathcal{T}$ let $\mathbf{a} \otimes \mathbf{a}$ be the operator $\mathring{\omega} \mapsto \langle\mathbf{a}, \mathring{\omega}\rangle\mathbf{a}$.

Proposition 26 (Natural-gradient recognition flow of the generators)**.** *(a) The laws R form a minimal regular exponential family with natural parameter $\mathring{\vartheta} = (\mathring{\Lambda}\mathring{\mu}, -\frac{1}{2}\mathring{\Lambda})$, sufficient statistic*

$(\mathring{\omega}, \mathrm{pr}^{\circ}_{\mathring{\Xi}}(\mathring{\omega}\otimes\mathring{\omega}))$, *expectation parameter* $\mathring{\varpi} = (\mathring{\mu}, \mathring{\Lambda}^{-1} + \mathrm{pr}^{\circ}_{\mathring{\Xi}}(\mathring{\mu}\otimes\mathring{\mu}))$ *and log-partition function* $\mathring{\Gamma}$. *Its Fisher metric is smooth and positive definite, and in the coordinates* $(\mathring{\mu}, \mathring{\Lambda})$

$$g^F\big((\delta\mathring{\mu}, \delta\mathring{\Lambda}), (\delta\mathring{\mu}, \delta\mathring{\Lambda})\big) = \langle \delta\mathring{\mu}, \mathring{\Lambda}\delta\mathring{\mu}\rangle + \tfrac{1}{2}\operatorname{tr}(\mathring{\Lambda}^{-1}\delta\mathring{\Lambda}\mathring{\Lambda}^{-1}\delta\mathring{\Lambda}). \tag{408}$$

(b) The field $-\operatorname{grad}^{\mathsf{G}} F$, *defined by* $\mathsf{G}(\operatorname{grad}^{\mathsf{G}} F, Z) = dF[Z]$ *for every tangent vector* Z, *is* C^1*; its maximal integral curves exist, are unique, and satisfy* $\frac{d}{dt}F = -\|\operatorname{grad}^{\mathsf{G}} F\|^2_{\mathsf{G}}$. *(c) The flow is of first order: its velocity is a function of the configuration alone and vanishes exactly at the critical points of* F. *It obeys the force balance* $\mathsf{G}(\dot{Q}, \cdot) = -dF$, *in which* G *acts as a friction tensor.* F *strictly decreases along every nonconstant solution, so no nonconstant solution is periodic. (d) In the chart* $(\mathring{\vartheta}, \lambda)$ *the generator block of the explicit Euler step with step* $\Delta t > 0$ *is Eq.* (46), *whenever* $\mathring{\Lambda}'$ *is positive definite, as when* $0 < \Delta t < 1$ *and* $\mathrm{pr}^{\circ}_{\mathring{\Xi}}\mathbb{E}_R\nabla^2\mathsf{i}_\lambda \succeq 0$. *(e) Let* $R' = \mathcal{N}(\mathring{\mu}', \mathring{\Lambda}'^{-1})$ *have expectation parameter* $\mathring{\varpi}'$, *let* $\mathring{\Gamma}^\star$ *be the convex conjugate of* $\mathring{\Gamma}$, *and let* $\widetilde{F}$ *be* F *at fixed* λ *as a function of* $\mathring{\varpi}$. *If* $\mathring{\Lambda}' \succ 0$ *and* $\nabla^2\widetilde{F} \preceq c_{\mathsf{G}}\nabla^2\mathring{\Gamma}^\star$ *on the segment from* $\mathring{\varpi}$ *to* $\mathring{\varpi}'$, *then*

$$F(R') \le F(R) - \Delta t^{-1} D_{\mathrm{KL}}(R\|R') - (\Delta t^{-1} - c_{\mathsf{G}}) D_{\mathrm{KL}}(R'\|R), \tag{409}$$

so a step with $0 < \Delta t \le 1/c_{\mathsf{G}}$ *decreases* F *unless* $R' = R$. *For the full family and* $\mathsf{i}_\lambda(\mathring{\omega}) = \frac{1}{2}\langle\mathring{\omega}, \mathbf{Q}\mathring{\omega}\rangle - \langle\mathbf{b}, \mathring{\omega}\rangle + c_3$ *with* $\mathbf{Q} \succ 0$, *the hypothesis holds with* $c_{\mathsf{G}} = 1$, *and the step* $\Delta t = 1$ *gives* $R' = \mathcal{N}(\mathbf{Q}^{-1}\mathbf{b}, \mathbf{Q}^{-1})$, *the minimizer of* F *over Gaussian* R.

Proof. (a) The Lebesgue density of R is $\exp(\langle\mathring{\vartheta}_1, \mathring{\omega}\rangle + \langle\mathring{\omega}, \mathring{\vartheta}_2\mathring{\omega}\rangle - \mathring{\Gamma}(\mathring{\vartheta}))$ with $\mathring{\Gamma}(\mathring{\vartheta}) = -\frac{1}{4}\langle\mathring{\vartheta}_1, \mathring{\vartheta}_2^{-1}\mathring{\vartheta}_1\rangle - \frac{1}{2}\log\det(-2\mathring{\vartheta}_2) + \frac{1}{2}\dim\mathcal{T}\log(2\pi)$. For $\mathring{\vartheta}_2 \in \mathring{\Xi}$, $\langle\mathring{\omega}, \mathring{\vartheta}_2\mathring{\omega}\rangle = \operatorname{tr}(\mathring{\vartheta}_2(\mathring{\omega}\otimes\mathring{\omega})) = \operatorname{tr}(\mathring{\vartheta}_2\,\mathrm{pr}^{\circ}_{\mathring{\Xi}}(\mathring{\omega}\otimes\mathring{\omega}))$, which identifies the sufficient statistic. The defining integral is finite exactly when $\mathring{\vartheta}_2$ is negative definite, an open condition, so the family is regular, and it is minimal because a nonzero polynomial of degree at most two vanishes only on a Lebesgue-null set. Since $\mathring{\Lambda}^{-1} \in \mathring{\Xi}$, $\nabla\mathring{\Gamma}$, the mean of the sufficient statistic, is $\mathring{\varpi}$ [63]. The score of R in direction $(\delta\mathring{\mu}, \delta\mathring{\Lambda})$ is $\langle\delta\mathring{\mu}, \mathring{\Lambda}(\mathring{\omega} - \mathring{\mu})\rangle - \frac{1}{2}(\langle\mathring{\omega} - \mathring{\mu}, \delta\mathring{\Lambda}(\mathring{\omega} - \mathring{\mu})\rangle - \operatorname{tr}(\mathring{\Lambda}^{-1}\delta\mathring{\Lambda}))$. Its two parts are uncorrelated because odd central moments vanish; the first has variance $\langle\delta\mathring{\mu}, \mathring{\Lambda}\delta\mathring{\mu}\rangle$, and by Isserlis' identity $\langle\mathring{\omega} - \mathring{\mu}, \Delta(\mathring{\omega} - \mathring{\mu})\rangle$ has variance $2\operatorname{tr}(\Delta\mathring{\Lambda}^{-1}\Delta\mathring{\Lambda}^{-1})$ for self-adjoint Δ, which gives Eq. (408). The form vanishes only when $\delta\mathring{\mu} = 0$ and $\mathring{\Lambda}^{-1/2}\delta\mathring{\Lambda}\mathring{\Lambda}^{-1/2} = 0$, and its coefficients are smooth. (b) G is C^1 and positive definite and F is C^2, so $\operatorname{grad}^{\mathsf{G}} F$ is C^1, and the Picard–Lindelöf theorem gives existence and uniqueness of maximal integral curves. Along one, $\frac{d}{dt}F = dF[\dot{Q}] = -\mathsf{G}(\operatorname{grad}^{\mathsf{G}} F, \operatorname{grad}^{\mathsf{G}} F)$. (c) The field depends on the configuration only, and its zeros are those of dF. The defining relation with $\dot{Q} = -\operatorname{grad}^{\mathsf{G}} F$ is the force balance. If the gradient vanished at some time, the constant curve through that point would be a solution, and uniqueness would make the solution constant. On a nonconstant solution the gradient therefore never vanishes and F strictly decreases, so the solution cannot return to a point it has left. (d) In the natural chart the Fisher metric is $\nabla^2\mathring{\Gamma}$ [63], and the chain rule through $\mathring{\varpi} = \nabla\mathring{\Gamma}(\mathring{\vartheta})$ gives $\nabla_{\mathring{\vartheta}}F = \nabla^2\mathring{\Gamma}\nabla_{\mathring{\varpi}}F$. Since G is block diagonal, the generator block of the flow is $\dot{\mathring{\vartheta}} = -\nabla_{\mathring{\varpi}}F$, with Euler step $\mathring{\vartheta}' = \mathring{\vartheta} - \Delta t\nabla_{\mathring{\varpi}}F$. By Lemma 18, $\nabla_{\mathring{\mu}}\mathbb{E}_R\mathsf{i}_\lambda = \mathbb{E}_R\nabla\mathsf{i}_\lambda$, and the gradient of $\mathbb{E}_R\mathsf{i}_\lambda$ in the covariance $S = \mathring{\Lambda}^{-1}$ within $\mathring{\Xi}$ is $\frac{1}{2}\mathrm{pr}^{\circ}_{\mathring{\Xi}}\mathbb{E}_R\nabla^2\mathsf{i}_\lambda$, while that of $\frac{1}{2}\log\det\mathring{\Lambda} = -\frac{1}{2}\log\det S$ is $-\frac{1}{2}\mathring{\Lambda}$. Put $\Upsilon = \frac{1}{2}\mathrm{pr}^{\circ}_{\mathring{\Xi}}\mathbb{E}_R\nabla^2\mathsf{i}_\lambda - \frac{1}{2}\mathring{\Lambda} \in \mathring{\Xi}$. Since $\mathring{\mu} = \mathring{\varpi}_1$ and $S = \mathring{\varpi}_2 - \mathrm{pr}^{\circ}_{\mathring{\Xi}}(\mathring{\varpi}_1\otimes\mathring{\varpi}_1)$, and since $\operatorname{tr}(\Upsilon\,\mathrm{pr}^{\circ}_{\mathring{\Xi}}\Delta) = \operatorname{tr}(\Upsilon\Delta)$ and $\operatorname{tr}(\Upsilon(\eta\otimes\mathring{\mu} + \mathring{\mu}\otimes\eta)) = 2\langle\Upsilon\mathring{\mu}, \eta\rangle$, the chain rule gives $\nabla_{\mathring{\varpi}_2}F = \Upsilon$ and $\nabla_{\mathring{\varpi}_1}F = \mathbb{E}_R\nabla\mathsf{i}_\lambda - 2\Upsilon\mathring{\mu}$. The second component of the Euler step is $-\frac{1}{2}\mathring{\Lambda}' = -\frac{1}{2}\mathring{\Lambda} - \Delta t\Upsilon$, the first equation of Eq. (46). The first component is $\mathring{\Lambda}'\mathring{\mu}' = \mathring{\Lambda}\mathring{\mu} - \Delta t\mathbb{E}_R\nabla\mathsf{i}_\lambda + 2\Delta t\Upsilon\mathring{\mu} = \mathring{\Lambda}'\mathring{\mu} - \Delta t\mathbb{E}_R\nabla\mathsf{i}_\lambda$, the second. If $0 < \Delta t < 1$ and the projected expected Hessian is positive semidefinite, $\mathring{\Lambda}' \succeq (1 - \Delta t)\mathring{\Lambda} \succ 0$. (e) The Lebesgue density r of R gives $\mathbb{E}_R\log r = \langle\mathring{\vartheta}, \mathring{\varpi}\rangle - \mathring{\Gamma}(\mathring{\vartheta}) = \mathring{\Gamma}^\star(\mathring{\varpi})$ [64], so Eq. (407) reads $\widetilde{F} = \mathbb{E}_R\mathsf{i}_\lambda + \mathring{\Gamma}^\star + \text{const}$. Taking the R-expectation of $\log(r/r')$ gives $D_{\mathrm{KL}}(R\|R') = \mathring{\Gamma}(\mathring{\vartheta}') - \mathring{\Gamma}(\mathring{\vartheta}) - \langle\mathring{\varpi}, \mathring{\vartheta}' - \mathring{\vartheta}\rangle$, and the Bregman divergence of

$\mathring{\Gamma}^\star$ is $\mathring{\Gamma}^\star(\mathring{\varpi}') - \mathring{\Gamma}^\star(\mathring{\varpi}) - \langle \mathring{\vartheta}, \mathring{\varpi}' - \mathring{\varpi}\rangle = D_{\mathrm{KL}}(R'\|R)$. The expectation domain $\{\mathring{\varpi}_2 - \mathrm{pr}_{\mathring{\Xi}}(\mathring{\varpi}_1 \otimes \mathring{\varpi}_1) \succ 0\}$ is convex: along a segment with weights $1-s$ and s, this difference equals the convex combination of its endpoint values plus $s(1-s)\,\mathrm{pr}_{\mathring{\Xi}}((\mathbf{a}-\mathbf{a}')\otimes(\mathbf{a}-\mathbf{a}'))$, where $\mathbf{a}, \mathbf{a}'$ are the endpoint means, and $\mathrm{pr}_{\mathring{\Xi}}$ preserves positive semidefiniteness. Taylor's formula with integral remainder along the segment and the curvature hypothesis give $\widetilde{F}(\mathring{\varpi}') \le \widetilde{F}(\mathring{\varpi}) + \langle \nabla\widetilde{F}(\mathring{\varpi}), \mathring{\varpi}' - \mathring{\varpi}\rangle + c_{\mathsf{G}} D_{\mathrm{KL}}(R'\|R)$, since the same formula applied to $\mathring{\Gamma}^\star$ produces its Bregman divergence. By (d), $\nabla\widetilde{F}(\mathring{\varpi}) = \Delta t^{-1}(\mathring{\vartheta} - \mathring{\vartheta}')$, and adding the two divergence formulas gives $\langle \mathring{\vartheta}' - \mathring{\vartheta}, \mathring{\varpi}' - \mathring{\varpi}\rangle = D_{\mathrm{KL}}(R\|R') + D_{\mathrm{KL}}(R'\|R)$; substitution proves Eq. (409). For the quadratic integrand in the full family, $\mathbb{E}_R \mathfrak{i}_\lambda = \frac{1}{2}\operatorname{tr}(\mathbf{Q}\mathring{\varpi}_2) - \langle \mathbf{b}, \mathring{\varpi}_1\rangle + c_3$ is affine in $\mathring{\varpi}$, so $\nabla^2\widetilde{F} = \nabla^2\mathring{\Gamma}^\star$. Moreover $\mathbb{E}_R\nabla^2\mathfrak{i}_\lambda = \mathbf{Q}$ and $\mathbb{E}_R\nabla\mathfrak{i}_\lambda = \mathbf{Q}\mathring{\mu} - \mathbf{b}$, so Eq. (46) with $\Delta t = 1$ gives $\mathring{\Lambda}' = \mathbf{Q}$ and $\mathring{\mu}' = \mathbf{Q}^{-1}\mathbf{b}$. Then $R' \propto e^{-\mathfrak{i}_\lambda}$, and $F = D_{\mathrm{KL}}(R\|R') + \text{const}$ is minimal at $R = R'$. □

K.7 The episode-history prior and history inertia

Use the history law (47) with $\mathcal{A}_{\mathrm{loop}}$ finite, measurable and nonnegative, $\mathbf{m} > 0$, and either kinetic term: the Gaussian increment $\mathfrak{U}(\mathring{\omega}', \mathring{\omega}) = \frac{1}{2}\|\mathring{\omega}' - \mathring{\omega}\|^2$, or the time-like defect $\mathfrak{U}(\mathring{\omega}', \mathring{\omega}) = \sum_{e\in E_+} \mathfrak{Q}(e^{-\mathrm{d}\rho\mathring{\omega}'_e} e^{\mathrm{d}\rho\mathring{\omega}_e})$ with $\mathfrak{Q}(\mathbf{H}) = \frac{1}{2}\|\mathbf{H} - I\|^2_{\mathsf{G}}$. The matrix $e^{-\mathrm{d}\rho\mathring{\omega}'_e} e^{\mathrm{d}\rho\mathring{\omega}_e}$ is the represented reference holonomy of the time-like loop formed by e at one episode, $\bar{e}$ at the next and the identity links of its endpoints between them.

Proposition 27 (The generator history prior)**.** *(a)* $0 < Z_{\mathrm{hist}} \le Z_{\mathrm{loop}}^{n_1+1} \le 1$*, so* $\mathbb{P}^{\mathrm{hist}}$ *is a probability law with density at most* Z_{hist}^{-1} *with respect to* $\bigotimes_n \nu^{\mathcal{T}}$*, and every function bounded by* $c_1 e^{c_2\|\mathring{\omega}^{(0:n_1)}\|}$ *is integrable under it. (b) Put* $Z^{\leftarrow}_{n_1} \equiv 1$ *and, for* $n < n_1$*,* $Z^{\leftarrow}_n(\mathring{\omega}) = \int e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega}') - \mathbf{m}\mathfrak{U}(\mathring{\omega}', \mathring{\omega})} Z^{\leftarrow}_{n+1}(\mathring{\omega}')\nu^{\mathcal{T}}(d\mathring{\omega}')$*. Then* $\mathbb{P}^{\mathrm{hist}}$ *is the law of the Markov chain with initial law* $Z_{\mathrm{hist}}^{-1} e^{-\mathcal{A}_{\mathrm{loop}}} Z^{\leftarrow}_0 \nu^{\mathcal{T}}$ *and transition kernels*

$$\mathbb{P}^{(n+1\leftarrow n)}(d\mathring{\omega}' \mid \mathring{\omega}) = \frac{e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega}') - \mathbf{m}\mathfrak{U}(\mathring{\omega}', \mathring{\omega})} Z^{\leftarrow}_{n+1}(\mathring{\omega}')}{Z^{\leftarrow}_n(\mathring{\omega})} \nu^{\mathcal{T}}(d\mathring{\omega}'), \tag{410}$$

which depend on n *and on the horizon through the backward normalizers. The chain whose kernels normalize each increment factor separately, started from the law proportional to* $e^{-\mathcal{A}_{\mathrm{loop}}}\nu^{\mathcal{T}}$*, has the law* $\mathbb{P}^{\mathrm{hist}}$ *exactly when* $Z^{\leftarrow}_{n_1-1}$ *is constant* $\nu^{\mathcal{T}}$*-almost everywhere; this fails for one scalar generator with* $\nu^{\mathcal{T}} = \mathcal{N}(0,1)$*,* $\mathcal{A}_{\mathrm{loop}} = 0$*, the Gaussian increment and* $n_1 = 1$*. (c) With static frames, the identity link of agent* i *between two episodes has the constant twist* $U_i^{-1}U_i = I$*. The time-like links therefore carry no variable,* $\mathbb{P}^{\mathrm{hist}}$ *is a law on the spatial generator histories alone, and each kinetic term is a function of the generators of one edge at two consecutive episodes.*

Proof. (a) Since $\mathcal{A}_{\mathrm{loop}} \ge 0$, $\mathfrak{U} \ge 0$ and $\mathbf{m} > 0$, the integrand of Z_{hist} is positive and at most $\prod_n e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega}^{(n)})}$, whose integral is $Z_{\mathrm{loop}}^{n_1+1} \le 1$; a product of Gaussian laws integrates $e^{c_2\|\cdot\|}$. (b) By Tonelli's theorem and induction from n_1, each $Z^{\leftarrow}_n$ is measurable with values in $(0,1]$, each kernel has total mass one, and $\int e^{-\mathcal{A}_{\mathrm{loop}}} Z^{\leftarrow}_0 d\nu^{\mathcal{T}} = Z_{\mathrm{hist}}$. Along a history the ratios $Z^{\leftarrow}_{n+1}/Z^{\leftarrow}_n$ telescope and $Z^{\leftarrow}_{n_1} \equiv 1$, so the chain density is that of $\mathbb{P}^{\mathrm{hist}}$, and the kernels are its conditional laws. The kernels of the separately normalized chain are the increment factors divided by $Z^{\leftarrow}_{n_1-1}(\mathring{\omega})$, the normalizer of one factor, so that chain has density proportional to $\prod_{n<n_1} Z^{\leftarrow}_{n_1-1}(\mathring{\omega}^{(n)})^{-1}$ relative to $\mathbb{P}^{\mathrm{hist}}$; by Fubini's theorem this product is almost everywhere constant exactly when $Z^{\leftarrow}_{n_1-1}$ is. In the scalar case $Z^{\leftarrow}_0(\mathring{\omega}) = (1+\mathbf{m})^{-1/2}\exp[-\mathbf{m}\mathring{\omega}^2/(2(1+\mathbf{m}))]$, which is not constant. (c) With a static frame U_i, the twist of the identity link does not depend on any variable, and Eq. (47) contains no other link. □

Proposition 28 (Inertia of the most probable history)**.** *Let $\mathfrak{V}_n : \mathcal{T} \to \mathbb{R}$ be differentiable and $\mathcal{A}^{\mathrm{hist}} = \sum_{n=0}^{n_1} \mathfrak{V}_n(\mathring{\omega}^{(n)}) + \mathbf{m} \sum_{n<n_1} \mathfrak{U}(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)})$. With $\mathfrak{V}_n = \mathcal{A}_{\mathrm{loop}} + \frac{1}{2}\langle \cdot - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\cdot - \mathring{\alpha})\rangle$, $\mathcal{A}^{\mathrm{hist}}$ is the negative log Lebesgue density of $\mathbb{P}^{\mathrm{hist}}$ up to a constant, and adding an episode matter term gives the corresponding posterior or conditional objective. A most probable history minimizes $\mathcal{A}^{\mathrm{hist}}$ with free or held ends. (a) For the Gaussian increment every critical point satisfies Eq.* (48)*, and with free ends $\mathbf{m}(\mathring{\omega}^{(0)} - \mathring{\omega}^{(1)}) + \nabla\mathfrak{V}_0(\mathring{\omega}^{(0)}) = 0$ and $\mathbf{m}(\mathring{\omega}^{(n_1)} - \mathring{\omega}^{(n_1-1)}) + \nabla\mathfrak{V}_{n_1}(\mathring{\omega}^{(n_1)}) = 0$. If each $\mathfrak{V}_n$ is continuous and $\sum_n \mathfrak{V}_n(\mathring{\omega}^{(n)}) \to \infty$ as $\|\mathring{\omega}^{(0:n_1)}\| \to \infty$, as when each $\mathfrak{V}_n$ is at least the anchor quadratic minus a constant, a most probable history exists. (b) For the time-like defect let $\mathrm{d}\rho$ be injective, let $\mathfrak{g}$ carry the pulled-back inner product, let $\mathrm{pr}_{\mathfrak{g}}$ be the $\langle\cdot,\cdot\rangle_{\mathsf{G}}$-orthogonal projection of matrices onto $\mathrm{d}\rho(\mathfrak{g})$, read in $\mathfrak{g}$, and write $\Theta_e^{\leftarrow} = e^{-\mathrm{d}\rho\mathring{\omega}'_e} e^{\mathrm{d}\rho\mathring{\omega}_e}$ for a later family $\mathring{\omega}'$ and an earlier family $\mathring{\omega}$. Then*

$$\begin{aligned}(\partial_{\mathring{\omega}}\mathfrak{U})_e &= \mathrm{pr}_{\mathfrak{g}}\,\mathbf{L}\big((\mathrm{d}\rho\mathring{\omega}_e)^*, (e^{-\mathrm{d}\rho\mathring{\omega}'_e})^*(\Theta_e^{\leftarrow} - I)\big),\\ (\partial_{\mathring{\omega}'}\mathfrak{U})_e &= -\mathrm{pr}_{\mathfrak{g}}\,\mathbf{L}\big(-(\mathrm{d}\rho\mathring{\omega}'_e)^*, (\Theta_e^{\leftarrow} - I)(e^{\mathrm{d}\rho\mathring{\omega}_e})^*\big),\end{aligned} \tag{411}$$

and every critical point satisfies, for $0 < n < n_1$,

$$\mathbf{m}\big[\partial_{\mathring{\omega}}\mathfrak{U}(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}) + \partial_{\mathring{\omega}'}\mathfrak{U}(\mathring{\omega}^{(n)}, \mathring{\omega}^{(n-1)})\big] + \nabla\mathfrak{V}_n(\mathring{\omega}^{(n)}) = 0. \tag{412}$$

For $r > 0$ let $c_r = \sqrt{2}\sup\|D^3\mathfrak{U}\|$ over $\|\mathring{\omega}\|, \|\mathring{\omega}'\| \le r$. On that set $\|\partial_{\mathring{\omega}}\mathfrak{U} - (\mathring{\omega} - \mathring{\omega}')\|$ and $\|\partial_{\mathring{\omega}'}\mathfrak{U} - (\mathring{\omega}' - \mathring{\omega})\|$ are at most $c_r r\|\mathring{\omega}' - \mathring{\omega}\|$, so at a history in the ball of radius r the left side of Eq. (412) *differs from that of Eq.* (48) *by at most $\mathbf{m}c_r r(\|\mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)}\| + \|\mathring{\omega}^{(n)} - \mathring{\omega}^{(n-1)}\|)$. At fixed $\mathring{\omega}$, $\mathfrak{U}(\mathring{\omega} + \delta\mathring{\omega}, \mathring{\omega}) = \frac{1}{2}\sum_e \|\mathbf{L}(-\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e)e^{\mathrm{d}\rho\mathring{\omega}_e}\|_{\mathsf{G}}^2 + O(\|\delta\mathring{\omega}\|^3)$, a generator-dependent increment tensor that reduces to $\frac{1}{2}\|\delta\mathring{\omega}\|^2$ at $\mathring{\omega} = 0$.*

Proof. (a) $\mathring{\omega}^{(n)}$ enters the increments to $n-1$ and $n+1$ in the interior and one increment at each free end, so differentiation gives the equations. Under the growth condition $\mathcal{A}^{\mathrm{hist}}$ is continuous and coercive on a finite-dimensional space, or on the affine subspace of held ends, and attains its minimum. (b) By Lemma 17(a), the differential of $\Theta_e^{\leftarrow}$ in direction $\delta\mathring{\omega}_e$ is $e^{-\mathrm{d}\rho\mathring{\omega}'_e}\mathbf{L}(\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e)$ in the earlier argument and $-\mathbf{L}(-\mathrm{d}\rho\mathring{\omega}'_e, \mathrm{d}\rho\delta\mathring{\omega}_e)e^{\mathrm{d}\rho\mathring{\omega}_e}$ in the later one, and the differential of $\mathfrak{Q}$ at $\mathbf{H}$ is $\mathbf{T} \mapsto \langle \mathbf{H} - I, \mathbf{T}\rangle_{\mathsf{G}}$. Moving the factors across the inner product with Lemmas 15(a) and 17(b), and projecting onto $\mathrm{d}\rho(\mathfrak{g})$, gives Eq. (411), and Eq. (412) is the stationarity of $\mathcal{A}^{\mathrm{hist}}$. The function $\mathfrak{U}$ is smooth, nonnegative and zero on the diagonal, so its gradient vanishes there. Since $\Theta_e^{\leftarrow} = I + \mathrm{d}\rho(\mathring{\omega}_e - \mathring{\omega}'_e) + O(\|(\mathring{\omega}, \mathring{\omega}')\|^2)$ and the inner product is pulled back, $\mathfrak{U} = \frac{1}{2}\|\mathring{\omega} - \mathring{\omega}'\|^2 + O(\|(\mathring{\omega}, \mathring{\omega}')\|^3)$, so the mixed second derivative at the origin is $-I$. Let $\mathrm{R}_1(\mathring{\omega}', \mathring{\omega}) = \partial_{\mathring{\omega}}\mathfrak{U} - (\mathring{\omega} - \mathring{\omega}')$. Then $\mathrm{R}_1(\mathring{\omega}, \mathring{\omega}) = 0$ and $\mathrm{R}_1(\mathring{\omega}', \mathring{\omega}) = \int_0^1 D_{\mathring{\omega}'}\mathrm{R}_1(\mathring{\omega} + t(\mathring{\omega}' - \mathring{\omega}), \mathring{\omega})[\mathring{\omega}' - \mathring{\omega}]dt$, where $D_{\mathring{\omega}'}\mathrm{R}_1$ vanishes at the origin and has derivative bounded by $\sup\|D^3\mathfrak{U}\|$ on the convex set $\{\|\mathring{\omega}\|, \|\mathring{\omega}'\| \le r\}$, whose points lie within $\sqrt{2}r$ of the origin. This gives the bound, and the same argument applies to $\partial_{\mathring{\omega}'}\mathfrak{U}$. For the increment tensor, the differential in the later argument gives $\Theta_e^{\leftarrow} = I - \mathbf{L}(-\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e)e^{\mathrm{d}\rho\mathring{\omega}_e} + O(\|\delta\mathring{\omega}\|^2)$ at $\mathring{\omega}' = \mathring{\omega} + \delta\mathring{\omega}$, and $\mathbf{L}(0, \mathbf{T}) = \mathbf{T}$. □

K.8 The linearized link equation

Assume the setting of Section 6.4: $\mathrm{d}\rho$ is injective and $\mathfrak{g}$ carries the pulled-back inner product; the history prior has zero anchor, prior covariance $\mathring{\Sigma}$ at every episode, action $\mathcal{A}_{\mathrm{loop}} = \sum_p c_p\mathfrak{Q}_p$, mass $\mathbf{m}$ and either kinetic term; and samples are written in the reference gauge. At episode n a finite set $\mathfrak{A}_n$ of records enters with generator-independent weights $\beta_a \in [0, 1]$. Record a has a scope $\{i, j\}$ joined by an edge copy $e : j \to i$ of E_+ or of its reversal, with $\mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e$, and the energy $E_a(k; \mathring{\omega}) = \frac{1}{2}(k_i - e^{\mathrm{d}\rho\mathring{\omega}_e}k_j)^\top W_a(k_i - e^{\mathrm{d}\rho\mathring{\omega}_e}k_j)$ with $W_a \succ 0$. Matter recognition is a fixed product

Gaussian law under which k_i at episode n has mean $\mu_i^{(n)}$ and covariance $\Sigma_i^{(n)}$, and

$$\begin{aligned}\mathcal{A}^{\mathrm{hist}}(\mathring{\omega}) = \sum_{n=0}^{n_1}\Big[\tfrac{1}{2}\langle\mathring{\omega}^{(n)},\mathring{\Sigma}^{-1}\mathring{\omega}^{(n)}\rangle + \mathcal{A}_{\mathrm{loop}}(\mathring{\omega}^{(n)}) + \sum_{a\in\mathfrak{A}_n}\beta_a\mathbb{E}[E_a(k;\mathring{\omega}^{(n)})]\Big] \\ + \mathbf{m}\sum_{n<n_1}\mathfrak{U}(\mathring{\omega}^{(n+1)},\mathring{\omega}^{(n)}).\end{aligned} \tag{413}$$

On $\mathcal{T}^{n_1+1}$, with $\langle\mathring{\omega},\mathring{\omega}'\rangle = \sum_n\langle\mathring{\omega}^{(n)},\mathring{\omega}'^{(n)}\rangle$, let $\mathbf{K}$, $\mathring{\Phi}$, $\mathbf{j}$ and $\mathbf{N}_0$ be defined by Eqs. (49), (50) and (51), where $B_{a,n} = \mathrm{d}\rho\mathring{\omega}_e^{(n)}$ for the copy e of record a, and let $c_{\mathring{\Sigma}}$ be the smallest eigenvalue of $\mathring{\Sigma}^{-1}$.

Proposition 29 (The linearized link equation)**.** *(a)* $\nabla\mathcal{A}^{\mathrm{hist}}(0) = -\mathbf{j}$ *and* $\nabla^2\mathcal{A}^{\mathrm{hist}}(0) = \mathbf{K}+\mathbf{N}_0$*, so to first order in the generators and the currents the stationarity condition is* $\mathbf{K}\mathring{\omega} = \mathbf{j}$*. (b)* $\mathbf{K}\succeq c_{\mathring{\Sigma}}I$*, so* $\mathring{\omega}_{\mathrm{lin}} = \mathbf{K}^{-1}\mathbf{j}$ *is the unique solution, with* $\|\mathring{\omega}_{\mathrm{lin}}\|\le\|\mathbf{j}\|/c_{\mathring{\Sigma}}$*. (c) Let* $c_0 = \|\mathbf{N}_0\|$ *and* $c_r = \sup_{\|\mathring{\omega}\|\le r}\|D^3\mathcal{A}^{\mathrm{hist}}(\mathring{\omega})\|$*. If* $c_0 + c_r r \le c_{\mathring{\Sigma}}/2$ *and* $\|\mathbf{j}\|\le c_{\mathring{\Sigma}}r/4$*, then* $\mathcal{A}^{\mathrm{hist}}$ *has exactly one critical point* $\mathring{\omega}_\star$ *in the ball of radius* r *about zero; it is the unique minimizer of* $\mathcal{A}^{\mathrm{hist}}$ *on that ball,* $\|\mathring{\omega}_\star\|\le 2\|\mathbf{j}\|/c_{\mathring{\Sigma}}$*, and*

$$\|\mathring{\omega}_\star - \mathring{\omega}_{\mathrm{lin}}\| \le \frac{2\|\mathbf{j}\|}{c_{\mathring{\Sigma}}^2}\Big(c_0 + \frac{c_r\|\mathbf{j}\|}{c_{\mathring{\Sigma}}}\Big). \tag{414}$$

Proof. (a) The anchor term has gradient zero and Hessian $\mathring{\Sigma}^{-1}$ at zero. Each $\mathfrak{Q}_p$ is smooth, being a polynomial in the entries of matrix exponentials, and $\mathfrak{Q}_p = \frac{1}{2}\|\mathbf{C}_p\mathring{\omega}\|^2 + O(\|\mathring{\omega}\|^3)$ by Proposition 18(c), so by uniqueness of Taylor coefficients its gradient at zero vanishes and its Hessian is $\mathbf{C}_p^\top\mathbf{C}_p$. The Gaussian increment is the quadratic form of $\mathbf{m}(-\Delta_n)$, and the time-like defect has the same second-order part and zero gradient at the origin by the expansion in the proof of Proposition 28(b). For a record a with $\Omega = \mathrm{d}\rho\mathring{\omega}_e$, dropping the episode index, independence of k_i and k_j gives $\mathbb{E}[E_a] = \frac{1}{2}[\|\mu_i - e^\Omega\mu_j\|^2_{W_a} + \mathrm{tr}(W_a\Sigma_i) + \mathrm{tr}(W_a e^\Omega\Sigma_j e^{\Omega^\top})]$. With $e^\Omega = I + \Omega + \frac{1}{2}\Omega^2 + O(\|\Omega\|^3)$ and $M_j = \Sigma_j + \mu_j\mu_j^\top$, the first-order part is $\mathrm{tr}(W_a\Omega\Sigma_j) - (\mu_i-\mu_j)^\top W_a\Omega\mu_j = -\mathrm{tr}(\mathbf{J}_a^\top\Omega)$, and the second-order part is

$$\tfrac{1}{2}\big[\mathrm{tr}(W_a\Omega M_j\Omega^\top) + \mathrm{tr}(W_a\Omega^2\Sigma_j) - (\mu_i-\mu_j)^\top W_a\Omega^2\mu_j\big] = \tfrac{1}{2}\big[\mathrm{tr}(W_a\Omega M_j\Omega^\top) - \mathrm{tr}(\mathbf{J}_a^\top\Omega^2)\big],$$

because $\mathbf{J}_a^\top = (\mu_j(\mu_i-\mu_j)^\top - \Sigma_j)W_a$. For a record on a reversed copy, $\Omega = -\mathrm{d}\rho\mathring{\omega}_{\bar{e}}$ with $\bar{e}\in E_+$, which changes the sign of the first-order part only. Summing with the weights gives the gradient and the Hessian. (b) The loop, episode and matter forms are nonnegative, the last because $\mathrm{tr}(W_a\Omega M_j\Omega^\top) = \mathbb{E}\|W_a^{1/2}\Omega k_j\|^2$, so $\langle\mathring{\omega},\mathbf{K}\mathring{\omega}\rangle \ge c_{\mathring{\Sigma}}\|\mathring{\omega}\|^2$ and $\|\mathbf{K}^{-1}\|\le 1/c_{\mathring{\Sigma}}$. (c) Write $\nabla\mathcal{A}^{\mathrm{hist}}(\mathring{\omega}) = -\mathbf{j} + \mathbf{K}\mathring{\omega} + \nabla\mathrm{R}(\mathring{\omega})$, so that $\nabla\mathrm{R}(0)=0$, $\nabla^2\mathrm{R}(0)=\mathbf{N}_0$, and $\|\nabla^2\mathrm{R}(\mathring{\omega})\|\le c_0 + c_r\|\mathring{\omega}\|$ for $\|\mathring{\omega}\|\le r$. The critical points are the fixed points of $\mathring{\omega}\mapsto\mathbf{K}^{-1}(\mathbf{j}-\nabla\mathrm{R}(\mathring{\omega}))$. On the ball of radius $r_1 = 2\|\mathbf{j}\|/c_{\mathring{\Sigma}}\le r/2$, $\|\nabla\mathrm{R}(\mathring{\omega})\|\le(c_0+c_r r)\|\mathring{\omega}\|\le\frac{1}{2}c_{\mathring{\Sigma}}r_1$, so the map sends this ball into itself, its values having norm at most $\|\mathbf{j}\|/c_{\mathring{\Sigma}} + r_1/2 = r_1$, and it has Lipschitz constant at most $(c_0+c_r r)/c_{\mathring{\Sigma}}\le\frac{1}{2}$ there; it therefore has exactly one fixed point $\mathring{\omega}_\star$ in that ball. On the ball of radius r, $\nabla^2\mathcal{A}^{\mathrm{hist}}\succeq(c_{\mathring{\Sigma}} - c_0 - c_r r)I\succeq\frac{1}{2}c_{\mathring{\Sigma}}I$, so $\mathcal{A}^{\mathrm{hist}}$ is strongly convex on that convex set and has at most one critical point there, which is its unique minimizer on the set. Finally $\mathring{\omega}_\star - \mathring{\omega}_{\mathrm{lin}} = -\mathbf{K}^{-1}\nabla\mathrm{R}(\mathring{\omega}_\star)$ and $\|\nabla\mathrm{R}(\mathring{\omega}_\star)\|\le(c_0+\frac{1}{2}c_r\|\mathring{\omega}_\star\|)\|\mathring{\omega}_\star\|$, which with $\|\mathring{\omega}_\star\|\le r_1$ gives Eq. (414). □

K.9 Induced link action

For the undirected alternative, let each link family Θ carry a nonnegative potential $\widetilde{p}_{X,\Theta}(o,y)$, jointly measurable in (Θ,o,y), with matter partition function $Z_{X,\Theta} = \int\widetilde{p}_{X,\Theta}d(\nu^Y\otimes\nu^O)\in[0,+\infty]$. When $0<Z_{X,\Theta}<\infty$ the undirected model at Θ is $\widetilde{p}_{X,\Theta}/Z_{X,\Theta}$.

Proposition 30 (Induced link action)**.** *(a) The generator marginal of the joint* (25) *is* P_{gen}*, and with record densities the link marginal of the posterior is proportional to* $p(o \mid \mathring{\omega})P_{\mathrm{gen}}(d\mathring{\omega})$*. (b) Let* $\mathfrak{Z}$ *be as in Eq.* (164) *and assume* $0 < \mathfrak{Z} < \infty$*. The law with density* $\mathfrak{Z}^{-1}e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega})}\widetilde{p}_{X,\Theta(\mathring{\omega})}(o,y)$ *with respect to* $\nu^{\mathcal{T}} \otimes \nu^{Y} \otimes \nu^{O}$ *is a probability law;* $Z_{X,\Theta(\mathring{\omega})} < \infty$ *for* $\nu^{\mathcal{T}}$*-almost every* $\mathring{\omega}$*; the generator marginal is* $\mathfrak{Z}^{-1}e^{-\mathcal{A}_{\mathrm{loop}}-\mathcal{A}_{\mathrm{ind}}}\nu^{\mathcal{T}}$ *with* $\mathcal{A}_{\mathrm{ind}} = -\log Z_{X,\Theta(\cdot)} \in (-\infty, +\infty]$ *almost everywhere; and on* $\{0 < Z_{X,\Theta(\mathring{\omega})} < \infty\}$ *the conditional law given* $\mathring{\omega}$ *is the undirected model at* $\Theta(\mathring{\omega})$*. A record extension with densities is the case* $Z_{X,\Theta} \equiv 1$*, in which* $\mathcal{A}_{\mathrm{ind}} = 0$*. (c) A proper generator prior does not imply* $\mathfrak{Z} < \infty$*. For* $G = \mathrm{GL}^{+}(1)$*, trivial frames,* $\mathcal{A}_{\mathrm{loop}} = 0$*,* $\nu^{\mathcal{T}} = \mathcal{N}(0,1)$*, two scalar agents with standard Gaussian self potentials, no record, and the edge potential* $\exp(e^{\mathring{\omega}_e}k_1k_2)$*, one has* $Z_{X,\Theta} = (1 - e^{2\mathring{\omega}_e})^{-1/2}$ *for* $\mathring{\omega}_e < 0$ *and* $Z_{X,\Theta} = +\infty$ *for* $\mathring{\omega}_e \geq 0$*, so* $\mathfrak{Z} = +\infty$*. (d) If the potential is a centered Gaussian,* $\widetilde{p}_{X,\Theta}(y) = \exp(-\frac{1}{2}y^{\top}\mathbf{P}(\Theta)y)$ *on* $\mathbb{R}^d$ *with symmetric* $\mathbf{P}(\Theta)$*, Lebesgue reference and no record, then* $Z_{X,\Theta} = (2\pi)^{d/2}(\det \mathbf{P}(\Theta))^{-1/2}$ *when* $\mathbf{P}(\Theta) \succ 0$ *and* $Z_{X,\Theta} = +\infty$ *otherwise. Hence* $\mathcal{A}_{\mathrm{ind}} = \frac{1}{2}\log\det \mathbf{P}(\Theta(\mathring{\omega})) - \frac{d}{2}\log(2\pi)$*, and* $\mathfrak{Z} < \infty$ *requires* $\mathbf{P}(\Theta(\mathring{\omega})) \succ 0$ *for* $\nu^{\mathcal{T}}$*-almost every* $\mathring{\omega}$ *and* $\int e^{-\mathcal{A}_{\mathrm{loop}}}(\det \mathbf{P}(\Theta(\cdot)))^{-1/2}d\nu^{\mathcal{T}} < \infty$*.*

Proof. (a) By Tonelli's theorem, integrate the records, each a kernel of unit mass, and then the matter against the probability kernel $P^{Y}_{\mathring{\omega}}$; the posterior marginal is that of Proposition 21(c). (b) Tonelli's theorem gives $\int e^{-\mathcal{A}_{\mathrm{loop}}}\widetilde{p}_{X,\Theta(\mathring{\omega})}d(\nu^{Y} \otimes \nu^{O}) = e^{-\mathcal{A}_{\mathrm{loop}}(\mathring{\omega})}Z_{X,\Theta(\mathring{\omega})}$, whose $\nu^{\mathcal{T}}$-integral is $\mathfrak{Z}$, so the density integrates to one, and finiteness of $\mathfrak{Z}$ with $e^{-\mathcal{A}_{\mathrm{loop}}} > 0$ forces $Z_{X,\Theta(\mathring{\omega})} < \infty$ almost everywhere. The generator marginal is $\mathfrak{Z}^{-1}e^{-\mathcal{A}_{\mathrm{loop}}}Z_{X,\Theta(\mathring{\omega})}\nu^{\mathcal{T}}$, with $\mathcal{A}_{\mathrm{ind}} = +\infty$ where $Z_{X,\Theta} = 0$, and dividing the joint density by the marginal density gives the conditional law. The densities of a record extension are conditional densities of unit mass in (y, o) for every link family, so as potentials they have $Z_{X,\Theta} = 1$. (c) Put $s = e^{\mathring{\omega}_e}$ and $u = (k_1 + k_2)/\sqrt{2}$, $v = (k_1 - k_2)/\sqrt{2}$. The exponent $-(k_1^2 + k_2^2)/2 + sk_1k_2$ equals $-(1-s)u^2/2 - (1+s)v^2/2$, so the integral of $(2\pi)^{-1}$ times its exponential is $((1-s)(1+s))^{-1/2}$ for $s < 1$ and diverges in u for $s \geq 1$; and $\nu^{\mathcal{T}}\{\mathring{\omega}_e \geq 0\} = \frac{1}{2}$. (d) Diagonalizing $\mathbf{P}(\Theta)$ factorizes the integral into one-dimensional Gaussian integrals, each equal to $(2\pi/\lambda_k)^{1/2}$ when the eigenvalue λ_k is positive and infinite otherwise. The conditions on $\mathfrak{Z}$ are those of (b). □

K.10 Coarse link laws

Let G be a closed subgroup of $\mathrm{GL}(K, \mathbb{R})$ acting by inclusion, so that $\mathfrak{g} \subseteq \mathfrak{gl}(K, \mathbb{R})$ and the exponential is the matrix exponential; extend $\langle \cdot, \cdot \rangle_{\mathsf{G}}$ to $\mathfrak{gl}(K, \mathbb{R})$ by the same formula, and let $\| \cdot \|$ be the operator norm and E_{kl} the matrix units. Let the fine generators $\mathring{\omega}_{e'}$ and $\mathring{\omega}_e$ of Eq. (165) be independent Gaussian random elements of $\mathfrak{g}$ with means $\mathring{\mu}_{e'}, \mathring{\mu}_e$ and positive-definite precisions $\mathring{\Lambda}_{e'}, \mathring{\Lambda}_e$, self-adjoint for $\langle \cdot, \cdot \rangle_{\mathsf{G}}$, as under a product Gaussian recognition law of the generators. A coarse generator is a $\mathfrak{g}$-valued random variable $\mathring{\omega}^c$ with $\exp(\mathring{\omega}^c) = \mathring{\Theta}^c_e$ almost surely. For $\upsilon \in \mathfrak{g}$ put $\upsilon_{e'} = \mathring{\Lambda}^{-1}_{e'}\upsilon$ and $\upsilon_e = \mathring{\Lambda}^{-1}_{e}\upsilon$, let $\zeta_{e'} \sim \mathcal{N}(0, \mathring{\Lambda}^{-1}_{e'})$ and $\zeta_e \sim \mathcal{N}(0, \mathring{\Lambda}^{-1}_{e})$ be independent, and define the cubic and quartic commutator forms

$$\begin{aligned} \kappa_3(\upsilon) &= \langle \upsilon, [\upsilon_{e'}, \upsilon_e] \rangle_{\mathsf{G}}, \\ \kappa_4(\upsilon) &= 3\mathbb{E}\langle \upsilon, [\upsilon_{e'}, \zeta_e] \rangle^2_{\mathsf{G}} + 3\mathbb{E}\langle \upsilon, [\zeta_{e'}, \upsilon_e] \rangle^2_{\mathsf{G}} \\ &\qquad + 2\langle \upsilon, [\upsilon_{e'}, [\upsilon_{e'}, \upsilon_e]] + [\upsilon_e, [\upsilon_e, \upsilon_{e'}]] \rangle_{\mathsf{G}}. \end{aligned} \tag{415}$$

Proposition 31 (Gaussian generator laws are not closed under coarse links)**.** *(i) If* $\mathfrak{g}$ *is abelian,* $\mathring{\omega}_{e'} + \mathring{\omega}_e$ *is a Gaussian coarse generator. (ii) Let* $G = \mathrm{GL}^{+}(K)$ *with* $K \geq 2$*, and let* $\mathbb{X}_K$ *be the set of real* $K \times K$ *matrices with* K *distinct real eigenvalues, exactly two of them negative. Then* $\mathbb{X}_K$ *is a nonempty open subset of* $\mathrm{GL}^{+}(K)$*, no element of* $\mathbb{X}_K$ *has a real logarithm, and* $\Pr(\mathring{\Theta}^c_e \in \mathbb{X}_K) > 0$*.*

Consequently no coarse generator exists, and the law of $\mathring{\Theta}_e^c$ is the image under the exponential of no law on $\mathfrak{g}$. (iii) Let the laws depend on a width $\tau \in (0,1]$ through the means $\tau\mathring{\mu}_{e'}, \tau\mathring{\mu}_e$ and the precisions $\tau^{-2}\mathring{\Lambda}_{e'}, \tau^{-2}\mathring{\Lambda}_e$, with $\mathring{\mu}_{e'}, \mathring{\mu}_e, \mathring{\Lambda}_{e'}, \mathring{\Lambda}_e$ held. On the event $\mathsf{E}_\tau = \{\|\mathring{\omega}_{e'}\| + \|\mathring{\omega}_e\| \le \frac{1}{2}\}$ the principal logarithm of $\mathring{\Theta}_e^c$ exists and lies in $\mathfrak{g}$, and $\Pr(\mathsf{E}_\tau) \ge 1 - c_1 e^{-c_2/\tau^2}$. If κ_3 is not identically zero, or if $\mathring{\mu}_{e'} = \mathring{\mu}_e = 0$ and κ_4 is not identically zero, there is $\tau_0 > 0$ such that for every $\tau \in (0, \tau_0)$ no Gaussian random element of $\mathfrak{gl}(K,\mathbb{R})$ coincides with $\operatorname{Log}\mathring{\Theta}_e^c$ almost surely on E_τ. (iv) For $\mathfrak{g} = \mathfrak{gl}(K,\mathbb{R})$ with $K \ge 2$, $\mathsf{G} = I_K$ and $\mathring{\Lambda}_{e'} = \mathring{\Lambda}_e = \mathrm{Id}$, the form κ_3 vanishes identically and $\kappa_4(E_{12}) = 12$. With $\mathring{\Lambda}_{e'} = \mathrm{Id}$ and $\mathring{\Lambda}_e^{-1}$ the identity plus the orthogonal projection onto the line of E_{12}, $\kappa_3(E_{11} + E_{12}) = 1$. For an inner product on $\mathfrak{g}$ invariant under the adjoint action of $\mathfrak{g}$ and precisions that are multiples of the identity, both forms vanish identically.

Proof. (i) Commuting matrices satisfy $e^{\mathring{\omega}_{e'}} e^{\mathring{\omega}_e} = e^{\mathring{\omega}_{e'} + \mathring{\omega}_e}$, and a sum of independent Gaussian random elements is Gaussian.

(ii) The determinant of a matrix in $\mathbb{X}_K$ is the product of its eigenvalues, hence positive. Near such a matrix the eigenvalues depend continuously on the entries; a simple real eigenvalue stays real, because a nonreal eigenvalue nearby would bring its complex conjugate into the same small disk, and signs and distinctness persist, so $\mathbb{X}_K$ is open. Each negative eigenvalue is simple, so its single Jordan block occurs once, while Culver's criterion requires every Jordan block of a negative eigenvalue to occur an even number of times [67]; hence no element of $\mathbb{X}_K$ has a real logarithm. With $\mathfrak{J} = E_{21} - E_{12}$, $\exp(\pi\mathfrak{J})\exp(\operatorname{diag}(0, \log 2, \log 3, \dots, \log K)) = \operatorname{diag}(-1, -2, 3, \dots, K) \in \mathbb{X}_K$, so the preimage of $\mathbb{X}_K$ under the continuous map $(\xi_1, \xi_2) \mapsto e^{\xi_1} e^{\xi_2}$ is a nonempty open subset of $\mathfrak{g} \times \mathfrak{g}$, which the joint law of the generators charges because it has an everywhere positive density. A coarse generator would place $\mathring{\Theta}_e^c$ almost surely in $\exp(\mathfrak{g})$, which misses $\mathbb{X}_K$, and the exponential image of any law on $\mathfrak{g}$ gives $\mathbb{X}_K$ mass zero.

(iii) Expansion of the logarithm. Let $\xi_1, \xi_2 \in \mathfrak{gl}(K,\mathbb{R})$ with $s = \|\xi_1\| + \|\xi_2\| \le \frac{1}{2}$, and let $\Pi_l = \sum_{a+b=l} \xi_1^a \xi_2^b / (a!b!)$ be the part of degree l of $e^{\xi_1} e^{\xi_2}$. Then $\|e^{\xi_1} e^{\xi_2} - I\| \le e^s - 1 \le e^{1/2} - 1 < 1$, so $\sum_{k \ge 1} (-1)^{k+1} (e^{\xi_1} e^{\xi_2} - I)^k / k$ converges to the principal logarithm [66]. Substituting $\sum_{l \ge 1} \Pi_l$ and grouping by total degree is legitimate, because replacing every coefficient by its absolute value and every word in ξ_1, ξ_2 by the product of the norms of its letters gives the expansion of $-\log(2 - e^s)$, which converges since $\frac{1}{2} < \log 2$. Hence $\operatorname{Log}(e^{\xi_1} e^{\xi_2}) = \sum_{l \ge 1} \operatorname{BCH}_l(\xi_1, \xi_2)$ with homogeneous polynomials of degree l satisfying $\|\operatorname{BCH}_l\| \le b_l s^l$, where $b_l \ge 0$ are the Taylor coefficients of $u \mapsto -\log(2 - e^u)$ at zero. The first three are $\operatorname{BCH}_1 = \Pi_1$, $\operatorname{BCH}_2 = \Pi_2 - \frac{1}{2}\Pi_1^2 = \frac{1}{2}[\xi_1, \xi_2]$ and $\operatorname{BCH}_3 = \Pi_3 - \frac{1}{2}(\Pi_1\Pi_2 + \Pi_2\Pi_1) + \frac{1}{3}\Pi_1^3 = \frac{1}{12}([\xi_1, [\xi_1, \xi_2]] + [\xi_2, [\xi_2, \xi_1]])$, the Baker–Campbell–Hausdorff terms [65], so

$$\begin{aligned} \operatorname{Log}(e^{\xi_1} e^{\xi_2}) &= \xi_1 + \xi_2 + \tfrac{1}{2}[\xi_1, \xi_2] + \tfrac{1}{12}\big([\xi_1, [\xi_1, \xi_2]] + [\xi_2, [\xi_2, \xi_1]]\big) + \operatorname{BCH}_{\ge 4}(\xi_1, \xi_2), \\ &\|\operatorname{BCH}_{\ge 4}(\xi_1, \xi_2)\| \le c_3 s^4, \qquad c_3 = \sum_{l \ge 4} b_l 2^{4-l} < \infty. \end{aligned} \tag{416}$$

The logarithm lies in $\mathfrak{g}$ when $\xi_1, \xi_2 \in \mathfrak{g}$. For $t \in [0,1]$ put $\gamma(t) = \operatorname{Log}(e^{t\xi_1} e^{\xi_2})$. By the bound above, γ is real-analytic with $\|\gamma(t)\| \le -\log(2 - e^{1/2}) < \frac{\pi}{2}$, and $\gamma(0) = \xi_2$, because the grouped series of $\operatorname{Log}(e^{\xi_2})$ is that of the formal identity $\log(e^u) = u$. Differentiating $e^{\gamma(t)} = e^{t\xi_1} e^{\xi_2}$ with Lemma 17 gives $\mathbf{L}(\gamma, \gamma') = \xi_1 e^{\gamma}$. Since $u \mapsto e^{u\gamma} \mathbf{T} e^{-u\gamma}$ and $u \mapsto e^{u \operatorname{ad}_\gamma} \mathbf{T}$, with $\operatorname{ad}_\gamma \mathbf{T} = [\gamma, \mathbf{T}]$, solve the same linear equation with the same initial value, $\mathbf{L}(\gamma, \mathbf{T}) = (\int_0^1 e^{u \operatorname{ad}_\gamma} du)(\mathbf{T}) e^{\gamma}$. As $\|\operatorname{ad}_\gamma\| \le 2\|\gamma\| < \pi$, the operator $\int_0^1 e^{u \operatorname{ad}_\gamma} du = \sum_{k \ge 0} \operatorname{ad}_\gamma^k / (k+1)!$ is inverted by the power series of $w \mapsto w/(e^w - 1)$, which converges for $|w| < 2\pi$. Hence γ solves the differential equation whose right side at $\mathbf{T}$ is that power series in $\operatorname{ad}_{\mathbf{T}}$ applied to ξ_1, an analytic vector field on the ball $\|\mathbf{T}\| < \pi$ that is tangent to $\mathfrak{g}$ at the points of $\mathfrak{g}$, because $\mathfrak{g}$ is a Lie subalgebra. The solution of this equation within $\mathfrak{g}$ from ξ_2

coincides with γ by uniqueness while it exists, and it exists on $[0,1]$ because γ stays in a compact subset of the ball; so $\operatorname{Log}(e^{\xi_1}e^{\xi_2}) = \gamma(1) \in \mathfrak{g}$.

Probability of the event. Write $\mathring{\omega}_{e'} = \tau\chi_{e'}$ and $\mathring{\omega}_e = \tau\chi_e$ with $\chi_{e'} = \mathring{\mu}_{e'} + \zeta_{e'}$ and $\chi_e = \mathring{\mu}_e + \zeta_e$, independent with laws that do not depend on τ. A Gaussian random element χ of a finite-dimensional space satisfies $\mathbb{E}e^{c\|\chi\|^2} < \infty$ for some $c > 0$, and Markov's inequality bounds $\Pr(\|\chi\| > 1/(4\tau))$ by a multiple of $e^{-c/(16\tau^2)}$, which bounds the probability of the complement of E_τ.

Polynomial part. Put $\psi_1 = \langle \upsilon, \chi_{e'} + \chi_e\rangle_{\mathsf{G}}$, $\psi_2 = \frac{1}{2}\langle \upsilon, [\chi_{e'}, \chi_e]\rangle_{\mathsf{G}}$, $\psi_3 = \frac{1}{12}\langle \upsilon, [\chi_{e'}, [\chi_{e'}, \chi_e]] + [\chi_e, [\chi_e, \chi_{e'}]]\rangle_{\mathsf{G}}$, $\psi_4 = \tau^{-4}\langle \upsilon, \operatorname{BCH}_{\geq 4}(\mathring{\omega}_{e'}, \mathring{\omega}_e)\rangle_{\mathsf{G}}$ on E_τ and $\psi_4 = 0$ off it, and $\Psi_\tau = \psi_1 + \tau\psi_2 + \tau^2\psi_3 + \tau^3\psi_4$. By Eq. (416) and homogeneity, $\Psi_\tau = \tau^{-1}\langle \upsilon, \operatorname{Log}\mathring{\Theta}^c_e\rangle_{\mathsf{G}}$ on E_τ, and $|\psi_4| \leq c_4(\|\chi_{e'}\| + \|\chi_e\|)^4$ with c_4 independent of τ, so every moment of $\psi_1, \dots, \psi_4$ is bounded uniformly in τ. Joint cumulants are multilinear and are polynomials in joint moments, and the cumulants of order three and four of the Gaussian variable ψ_1 vanish, so, with cum_l the cumulant of order l and cum the joint cumulant,

$$\begin{aligned} \operatorname{cum}_3 \Psi_\tau &= 3\tau \operatorname{cum}(\psi_1, \psi_1, \psi_2) + O(\tau^2), \\ \operatorname{cum}_4 \Psi_\tau &= 4\tau \operatorname{cum}(\psi_1, \psi_1, \psi_1, \psi_2) \\ &\qquad + \tau^2\big[6\operatorname{cum}(\psi_1, \psi_1, \psi_2, \psi_2) + 4\operatorname{cum}(\psi_1, \psi_1, \psi_1, \psi_3)\big] + O(\tau^3). \end{aligned} \tag{417}$$

Gaussian integration by parts. Let ∂_υ be the derivative of a function of $(\zeta_{e'}, \zeta_e)$ along $(\upsilon_{e'}, \upsilon_e)$. Every polynomial φ of $(\zeta_{e'}, \zeta_e)$ satisfies $\mathbb{E}[(\psi_1 - \mathbb{E}\psi_1)\varphi] = \mathbb{E}[\partial_\upsilon\varphi]$, by Eq. (403) applied to the two summands of $\psi_1 - \mathbb{E}\psi_1 = \langle \upsilon, \zeta_{e'}\rangle_{\mathsf{G}} + \langle \upsilon, \zeta_e\rangle_{\mathsf{G}}$. Two applications give $\operatorname{cum}(\psi_1, \psi_1, \psi_2) = \mathbb{E}[(\psi_1 - \mathbb{E}\psi_1)^2(\psi_2 - \mathbb{E}\psi_2)] = \partial_\upsilon^2\psi_2 = \kappa_3(\upsilon)$ for all means, because ψ_2 is a quadratic polynomial whose second derivative along $(\upsilon_{e'}, \upsilon_e)$ is $\langle \upsilon, [\upsilon_{e'}, \upsilon_e]\rangle_{\mathsf{G}}$. If the means vanish, the map $(\zeta_{e'}, \zeta_e) \mapsto (-\zeta_{e'}, -\zeta_e)$ preserves the law and changes the signs of ψ_1 and ψ_3 but not of ψ_2, so every joint moment and cumulant with an odd number of entries from $\{\psi_1, \psi_3\}$ vanishes; in particular $\operatorname{cum}(\psi_1, \psi_1, \psi_1, \psi_2) = 0$, while $\mathbb{E}\psi_2 = 0$ because the generators are independent and centered. Repeated integration by parts gives $\mathbb{E}[\psi_1^2\psi_2^2] = \mathbb{E}[\psi_1^2]\mathbb{E}[\psi_2^2] + 2\mathbb{E}[(\partial_\upsilon\psi_2)^2]$ and $\mathbb{E}[\psi_1^3\psi_3] = 3\mathbb{E}[\psi_1^2]\mathbb{E}[\psi_1\psi_3] + \partial_\upsilon^3\psi_3$, hence $\operatorname{cum}(\psi_1, \psi_1, \psi_2, \psi_2) = 2\mathbb{E}[(\partial_\upsilon\psi_2)^2]$ and $\operatorname{cum}(\psi_1, \psi_1, \psi_1, \psi_3) = \partial_\upsilon^3\psi_3$, which is six times the value of the homogeneous cubic ψ_3 at $(\upsilon_{e'}, \upsilon_e)$. Since $\partial_\upsilon\psi_2 = \frac{1}{2}\langle \upsilon, [\upsilon_{e'}, \zeta_e]\rangle_{\mathsf{G}} + \frac{1}{2}\langle \upsilon, [\zeta_{e'}, \upsilon_e]\rangle_{\mathsf{G}}$ has independent centered summands, the coefficient of τ^2 in Eq. (417) is $\kappa_4(\upsilon)$ in the centered case.

Gaussian rigidity. Fix υ with $\kappa_3(\upsilon) \neq 0$, or, in the centered case, with $\kappa_4(\upsilon) \neq 0$, and suppose that along a sequence of widths tending to zero some Gaussian random element of $\mathfrak{gl}(K, \mathbb{R})$ coincides with $\operatorname{Log}\mathring{\Theta}^c_e$ almost surely on E_τ. Pairing it with υ and dividing by τ gives real Gaussian variables $\mathrm{X}_\tau \sim \mathcal{N}(m_\tau, s_\tau^2)$ equal to Ψ_τ outside an event of probability at most $c_1 e^{-c_2/\tau^2}$. Since $\Psi_\tau \to \psi_1$ in mean square, $\mathrm{X}_\tau \to \psi_1$ in probability, hence in law. The moduli $e^{-s_\tau^2 t^2/2}$ of the characteristic functions then converge, so $s_\tau^2 \to \operatorname{Var}\psi_1$, and the phases $e^{im_\tau t}$ converge to $e^{it\mathbb{E}\psi_1}$ for every t; integrating $e^{i(m_\tau - \mathbb{E}\psi_1)t}$ against the standard normal density gives $e^{-(m_\tau - \mathbb{E}\psi_1)^2/2} \to 1$, so $m_\tau \to \mathbb{E}\psi_1$. The moments of order eight of X_τ are therefore bounded, and the Cauchy–Schwarz inequality on the exceptional event shows that the moments of order at most four of X_τ and Ψ_τ differ by $O(e^{-c_2/(2\tau^2)})$, and so do their cumulants of order three and four. Those of X_τ vanish, so by Eq. (417) the quantity $3\tau\kappa_3(\upsilon) + O(\tau^2)$, and in the centered case $\tau^2\kappa_4(\upsilon) + O(\tau^3)$, would tend to zero faster than every power of τ, which is impossible. Hence such a sequence does not exist, and τ_0 does.

(iv) For $\mathsf{G} = I_K$ the inner product is $\operatorname{tr}(\mathbf{S}^\top\mathbf{T})$. With unit precisions $\upsilon_{e'} = \upsilon_e = \upsilon$, so $\kappa_3(\upsilon) = \langle \upsilon, [\upsilon, \upsilon]\rangle = 0$ and the double brackets of κ_4 vanish. For $\upsilon = E_{12}$, $\langle \upsilon, [\upsilon, \zeta]\rangle = \zeta_{22} - \zeta_{11}$, of variance 2 when ζ has independent standard normal entries, and $\langle \upsilon, [\zeta, \upsilon]\rangle$ is its negative, so $\kappa_4(\upsilon) = 3{\cdot}2 + 3{\cdot}2 = 12$. In the second instance $\upsilon_{e'} = \upsilon = E_{11} + E_{12}$ and $\upsilon_e = E_{11} + 2E_{12}$, whose bracket $[\upsilon_{e'}, \upsilon_e] = E_{12}$ pairs with υ to 1. For an invariant inner product, $\langle \upsilon, [\upsilon, \mathbf{T}]\rangle_{\mathsf{G}} = -\langle [\upsilon, \upsilon], \mathbf{T}\rangle_{\mathsf{G}} = 0$ for every $\mathbf{T} \in \mathfrak{g}$, and with precisions that are multiples of the identity $\upsilon_{e'}$ and υ_e are multiples of υ, so every term of both forms vanishes. □

When the logarithm exists, the commutator term of the expansion thus makes the coarse generator non-Gaussian at first order in the width whenever κ_3 is not identically zero and, for centered laws, at second order whenever κ_4 is not. For an invariant inner product with isotropic precisions both leading coefficients vanish, and this test gives no conclusion.

Remark 7 (A finite-dimensional description of coarse link laws)**.** *The law of the coarse link always exists, as the pushforward of the fine law through the ordered product* (165)*, and the exact coarse-graining of Section 12 applies to it. For independent links in series the step sends laws ν' and ν on G to the law $\nu' * \nu$ of their product, so a finite-dimensional family of link laws that survives the step must be closed under this convolution. For $G = \mathrm{GL}^+(K)$ with $K \geq 2$ the question is open. It would be settled by a weakly continuous injective map from a connected open subset of some $\mathbb{R}^d$ into the Borel probability laws on $\mathrm{GL}^+(K)$ whose image contains the exponential image of every nondegenerate Gaussian law on $\mathfrak{gl}(K, \mathbb{R})$ and contains $\nu' * \nu$ whenever it contains ν' and ν, or by a proof that for some $K \geq 2$ no such map exists.*

L Supporting Proofs

This appendix proves statements of the main text and of the preceding appendices whose arguments are not given where the statements are made. Each subsection restates the objects it uses, in the notation of the section that cites it, and derives its statement from the definitions given there together with standard measure theory and linear algebra. Relative entropies take values in $[0, \infty]$, with the value $+\infty$ whenever absolute continuity fails.

L.1 Coordinate actions on latent generators

Fix the base point c_*, a matrix group $G \subseteq \mathrm{GL}(K)$ with Lie algebra $\mathfrak{g}$, frames $U_i \in G$, and for each oriented copy $e : j \to i$ in E_+ a generator $\mathring{\omega}_e \in \mathfrak{g}$ with link $\Theta_e = U_i \exp(\mathring{\omega}_e) U_j^{-1}$ and reversed generator $\mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e$, as in Eq. (19). Then $\Theta_{\bar{e}} = U_j \exp(-\mathring{\omega}_e) U_i^{-1} = \Theta_e^{-1}$.

Lemma 19 (Coordinate actions on generators)**.** *(i) Under a passive change $U_i' = g_i U_i$, the links $\Theta_e' = g_i \Theta_e g_j^{-1}$ of Eq. (17) are the links of the same generators $\mathring{\omega}_e$ relative to the frames U_i'. (ii) For $h \in G$, the unchanged links Θ_e are the links of the generators $\mathrm{Ad}_{h^{-1}} \mathring{\omega}_e = h^{-1} \mathring{\omega}_e h$ relative to the frames $U_i h$. (iii) Both operations produce the frames $U_i h$ when the passive change is $g_i = U_i h U_i^{-1}$. The resulting frames, links and generators then coincide if and only if $h^{-1} \mathring{\omega}_e h = \mathring{\omega}_e$ for every $e \in E_+$.*

Proof. (i) Substitution gives $g_i \Theta_e g_j^{-1} = (g_i U_i) \exp(\mathring{\omega}_e) (g_j U_j)^{-1}$. (ii) Conjugating the absolutely convergent exponential series term by term gives $h^{-1} \exp(X) h = \exp(h^{-1} X h)$ for every $X \in \mathfrak{g}$, and $h^{-1} X h \in \mathfrak{g}$ because $\mathfrak{g}$ is invariant under the adjoint action of G. Hence

$$(U_i h) \exp(h^{-1} \mathring{\omega}_e h) (U_j h)^{-1} = U_i h h^{-1} \exp(\mathring{\omega}_e) h h^{-1} U_j^{-1} = \Theta_e.$$

(iii) With $g_i = U_i h U_i^{-1}$ the frames of (i) are $U_i h$. Operation (i) keeps every generator and replaces each link by $U_i h \exp(\mathring{\omega}_e) h^{-1} U_j^{-1}$, whereas operation (ii) keeps every link and replaces each generator by $h^{-1} \mathring{\omega}_e h$. If the generators agree, the links agree as well, since a link is determined by its generator and the frames. Conversely, agreement of the data includes agreement of the generators. □

The passive vertex group therefore acts on frames and link presentations with every generator fixed, while a change of common reference fixes every link and acts on all generators through the single linear map $\mathrm{Ad}_{h^{-1}}$ of $\mathfrak{g}$.

L.2 Evidence identity and restriction principle

Let a normalized generative law have positive finite evidence Z_o at the record o and posterior P_o on the latent space, and define the variational free energy of a recognition law Q on that space by

$$\mathcal{F}(Q) = D_{\mathrm{KL}}(Q \| P_o) - \log Z_o \in (-\infty, \infty], \tag{418}$$

which agrees with Eqs. (22) and (68) whenever their split forms are finite.

Proposition 32 (Restriction principle)**.** *For every recognition law Q, $\mathcal{F}(Q) \geq -\log Z_o$, with equality if and only if $Q = P_o$. For arbitrary sets of laws $\mathcal{Q}' \subseteq \mathcal{Q}$,*

$$\inf_{Q \in \mathcal{Q}'} \mathcal{F}(Q) = -\log Z_o + \inf_{Q \in \mathcal{Q}'} D_{\mathrm{KL}}(Q \| P_o), \tag{419}$$

the infimum over $\mathcal{Q}'$ is at least the infimum over $\mathcal{Q}$, and when both divergence infima are finite,

$$\inf_{Q \in \mathcal{Q}'} \mathcal{F}(Q) - \inf_{Q \in \mathcal{Q}} \mathcal{F}(Q) = \inf_{Q \in \mathcal{Q}'} D_{\mathrm{KL}}(Q \| P_o) - \inf_{Q \in \mathcal{Q}} D_{\mathrm{KL}}(Q \| P_o) \geq 0. \tag{420}$$

In particular, if $\mathcal{Q}$ contains P_o, the restricted optimum exceeds $-\log Z_o$ by exactly the divergence infimum over $\mathcal{Q}'$ in Eq. (419). None of these identities requires an infimum to be attained, and the infimum over a restricted family need not be attained.

Proof. If $Q \ll P_o$ with density f, then $D_{\mathrm{KL}}(Q\|P_o) = \int \varphi(f)dP_o$ for the strictly convex function $\varphi(x) = x\log x$, whose negative part is bounded by e^{-1}. Jensen's inequality gives $\int \varphi(f)dP_o \geq \varphi(\int f dP_o) = \varphi(1) = 0$, with equality exactly when $f = 1$ almost everywhere. Otherwise the divergence is $+\infty$ and $Q \neq P_o$. This proves the bound and its equality case. The infimum over a set of a finite constant plus a $[0,\infty]$-valued function is that constant plus the infimum of the function, including the value $+\infty$, which is Eq. (419). Inclusion of sets reverses the order of infima, and the subtraction in Eq. (420) is legitimate when both infima are finite. For the last clause take $P_o = \mathcal{N}(0,1)$ and $\mathcal{Q}' = \{\mathcal{N}(\mu,1) : \mu > 0\}$. By Eq. (15) the divergences are $\mu^2/2$, whose infimum 0 is approached as $\mu \downarrow 0$ and attained by no member. □

A positive restricted gap is the condition $\inf_{Q\in\mathcal{Q}'} D_{\mathrm{KL}}(Q\|P_o) > 0$. Proposition 38 evaluates this gap for block-factorized Gaussian recognition of a correlated Gaussian posterior.

L.3 Memory residual, sufficiency and slow spectrum

This subsection uses the notation of Section 8.5. A finite irreducible generator L, reversible under $\pi > 0$, acts on densities $r = dQ/d\pi$ in $L^2(\pi)$ with inner product $\langle f, g\rangle = \sum_x \pi_x f_x g_x$; $\mathsf{P} = \mathbb{E}_\pi[\cdot \mid T]$ for a partition statistic T, $\mathsf{R} = I - \mathsf{P}$, and $\mathsf{A}, \mathsf{B}, \mathsf{C}, \mathsf{D}$ are the blocks of L. Closure of the retained label for every initial law is the condition $\mathsf{C} = 0$, and the residual of the local memory replacement obeys Eq. (84) whenever $\mathsf{D} \preceq -\gamma I$ on the range of R.

Consider four states with symmetric jump rates $u > 0$ on $1 \leftrightarrow 2$ and on $3 \leftrightarrow 4$, $\epsilon > 0$ on $1 \leftrightarrow 3$, $\epsilon' > 0$ on $2 \leftrightarrow 4$, and no other jumps, so that $(\mathsf{L}f)_x = \sum_y \mathsf{L}_{xy}(f_y - f_x)$, the uniform law is reversible, and T labels the cells $\{1,2\}$ and $\{3,4\}$. For $\epsilon = \epsilon'$ and $u = 1$ this is the generator of Section 8.4. Put $\delta = \epsilon' - \epsilon$.

Proposition 33 (Distinct conditions for an autonomous retained model)**.** *In the orthonormal basis $\mathbf{1} = (1,1,1,1)$, $h = (1,1,-1,-1)$, $v_+ = (1,-1,1,-1)$, $v_- = (1,-1,-1,1)$ of $L^2(\pi)$,*

$$\mathsf{L}\mathbf{1} = 0, \qquad \mathsf{L}v_+ = -2uv_+, \qquad \mathsf{L}h = -(\epsilon+\epsilon')h + \delta v_-, \qquad \mathsf{L}v_- = \delta h - (2u+\epsilon+\epsilon')v_-. \tag{421}$$

Hence the spectrum of L is $\{0, -2u, \lambda_+, \lambda_-\}$ with $\lambda_\pm = -(u+\epsilon+\epsilon') \pm \sqrt{u^2+\delta^2}$, and

$$\|\mathsf{A}\| = \epsilon + \epsilon', \qquad \|\mathsf{B}\| = \|\mathsf{C}\| = |\delta|, \qquad \mathsf{D} \preceq -2uI. \tag{422}$$

Consequently the following hold. (i) For $\epsilon = \epsilon' > u$ the label is exactly closed, $\mathsf{C} = 0$, so the memory residual vanishes, while the label relaxes at rate 2ϵ, faster than the within-cell rate $2u$; sufficiency holds without a slow retained spectrum. (ii) For $\epsilon \neq \epsilon'$ the label is not closed for every initial law, while $-(\epsilon+\epsilon') \leq \lambda_+ \leq -(\epsilon+\epsilon') + \delta^2/(2u)$, the eigenvector of λ_+ is $h + \kappa v_-$ with $|\kappa| \leq |\delta|/(2u)$, and $\lambda_- \leq -2u$; a slow retained mode exists without sufficiency once $\epsilon+\epsilon'$ is small compared with u. (iii) For fixed $\epsilon \neq \epsilon'$, the right side of Eq. (84) is at most

$$\delta^2\|r_0\|\left(\frac{e^{-2ut}}{2u} + \frac{\max(\epsilon,\epsilon')}{2u^2}\right), \tag{423}$$

which tends to zero as $u \to \infty$, although $\mathsf{C} \neq 0$ for every u. Thus a small memory residual, statistical sufficiency of the retained label, and a slow retained spectrum are pairwise distinct conditions.

Proof. Equation (421) follows by evaluating $(\mathsf{L}f)_x$ at the four states; for example $\mathsf{L}h = (-2\epsilon, -2\epsilon', 2\epsilon, 2\epsilon') = -(\epsilon+\epsilon')h + \delta v_-$. The four vectors are orthonormal for the uniform law. On $\mathrm{span}\{h, v_-\}$ the operator is the symmetric matrix with diagonal entries $-(\epsilon+\epsilon')$ and $-(2u+\epsilon+\epsilon')$ and off-diagonal entry δ, whose eigenvalues are $\lambda_\pm$; together with $\mathsf{L}\mathbf{1} = 0$ and $\mathsf{L}v_+ = -2uv_+$ this gives the spectrum. The range of P is $\mathrm{span}\{\mathbf{1}, h\}$, the functions constant on cells, and the range of R is $\mathrm{span}\{v_+, v_-\}$. Reading off Eq. (421), A annihilates $\mathbf{1}$ and multiplies h by $-(\epsilon+\epsilon')$; C annihilates $\mathbf{1}$ and sends h to δv_-; $\mathsf{B} = \mathsf{C}^*$ annihilates v_+ and sends v_- to δh; and D has eigenvalues $-2u$ on v_+ and $-(2u+\epsilon+\epsilon')$ on v_-. This is Eq. (422). For (i), $\delta = 0$ gives $\mathsf{C} = \mathsf{B} = 0$ and the spectrum $\{0, -2\epsilon, -2u, -2u-2\epsilon\}$, with eigenvector h for -2ϵ. For (ii), $\delta \neq 0$ gives $\mathsf{C} \neq 0$; the bounds $u \le \sqrt{u^2+\delta^2} \le u + \delta^2/(2u)$ give the bounds on λ_+ and on $\kappa = (\sqrt{u^2+\delta^2} - u)/\delta$, and $\lambda_- \le -(u+\epsilon+\epsilon') - u$. For (iii), L is self-adjoint and negative semidefinite on $L^2(\pi)$, so $e^{t\mathsf{L}}$ is a contraction and $\|a(t)\|, \|b(t)\| \le \|r_0\|$. Then $\|\dot a\| = \|\mathsf{A}a + \mathsf{B}b\| \le (\epsilon + \epsilon' + |\delta|)\|r_0\| = 2\max(\epsilon, \epsilon')\|r_0\|$, so $M_t \le 2\max(\epsilon, \epsilon')\|r_0\|$. Substituting $\gamma = 2u$, $\|\mathsf{B}\|\|\mathsf{C}\| = \delta^2$ and these bounds into Eq. (84) gives Eq. (423). □

L.4 Block descent and the Fisher clock

Let $\xi = (\xi_b)_b$ range over an open subset of a finite-dimensional coordinate space divided into blocks b, let $\mathcal{V}$ be a C^1 potential, either the consensus potential $\mathcal{E}$ or a collective free energy $\mathcal{F}_V$, let $G_b(\xi)$ be continuous fields of symmetric positive-definite block metrics, and let $G = \bigoplus_b G_b$. For rates $\eta_b > 0$ consider a C^1 solution of

$$\eta_b^{-1} G_b \dot\xi_b = -\nabla_{\xi_b}\mathcal{V}, \tag{424}$$

which is Eq. (136); blocks with zero rate are held fixed and omitted. Write $\mathrm{grad}^G \mathcal{V} = G^{-1}\nabla\mathcal{V}$, $\|w\|_G^2 = G(w,w)$, $\|\nabla_{\xi_b}\mathcal{V}\|^2_{G_b^{-1}} = (\nabla_{\xi_b}\mathcal{V})^\top G_b^{-1}\nabla_{\xi_b}\mathcal{V}$, and $G_\eta = \bigoplus_b \eta_b^{-1} G_b$ for the rate-weighted metric. Along the solution, the Fisher clock of Eq. (176) and the block clocks are $\vartheta(t) = \int_0^t \|\dot\xi(s)\|_G ds$ and $\vartheta_b(t) = \int_0^t \|\dot\xi_b(s)\|_{G_b} ds$.

Proposition 34 (Block descent and Fisher clock)**.** *(i) For all positive rates, equal or not, every solution of Eq.* (424) *satisfies*

$$\frac{d\mathcal{V}}{dt} = -\sum_b \eta_b \|\nabla_{\xi_b}\mathcal{V}\|^2_{G_b^{-1}} = -2\mathcal{R} \le 0, \qquad \mathcal{R} = \sum_b \frac{1}{2\eta_b} G_b(\dot\xi_b, \dot\xi_b). \tag{425}$$

(ii) For an orientation-preserving C^1 diffeomorphism ϕ of parameter intervals, the curve $\xi \circ \phi$ has the clock $\vartheta \circ \phi$ from the corresponding origin, and corresponding subarcs have equal Fisher length. The clock is strictly increasing on intervals where $\dot\xi \neq 0$ and constant on stationary intervals. (iii) The squared clock speed is additive, $\dot\vartheta^2 = \sum_b \dot\vartheta_b^2$ with $\dot\vartheta_b = \eta_b\|\nabla_{\xi_b}\mathcal{V}\|_{G_b^{-1}}$, and over a common interval $\max_b \vartheta_b \le (\sum_b \vartheta_b^2)^{1/2} \le \vartheta \le \sum_b \vartheta_b$. (iv) For a common rate $\eta_b = \eta$, $\dot\vartheta = \eta\|\mathrm{grad}^G \mathcal{V}\|_G$ and $\dot\vartheta^2 = -\eta\dot{\mathcal{V}} = 2\eta\mathcal{R}$, and where $\nabla\mathcal{V} \neq 0$, $d\mathcal{V}/d\vartheta = -\|\mathrm{grad}^G \mathcal{V}\|_G$. (v) For arbitrary rates, wherever $\dot\xi \neq 0$,

$$\left|\frac{d\mathcal{V}}{d\vartheta}\right| = \frac{|d\mathcal{V}(\dot\xi)|}{\|\dot\xi\|_G} \le \|\mathrm{grad}^G \mathcal{V}\|_G, \tag{426}$$

with equality exactly when $d\mathcal{V} = 0$ or $\dot\xi$ is parallel to $\mathrm{grad}^G \mathcal{V}$. The flow is the unit-rate natural-gradient flow of G_η, and the identities of (iv) hold with $\eta = 1$ for the clock read with G_η.

Proof. (i) By the chain rule $d\mathcal{V}/dt = \sum_b (\nabla_{\xi_b}\mathcal{V})^\top \dot\xi_b$, and Eq. (424) gives $\dot\xi_b = -\eta_b G_b^{-1}\nabla_{\xi_b}\mathcal{V}$. Hence $d\mathcal{V}/dt = -\sum_b \eta_b \|\nabla_{\xi_b}\mathcal{V}\|^2_{G_b^{-1}}$ and $G_b(\dot\xi_b, \dot\xi_b) = \eta_b^2 \|\nabla_{\xi_b}\mathcal{V}\|^2_{G_b^{-1}}$, so that $2\mathcal{R}$ equals the negative of

$d\mathcal{V}/dt$. (ii) For $\tilde{\xi} = \xi \circ \phi$ with $\phi' > 0$, $\|\tilde{\xi}'(s)\|_G = \|\dot{\xi}(\phi(s))\|_G \phi'(s)$, and the substitution $t = \phi(s)$ in the length integral proves the reparameterization law. The integrand $\|\dot{\xi}\|_G$ is continuous, nonnegative, and zero exactly where $\dot{\xi} = 0$. (iii) Block diagonality gives $\|\dot{\xi}\|_G^2 = \sum_b \|\dot{\xi}_b\|_{G_b}^2$, and $\|\dot{\xi}_b\|_{G_b} = \eta_b \|\nabla_{\xi_b} \mathcal{V}\|_{G_b^{-1}}$ by the computation in (i). Pointwise $\dot{\vartheta}_b \le \dot{\vartheta} \le \sum_b \dot{\vartheta}_b$, and integrating the right inequality gives $\vartheta \le \sum_b \vartheta_b$. Minkowski's inequality for integrals of the vector $(\dot{\vartheta}_b)_b$ gives $(\sum_b \vartheta_b^2)^{1/2} \le \int (\sum_b \dot{\vartheta}_b^2)^{1/2} ds = \vartheta$, and the first inequality is elementary. (iv) With $\eta_b = \eta$, $\dot{\xi} = -\eta \operatorname{grad}^G \mathcal{V}$, so $\dot{\vartheta} = \eta \| \operatorname{grad}^G \mathcal{V}\|_G$, and (i) gives $\dot{\mathcal{V}} = -\eta \| \operatorname{grad}^G \mathcal{V}\|_G^2 = -2\mathcal{R}$. Dividing $\dot{\mathcal{V}}$ by $\dot{\vartheta} > 0$ gives the last identity. (v) The defining relation of the gradient gives $d\mathcal{V}(\dot{\xi}) = G(\operatorname{grad}^G \mathcal{V}, \dot{\xi})$, and the Cauchy–Schwarz inequality for G gives Eq. (426) with its equality case. Equation (424) reads $G_\eta \dot{\xi} = -\nabla \mathcal{V}$, so (iv) applies with G_η in place of G and $\eta = 1$. □

With unequal rates the velocity $-(\eta_b G_b^{-1} \nabla_{\xi_b} \mathcal{V})_b$ is generally not parallel to $\operatorname{grad}^G \mathcal{V}$, and the inequality in Eq. (426) is then strict. This is the precise sense in which the single-rate identity of Eq. (177) cannot be applied unchanged to a rate-weighted flow.

L.5 Conditional and link-marginal statements

Use the latent-link joint of Eq. (25), written $\mathbb{P}(d\mathring{\omega}, dy, do) = P_{\mathrm{gen}}(d\mathring{\omega}) \mathbb{P}_{\mathring{\omega}}(dy, do)$ with any source labels absorbed into y and the structural data suppressed. Let $\ell_o(y; \mathring{\omega})$ be the record density, assume that the held-link evidence $Z_o(\mathring{\omega}) = \int \ell_o(y; \mathring{\omega}) \mathbb{P}_{\mathring{\omega}}(dy)$ is positive and finite for P_{gen}-almost every $\mathring{\omega}$ and that $Z_o = \int Z_o(\mathring{\omega}) P_{\mathrm{gen}}(d\mathring{\omega}) \in (0, \infty)$, and write $\Pi_{\mathring{\omega}}$ for the held-link posterior at $\Theta(\mathring{\omega})$.

Proposition 35 (Conditional transfer and its limit)**.** *(i) The posterior is* $\Pi(d\mathring{\omega}, dy) = \Pi^{\mathcal{T}}(d\mathring{\omega}) \Pi_{\mathring{\omega}}(dy)$ *with* $\Pi^{\mathcal{T}}(d\mathring{\omega}) = Z_o(\mathring{\omega}) P_{\mathrm{gen}}(d\mathring{\omega}) / Z_o$*. (ii) The matter marginal* $\int \Pi_{\mathring{\omega}} \Pi^{\mathcal{T}}(d\mathring{\omega})$ *of the posterior, the matter marginal* $\int Q_{\mathring{\omega}} R(d\mathring{\omega})$ *of a recognition law* $R(d\mathring{\omega}) Q_{\mathring{\omega}}(dy)$*, and their images under a normalized coarse channel* C *that does not read the links are mixtures of the corresponding conditional laws. A property of laws on the matter space that holds for almost every conditional law therefore holds for these marginals whenever the set of laws with the property is closed under such mixtures. (iii) Gaussianity is not closed under such mixtures. Let* $G = \mathrm{GL}^+(1)$ *act on* $\mathbb{R}$ *by multiplication, and take two agents with one reversal pair* $e : j \to i$*, trivial frames, no loop terms,* $\nu^{\mathcal{T}} = \mathcal{N}(0, 1)$*, no record, and the held-link matter law* $k_j \sim \mathcal{N}(0, 1)$*,* $k_i \mid k_j \sim \mathcal{N}(e^{\mathring{\omega}_e} k_j, 1)$*. Every conditional matter law is Gaussian, while the posterior marginal of* k_i *is centered with* $\mathbb{E} k_i^4 = 3(1 + 2e^2 + e^8) > 3(\mathbb{E} k_i^2)^2$*, so it is not Gaussian.*

Proof. (i) The posterior has density $\ell_o(y; \mathring{\omega}) / Z_o$ with respect to $P_{\mathrm{gen}}(d\mathring{\omega}) \mathbb{P}_{\mathring{\omega}}(dy)$, and the factorization of this density as $[Z_o(\mathring{\omega}) / Z_o][\ell_o(y; \mathring{\omega}) / Z_o(\mathring{\omega})]$ gives the disintegration, the second factor being the held-link posterior density. (ii) Integrating out $\mathring{\omega}$ in a disintegration gives the mixture of its conditional laws, and for a link-independent channel Fubini's theorem gives $(\int Q_{\mathring{\omega}} R(d\mathring{\omega})) C = \int (Q_{\mathring{\omega}} C) R(d\mathring{\omega})$. (iii) With no loop terms the action $\mathcal{A}_{\mathrm{loop}}$ vanishes and $P_{\mathrm{gen}} = \nu^{\mathcal{T}}$, and with no record the posterior is the prior. Given $\mathring{\omega} = \mathring{\omega}_e$, (k_i, k_j) is a centered Gaussian vector with $\mathbb{E}[k_i^2 \mid \mathring{\omega}] = 1 + e^{2\mathring{\omega}}$, so that $\mathbb{E}[k_i^4 \mid \mathring{\omega}] = 3(1 + e^{2\mathring{\omega}})^2$. Since $\mathbb{E} e^{s\mathring{\omega}} = e^{s^2/2}$ for a standard normal generator, $\mathbb{E} k_i^2 = 1 + e^2$ and $\mathbb{E} k_i^4 = 3(1 + 2e^2 + e^8)$. A centered Gaussian variable with this variance would have fourth moment $3(1 + 2e^2 + e^4)$, and $e^8 > e^4$. □

A Gaussian law fitted to such a marginal is therefore a projection of it rather than the marginal itself.

L.6 Noncompact latent links and the holonomy-fixed Gaussian family

Let $G = \mathrm{GL}^+(K)$ with $K \geq 1$ act on $\mathbb{R}^K$ by its defining representation, with links $\Theta_e = U_i \exp(\mathring{\omega}_e) U_j^{-1}$ as in Appendix L.1 and generator space $\mathcal{T} = \mathfrak{gl}(K, \mathbb{R})^{E_+}$, a real vector space carrying Lebesgue measure. A closed walk $p = (e_0, \ldots, e_{r-1})$ of a connected block, based at i_0, consists of copies $e_a : i_{a+1} \to i_a$ with $i_r = i_0$, each a member of E_+ or a reversal of one, and has holonomy $H_p = \Theta_{e_0} \Theta_{e_1} \cdots \Theta_{e_{r-1}}$ in the ordering of Eq. (18). For $e \in E_+$ let n_e be the number of indices with $e_a = e$ minus the number with $e_a = \bar{e}$. For a vertex ϱ of the block, the holonomy-fixed nondegenerate Gaussian family $\mathcal{G}_\varrho$ consists of the laws $\mathcal{N}(m, \Sigma)$ on $\mathbb{R}^K$ with $\Sigma \succ 0$ that are fixed by the pushforward of every holonomy of a closed walk of the block based at ϱ. A coherence score of the block is an infimum over $\mathcal{G}_\varrho$ of a weighted sum of relative entropies from the transported member marginals, with the convention that an infimum over the empty set is $+\infty$.

Proposition 36 (Latent noncompact links empty the Gaussian parent family)**.** *Suppose the block contains a closed walk p with $n_e \neq 0$ for some $e \in E_+$. (i) For every choice of frames,* $\det H_p = \exp(\sum_{e \in E_+} n_e \operatorname{tr} \mathring{\omega}_e)$*. (ii) The generator families with $\sum_{e \in E_+} n_e \operatorname{tr} \mathring{\omega}_e = 0$ form a hyperplane of $\mathcal{T}$, a Lebesgue-null set, and for every generator family off this hyperplane $\mathcal{G}_\varrho$ is empty at every vertex ϱ of the block. (iii) Under every law of the generators that is absolutely continuous with respect to Lebesgue measure on $\mathcal{T}$, including the generator prior of Eq.* (23) *and every recognition law absolutely continuous with respect to it, $\mathcal{G}_\varrho$ is empty and every coherence score of the block is $+\infty$ almost surely.*

Proof. (i) Intermediate frames cancel in the product, so $H_p = U_{i_0} \exp(\mathring{\omega}_{e_0}) \cdots \exp(\mathring{\omega}_{e_{r-1}}) U_{i_0}^{-1}$. The determinant is multiplicative, and $\det \exp(B) = \exp(\operatorname{tr} B)$ for every real square matrix B, as one sees by triangularizing B over $\mathbb{C}$. With $\mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e$ the sum of the traces collects into $\sum_e n_e \operatorname{tr} \mathring{\omega}_e$. (ii) The map $\ell(\mathring{\omega}) = \sum_e n_e \operatorname{tr} \mathring{\omega}_e$ is linear on $\mathcal{T}$ and nonzero, since on the family with $\mathring{\omega}_e = I_K$ for one e with $n_e \neq 0$ and every other generator zero it equals $n_e K \neq 0$. Its kernel is therefore a hyperplane and has Lebesgue measure zero. If $\mathcal{N}(m, \Sigma)$ with $\Sigma \succ 0$ is fixed by a matrix H, then $H_\# \mathcal{N}(m, \Sigma) = \mathcal{N}(Hm, H\Sigma H^\top)$ forces $H\Sigma H^\top = \Sigma$, hence $(\det H)^2 = 1$. Off the hyperplane $|\det H_p| = e^{\ell(\mathring{\omega})} \neq 1$. For another base vertex ϱ choose a walk w in the block from ϱ to i_0 with path product Θ_w. The closed walk that follows w, then p, then the reversal of w has holonomy $\Theta_w H_p \Theta_w^{-1}$, because reversed copies carry inverse links, and its determinant equals $\det H_p$. No nondegenerate Gaussian law at ϱ is therefore fixed by every based holonomy, and $\mathcal{G}_\varrho = \varnothing$. (iii) An absolutely continuous law gives probability zero to the null hyperplane. The generator prior has a Lebesgue density because its Gaussian base law has positive-definite covariances on every summand and its loop factor is positive. On the complement of the hyperplane, (ii) gives an empty family, and the convention gives the value $+\infty$. □

The exact retained law of Eq. (145) is unaffected; the statement concerns the invariant Gaussian family alone.

L.7 Gaussian relative entropy and Fisher metric

Proposition 37 (Gaussian relative entropy and Fisher metric)**.** *(i) For $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$ on $\mathbb{R}^K$ with $\Sigma_q, \Sigma_p \succ 0$,*

$$D_{\mathrm{KL}}(q \| p) = \frac{1}{2} \left[\log \frac{\det \Sigma_p}{\det \Sigma_q} + \operatorname{tr}(\Sigma_p^{-1} \Sigma_q) + (\mu_q - \mu_p)^\top \Sigma_p^{-1} (\mu_q - \mu_p) - K \right]. \tag{427}$$

(ii) In the moment chart (μ, Σ), with $\Lambda = \Sigma^{-1}$ and symmetric covariance tangents, the score of the Gaussian family in the direction $(\dot\mu, \dot\Sigma)$ is

$$s_{(\dot\mu,\dot\Sigma)}(x) = \dot\mu^\top \Lambda(x-\mu) + \tfrac{1}{2}(x-\mu)^\top \Lambda\dot\Sigma\Lambda(x-\mu) - \tfrac{1}{2}\operatorname{tr}(\Lambda\dot\Sigma), \tag{428}$$

and the Fisher metric is

$$G\big((\dot\mu_1,\dot\Sigma_1),(\dot\mu_2,\dot\Sigma_2)\big) = \dot\mu_1^\top \Lambda\dot\mu_2 + \tfrac{1}{2}\operatorname{tr}(\Lambda\dot\Sigma_1\Lambda\dot\Sigma_2), \tag{429}$$

with vanishing mean–covariance block. (iii) Every affine map $(\mu,\Sigma) \mapsto (Q\mu + v, Q\Sigma Q^\top)$ with $Q \in \mathrm{GL}(K)$ and $v \in \mathbb{R}^K$ is an isometry of Eq. (429)*.*

Proof. (i) For $X \sim q$, a symmetric matrix M and a vector m, expanding $X - m = (X-\mu_q) + (\mu_q - m)$ gives $\mathbb{E}_q[(X-m)^\top M(X-m)] = \operatorname{tr}(M\Sigma_q) + (\mu_q-m)^\top M(\mu_q-m)$. With $\log q(x) = -\frac{1}{2}(x-\mu_q)^\top\Sigma_q^{-1}(x-\mu_q) - \frac{1}{2}\log\det(2\pi\Sigma_q)$ and the analogous formula for p, this gives $\mathbb{E}_q \log q = -\frac{K}{2} - \frac{1}{2}\log\det(2\pi\Sigma_q)$ and

$$\mathbb{E}_q \log p = -\tfrac{1}{2}\big[\operatorname{tr}(\Sigma_p^{-1}\Sigma_q) + (\mu_q-\mu_p)^\top\Sigma_p^{-1}(\mu_q-\mu_p)\big] - \tfrac{1}{2}\log\det(2\pi\Sigma_p),$$

and subtraction gives Eq. (427). (ii) Write $e = x-\mu$, so that $\log q(x) = -\frac{1}{2}e^\top\Lambda e + \frac{1}{2}\log\det\Lambda - \frac{K}{2}\log 2\pi$. A tangent vector $(\dot\mu,\dot\Sigma)$ moves e by $-\dot\mu$ and Λ by $-\Lambda\dot\Sigma\Lambda$, and $d\log\det\Lambda = \operatorname{tr}(\Lambda^{-1}d\Lambda)$; differentiating gives Eq. (428). Under $e \sim \mathcal{N}(0,\Sigma)$ the linear term has variance $\dot\mu^\top\Lambda\Sigma\Lambda\dot\mu = \dot\mu^\top\Lambda\dot\mu$, and its covariance with the quadratic term is a third moment of a centered Gaussian vector and vanishes. For a symmetric M, write $e = \Sigma^{1/2}z$ with $z \sim \mathcal{N}(0,I)$ and diagonalize $\Sigma^{1/2}M\Sigma^{1/2} = O\,\mathrm{diag}(m_k)O^\top$ with O orthogonal. Then $e^\top Me = \sum_k m_k w_k^2$ with $w = O^\top z$ standard normal, so $\mathbb{E}[e^\top Me] = \operatorname{tr}(M\Sigma)$ and $\mathrm{Var}(e^\top Me) = 2\sum_k m_k^2 = 2\operatorname{tr}(M\Sigma M\Sigma)$. For $M = \Lambda\dot\Sigma\Lambda$ the mean of the quadratic term cancels the constant term, so the score is centered, and the quadratic term has variance $\frac{1}{4}\cdot 2\operatorname{tr}(\Lambda\dot\Sigma\Lambda\dot\Sigma)$. Hence $\mathbb{E}s^2_{(\dot\mu,\dot\Sigma)} = \dot\mu^\top\Lambda\dot\mu + \frac{1}{2}\operatorname{tr}(\Lambda\dot\Sigma\Lambda\dot\Sigma)$, and polarization gives Eq. (429) with its vanishing cross block. (iii) The differential of the affine map sends $(\dot\mu,\dot\Sigma)$ to $(Q\dot\mu, Q\dot\Sigma Q^\top)$, and the precision at the image point is $Q^{-\top}\Lambda Q^{-1}$. Substitution leaves the first term of Eq. (429) unchanged, and cyclicity of the trace leaves the second unchanged. □

This is the Fisher metric of the multivariate normal model [38]. Its covariance block is the member $a = \frac{1}{2}$, $b = 0$ of the two-parameter family of congruence-invariant metrics quoted in Appendix A, and Eq. (427) is Eq. (15).

L.8 Block-factorized Gaussian recognition

Let the target be a nondegenerate Gaussian law $p^* = \mathcal{N}(\mu^*, \Lambda^{-1})$ on $\mathbb{R}^n$, with $\Lambda \succ 0$ and $\mathsf{h} = \Lambda\mu^*$, for instance a Gaussian posterior or the normalized Gaussian law of a quadratic energy. Partition the coordinates into blocks b of dimensions n_b, write Λ_{bc} for the corresponding blocks of Λ, and let $\mathcal{Q}_{\mathcal{P}}$ be the Gaussian laws with unrestricted mean and block-diagonal covariance $\mathrm{blockdiag}_b(\Sigma_b)$, $\Sigma_b \succ 0$.

Proposition 38 (Block optimum and determinant gap)**.** *(i) Over $\mathcal{Q}_{\mathcal{P}}$, $D_{\mathrm{KL}}(Q\|p^*)$ has the unique minimizer $\mu_Q = \mu^*$, $\Sigma_b = \Lambda_{bb}^{-1}$ for every b. (ii) Let the other blocks carry a law Q_{-b} with finite second moments, finite differential entropy and block means μ_c. Among all laws Q_b of block b, $D_{\mathrm{KL}}(Q_b \otimes Q_{-b}\|p^*)$ is minimized uniquely by*

$$Q_b^* = \mathcal{N}\Big(\Lambda_{bb}^{-1}\big(\mathsf{h}_b - \textstyle\sum_{c\neq b}\Lambda_{bc}\mu_c\big), \Lambda_{bb}^{-1}\Big). \tag{430}$$

Its covariance is fixed by Λ_{bb} in one coordinate step and does not depend on the covariances of the other blocks. (iii) The restricted optimum is the determinant gap

$$\min_{Q\in\mathcal{Q}_{\mathcal{P}}} D_{\mathrm{KL}}(Q\|p^*) = \frac{1}{2}\Big[\sum_b \log\det\Lambda_{bb} - \log\det\Lambda\Big] \geq 0, \tag{431}$$

with equality if and only if $\Lambda_{bc}=0$ for every pair of distinct blocks. When p^* is the posterior of a normalized model with evidence Z_o, Proposition 32 identifies Eq. (431) *with the excess of the restricted free-energy optimum over $-\log Z_o$.*

Proof. By Eq. (427) with $\Sigma_p=\Lambda^{-1}$,

$$D_{\mathrm{KL}}(Q\|p^*) = \tfrac{1}{2}\big[\operatorname{tr}(\Lambda\Sigma_Q) + (\mu_Q-\mu^*)^\top\Lambda(\mu_Q-\mu^*) - n - \log\det\Lambda - \log\det\Sigma_Q\big].$$

On $\mathcal{Q}_{\mathcal{P}}$, $\operatorname{tr}(\Lambda\Sigma_Q)=\sum_b\operatorname{tr}(\Lambda_{bb}\Sigma_b)$ and $\log\det\Sigma_Q=\sum_b\log\det\Sigma_b$. Each Λ_{bb} is positive definite as a principal submatrix of Λ; let $m_{b,1},\dots,m_{b,n_b}>0$ be the eigenvalues of $\Lambda_{bb}^{1/2}\Sigma_b\Lambda_{bb}^{1/2}$. Then

$$D_{\mathrm{KL}}(Q\|p^*) = \tfrac{1}{2}(\mu_Q-\mu^*)^\top\Lambda(\mu_Q-\mu^*) + \tfrac{1}{2}\sum_b\sum_{k=1}^{n_b}(m_{b,k}-1-\log m_{b,k}) + \tfrac{1}{2}\Big[\sum_b\log\det\Lambda_{bb}-\log\det\Lambda\Big]. \tag{432}$$

Since $x-1-\log x\geq 0$ with equality only at $x=1$, the first two terms vanish exactly at $\mu_Q=\mu^*$ and $\Sigma_b=\Lambda_{bb}^{-1}$, which proves (i) and the value in (iii). That value is nonnegative because it is a relative entropy, and by Proposition 32 it vanishes exactly when the minimizer $\mathcal{N}(\mu^*,\operatorname{blockdiag}_b(\Lambda_{bb}^{-1}))$ equals p^*, that is, when $\Lambda=\operatorname{blockdiag}_b(\Lambda_{bb})$. For (ii), $-\log p^*(y) = \frac{1}{2}y^\top\Lambda y-\mathsf{h}^\top y+c_0$. The terms containing y_b are $\frac{1}{2}y_b^\top\Lambda_{bb}y_b+\sum_{c\neq b}y_b^\top\Lambda_{bc}y_c-\mathsf{h}_b^\top y_b$, so integration against Q_{-b} leaves $\frac{1}{2}y_b^\top\Lambda_{bb}y_b - y_b^\top(\mathsf{h}_b-\sum_{c\neq b}\Lambda_{bc}\mu_c)$ plus a finite constant. Completing the square shows that this equals $-\log Q_b^*(y_b)$ up to a constant. Hence $D_{\mathrm{KL}}(Q_b\otimes Q_{-b}\|p^*) = D_{\mathrm{KL}}(Q_b\|Q_b^*)+\text{const}$ in $(-\infty,\infty]$, where the constant collects the entropy of Q_{-b} and the terms free of y_b, and Proposition 32 gives the unique minimizer. □

Comparing Eq. (431) with its equality case recovers Fischer's inequality $\det\Lambda\leq\prod_b\det\Lambda_{bb}$ [69]. By (ii), a node whose diagonal precision block is the sum $2\Sigma_0^{-1}$ of a prior precision Σ_0^{-1} and an edge contribution Σ_0^{-1} has mean-field covariance $\Sigma_0/2$, whatever the covariances of its neighbors.

L.9 Exact receiver update in the comparison-record model

Fix a receiver i of the selective record model of Eq. (27), with source prior $\pi_{ij}>0$ on a finite source set, maps A_i and covariance $\tau_i\Gamma_i$ common to the row, and the record observed at $o_i=0$. Suppose that this record is the only factor reading J_i. Condition on every latent coordinate other than k_i and J_i, which fixes the source states k_j, and on every other record, and suppose that under this conditioning $k_i\sim\mathcal{N}(\mu_i,\Sigma_i)$ with $\Sigma_i\succ 0$. Put $\widetilde{W}_i=(\tau_i\Gamma_i)^{-1}$ and

$$\Sigma_i^+ = \big(\Sigma_i^{-1}+A_i^\top\widetilde{W}_iA_i\big)^{-1}, \qquad \Sigma_i^{\mathrm{pred}} = \widetilde{W}_i^{-1}+A_i\Sigma_iA_i^\top, \qquad r_{ij}=A_i\mu_i-B_{ij}k_j. \tag{433}$$

Proposition 39 (Exact receiver update)**.** *Given the conditioning variables and $o_i=0$, the source label has posterior weights $w_{ij}\propto\pi_{ij}\exp[-\frac{1}{2}r_{ij}^\top(\Sigma_i^{\mathrm{pred}})^{-1}r_{ij}]$, normalized over the source set. Conditionally on $J_i=j$, $k_i\sim\mathcal{N}(\Sigma_i^+(\Sigma_i^{-1}\mu_i+A_i^\top\widetilde{W}_iB_{ij}k_j),\Sigma_i^+)$, and*

$$\mathbb{E}[k_i\mid k_{-i},o] = \mu_i+\sum_j w_{ij}\mathsf{K}_i(B_{ij}k_j-A_i\mu_i), \qquad \mathsf{K}_i=\Sigma_i^+A_i^\top\widetilde{W}_i=\Sigma_iA_i^\top(\Sigma_i^{\mathrm{pred}})^{-1}, \tag{434}$$

where k_{-i} denotes the conditioning latent coordinates. In the full-state receiving form $A_i = I$, $B_{ij} = \Omega_{ij}$, the gain is the Kalman gain $\mathsf{K}_i = \Sigma_i^+ \widetilde{W}_i = \Sigma_i(\Sigma_i + \tau_i\Gamma_i)^{-1}$, set by the receiver's posterior covariance Σ_i^+, and it differs from the identity for every positive record covariance.

Proof. Under the conditioning, the joint law of (k_i, J_i) at $o_i = 0$ has density proportional to $\pi_{ij}\mathcal{N}(k_i; \mu_i, \Sigma_i)\mathcal{N}(0; A_i k_i - B_{ij}k_j, \tau_i\Gamma_i)$. For fixed j, the integral over k_i of the last two factors is the density at 0 of $A_i K - B_{ij}k_j + \varepsilon$ with independent $K \sim \mathcal{N}(\mu_i, \Sigma_i)$ and $\varepsilon \sim \mathcal{N}(0, \tau_i\Gamma_i)$, namely $\mathcal{N}(0; r_{ij}, \Sigma_i^{\mathrm{pred}})$. Its normalizing constant does not depend on j because A_i and Γ_i are common to the row, which gives the weights. For fixed j the exponent in k_i,

$$-\tfrac{1}{2}\big[(k_i - \mu_i)^\top \Sigma_i^{-1}(k_i - \mu_i) + (A_i k_i - B_{ij}k_j)^\top \widetilde{W}_i (A_i k_i - B_{ij}k_j)\big],$$

is a quadratic with Hessian $-(\Sigma_i^+)^{-1}$ whose gradient vanishes at $\Sigma_i^+(\Sigma_i^{-1}\mu_i + A_i^\top \widetilde{W}_i B_{ij}k_j)$, which gives the conditional components. Averaging their means with the weights w_{ij}, which sum to one, and using $\Sigma_i^+\Sigma_i^{-1} = I - \Sigma_i^+ A_i^\top \widetilde{W}_i A_i$ gives the first identity in Eq. (434). The second follows from $A_i^\top \widetilde{W}_i \Sigma_i^{\mathrm{pred}} = A_i^\top + A_i^\top \widetilde{W}_i A_i \Sigma_i A_i^\top = (\Sigma_i^+)^{-1}\Sigma_i A_i^\top$ after multiplication by Σ_i^+ on the left and by $(\Sigma_i^{\mathrm{pred}})^{-1}$ on the right. For $A_i = I$, $\Sigma_i^{\mathrm{pred}} = \Sigma_i + \tau_i\Gamma_i$, and $\Sigma_i(\Sigma_i + \tau_i\Gamma_i)^{-1} = I$ would require $\tau_i\Gamma_i = 0$. □

L.10 Total correlation and appended energies

Proposition 40 (Total-correlation chain identity)**.** *Let $Y = (Y_b)_b$ take values in a finite product of standard Borel spaces, let Q be a law with marginals Q_b, let ρ_b be arbitrary probability laws, and let $\mathrm{TC}(Q) = D_{\mathrm{KL}}(Q \| \bigotimes_b Q_b)$ be the total correlation. Then*

$$D_{\mathrm{KL}}\Big(Q \Big\| \bigotimes_b \rho_b\Big) = \mathrm{TC}(Q) + \sum_b D_{\mathrm{KL}}(Q_b \| \rho_b) \tag{435}$$

in $[0, \infty]$. For a product prior $\bigotimes_b \rho_b$, the complexity of a correlated recognition law is thus its total correlation plus its marginal divergences, and replacing Q by the product of its marginals lowers this complexity by exactly $\mathrm{TC}(Q)$. Equation (38) is the two-block case, in which the total correlation is the mutual information.

Proof. Write $\rho = \bigotimes_b \rho_b$ and $\bar{Q} = \bigotimes_b Q_b$. Suppose first that $Q \ll \rho$ with density f. By Fubini's theorem $Q_b \ll \rho_b$ with density $f_b(y_b)$, the integral of f over the other coordinates, so $\bar{Q}$ has density $\prod_b f_b$ relative to ρ. The set $N = \{y : \prod_b f_b(y_b) = 0\}$ satisfies $\bar{Q}(N) = 0$ and $Q(N) \le \sum_b Q_b(\{f_b = 0\}) = 0$. Hence $g = f / \prod_b f_b$ off N, with $g = 0$ on N, is a density of Q relative to $\bar{Q}$, and $\log f = \log g + \sum_b \log f_b(y_b)$ holds Q-almost surely. The Q-integrals of $\log f$, $\log g$ and $\log f_b(Y_b)$ are $D_{\mathrm{KL}}(Q\|\rho)$, $\mathrm{TC}(Q)$ and $D_{\mathrm{KL}}(Q_b\|\rho_b)$, and each integrand has a negative part with integral at most e^{-1}, because $x \log x \ge -e^{-1}$. Additivity of such integrals proves Eq. (435) in $(-\infty, \infty]$, and every term is nonnegative. If Q is not absolutely continuous with respect to ρ, the left side is infinite. Then either some Q_b is not absolutely continuous with respect to ρ_b, or $\bar{Q} \ll \rho$ and Q is not absolutely continuous with respect to $\bar{Q}$; in both cases the right side is infinite. □

Proposition 41 (Appended energy)**.** *Let a normalized joint $\mathbb{P}(dy, do)$ have density $p(o, y)$ relative to $\nu_Y \otimes \nu_O$ and positive finite evidence $p(o)$ at the record o, and let U be a real measurable function of the latent state that is bounded below. Then $Z_U = \int e^{-U(y)}\mathbb{P}(dy, do) \in (0, \infty)$, the tilted law $\widetilde{\mathbb{P}} = Z_U^{-1} e^{-U}\mathbb{P}$ is a normalized joint, and for every recognition law Q with finite free energy $\mathcal{F}_V[Q] = \mathbb{E}_Q[\log q(Y) - \log p(o, Y)]$ in its split form and $\mathbb{E}_Q|U| < \infty$,*

$$\mathcal{F}_V[Q] + \mathbb{E}_Q U = \widetilde{\mathcal{F}}_V[Q] - \log Z_U, \tag{436}$$

where $\widetilde{\mathcal{F}}_V$ is the variational free energy of $\widetilde{\mathbb{P}}$. The appended objective is therefore minimized exactly at the posterior of the tilted joint, with minimum $-\log \int e^{-U(y)} p(o,y)\nu_Y(dy)$. It is the free energy of a normalized joint, and hence an evidence bound in the sense of Eq. (22)*, only after the term $\log Z_U$, which depends on the generative parameters through $\mathbb{P}$, is restored.*

Proof. Since U is real and bounded below, $0 < e^{-U} \le e^{-\inf U}$, and integration against the probability law $\mathbb{P}$ gives $0 < Z_U < \infty$. The tilted joint has density $\tilde{p}(o,y) = p(o,y)e^{-U(y)}/Z_U$, so $\widetilde{\mathcal{F}}_V[Q] = \mathbb{E}_Q[\log q(Y) - \log \tilde{p}(o,Y)] = \mathcal{F}_V[Q] + \mathbb{E}_Q U + \log Z_U$, with every expectation finite. By Proposition 32 applied to $\widetilde{\mathbb{P}}$, $\widetilde{\mathcal{F}}_V[Q] \ge -\log \tilde{p}(o)$, with equality exactly at the tilted posterior, where $\tilde{p}(o) = Z_U^{-1} \int e^{-U(y)} p(o,y)\nu_Y(dy) \in (0,\infty)$. Subtracting $\log Z_U$ gives the stated minimum. □

L.11 Downward kernels and the conditional cross-scale divergence

Let a parent latent Z and child latents Y take values in standard Borel spaces, and let

$$\mathbb{P}(dz, dy, do) = P_Z(dz)\mathcal{K}_\downarrow(dy \mid z)\ell(o \mid y, z)\nu_O(do) \tag{437}$$

be a normalized hierarchical joint, with a proper parent prior P_Z, a normalized downward kernel $\mathcal{K}_\downarrow$, a record density normalized against ν_O, and positive finite evidence Z_o at the record o. A recognition law is disintegrated as $Q(dz, dy) = Q_Z(dz)Q_{Y|Z}(dy \mid z)$.

Proposition 42 (Downward coupling)**.** *If $Q \ll P_Z\mathcal{K}_\downarrow$ and $\mathbb{E}_Q|\log \ell(o \mid Y, Z)| < \infty$, then*

$$\begin{aligned} \mathcal{F}_V[Q] &= D_{\mathrm{KL}}(Q_Z \| P_Z) + \int D_{\mathrm{KL}}\big(Q_{Y|Z}(\cdot \mid z) \big\| \mathcal{K}_\downarrow(\cdot \mid z)\big) Q_Z(dz) - \mathbb{E}_Q \log \ell(o \mid Y, Z) \\ &= D_{\mathrm{KL}}\big(Q \| \mathbb{P}(\cdot \mid o)\big) - \log Z_o. \end{aligned} \tag{438}$$

The conditional cross-scale divergence is therefore the cross-scale term of an exact evidence bound, which equals $-\log Z_o$ exactly at the posterior. Its generative denominator reads the parent's latent coordinate z and never the parent's recognition law. For a product law $Q_Z \otimes Q_Y$ the cross-scale term is $\mathbb{E}_{Q_Z} D_{\mathrm{KL}}(Q_Y \| \mathcal{K}_\downarrow(\cdot \mid Z))$.

Proof. Let f be the density of Q relative to $P_Z\mathcal{K}_\downarrow$. Then $Q_Z \ll P_Z$ with density $f_Z(z) = \int f(z,y)\mathcal{K}_\downarrow(dy \mid z)$, the set $\{f_Z = 0\}$ is Q_Z-null, and off it $Q_{Y|Z}(dy \mid z) = [f(z,y)/f_Z(z)]\mathcal{K}_\downarrow(dy \mid z)$ is a version of the conditional law. Thus $\log f = \log f_Z(z) + \log[f(z,y)/f_Z(z)]$ holds Q-almost surely, each logarithm has a negative part with integral at most e^{-1}, and integration gives $D_{\mathrm{KL}}(Q \| P_Z\mathcal{K}_\downarrow) = D_{\mathrm{KL}}(Q_Z \| P_Z) + \int D_{\mathrm{KL}}(Q_{Y|Z}(\cdot \mid z) \| \mathcal{K}_\downarrow(\cdot \mid z)) Q_Z(dz)$ in $[0,\infty]$. Subtracting the integrable expected log record density gives the first line, which is the defining form of Eq. (22) for the joint (437). Substituting the posterior density $\ell(o \mid y, z)/Z_o$ relative to $P_Z\mathcal{K}_\downarrow$ gives the second line, and its equality case is that of Proposition 32. For a product law $Q_{Y|Z}(\cdot \mid z) = Q_Y$ for every z. □

In the augmented joint of Definition 3, $\mathcal{K}_\downarrow$ is the product of the link kernels of Eq. (327) over the children of one parent, and this term is the conditional coupling of the first reading of top-down influence in Appendix G.

M Notation and Conventions

An edge ij carries information from j to i. All probability divergences use natural logarithms. The base point c, inference time t, hierarchy level ℓ and source index j refer to different kinds of variables. Specialized appendices define their additional symbols locally.

Table 4: Principal notation of the monograph.

Symbol	Meaning
$\mathcal{C}$, c, c_*	Continuous base manifold, a base point, and the fixed point used for formation analysis.
$\mathcal{U}_i$, $V(c)$, χ_i	Domain of agent i, available agents at c, and an optional presence weight or indicator.
P, G, $\mathfrak{g}$	Principal frame bundle, structure group, and Lie algebra.
Y, $\mathcal{B}$, $\mathcal{S}$	Statistical sample space, regular family of laws, and associated statistical bundle $P \times_G \mathcal{B}$.
ζ_i	Joint statistical section of agent i. When a state–model split is used, $\zeta_i(dk, dm) = q_i(dk \mid m)s_i(dm)$.
k, m; q_i, s_i	State and model presentation; belief about the state and law over model presentations. A model becomes predictive through its normalized kernel.
p_i, r_i	Prior on states and prior on model presentations. A prior is not a synonym for a generative model.
$\mathbb{P}$, P_o, Q, Z_o	Full generative law, its posterior at the observed record, recognition law, and evidence. Some local proofs write P for a probability law when no principal bundle is involved.
U_i, ϕ_i, g_i	Frame presentation, a local Lie-algebra chart with $U_i = \exp \phi_i$, and passive coordinate change.
Ω_{ij}	Flat reference identification $U_i U_j^{-1}$. It does not encode a general selective message.
Θ_{ij}	Independent invertible interaction link; the latent exponential-chart case is $\Theta_e = U_i e^{\mathring{\omega}_e} U_j^{-1}$.
$\mathring{\omega}_e$, $\mathcal{T}$	Reference Lie-algebra generator and generator space $\mathfrak{g}^{E_+}$; a ring accent marks generator-sector data.
$\mathring{\alpha}$, $\mathring{\Sigma}$, $\mathring{\mu}$, $\mathring{\Lambda}$	Generator prior anchor/covariance and Gaussian recognition mean/precision.
P_{gen}, R, $\mathcal{A}_{\mathrm{loop}}$, Z_{loop}	Generator prior, its recognition law, loop action and prior normalizer; R_i still denotes a receiver map.
$\mathbf{m}$, $\mathbf{K}$, $\mathbf{N}_0$, $\mathbf{j}$	Episode-increment precision, Proca-type operator, current correction of its exact Hessian and matter source.
S_j, R_i, L_{ij}	Sender readout, receiver reconstruction and selective content map $L_{ij} = R_i S_j : E_j \to E_i$.
$\mathcal{K}_{ij}$, N_{ij}	Normalized communication kernel and the noise covariance in its Gaussian realization.
A_{ij}, B_{ij}, Γ_{ij}	Comparison maps into an observation space and its Gaussian noise shape. These need not equal the content maps.
J_i, π_{ij}, β_{ij}	Chosen source, source-prior probability, and variational or conditional source probability.
E_{ij}, τ, τ_γ	Comparison energy, source temperature, and optional model-row temperature. Context distinguishes a vector fiber E_i from a scalar comparison cost.
$\mathcal{F}_V$	Exact collective variational free energy of a declared normalized generative law.

Symbol	Meaning
$\mathcal{E}$	Channel-consensus objective. An exact generative identification requires a separate derivation.
γ_{ij}, (q), (s)	Optional product-model attention and state/model sector decorations in specialized calculations.
α_i	Optional gain multiplying a divergence in $\mathcal{E}$; it is not automatically a generative precision.
μ_i, Σ_i, Λ_i	Gaussian mean, covariance and precision Σ_i^{-1}.
G, g^F, h^ω	A Fisher matrix, Fisher metric, and its pullback using a declared connection. The structure group G is distinguished by context.
η, M	Mobility or mobility operator; additional kinetic tensors are defined where introduced.
C, Q^c, P_o^c	A normalized retained channel and the induced recognition and posterior laws. The base manifold is calligraphic $\mathcal{C}$.
H_o, H_o^c, ρ	Dimensionless fine and coarse negative log densities and their reference law.
T_{cfg}	Temperature of a separately declared Gibbs ensemble over belief configurations; distinct from source and physical temperatures.
t, ϑ_i	Evolution parameter and information clock, the Fisher length along an appropriate smooth belief history.
ℓ, I, U_I	Hierarchy level, parent label or cluster, and chosen parent frame.
W, $\mathcal{H}(W)$, X	Complex rank-two communication bundle, its real Hermitian tensor bundle, and a Hermitian tensor.
σ_μ, x^μ	Pauli basis including the identity and the four real components of $X = x^\mu \sigma_\mu$.
$e : T\mathcal{C} \to \mathcal{H}(W)$	Additional soldering map used to identify internal Lorentz geometry with tangent geometry.

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